(a) Wear of threads

1) Permitted no longer than one-fourth the length of a single thread after removal of the high metal.

2) Not repairable.

(b) Loose

1) Permitted if the keys are still in place.

2) Not repairable.

(c) One or both keys missing

1) Not permitted.

2) Not repairable.

(d) Missing stud

1) Not permitted.

(e) Distortion

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-013-G00

(11) Examine the AGB centering pins [41] that you can see for this condition (Figure 601 (Sheet 5), Figure 601 (Sheet 6), Figure 601 (Sheet 9), and Figure 601 (Sheet 10)):

(a) Loose or worn

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-014-G00

(12) Examine the ring-locked nipples [6] that you can see for these conditions (Figure 601 (Sheet 9) and Figure 601 (Sheet 17)):

(a) Worn thread

1) Not permitted.

2) Not repairable.

(b) Looseness

1) Not permitted.

2) Not repairable.

(c) Blocked hole

1) Not permitted.

2) Not repairable.

(d) Missing part

1) Not permitted.

(e) Missing wedge cone

1) Not permitted.

(f) Circumferential score marks on silver plated seating.

1) Permitted if the score mark does not penetrate parent metal.

2) Not repairable.

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(g) Circumferential scratches in Area A and C

1) Permitted if not more than 0.0051 in. (0.13 mm) in depth after removal of the high metal.

2) Not repairable.

(h) Axial scratches in Area A and C

1) Permitted if not more than 0.0051 in. (0.13 mm) in depth after removal of the high metal.

2) Not repairable.

(i) Nicks in Area C and B

1) Not permitted.

2) Not repairable.

(j) Circumferential scratches in Area B

1) Permitted if the scratch cannot be felt with a 0.04 in. (1 mm) radius scriber and up to 0.0100 in. (0.25 mm) in depth.

2) Not repairable.

(k) Axial scratches in Area B

1) Not permitted.

2) Not repairable.

(l) Nicks in Area A

1) Permitted if the nick cannot be felt with a 0.0300 in. (0.76 mm) radius scriber.

2) Not repairable.

(m) Missing locking

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-015-G00

(13) Examine all areas of plug [36] for this condition (Figure 601 (Sheet 6) and Figure 601 (Sheet 18)):

(a) Cracks

1) Not permitted.

SUBTASK 72-60-00-210-016-G00

(14) Examine Location 5 of plug [36] for (Figure 601 (Sheet 18)):

(a) Scratches, dents and nicks

1) Permitted if the hexagon can be maintained by the wrench tool.

2) Not repairable.

SUBTASK 72-60-00-210-017-G00

(15) Examine the safety wire holes of plug [36] for these conditions (Figure 601 (Sheet 18)):

(a) Deformation and wear

1) Permitted if metal around hole is not cracked or torn.

2) Not repairable.

(b) Tear-out

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1) Permitted if there is one remaining hole.

2) Not repairable.

SUBTASK 72-60-00-210-018-G00

(16) Examine the safety wires of the hexagon head plug [36] for these conditions (Figure 601 (Sheet 6)):

(a) Broken or missing

1) Not permitted.

SUBTASK 72-60-00-210-019-G00

(17) Examine area C of the AGB rotating seal assembly [10] for this condition (Figure 601 (Sheet 2), Figure 601 (Sheet 5), and Figure 601 (Sheet 13)):

(a) Fretting

1) Permitted if fretting cannot be felt with a 0.0300 in. (0.76 mm) radius scriber.

2) Not repairable.

SUBTASK 72-60-00-210-051-G00

(18) Examine all areas (except area C) of the AGB rotating seal assembly [10] for this condition (Figure 601 (Sheet 13)):

(a) Fretting

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-020-G00

(19) Examine the mating ring [11] of the AGB rotating seal assembly [10] for these conditions (Figure 601 (Sheet 13)):

(a) Wear

1) Not serviceable.

(b) Brownish discoloration

1) Permitted.

SUBTASK 72-60-00-210-052-G00

(20) Examine all areas of the seal housing carbon surface [42] of the AGB rotating seal assembly [10] that you can see for these conditions (Figure 601 (Sheet 13)):

(a) Wear

1) Not serviceable.

SUBTASK 72-60-00-210-021-G00

(21) Examine the O-ring [13] of the AGB rotating seal assembly [10] for this condition (Figure 601 (Sheet 13)):

(a) Brownish discoloration

1) Permitted.

SUBTASK 72-60-00-210-022-G00

(22) Examine the AGB seal housing [12] of the AGB rotating seal assembly [10] for this condition (Figure 601 (Sheet 13)):

(a) Red coloration

1) Permitted.

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SUBTASK 72-60-00-210-023-G00

(23) Examine the extraction pin of the AGB rotating seal assembly [10] for these conditions (Figure 601 (Sheet 13)):

(a) Scratches, nicks and score

1) Permitted if the extraction pin can be held by the extraction tool.

2) Not repairable.

SUBTASK 72-60-00-210-053-G00

(24) Examine all other areas of the AGB rotating seal assembly [10] for these conditions (Figure 601 (Sheet 13)):

(a) Scratches, nicks and score

1) Not permitted.

SUBTASK 72-60-00-210-054-G00

(25) Examine face sealing surfaces [8] and radial sealing surfaces [9] of the AGB assembly [1] that you can see for this condition (Figure 601 (Sheet 7)):

(a) Scratches, nicks dents and score

1) Not permitted.

SUBTASK 72-60-00-210-024-G00

(26) Examine all areas of the bevel gear assembly [44] for this condition (Figure 601 (Sheet 6)):

(a) Heat discoloration

1) Not permitted.

SUBTASK 72-60-00-210-025-G00

(27) Examine the gear shaft tail [38] for these conditions (Figure 601 (Sheet 6)):

(a) Dents, nicks and scratches

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-026-G00

(28) Examine the splines of the bevel gear assembly [44] and of the spur gears [48] for these conditions (Figure 601 (Sheet 5), Figure 601 (Sheet 6), Figure 601 (Sheet 9), and Figure 601 (Sheet 10)):

(a) Scratches

1) Permitted if the scratches cannot be felt with a 0.0300 in. (0.76 mm) radius scriber.

2) Not repairable.

(b) Nicks, dents and burrs

1) Permitted up to 0.0252 in. (0.64 mm) in depth after removal of the high metal.

2) Not repairable.

(c) Wear

1) Permitted, if wear that affects the spline flanks cannot be felt with a 0.0300 in. (0.76 mm) radius scriber.

2) Not repairable.

SUBTASK 72-60-00-210-027-G00

(29) Examine the driver of the spur gear shaft [15] location for these conditions (Figure 601 (Sheet 10) and Figure 601 (Sheet 19)):

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(a) Dents, nicks, scratches and score

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-028-G00

(30) Examine all areas of the OPT sensor support [5] for these conditions (Figure 601 (Sheet 5)):

(a) Fretting

1) Not serviceable.

(b) Scratches, nicks, dents

1) Not serviceable.

SUBTASK 72-60-00-210-030-G00

(31) Examine all areas of the coil [3] for these conditions (Figure 601 (Sheet 5)):

(a) Scratches, nicks, dents and score

1) Not permitted.

2) Not repairable.

(b) Worn surface treatment

1) Not permitted.

2) Not repairable.

(c) Missing or loose plug or restrictor

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-055-G00

(32) Examine all areas of the threads of the nipple [27] that you can see for this condition (Figure 601 (Sheet 5) and Figure 601 (Sheet 20)):

(a) Worn threads

1) Not permitted.

SUBTASK 72-60-00-210-031-G00

(33) Examine the nipple [27] that you can see for these conditions (Figure 601 (Sheet 20)):

(a) Circumferential scratches in Area A and Area C

1) Permitted up to 0.0051 in. (0.13 mm) in depth after removal of the high metal.

2) Not repairable.

(b) Circumferential scratches in Area B

1) Permitted if the scratches cannot be felt with a 0.04 in. (1 mm) radius scriber.

2) Not repairable.

(c) Axial scratches in Area A and Area C

1) Permitted up to 0.0051 in. (0.13 mm) in depth after removal of the high metal.

2) Not repairable.

(d) Axial scratches in Area B

1) Not permitted.

2) Not repairable.

(e) Nicks in Area A

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1) Permitted up to 0.031 in. (0.8 mm) in depth after removal of the high metal.

2) Not repairable.

(f) Nicks in Area B and C

1) Not permitted.

2) Not repairable.

(g) Scratches, dents, nicks on Location 4

1) Permitted if the hexagon can be maintained by the wrench tool.

2) Not repairable.

SUBTASK 72-60-00-210-032-G00

(34) Examine the bracket [7], bracket [29], bracket [30], bracket [31], bracket [32], and bracket [33] for these conditions (Figure 601 (Sheet 5) and Figure 601 (Sheet 6)):

(a) Dents on welded joints, bending radius of brackets, radius zones and zones near slots, grooves and holes and stiffening ribs.

1) Not permitted.

2) Not repairable.

(b) Dents on all areas except those listed above

1) Not permitted.

2) Not repairable.

(c) Scores, scratches, rub and burrs on welded joints, bending radius of brackets, radius zones and zones near slots, grooves and holes and stiffening ribs.

1) Not permitted.

2) Not repairable.

(d) Scores, scratches, rub and burrs on all areas except those listed above.

1) Not permitted.

2) Not repairable.

(e) Wear of the threads on riveted nut

1) Not permitted.

2) Not repairable.

(f) Looseness of riveted nut

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-033-G00

(35) Examine the identification plate support [16] for these conditions (Figure 601 (Sheet 9)):

(a) Looseness or loss of serrated nails

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-035-G00

(36) Examine the seal grooves [47] and all other areas of the manual control cover [23] for these conditions (Figure 601 (Sheet 10)):

(a) Scratches, nicks, dents and score

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1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-056-G00

(37) Examine the face sealing surface [8] and radial sealing surface [9] of the transfer housing [35] for these conditions (Figure 601 (Sheet 5) and Figure 601 (Sheet 7)):

(a) Scratches, nicks, dents and score

1) Not permitted.

2) Not repairable.

(b) Fretting

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-037-G00

(38) Examine all other areas of the transfer tube [34] for these conditions:

(a) Scratches, nicks, dents and score

1) Not permitted

2) Maximum repairable limit: In the limits of the applicable repair

3) Repair method: Refer to Blending and Spot Touch-Up of the Accessory Gearbox Module, TASK 72-63-00-300-802-G00.

SUBTASK 72-60-00-210-057-G00

(39) Examine all areas of spur gears [48] for this condition (Figure 601 (Sheet 5), Figure 601 (Sheet 9), and Figure 601 (Sheet 10)):

(a) Heat discoloration

1) Not permitted.

SUBTASK 72-60-00-210-058-G00

(40) Examine all other areas of the AGB assembly [1] for these conditions:

(a) Scratches, nicks, dents and score

1) Any amount if not felt with a 0.03 in. (0.76 mm) radius scriber

2) Maximum repairable limit: In the limits of the applicable repair

3) Repair method: Refer to Blending and Spot Touch-Up of the Manual Control Cover, TASK 72-63-03-300-802-G00.

H. Accessory Drives (Transfer Gearbox) Inspection

SUBTASK 72-60-00-210-038-G00

(1) Make sure that the TGB assembly [2] you will examine is clean.

SUBTASK 72-60-00-210-039-G00

(2) Examine all areas of the TGB assembly [2] that you can see for these conditions (Figure 602 (Sheet 2)):

(a) Cracks

1) Not permitted.

2) Not repairable.

(b) Corrosion

1) Not permitted.

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2) Not repairable.

(c) Lack of protection on all non-functional surfaces that you can see.

NOTE: A non-functional surface is not an interface area or sealing area.

1) Permitted, 5 percent of the checked surface.

2) Not repairable.

(d) Lack of protection on all other surfaces that you can see.

1) Not permitted.

SUBTASK 72-60-00-210-074-G00

(3) Examine the area E of the TGB assembly [2] (Figure 602 (Sheet 8)).

(a) Scratches, nicks, dents, and scores

1) Not permitted.

2) Maximum repairable limit: In the limit of the applicable repair.

3) Repair method: Refer to Blending and Spot Touch-Up of the Transfer Gearbox Module, TASK 72-62-00-300-802-G00.

SUBTASK 72-60-00-210-059-G00

(4) Examine all other areas of the TGB assembly [2] for these conditions:

(a) Scratches, nicks, dents, score or imprints

1) Not permitted.

2) Maximum repairable limit: In the limits of the applicable repair.

3) Repair method: Refer to Blending and Spot Touch-Up of the Transfer Gearbox Module, TASK 72-62-00-300-802-G00.

(b) Corrosion

1) Not permitted.

2) Maximum repairable limit: In the limit of the applicable repair.

3) Repair method: Refer to Blending and Spot Touch-Up of the Transfer Gearbox Module, TASK 72-62-00-300-802-G00.

SUBTASK 72-60-00-211-010-G00

(5) Examine screw holes of TGB assembly [2] (Figure 602 (Sheet 5)).

(a) Corrosion

1) Not permitted.

2) Maximum repairable limit: In the limit of the applicable repair.

3) Repair method: Refer to Transfer Gearbox Assembly Corrosion Removal, TASK 72-62-00-300-804-G00.

SUBTASK 72-60-00-210-060-G00

(6) Examine location 4 of the slider [19] for this condition (Figure 602 (Sheet 2) and Figure 602 (Sheet 9)):

(a) Scratches, nicks, dents and score

1) Not permitted.

2) Maximum repairable limit: In the limit of the applicable repair.

3) Repair method: Refer to Blending and Spot Touch-Up of the Transfer Gearbox Module, TASK 72-62-00-300-802-G00.

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(b) Fretting

1) Acceptable if the residual thickness is between 0.122 in. (3.1 mm) and 0.130 in. (3.3 mm).

2) Not repairable.

SUBTASK 72-60-00-210-061-G00

(7) Examine all other areas of the slider [19] for this condition (Figure 602 (Sheet 2) and Figure 602 (Sheet 9)):

(a) Scratches, nicks, dents and score

1) Not permitted

2) Maximum repairable limit: In the limits of the applicable repair

3) Repair method: Refer to Blending and Spot Touch-Up of the Accessory Gearbox Module, TASK 72-63-00-300-802-G00.

(b) Fretting

1) Not serviceable.

2) Not permitted.

SUBTASK 72-60-00-210-062-G00

(8) Examine all areas of the ball joint seal [71] that you can see for these conditions (Figure 602 (Sheet 7)):

(a) Cuts

1) Not permitted.

(b) Nicks, dents and scratches

1) Not permitted.

SUBTASK 72-60-00-210-063-G00

(9) Examine all areas of the seal grooves [53] of the bearing support for these conditions (Figure 602 (Sheet 2)):

(a) Nicks, dents and scratches

1) Not permitted.

SUBTASK 72-60-00-210-064-G00

(10) Examine all areas of the bearing support for these conditions (Figure 602 (Sheet 2)):

(a) Nicks, dents and scratches

1) Not permitted.

SUBTASK 72-60-00-210-065-G00

(11) Examine the tapped extraction holes [55] of the bearing support for these conditions (Figure 602 (Sheet 5)):

(a) Worn threads

1) Any amount if not less than one half entrance thread after removal of high metal.

SUBTASK 72-60-00-210-066-G00

(12) Examine all threads [46] of the nipple [57] that you can see for these conditions (Figure 602 (Sheet 6) and Figure 602 (Sheet 10)):

(a) Worn threads

1) Not permitted.

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SUBTASK 72-60-00-210-041-G00

(13) Examine the nipple [57] that you can see for these conditions (Figure 602 (Sheet 10)):

(a) Circumferential scratches in Area A and C

1) Permitted up to 0.0051 in. (0.13 mm) in depth after removal of the high metal.

2) Not repairable.

(b) Circumferential scratches in Area B

1) Permitted if the scratches cannot be felt with a 0.04 in. (1 mm) radius scriber.

2) Not repairable.

(c) Axial scratches in Area A and C

1) Permitted up to 0.0051 in. (0.13 mm) in depth after removal of the high metal.

2) Not repairable.

(d) Axial scratches in Area B

1) Not permitted.

2) Not repairable.

(e) Nicks in Area A

1) Permitted up to 0.031 in. (0.8 mm) in depth after removal of the high metal.

2) Not repairable.

(f) Nicks in Area B and C:

1) Not permitted.

2) Not repairable.

(g) Scratches, dents, nicks on location 6:

1) Permitted if the hexagon can be maintained by the wrench tool.

2) Not repairable.

SUBTASK 72-60-00-210-067-G00

(14) Examine all areas of the seal groove [58] of the nipple [57] for these conditions (Figure 602 (Sheet 10)):

(a) Scratches, nicks, dents

1) Not permitted.

SUBTASK 72-60-00-210-068-G00

(15) Examine all areas of the centering pins [59] of the TGB housing [52] for these conditions (Figure 602 (Sheet 2)):

(a) Loose, wear

1) Not permitted.

SUBTASK 72-60-00-210-069-G00

(16) Examine all areas of the gear shaft tail (except ring oil level) of the bevel gears [68] for these conditions (Figure 602 (Sheet 2)):

(a) Scratches, nicks, dents

1) Not permitted.

SUBTASK 72-60-00-210-043-G00

(17) Examine all areas of the transfer tube [18] for these conditions (Figure 602 (Sheet 2) and Figure 602 (Sheet 12)):

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(a) Fretting

1) Acceptable if the fretting is located only on area Y and does not reach area W, and if the depth is lower than 0.020 in. (0.5 mm).

a) Do a test for surface cracks with dye penetrant (Fluorescent Penetrant Inspection, TASK 70-11-06-230-802-G00).

(b) Scratches, nicks, dents and score

1) Not permitted

2) Maximum repairable limit: In the limits of the applicable repair

3) Repair method: Refer to Blending and Spot Touch-Up of the Transfer Gearbox Module, TASK 72-62-00-300-802-G00.

SUBTASK 72-60-00-210-047-G00

(18) Examine the splines [70] of the bevel gears [68] for these conditions (Figure 602 (Sheet 2)):

(a) Scratches

1) Any amount if not felt with a 0.03 in. (0.76 mm) radius scriber.

2) Not repairable.

(b) Nicks, dents and burrs

1) Permitted up to 0.0252 in. (0.64 mm) in depth after removal of high metal.

2) Not repairable.

(c) Wear

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-048-G00

(19) Examine the retainer sectors [64] for these conditions (Figure 602 (Sheet 3)):

(a) Missing protective coating

1) Not permitted.

SUBTASK 72-60-00-210-070-G00

(20) Examine area 1 of the retainer sectors [64] for these conditions (Figure 602 (Sheet 11)):

(a) Fretting

1) Acceptable if the residual thickness is between 0.094 in. (2.4 mm) and 0.102 in. (2.6 mm) after hand blending.

a) The minimum length and width of the blending must be 40 times the depth of the deviation.

(b) Nicks, dents, scratches and score

1) Acceptable if the residual thickness is between 0.094 in. (2.4 mm) and 0.102 in. (2.6 mm) after hand blending.

a) The minimum length and width of the blending must be 40 times the depth of the deviation.

SUBTASK 72-60-00-210-071-G00

(21) Examine all other areas of the retainer sectors [64] for these conditions (Figure 602 (Sheet 3) and Figure 602 (Sheet 11)):

(a) Nicks, dents, scratches and score

1) Not permitted.

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2) Not repairable.

(b) Fretting

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-049-G00

(22) Examine the identification plate [65] of the TGB housing [52] for this condition (Figure 602 (Sheet 2)):

(a) Loose or loss of serrated nails

1) Not permitted.

SUBTASK 72-60-00-210-072-G00

(23) Examine the inserts [39] (0.2500 - 28 UNJF-3B) of the TGB housing [52] for these conditions (Figure 602 (Sheet 4)):

(a) Wear of internal threads

1) Permitted, one full thread cumulative or continuous may be missing after removal of the worn thread if not less than one-half lead in thread remains.

2) Not repairable.

(b) Looseness

1) Not permitted.

(c) Missing inserts

1) Not permitted.

(d) Missing keys

1) Not permitted.

(e) Self-locking quality

1) Acceptable if running torque is between 3.540 in-lb (0.400 N·m) and 30.004 in-lb (3.390 N·m).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-60-00-080-001-G00

(1) Clean the work area of all tools and items you used to do the task.

SUBTASK 72-60-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

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1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-60-00-860-001-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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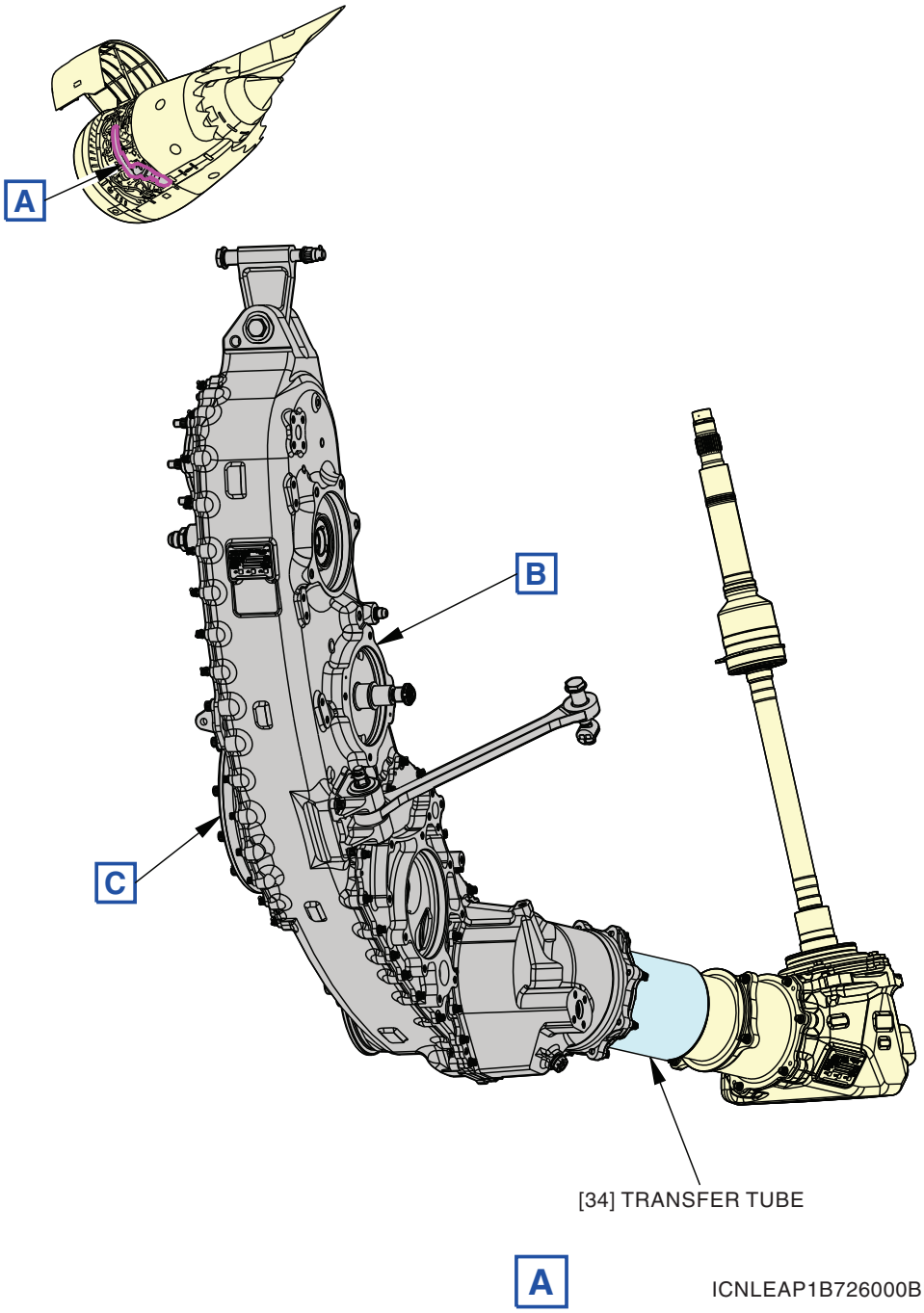

Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 1 of 20)

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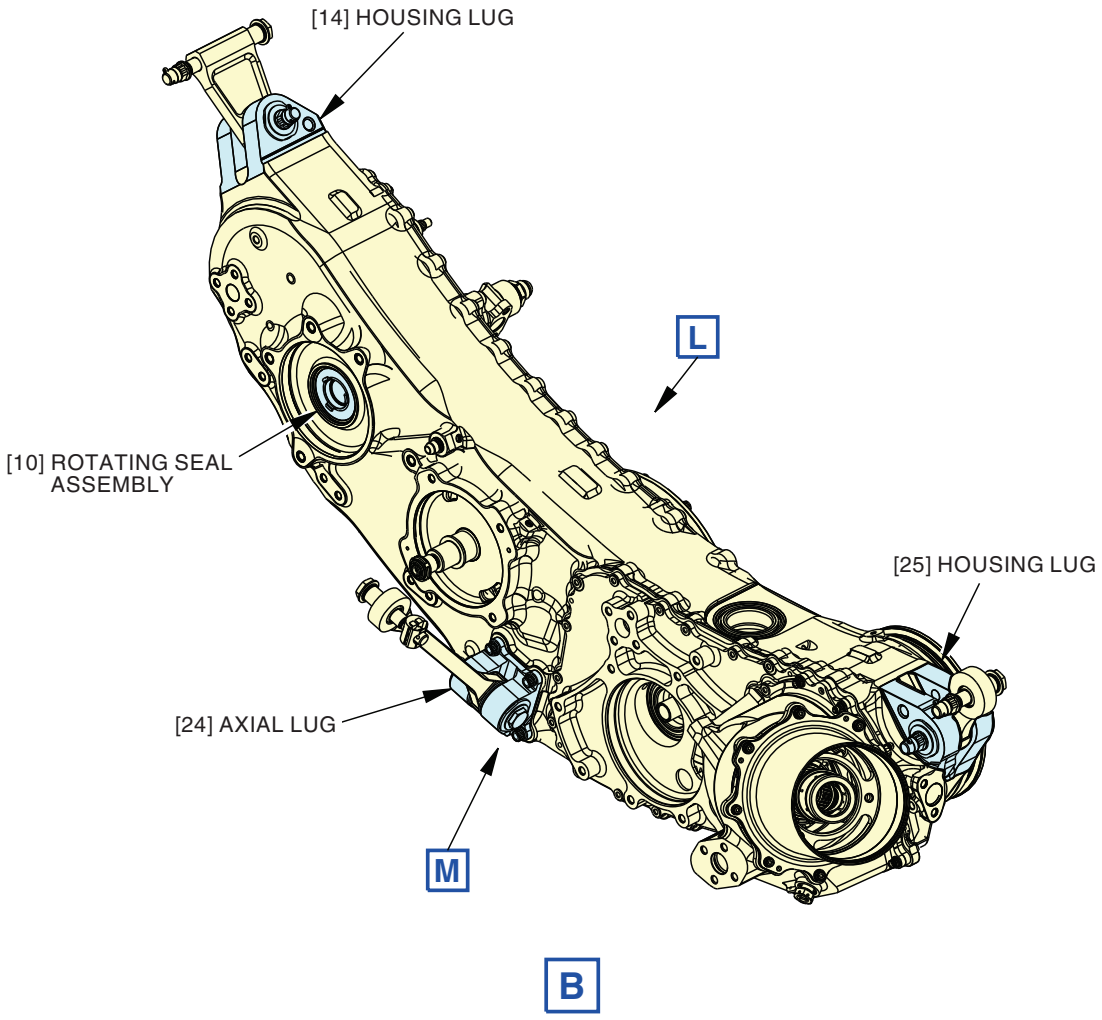

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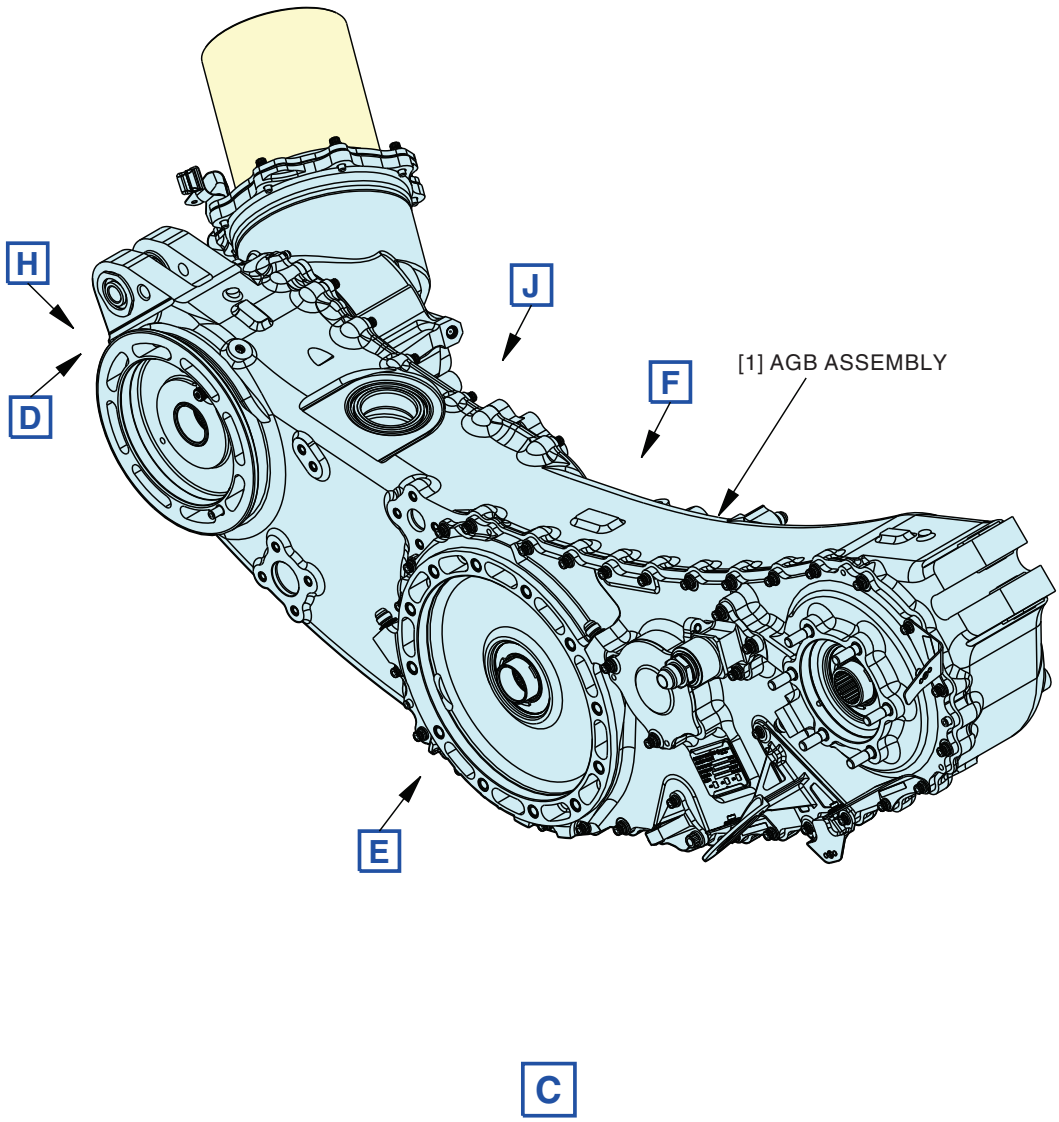
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 2 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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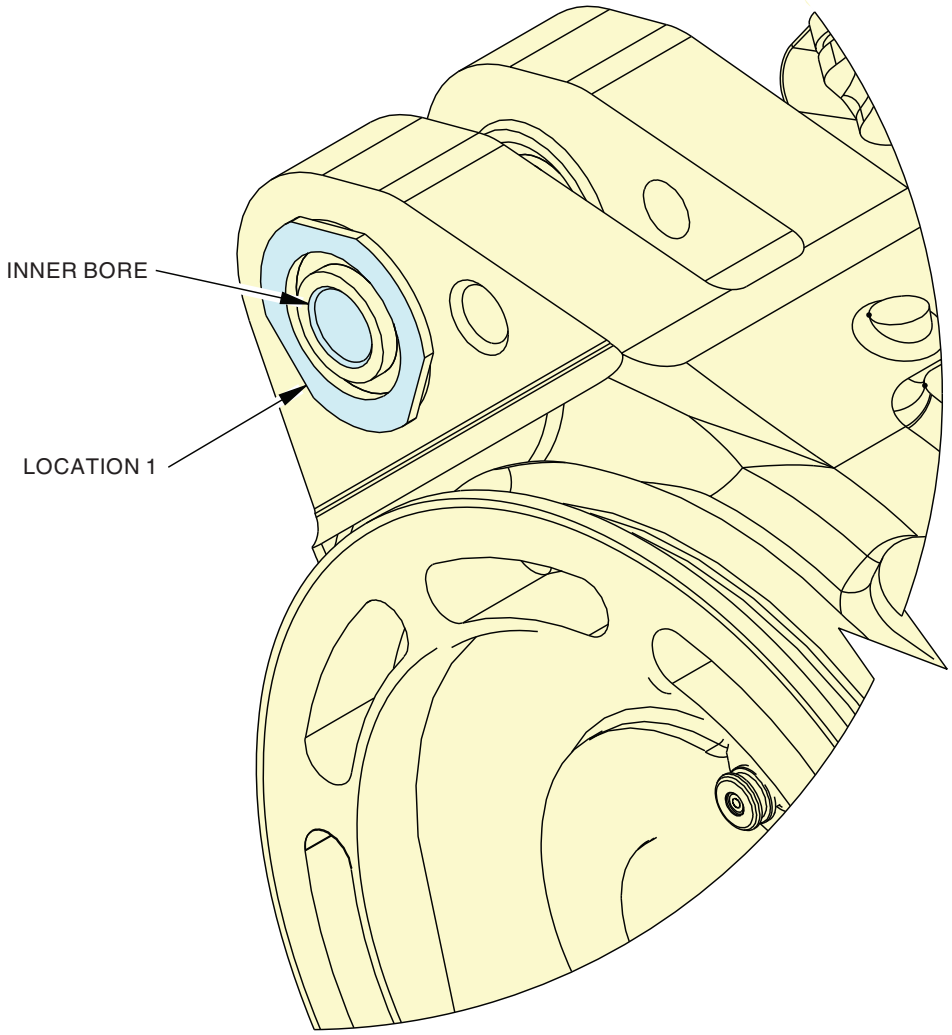
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 3 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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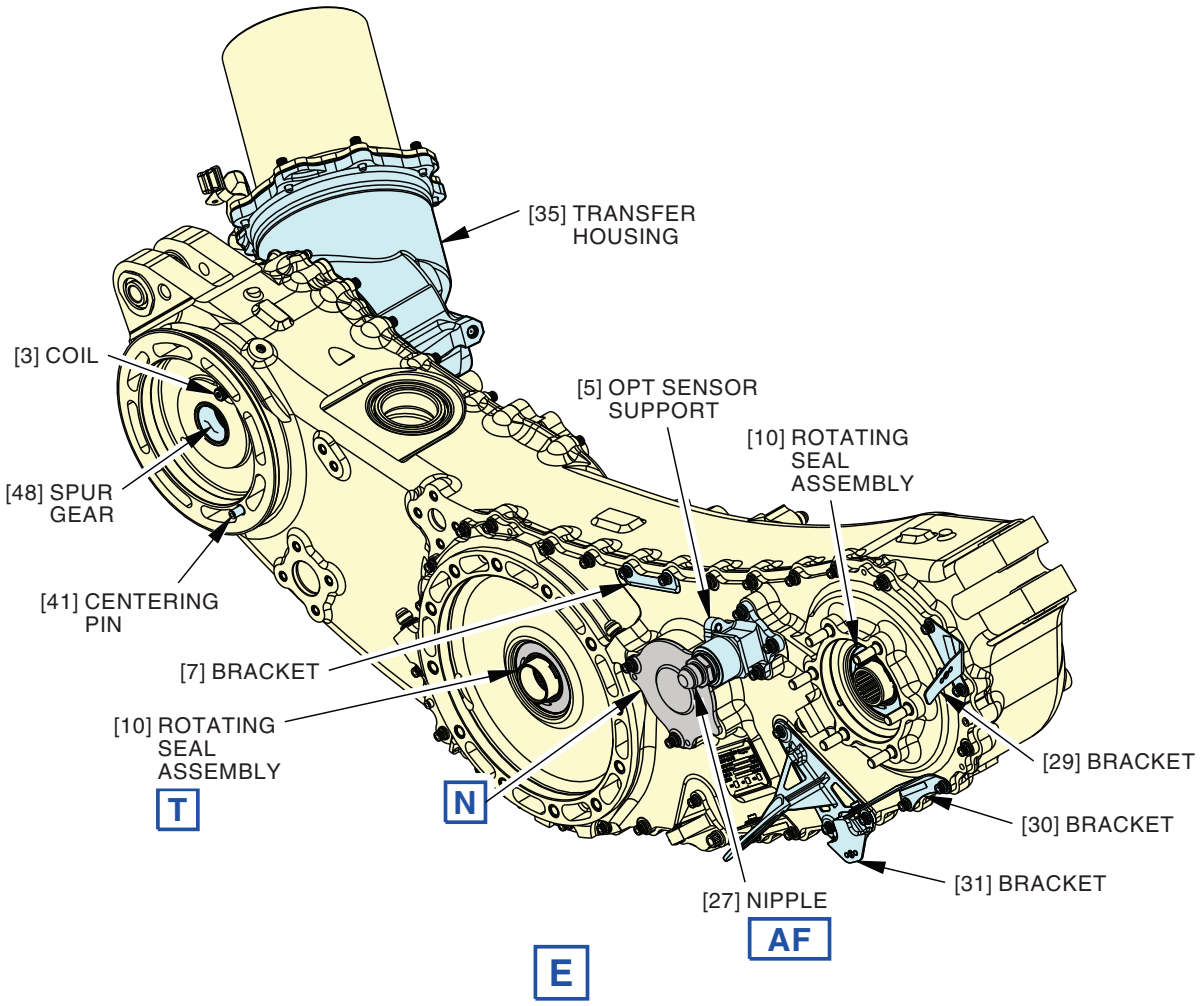
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 4 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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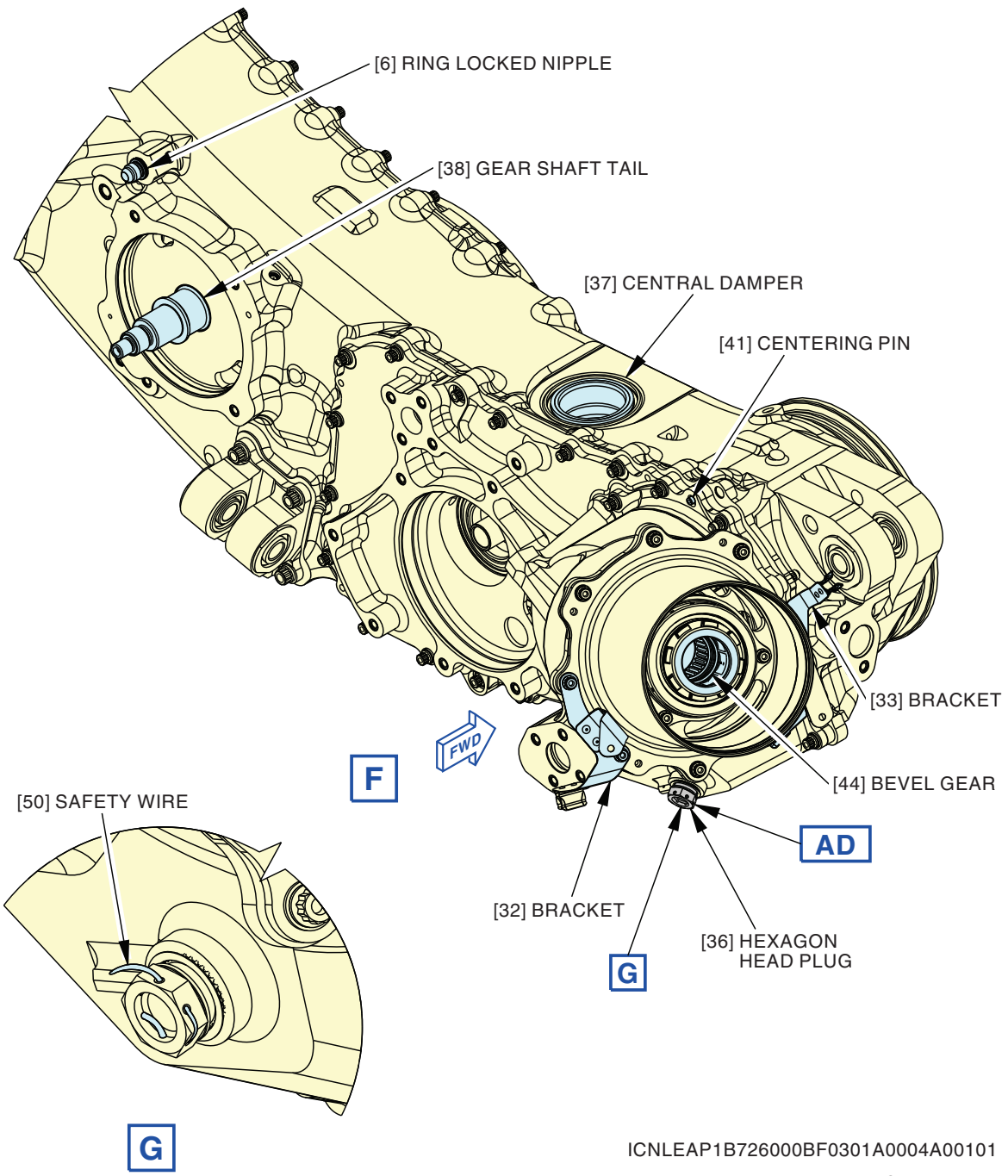
2575864 S0000617903_V2

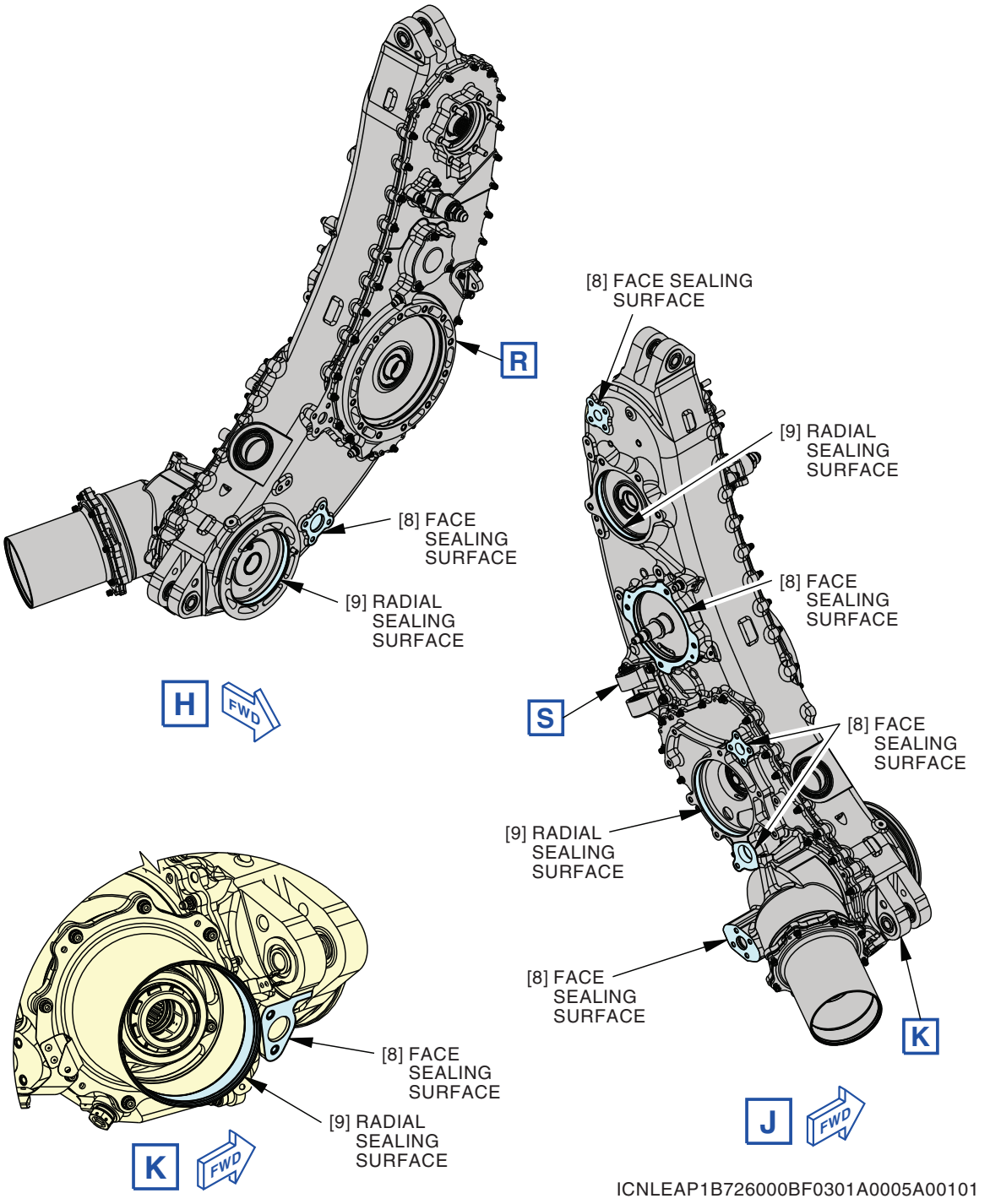
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 5 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



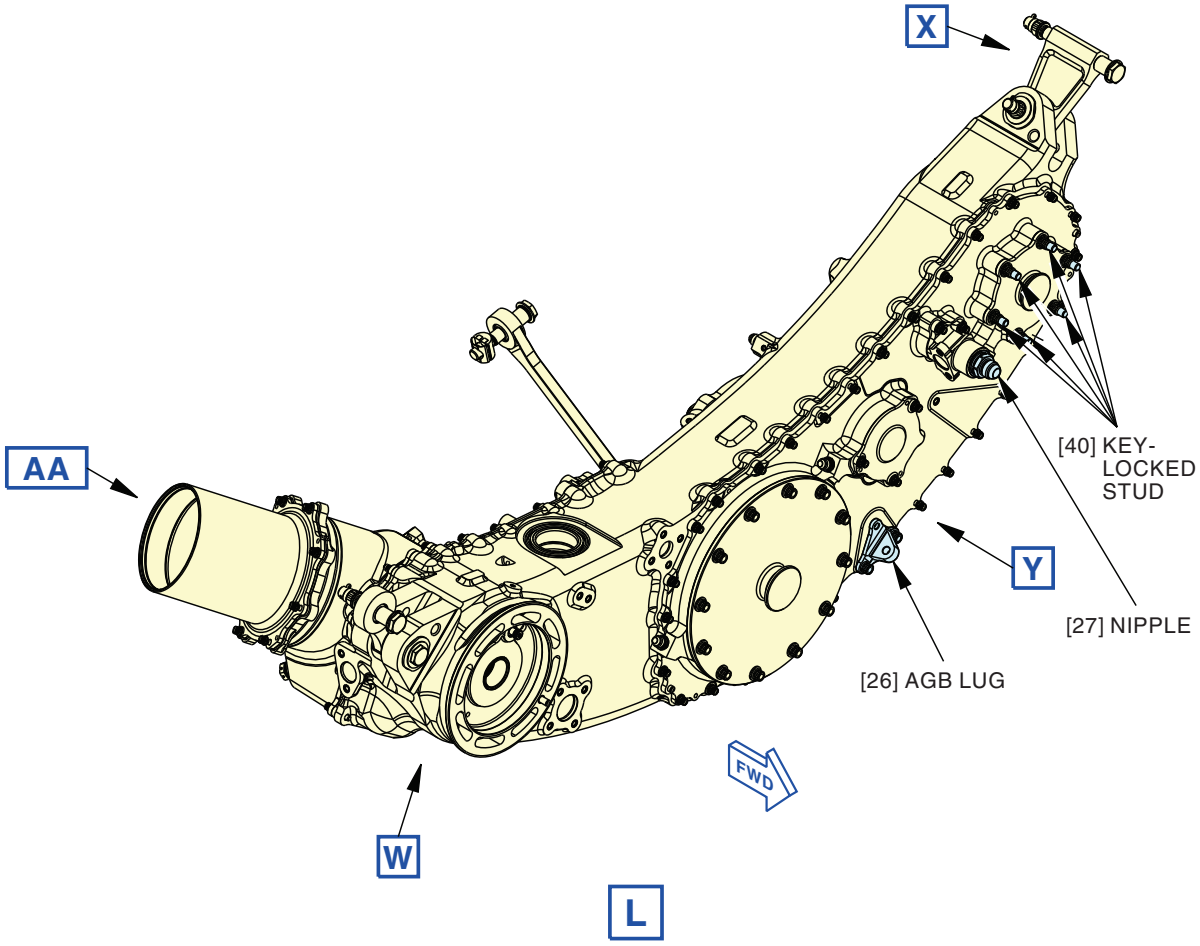
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 6 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 7 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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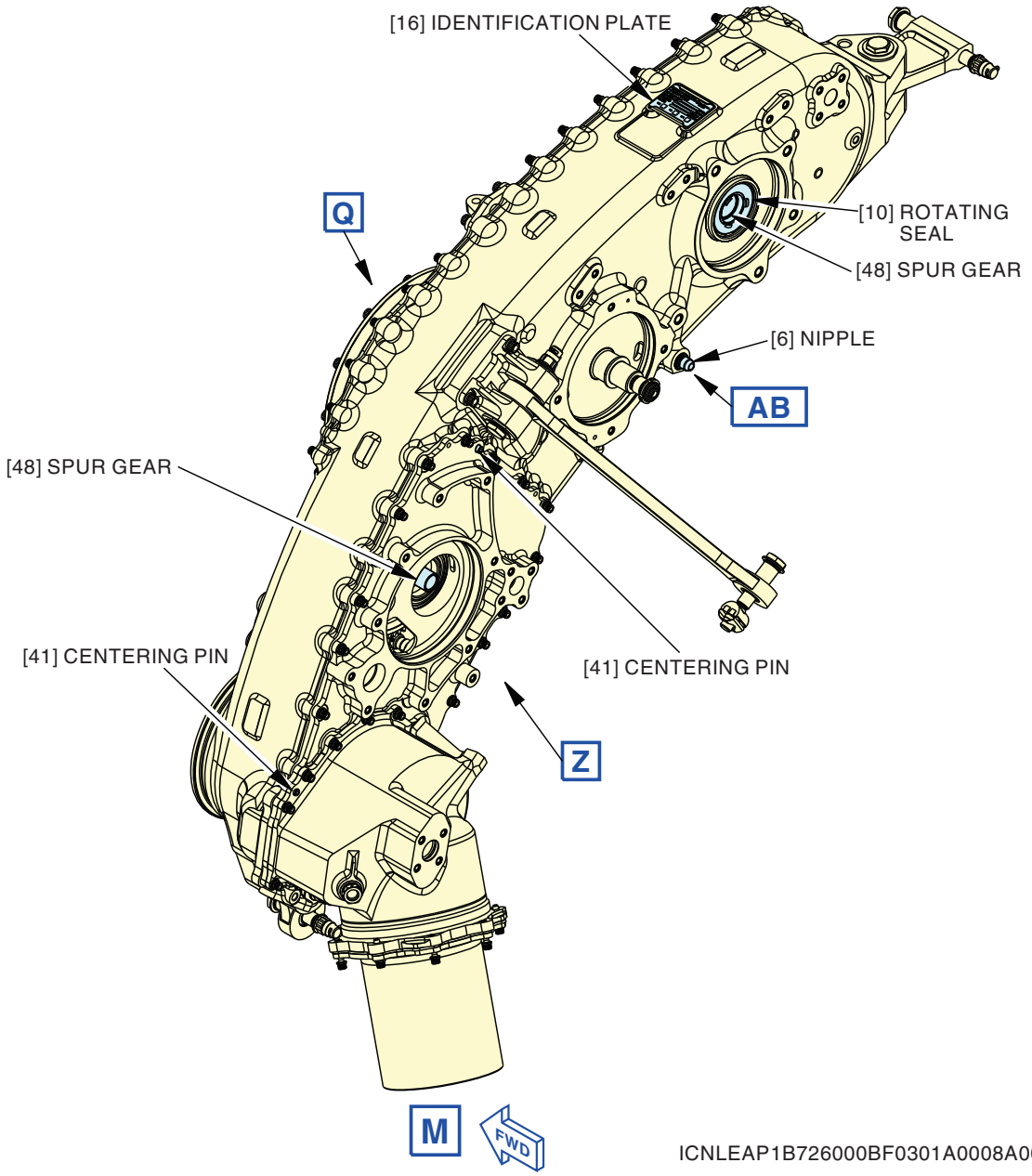
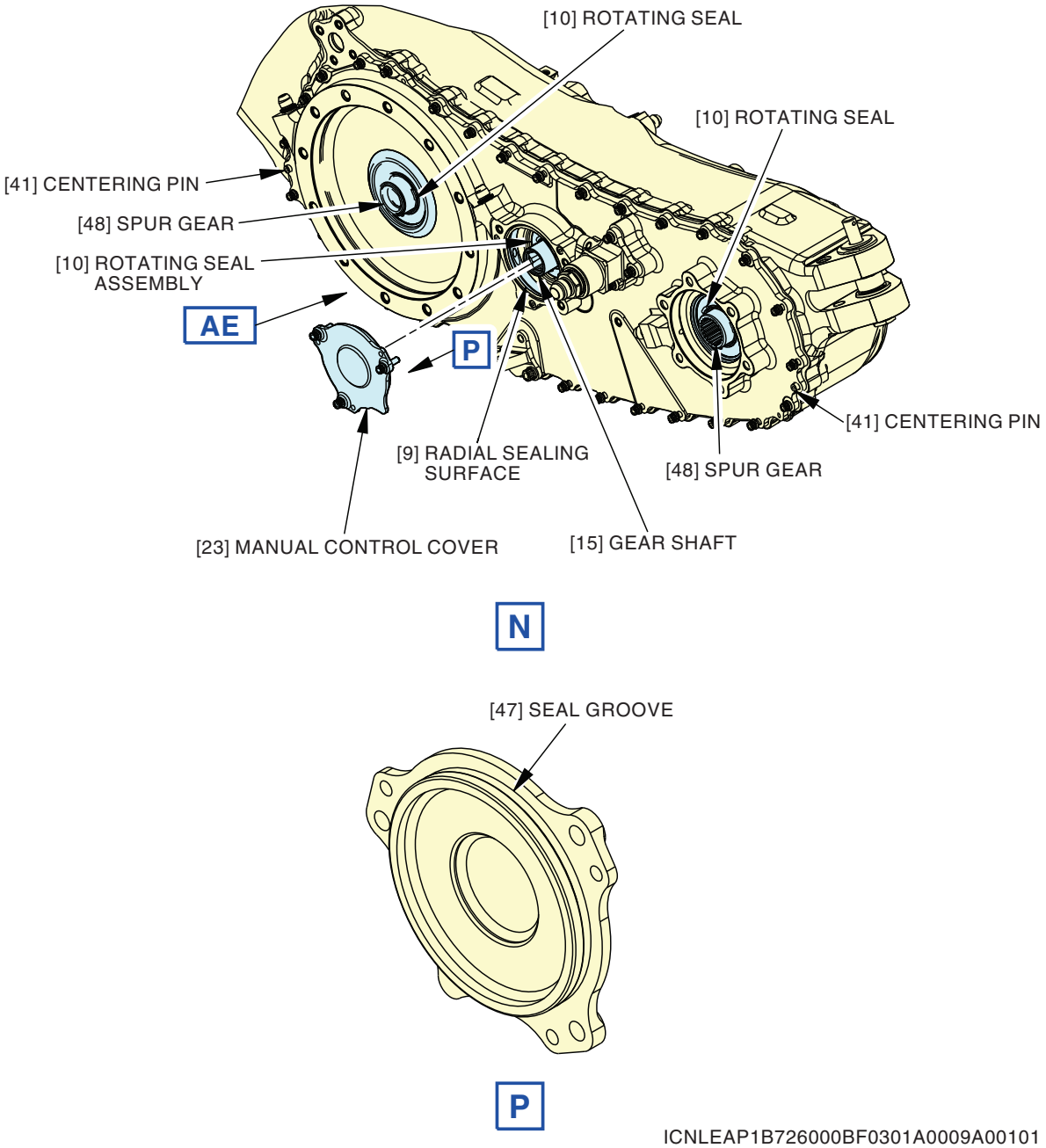
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 8 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



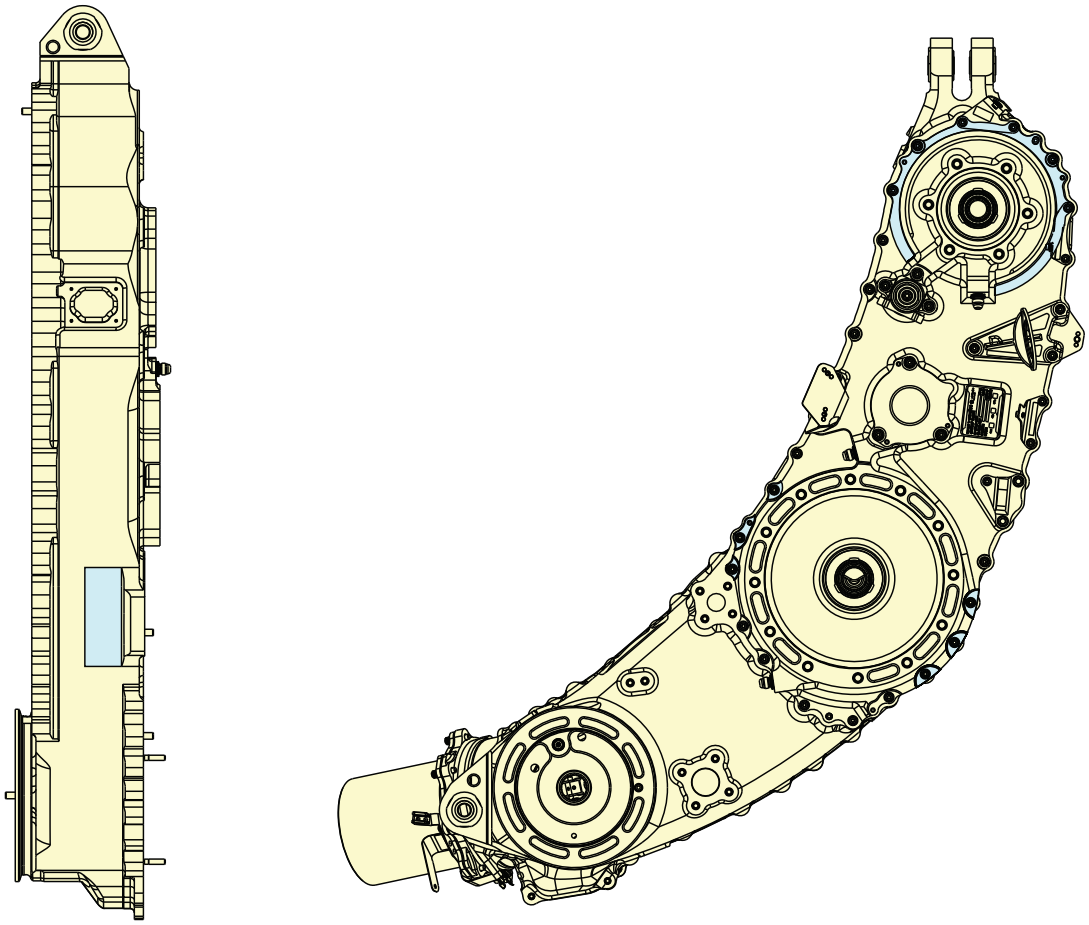
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 9 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 10 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AREAS WHERE SCRATCHES, NICKS, DENTS AND SCORE ARE NOT PERMITTED.

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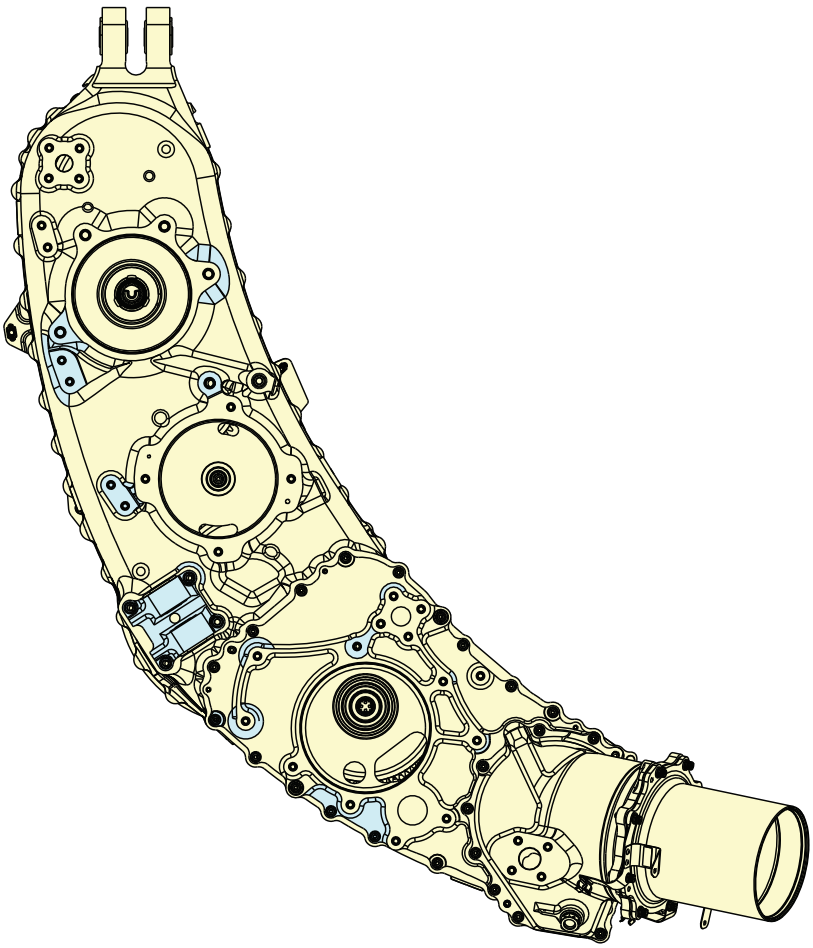
2575871 S0000617911_V1

Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 11 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



S

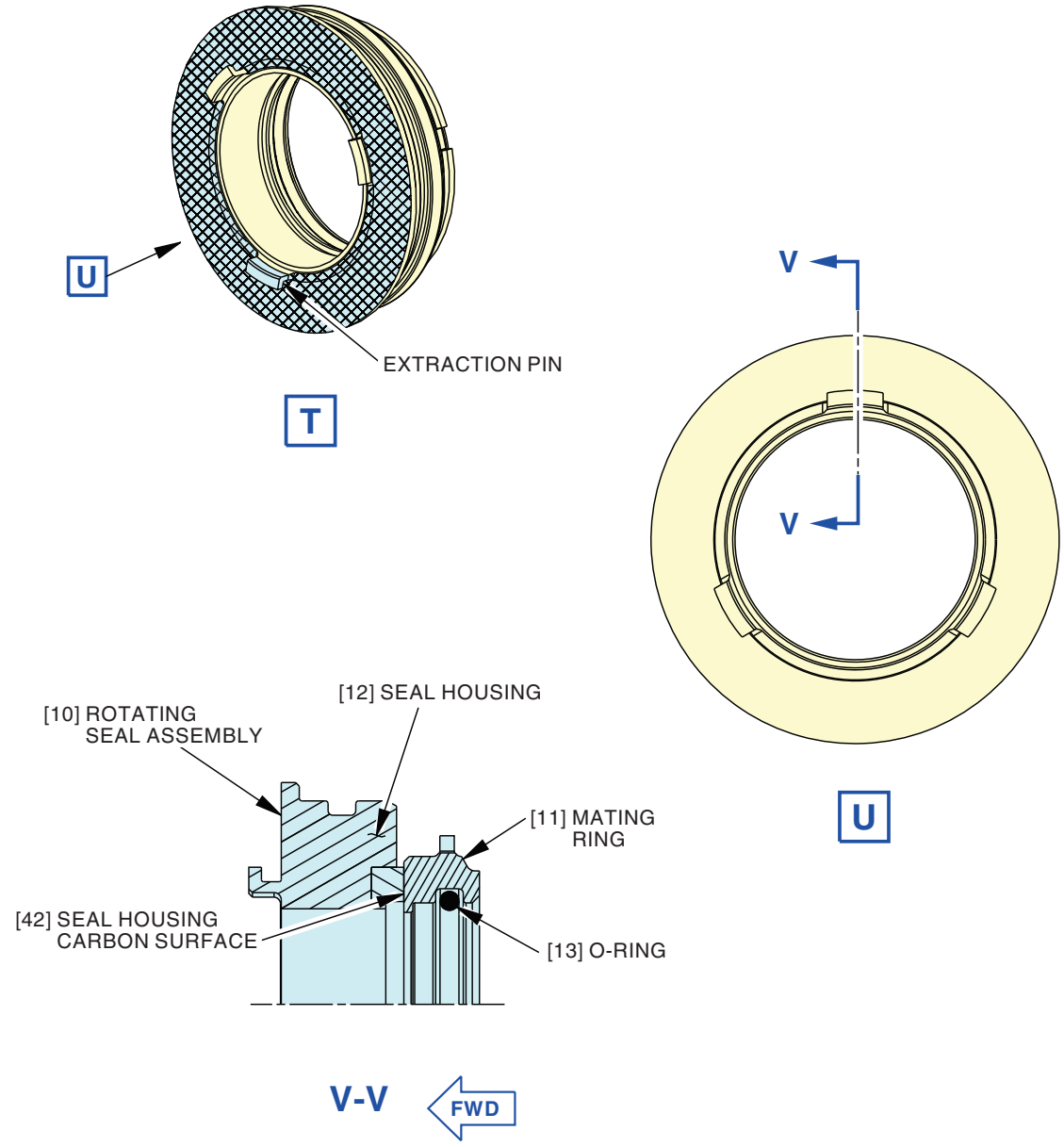
AREAS WHERE SCRATCHES, NICKS, DENTS AND SCORE ARE NOT PERMITTED.

ICNLEAP1B726000BF0301A0015A00101

2575872 S0000617912_V1

Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 12 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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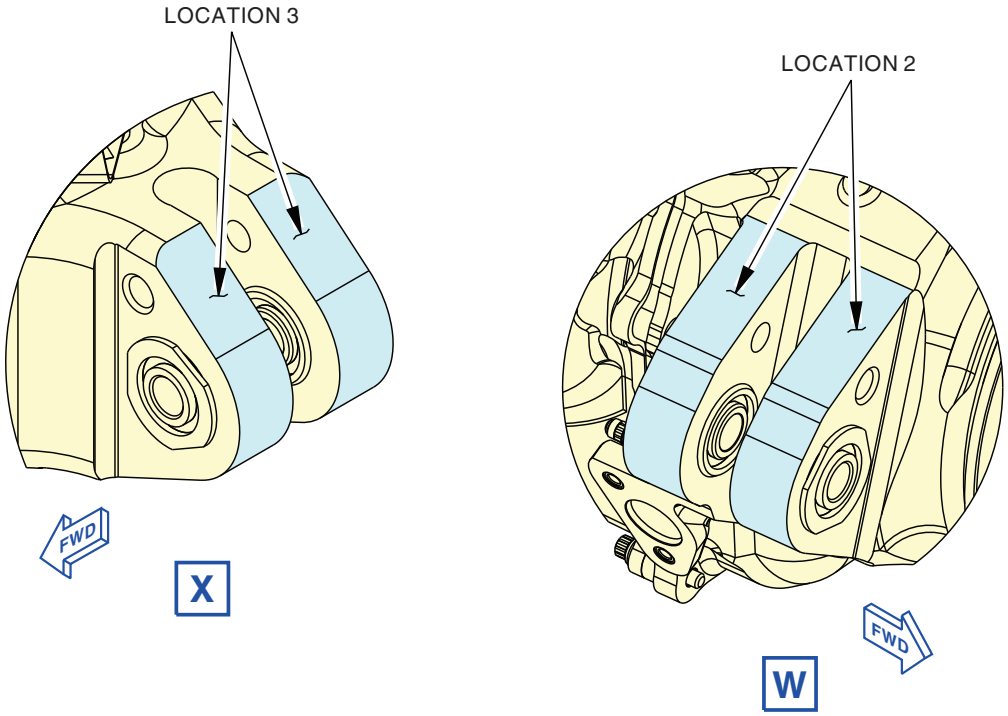
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 13 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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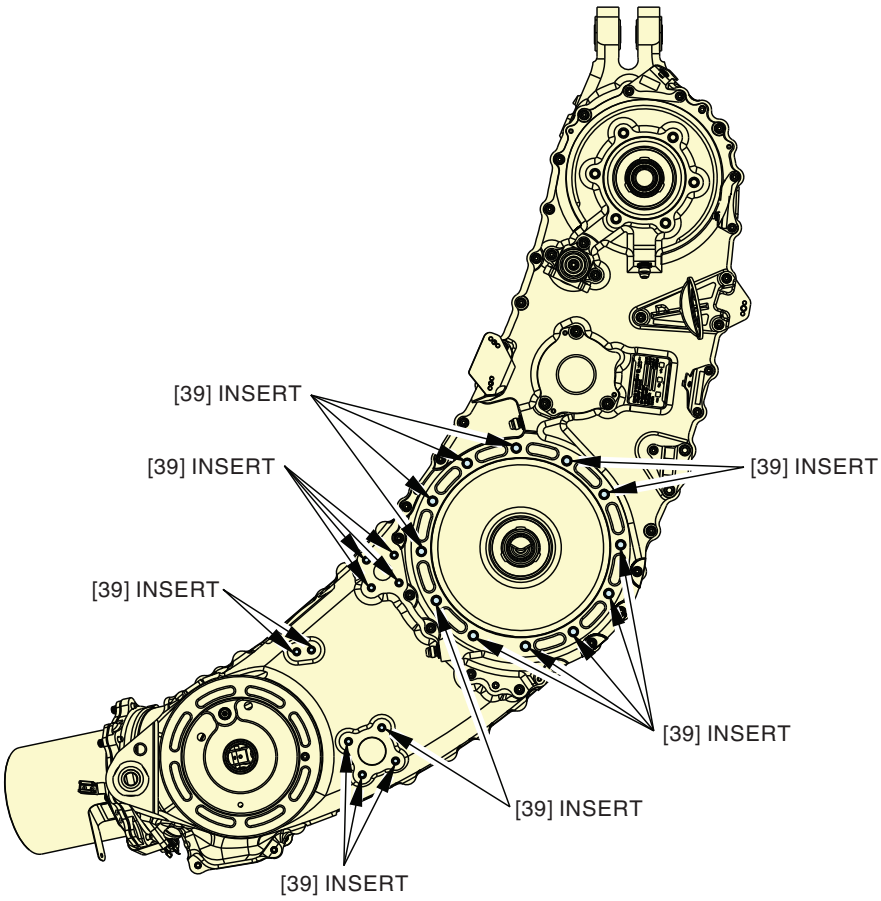
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 14 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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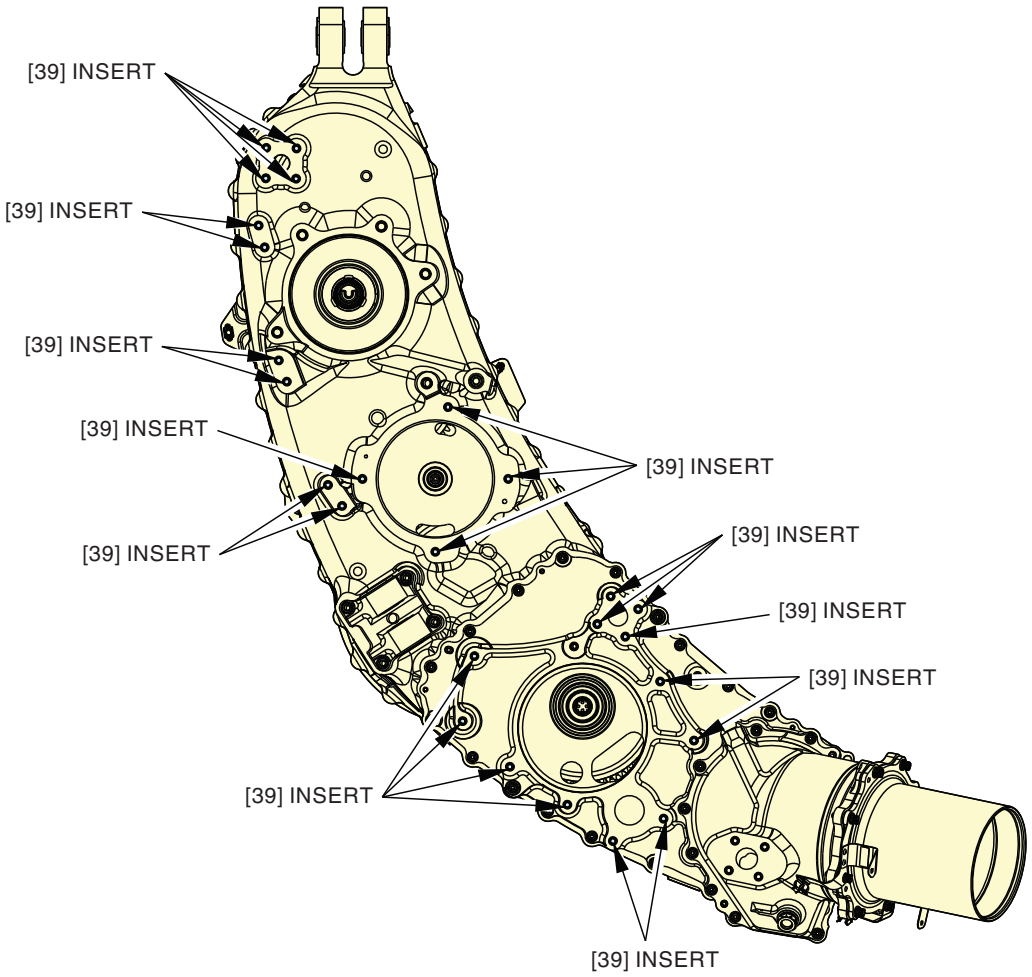
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 15 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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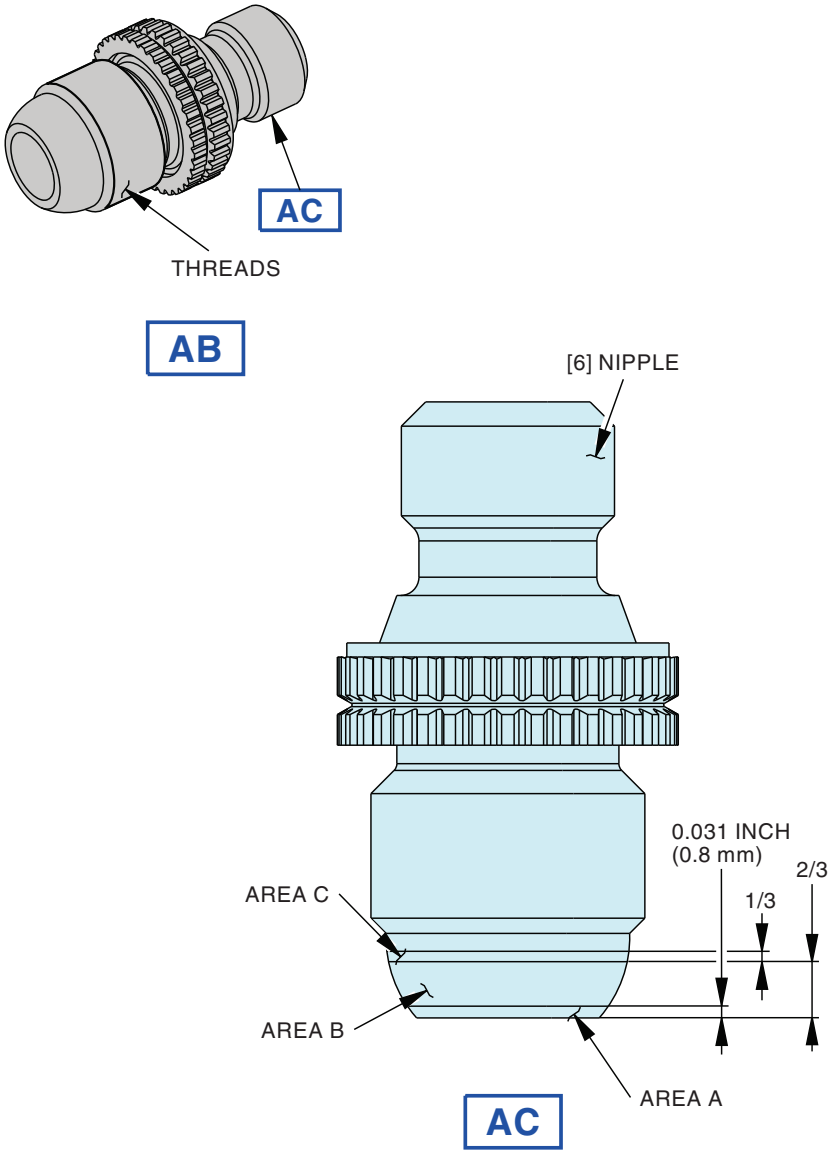
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 16 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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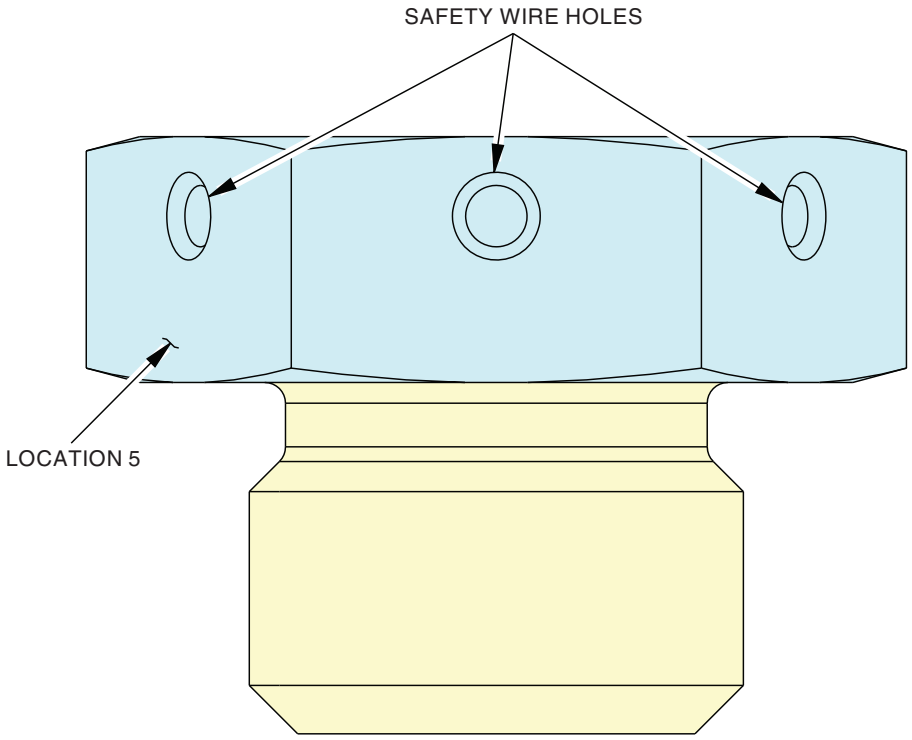
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 17 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AD

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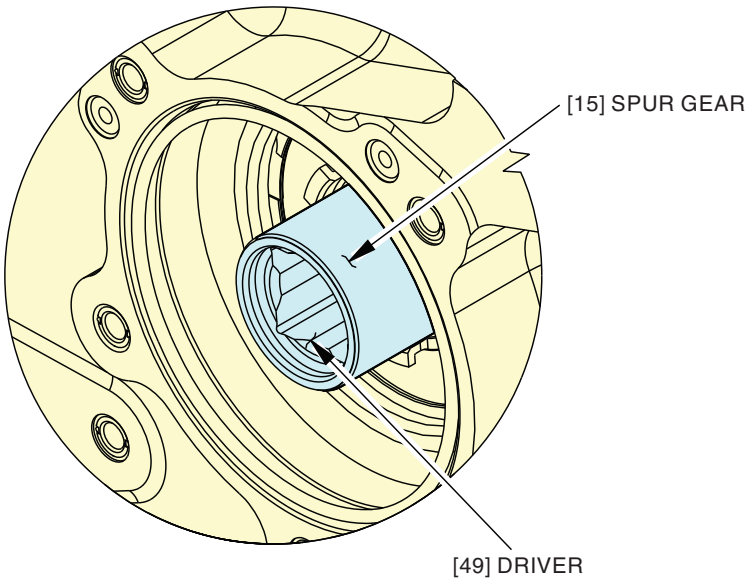
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 18 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AE

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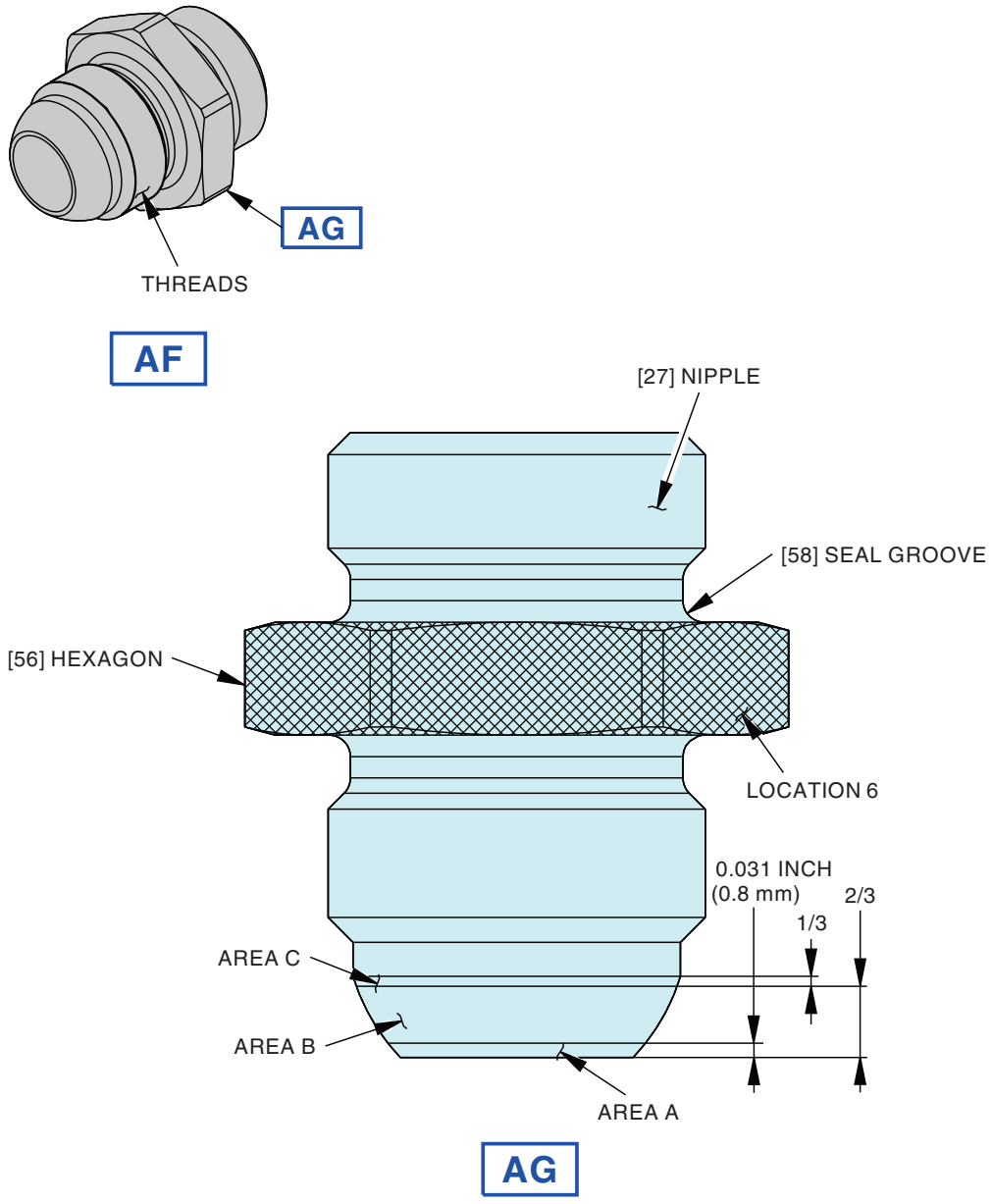
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 19 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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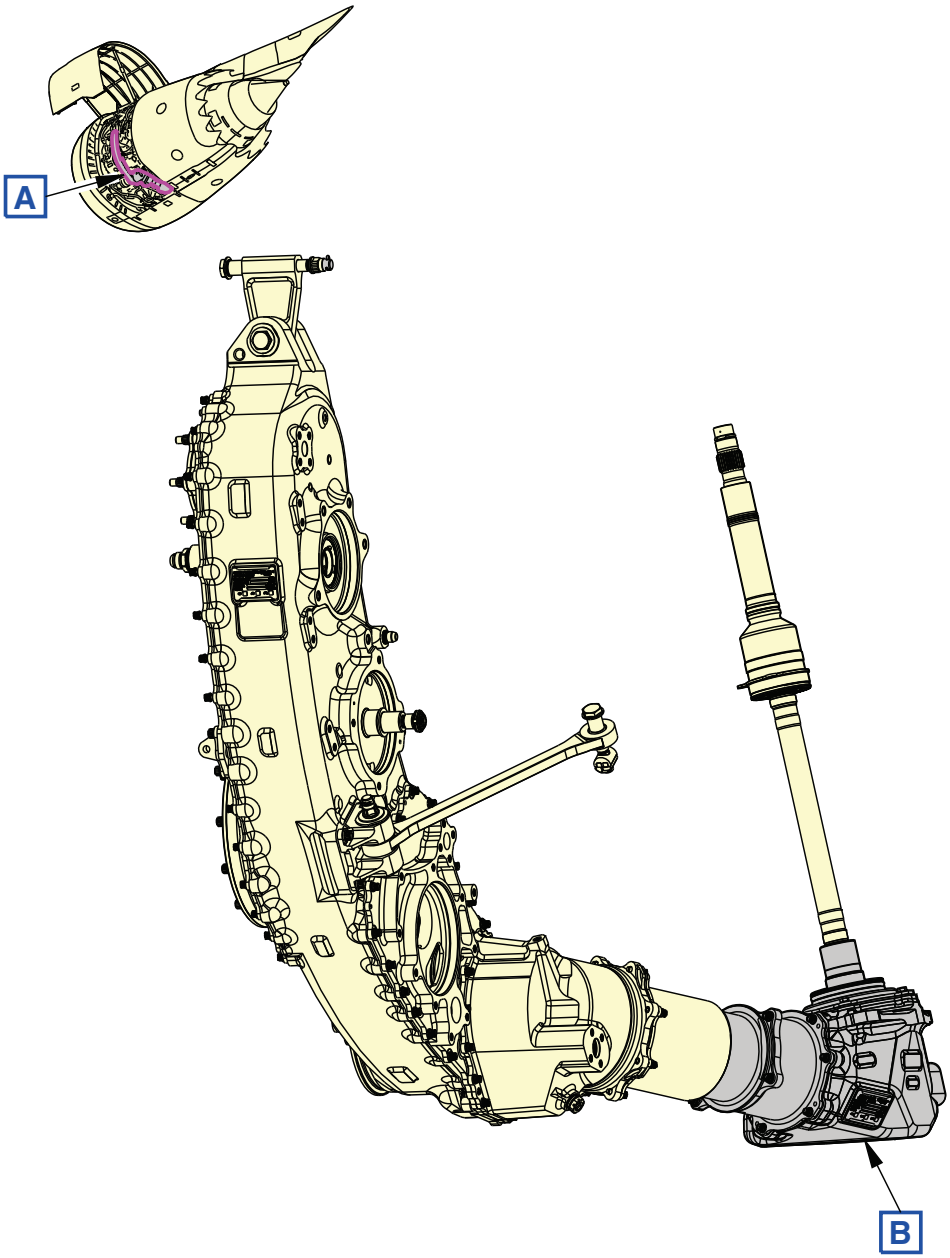
Accessory Gearbox Assembly Inspection
Figure 601/72-60-00-990-804-G00 (Sheet 20 of 20)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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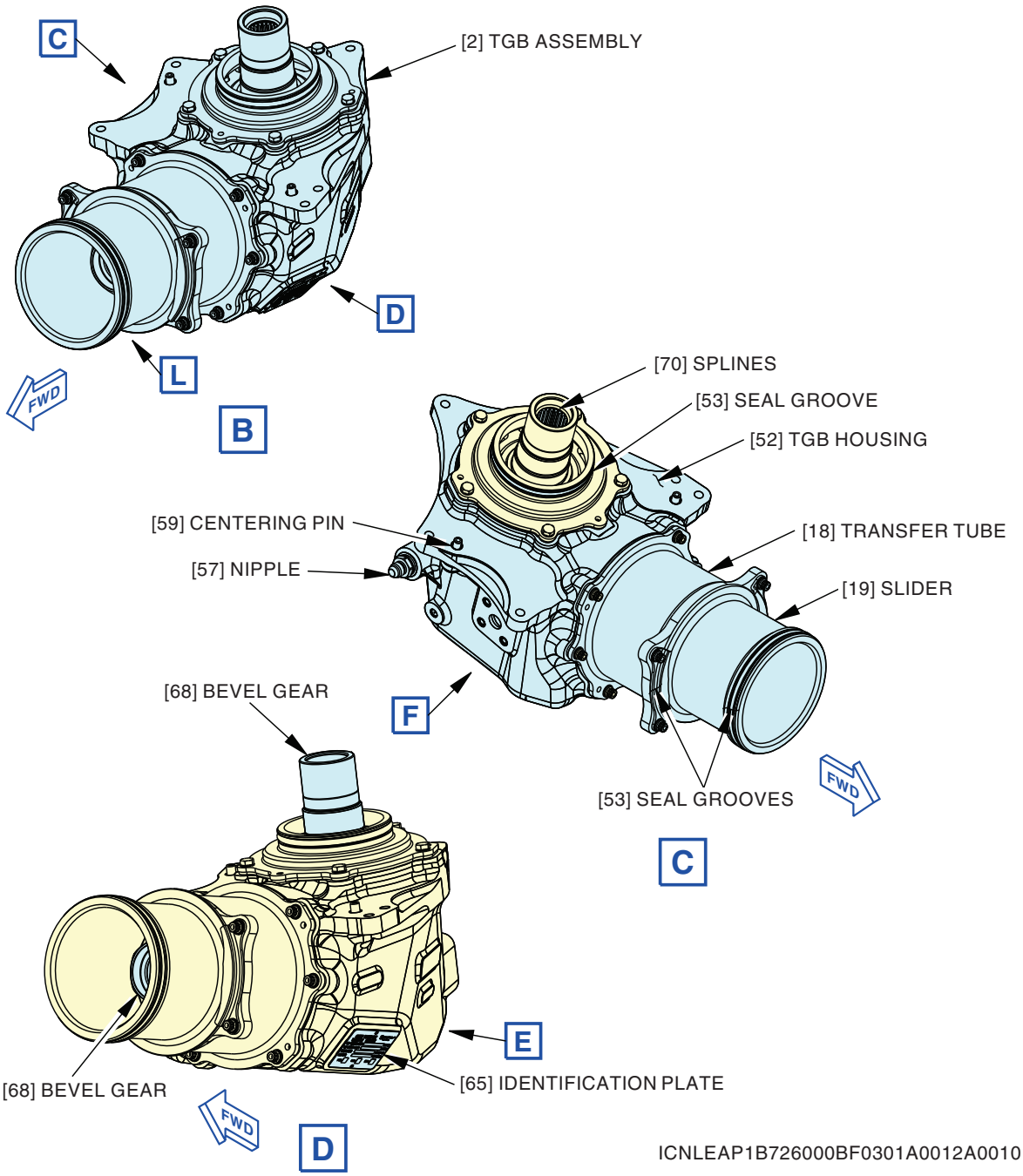
Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 1 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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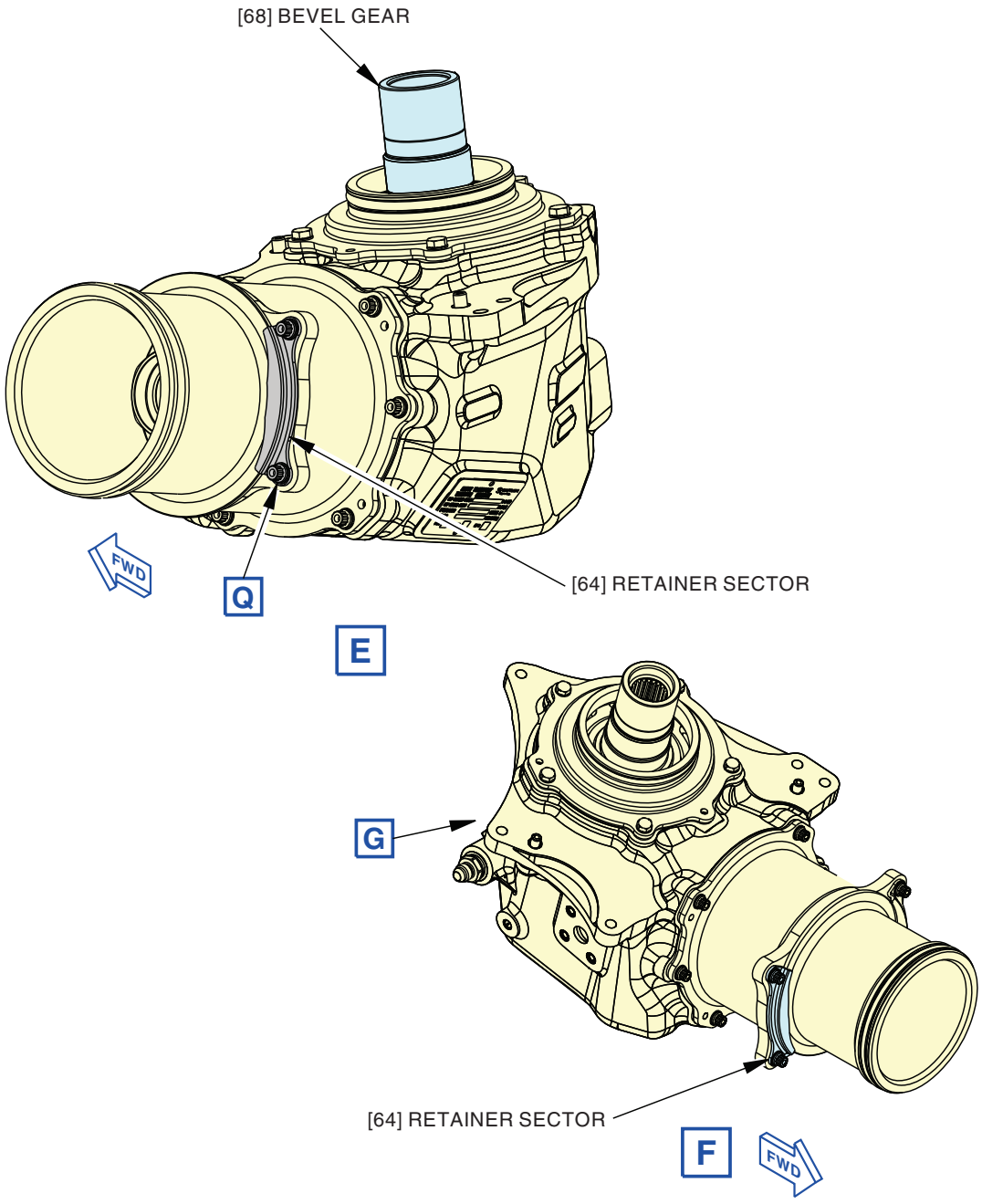
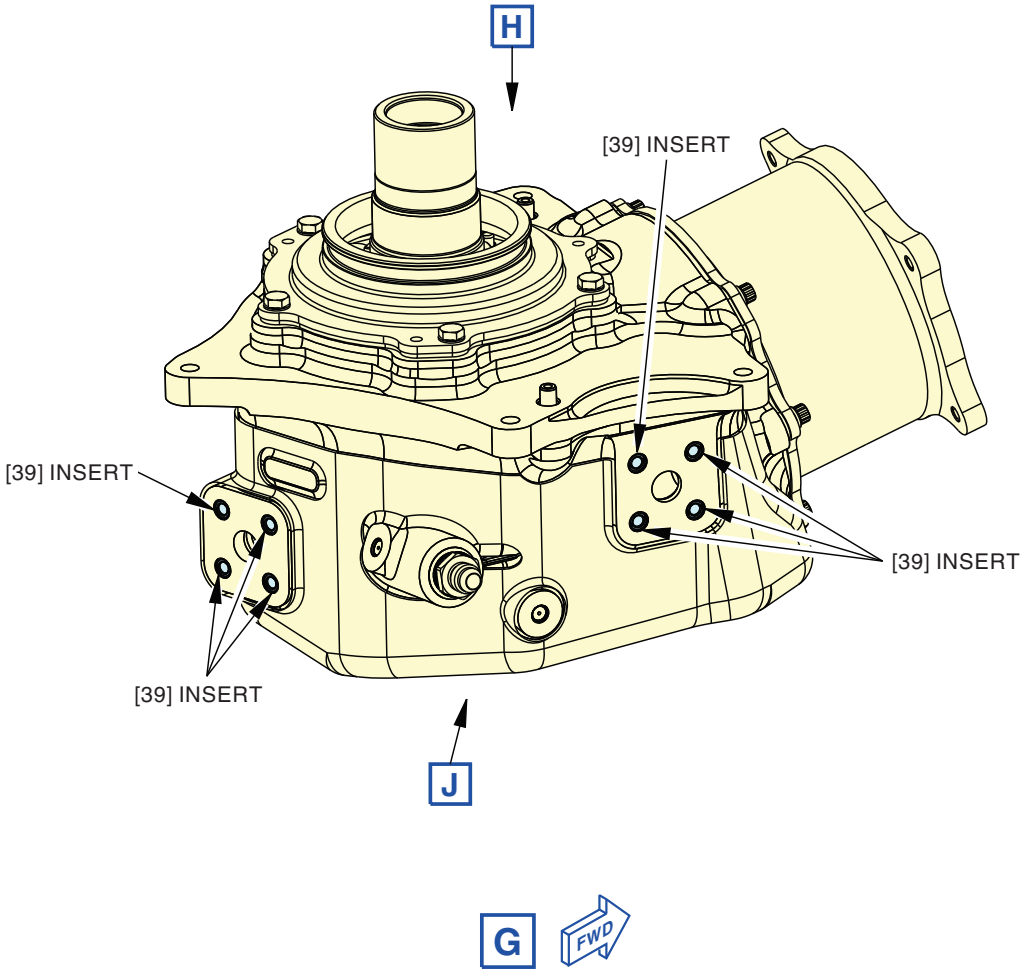
Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 2 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 3 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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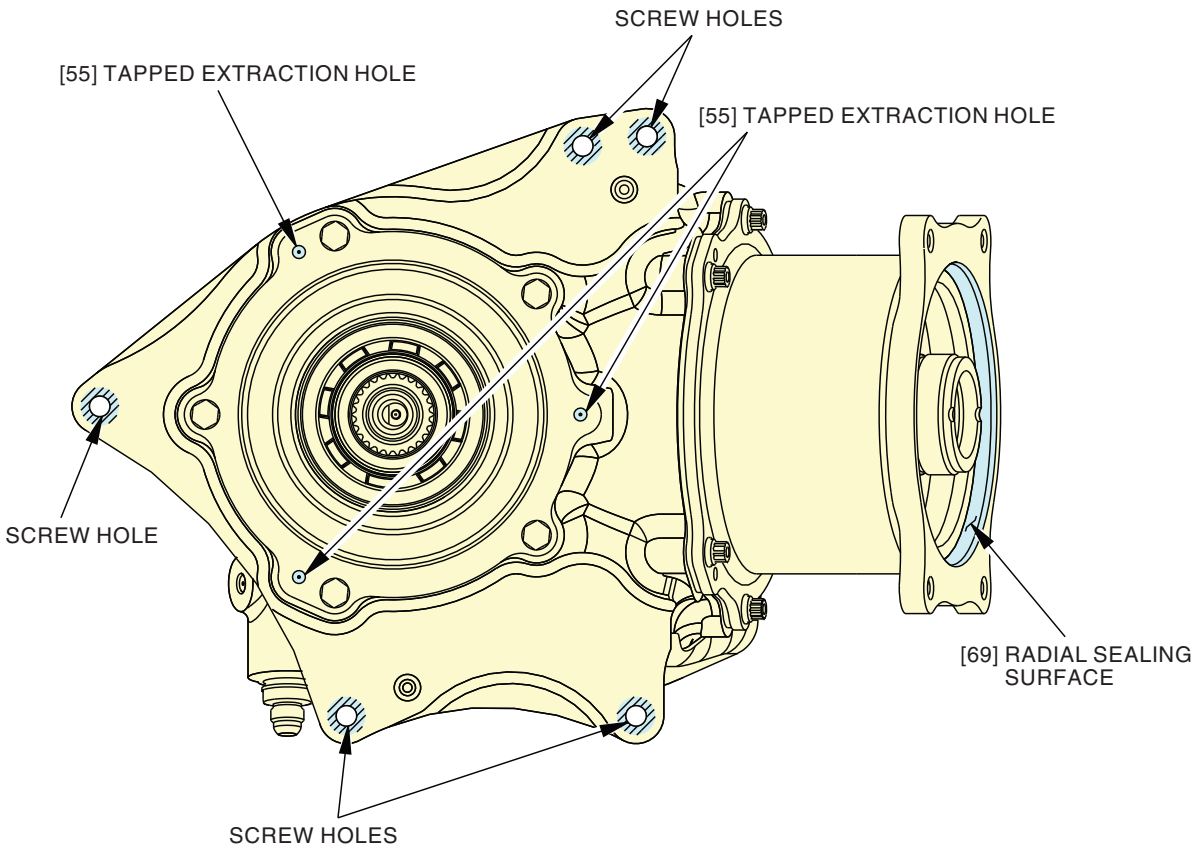
2575886 S0000617924_V1

Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 4 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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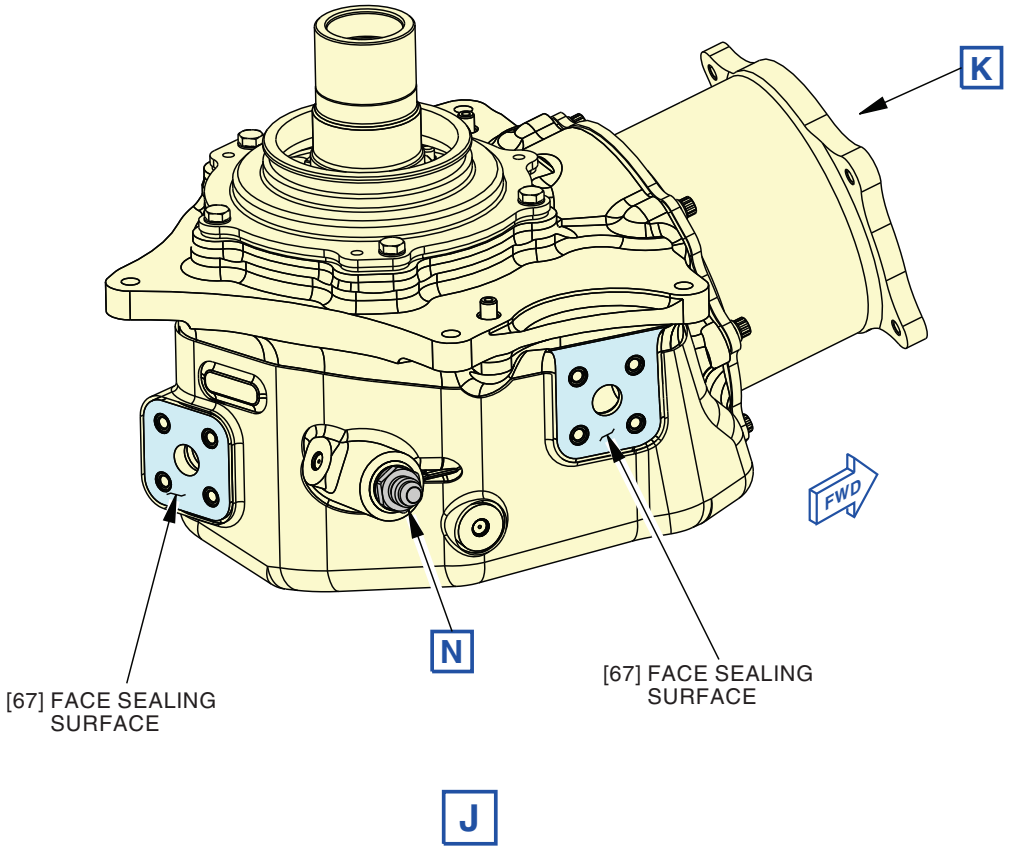
2575887 S0000617925_V2

Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 5 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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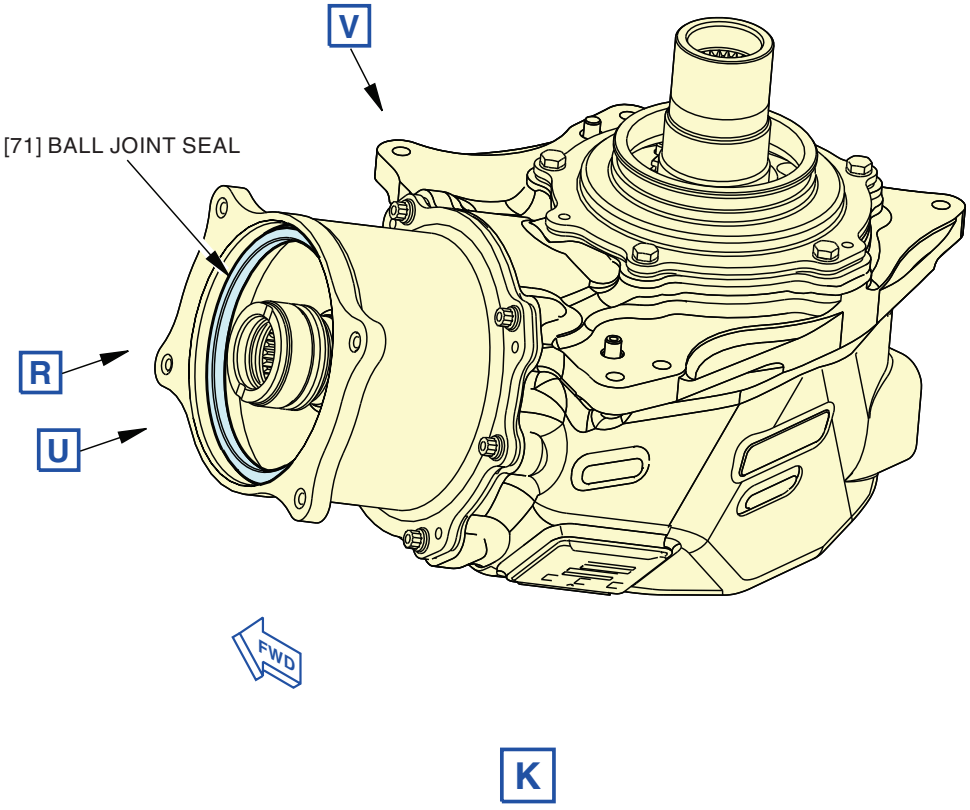
2575888 S0000617926_V1

Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 6 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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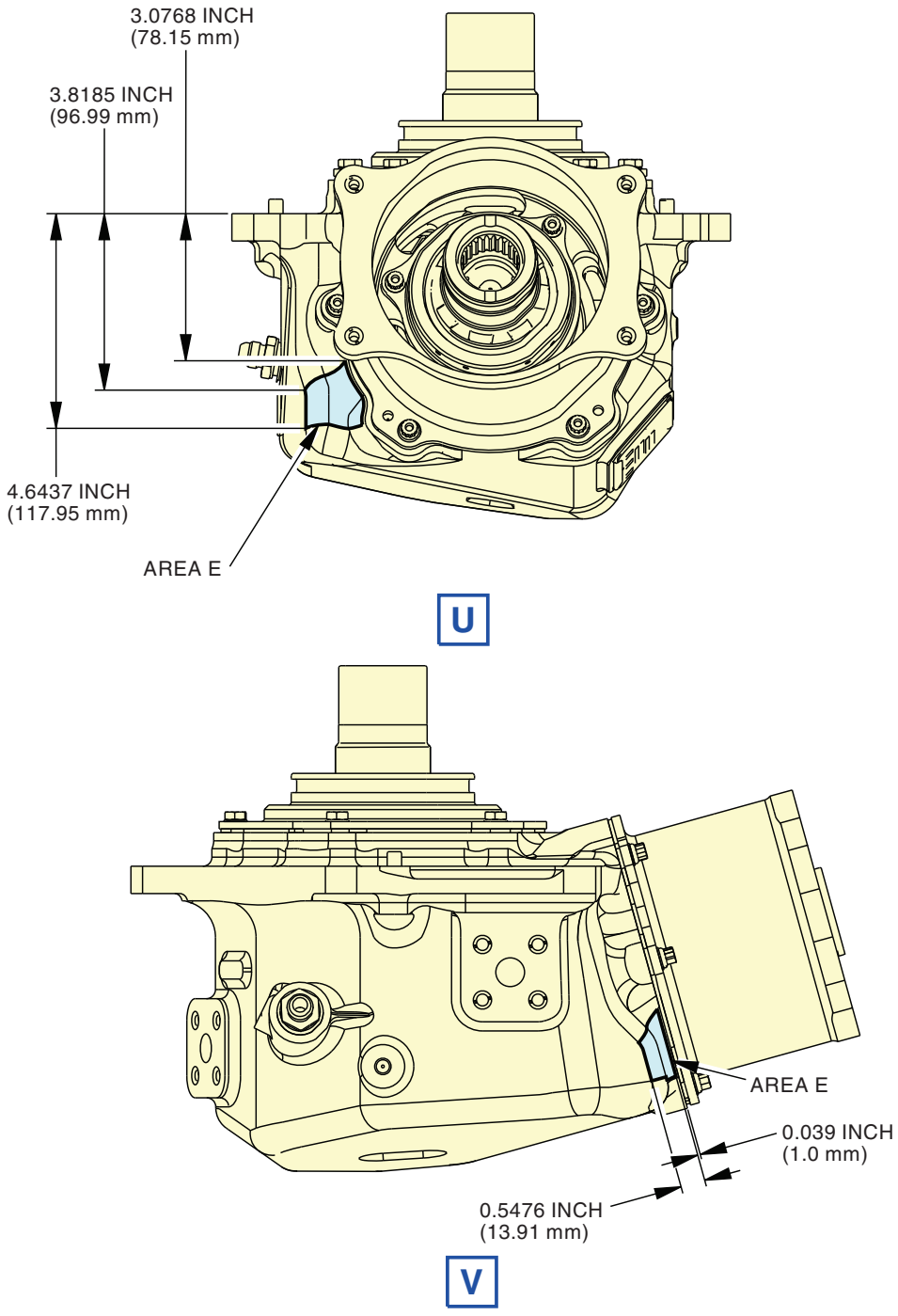
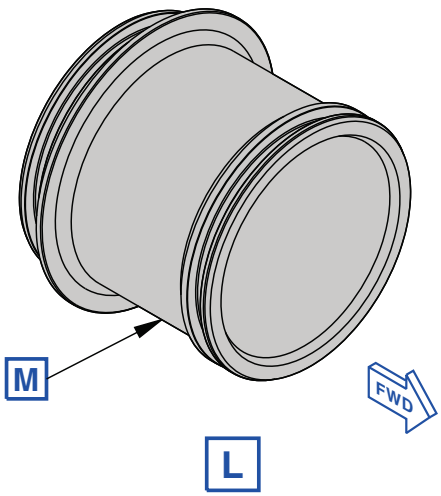
Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 7 of 12)

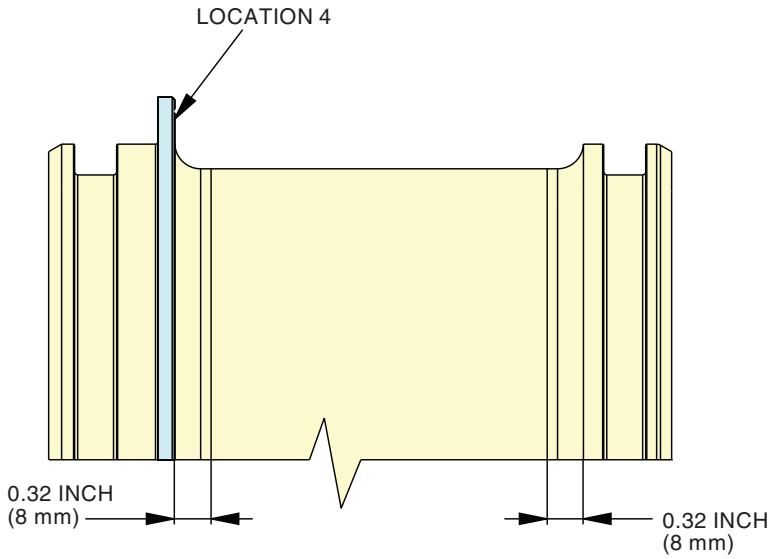
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 8 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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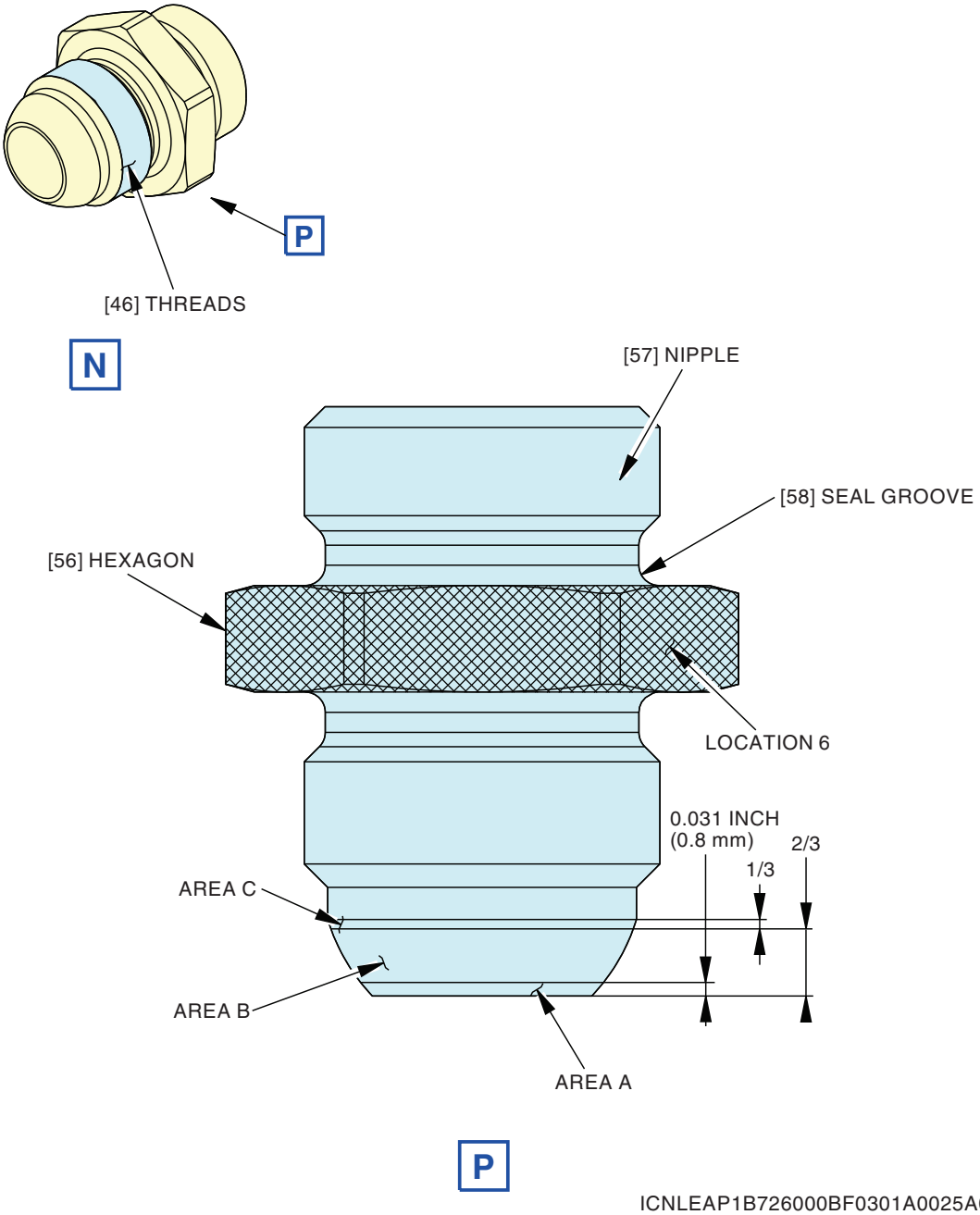
Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 9 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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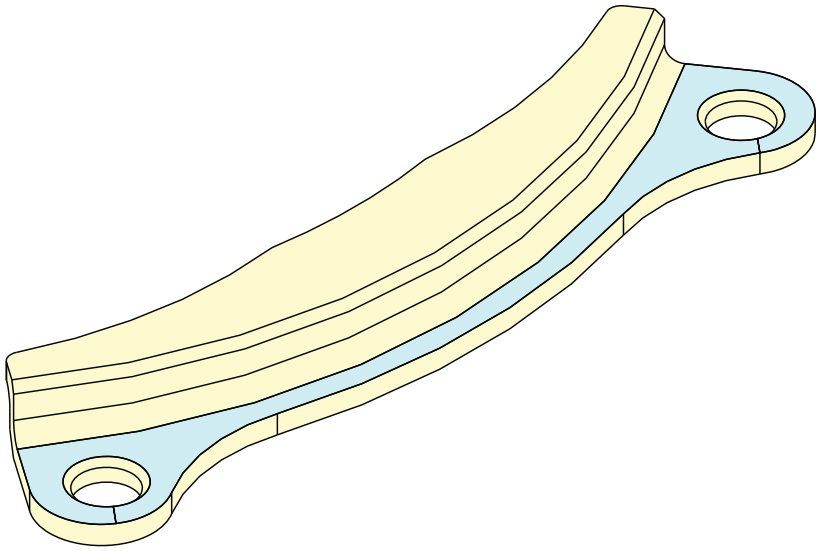
Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 10 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Q

AREA 1

ICNLEAP1B726000BF0301A0031A00101
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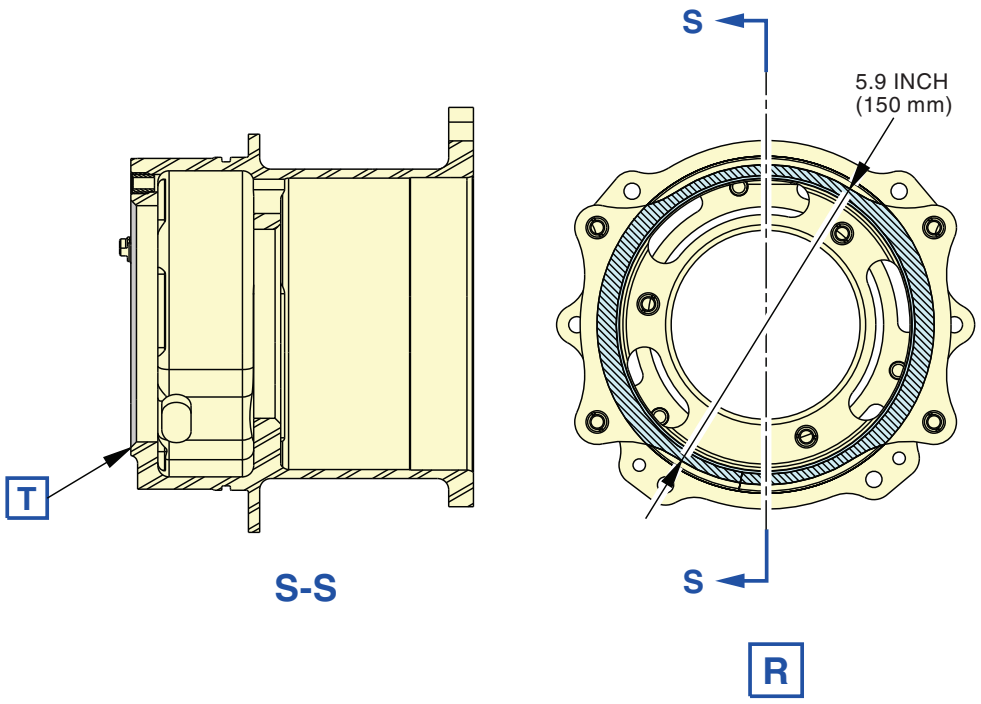
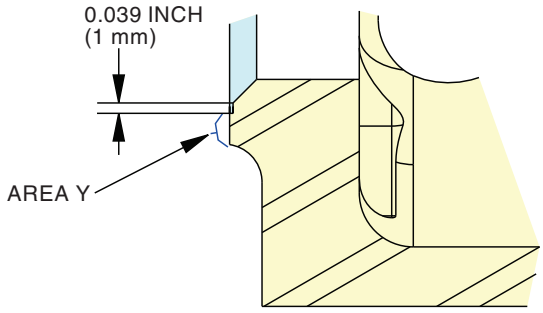
Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 11 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





AREA Y

AREA W

T

ICNLEAP1B726000BF0301A0032A00101

2575895 S0000617931_V1

Transfer Gearbox Assembly Inspection
Figure 602/72-60-00-990-805-G00 (Sheet 12 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-60-00-200-803-G00

3. Accessory Drives Damper Bearings Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to examine the accessory drives damper bearings.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-60-00-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-60-00-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Accessory Drives Damper Bearings Inspection

SUBTASK 72-60-00-210-073-G00

(1) Examine all areas of the AGB damper bearings, of AGB axial lugs [24], AGB housing lugs [14] and AGB housing lugs [25] that you can see for these conditions (Figure 603 (Sheet 2) and Figure 603 (Sheet 3)):

(a) Cracks

1) Not permitted.

2) Not repairable.

(b) Elastomer ruptured or sheared

1) Not permitted.

2) Not repairable.

(c) Corroded, pitted or galled

1) Not permitted.

2) Not repairable.

(d) Axial and radial play

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Not permitted.

2) Not repairable.

(e) Fretting on location 1:

1) Permitted up to 0.004 in. (0.1 mm) in depth.

2) Not repairable.

(f) Loose or missing

1) Not permitted.

2) Not repairable.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-60-00-080-002-G00

(1) Clean the work area of all tools and items you used to do the task.

SUBTASK 72-60-00-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-60-00-860-004-G00

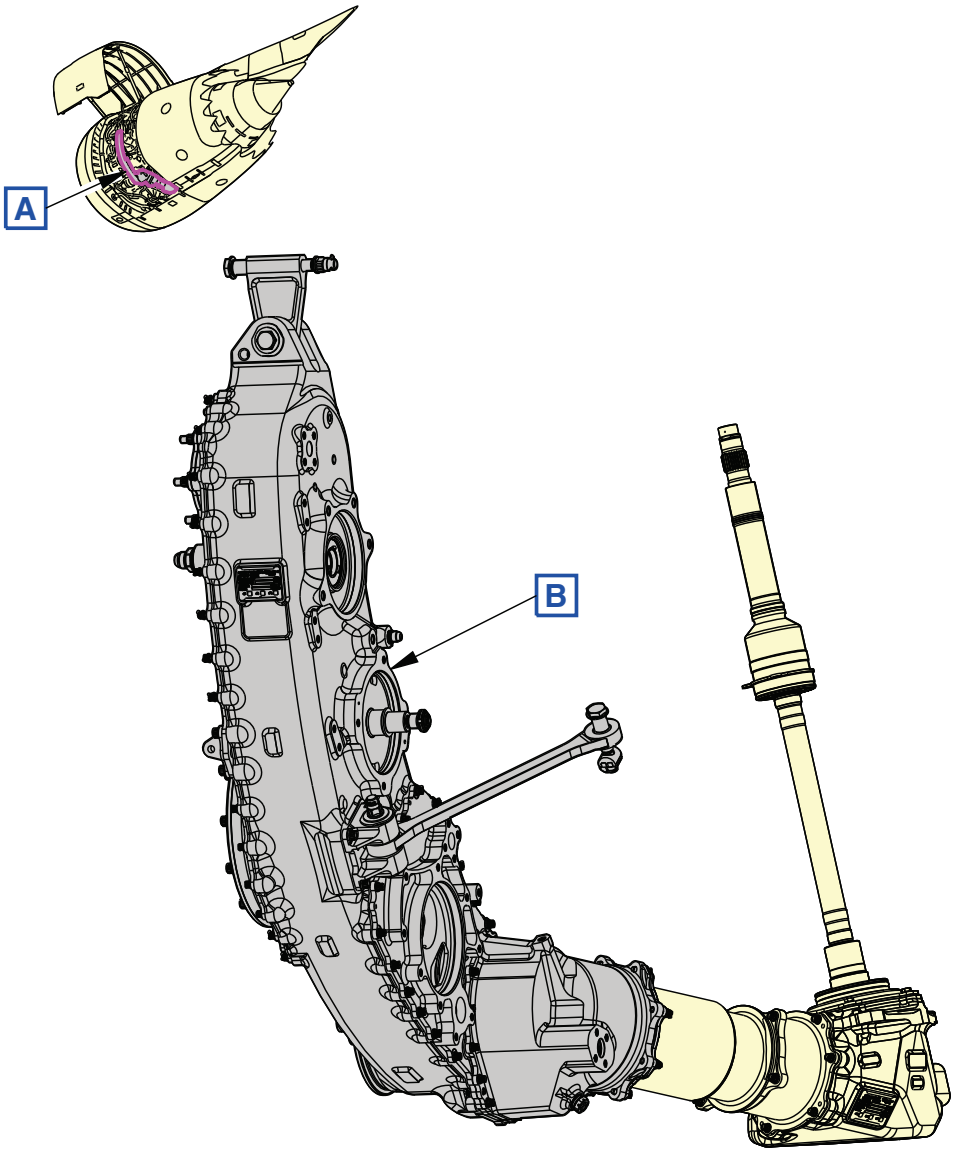
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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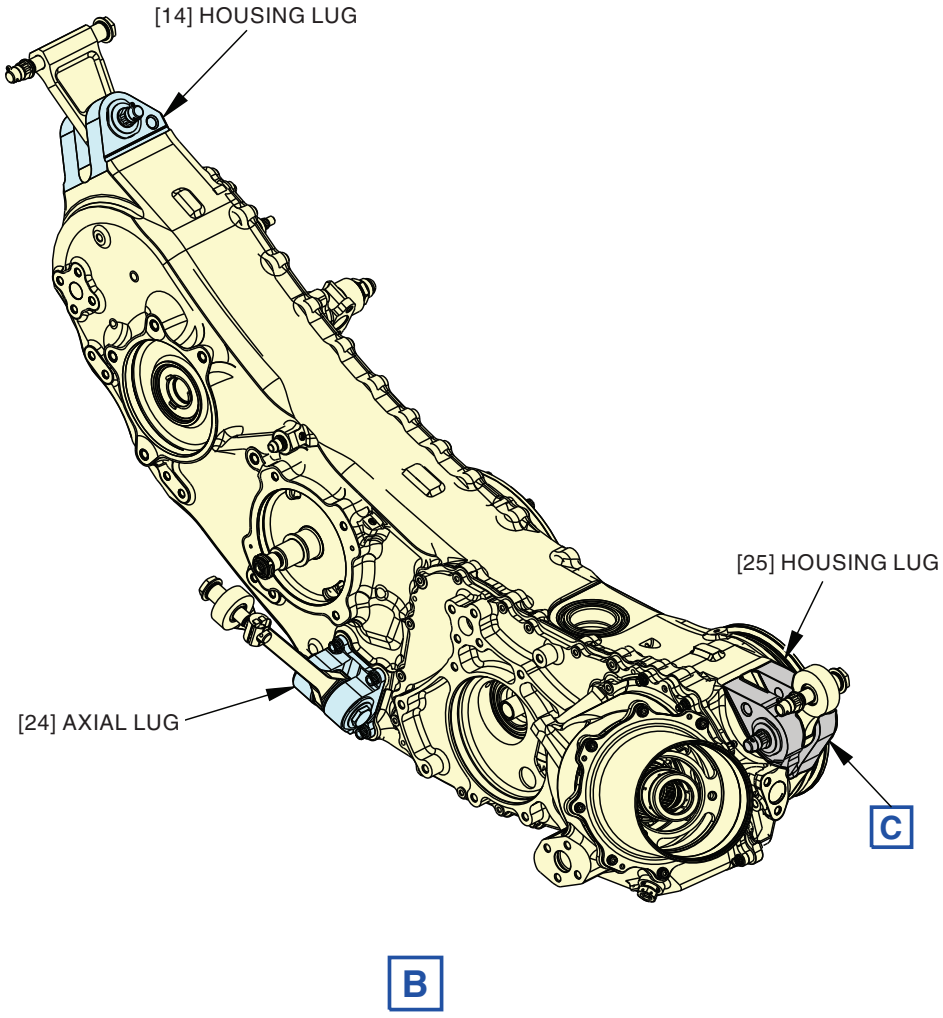
2812378 S0000647814_V1

Accessory Drives Damper Bearings Inspection
Figure 603/72-60-00-990-806-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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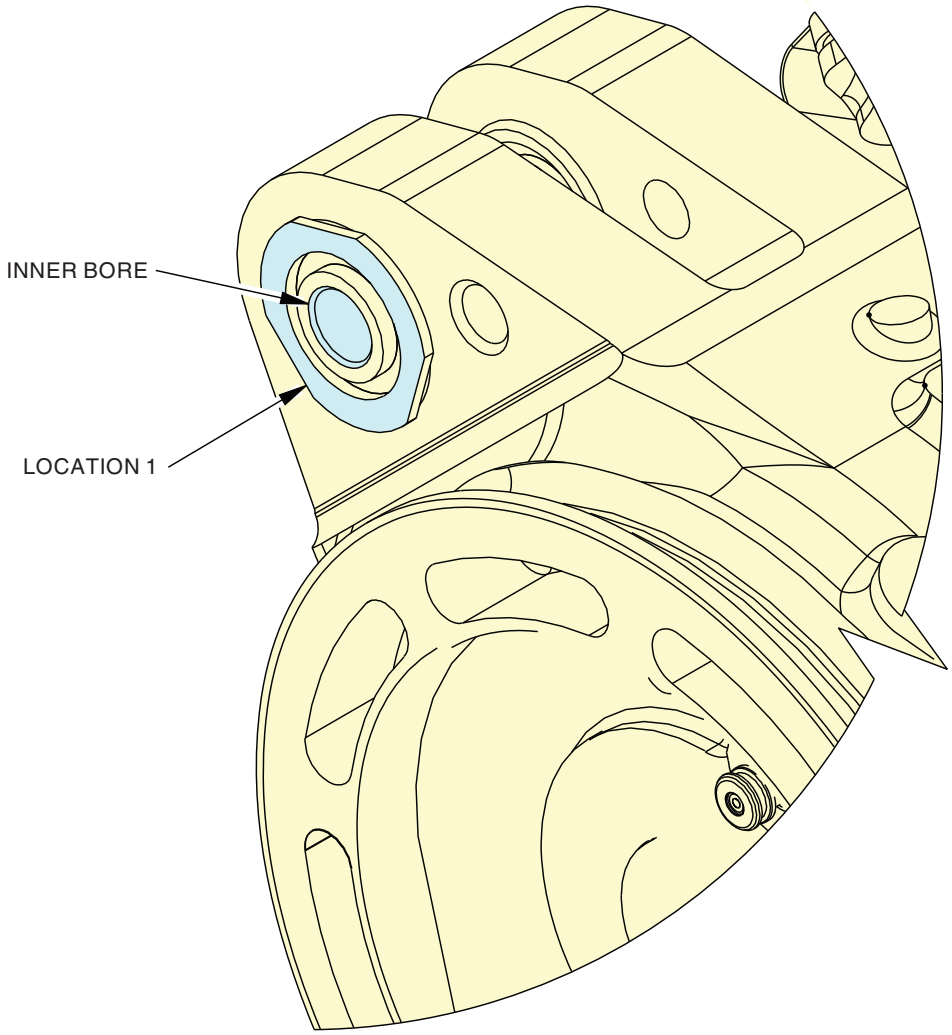
2812380 S0000647815_V1

Accessory Drives Damper Bearings Inspection
Figure 603/72-60-00-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Accessory Drives Damper Bearings Inspection
Figure 603/72-60-00-990-806-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRANSFER GEARBOX MODULE - REPAIR

1. General

A. This procedure has these tasks:

(1) A blending and spot touch-up of the transfer gearbox module

(2) A blending and spot touch-up on specific area of the transfer gearbox assembly

(3) A corrosion removal on the transfer gearbox assembly.

TASK 72-62-00-300-802-G00

2. Blending and Spot Touch-Up of the Transfer Gearbox Module

(Figure 801, Figure 802, Figure 803)

A. General

(1) This task gives the instructions to do the blending and spot touch-up of the Transfer Gearbox (TGB) Module, made up of TGB assembly only.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50295 [CP2010]	Solvent - cleaning, White Spirit	
C50203 [CP2640]	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	
C50208 [CP2659]	Coating - Chemical Conversion - Bonderite M-CR 1132 Aero Pen (Formerly Alodine 1132)	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Prepare for Blending and Spot Touch-Up

SUBTASK 72-62-00-860-004-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-62-00-860-005-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-00-010-002-G00

(3) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

H. Dimensional Information

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE CORRECT APPLICABLE GLOVES, FACEMASK, EYESHIELDS AND PROTECTIVE CLOTHES AT ALL TIMES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

WARNING ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE AN INJURY OR KILL YOU.

SUBTASK 72-62-00-320-012-G00

(1) Do not remove more parent material than necessary.

(a) Indications must be repaired without steps and sharp edges.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) The hand blending must cover all the damage area.

SUBTASK 72-62-00-320-013-G00

(2) Blend the area to repair as follows:

(a) Make sure that the quantity of blend repairs on:

1) The TGB housing [2] is less than or equal to 10.

2) The TGB bearing support [4] is less than or equal to10.

3) The TGB transfer tube [3] is less than or equal to 10.

(b) Make sure that the distance between two blends is equal to the width of maximum indication observed on repaired area.

1) If it is not, the indications must be included on one blend.

(c) Blending is not permitted on an area already blended.

(d) Obey the dimension limits of areas below:

1) On area H, the maximum finish dimension Z is equal to 0.0354 in. (0.90 mm).

a) The minimum description blending limit X is equal to 40 x dimension Z and minimum description blending limit Y is equal to 40 x dimension Z.

b) The maximum description blending limit X is equal to 45 x dimension Z and maximum description blending limit Y is equal to 45 x dimension Z.

2) On area K, the maximum finish dimension Z is equal to 0.0196 in. (0.50 mm).

a) The minimum description blending limit X is equal to 40 x dimension Z and minimum description blending limit Y is equal to 40 x dimension Z.

b) The maximum description blending limit X is equal to 45 x dimension Z and maximum description blending limit Y is equal to 45 x dimension Z.

3) On area J, the maximum finish dimension Z is equal to 0.0039 in. (0.10 mm).

a) The minimum description blending limit X is equal to 40 x dimension Z and minimum description blending limit Y is equal to 40 x dimension Z.

b) The maximum description blending limit X is equal to 45 x dimension Z and maximum description blending limit Y is equal to 45 x dimension Z.

(e) Blending is not allowed on the area G.

CAUTION

MAKE SURE THAT THE AREAS L, N AND P ON THE TRANSFER GEARBOX ARE SUBJECT TO HAVE A MINIMUM 0.118 INCH (3.0 MM) RESIDUAL THICKNESS WHEN YOU DO THE BLEND. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE ENGINE DAMAGE.

(f) Areas L, N, and P are subject to have a minimum 0.118 in. (3.0 mm) residual thickness after blending.

1) On area L the maximum finish dimension Z is equal to 0.0354 in. (0.90 mm), if only after blending the measured residual thickness of material is minimum 0.118 in. (3.0 mm).

a) The minimum description blending limit X is equal to 40 x dimension Z and minimum description blending limit Y is equal to 40 x dimension Z.

b) The maximum description blending limit X is equal to 45 x dimension Z and maximum description blending limit Y is equal to 45 x dimension Z.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) On area N the maximum finish dimension Z is equal to 0.0196 in. (0.50 mm), if only after blending the measured residual thickness of material is minimum 0.118 in. (3.0 mm).
   a) The minimum description blending limit X is equal to 40 x dimension Z and minimum description blending limit Y is equal to 40 x dimension Z.
   b) The maximum description blending limit X is equal to 45 x dimension Z and maximum description blending limit Y is equal to 45 x dimension Z.
3) On area P the maximum finish dimension Z is equal to 0.0039 in. (0.10 mm), if only after blending the measured residual thickness of material is minimum 0.118 in. (3.0 mm).
   a) The minimum description blending limit X is equal to 40 x dimension Z and the minimum description blending limit Y is equal to 40 x dimension Z.
   b) The maximum description blending limit X is equal to 45 x dimension Z and the maximum description blending limit Y is equal to 45 x dimension Z.
4) On locations edge M the maximum finish dimension Z is equal to 0.0236 in. (0.60 mm).
   a) The minimum description blending limit X on edge M is equal to 40 x dimension Z.
   b) The maximum description blending limit X on edge M is equal to 45 x dimension Z.
   c) The description blending limit Y on edge M is edge to edge.

I. Blending and Spot Touch-Up of the Transfer Gearbox Module

SUBTASK 72-62-00-320-019-G00

(1) Check for dimensions, locations, and quantities of indications on areas and compare with repair limits (Figure 801).

CAUTION

YOU MUST NOT BLEND THE PART IN NOT PERMITTED AREA. THIS CAN CAUSE DAMAGE TO THE PART.

(a) If the indications or repair area is more than the limits, the part is not repairable by this procedure.

(b) Blending is not allowed on the area G.

(c) Blending is not permitted on an area already blended.

(d) Make sure that the distance between two blends is equal to the width of maximum indication observed on repaired area.
   1) If it is not, the indications must be included on one blend.

SUBTASK 72-62-00-320-020-G00

(2) Remove the indications from the repair area by a smooth hand blending (Figure 802 and Figure 803, TASK 70-43-42-300-802-G00).

(a) If one blending impacts several areas, consider the lowest depth criteria.

(b) If the area to repair is impacted by matting imprints, make sure that the edges are blended into a smooth contour without increasing the depth of the matted area.

(c) If the area to repair is impacted by corrosion, make sure that the blended surface is less than or equal to the maximal dimensional limits detailed in Dimensional Information topic.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-62-00-120-002-G00

(3) Clean the repair area of the TGB module [1] with a brush or a cloth using Ardrox 5503, B50290 [CP2559], solvent, B00682 [CP2011], or White Spirit, B50295 [CP2010].

SUBTASK 72-62-00-020-001-G00

(4) In case of corrosion, do an inspection with, at least, a x3 binocular.

(a) If pitting is seen, refer to (D) (2).

SUBTASK 72-62-00-230-002-G00

(5) Do the applicable fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

(a) No cracks are permitted.

SUBTASK 72-62-00-350-002-G00

(6) Do a spot touch-up of oxide film coating on the repaired areas (TASK 70-43-07-300-801-G00).

(a) Use only Bonderite M-CR 1200S coating, C50203 [CP2640], and Bonderite M-CR 1132 Aero Pen, C50208 [CP2659].

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-00-410-003-G00

(1) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-62-00-410-004-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-00-860-006-G00

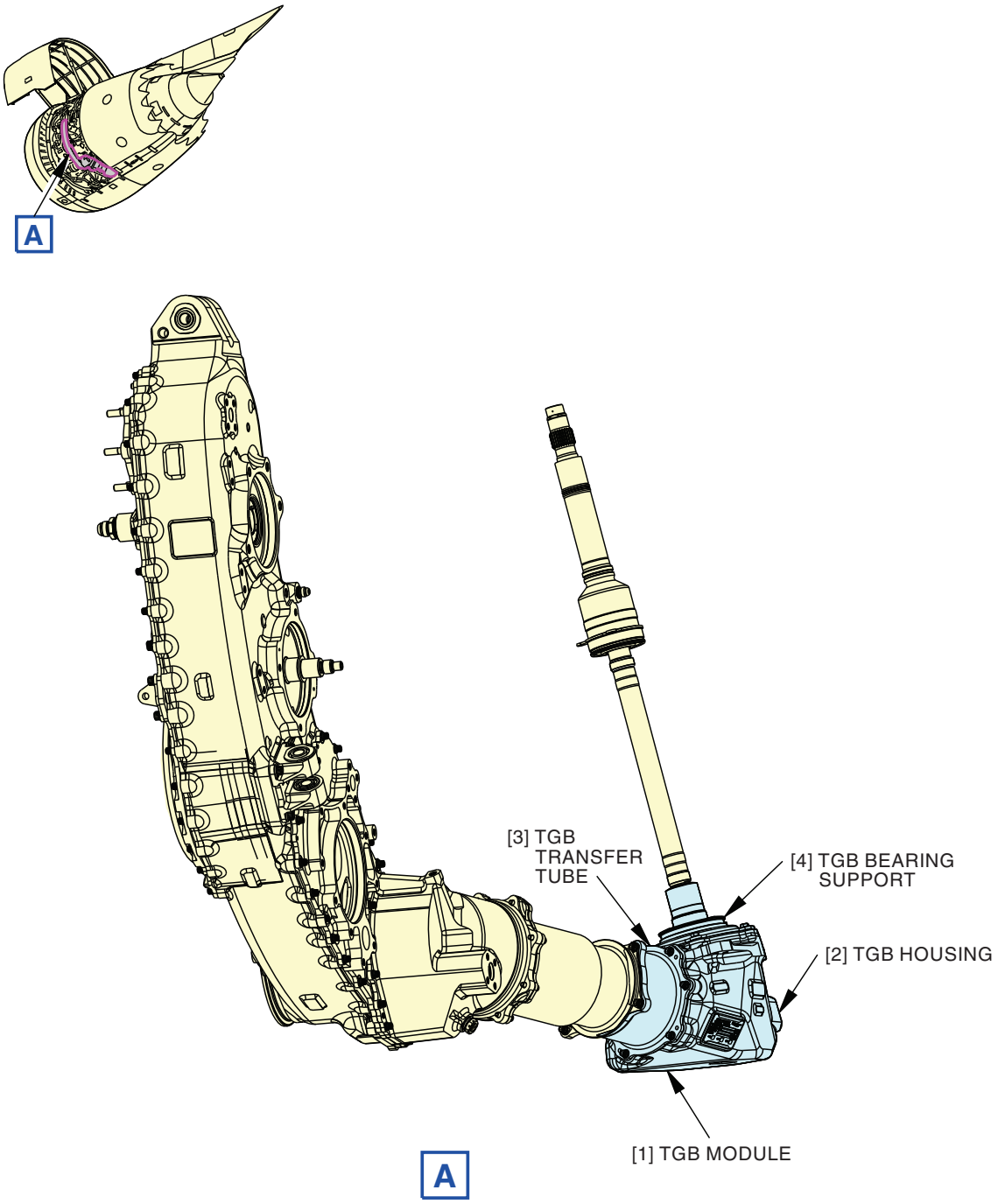
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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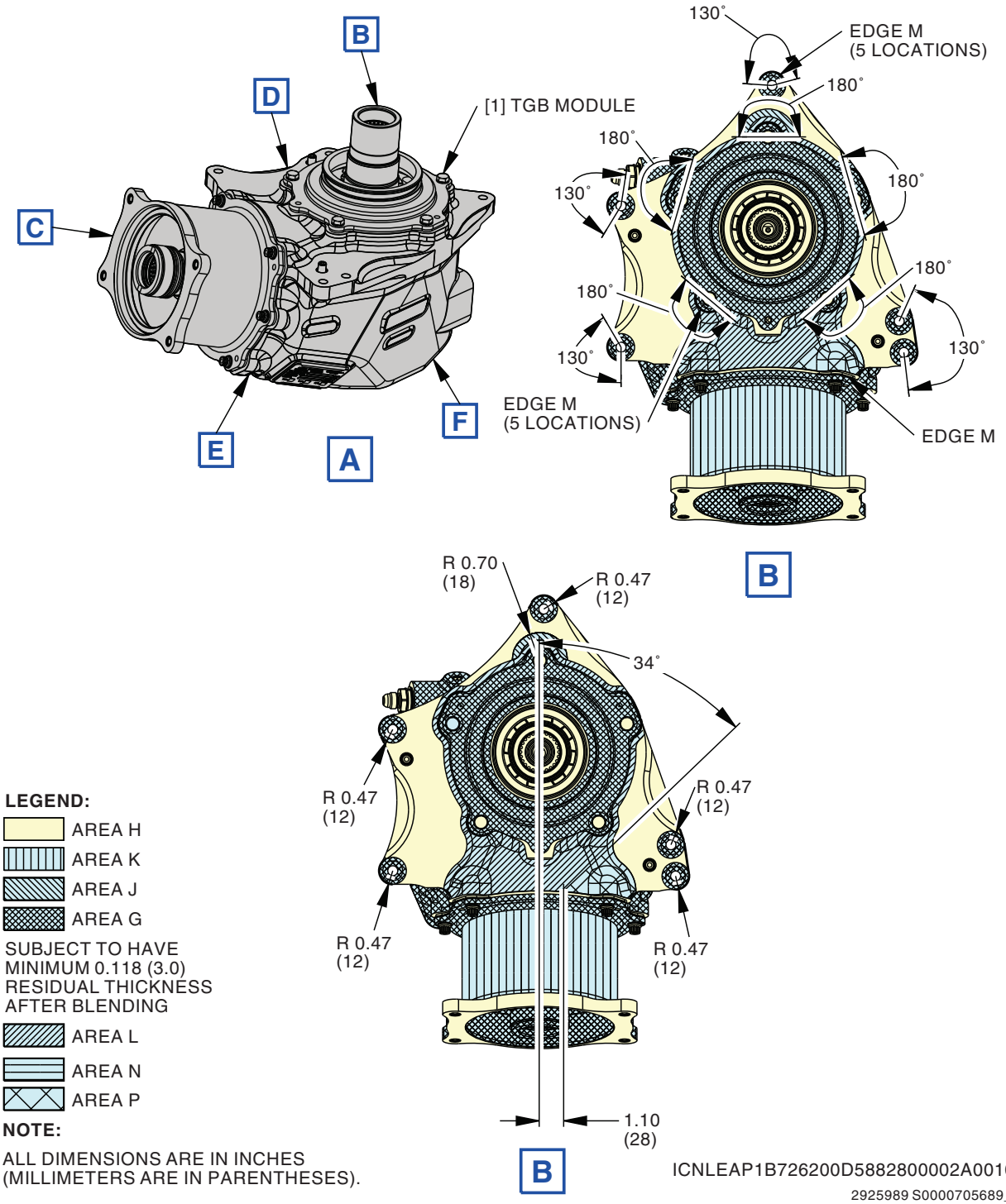
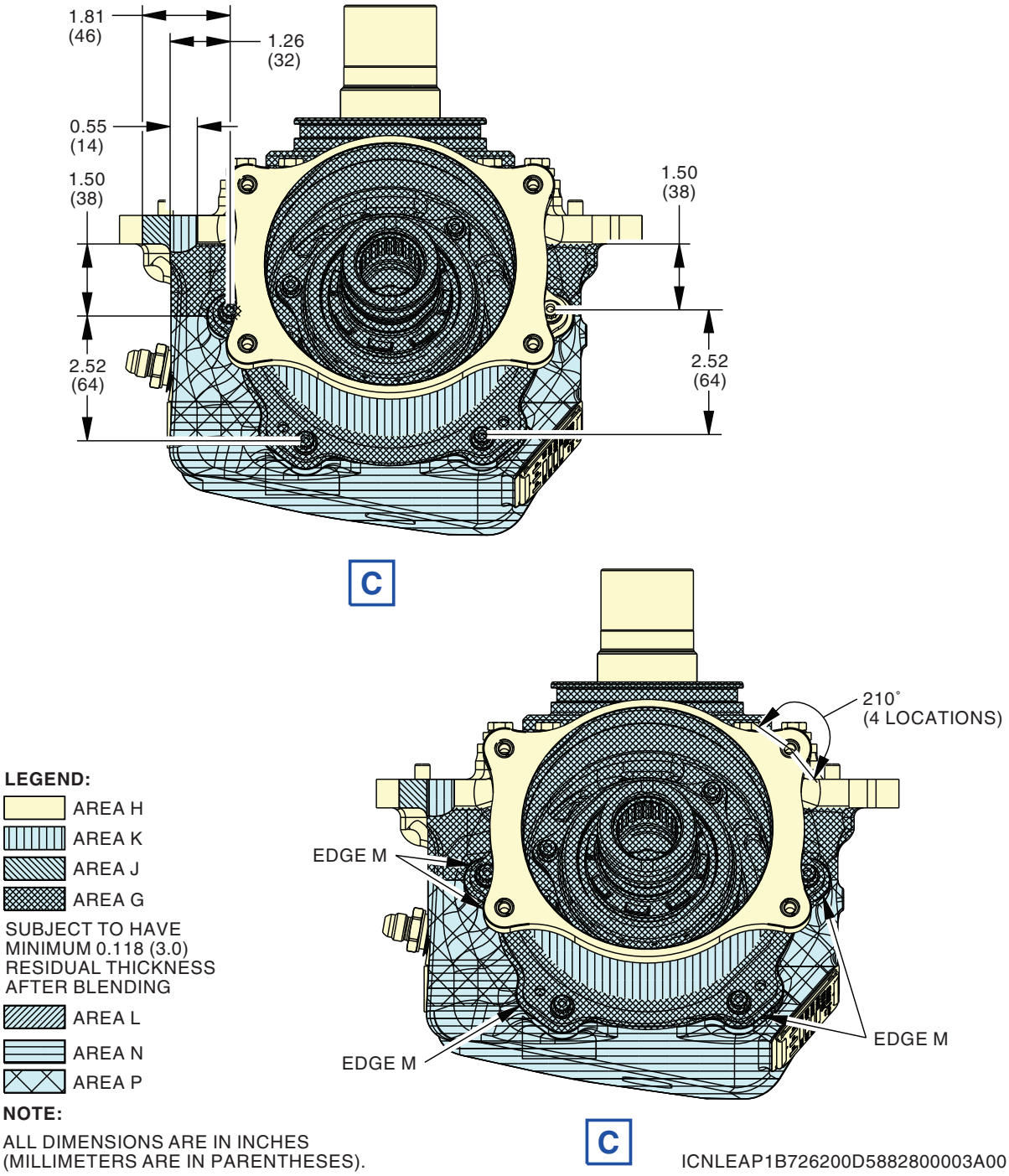
Blending and Spot Touch-Up of the Transfer Gearbox Module
Figure 801/72-62-00-990-804-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



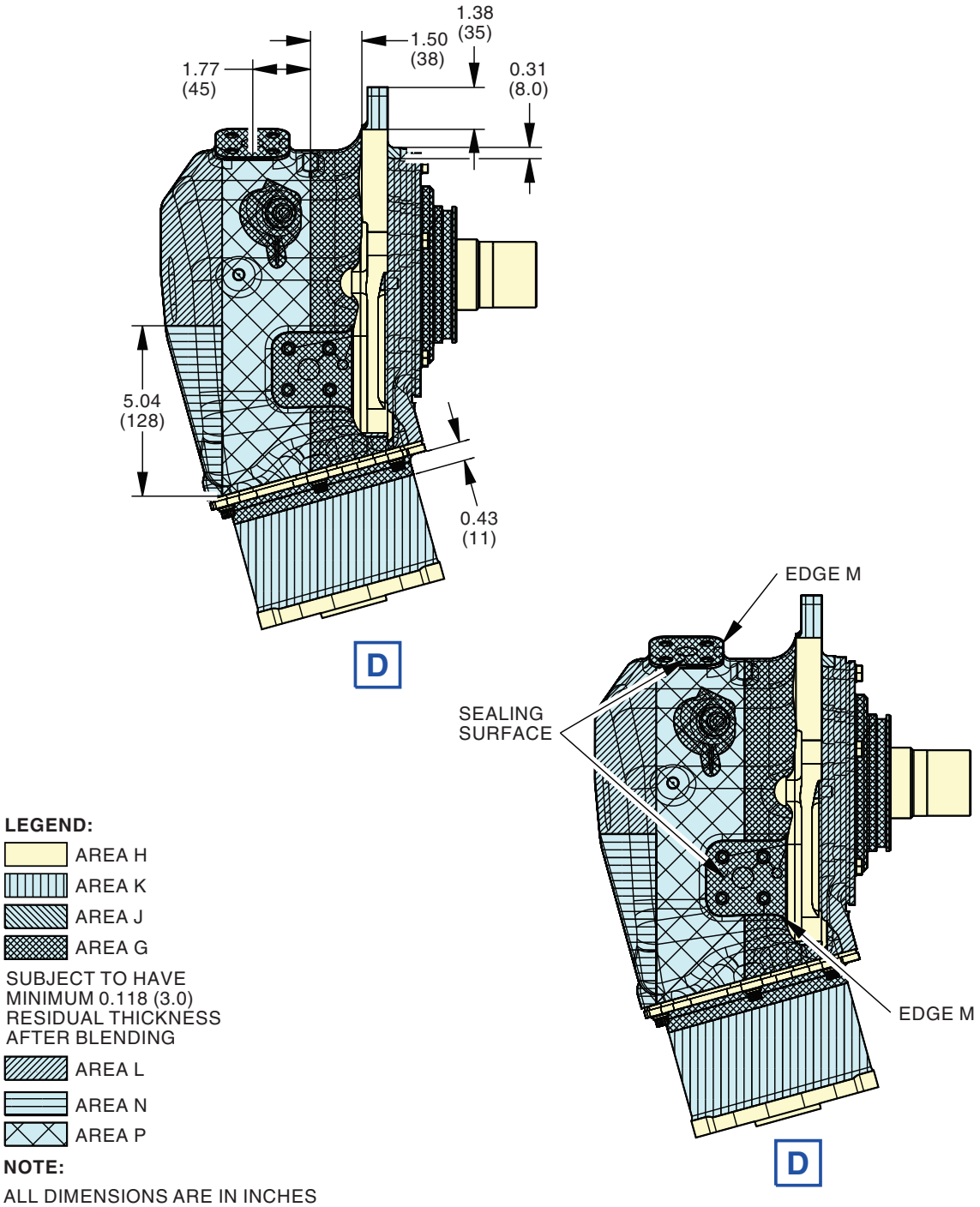
Blending and Spot Touch-Up of the Transfer Gearbox Module
Figure 801/72-62-00-990-804-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Blending and Spot Touch-Up of the Transfer Gearbox Module
Figure 801/72-62-00-990-804-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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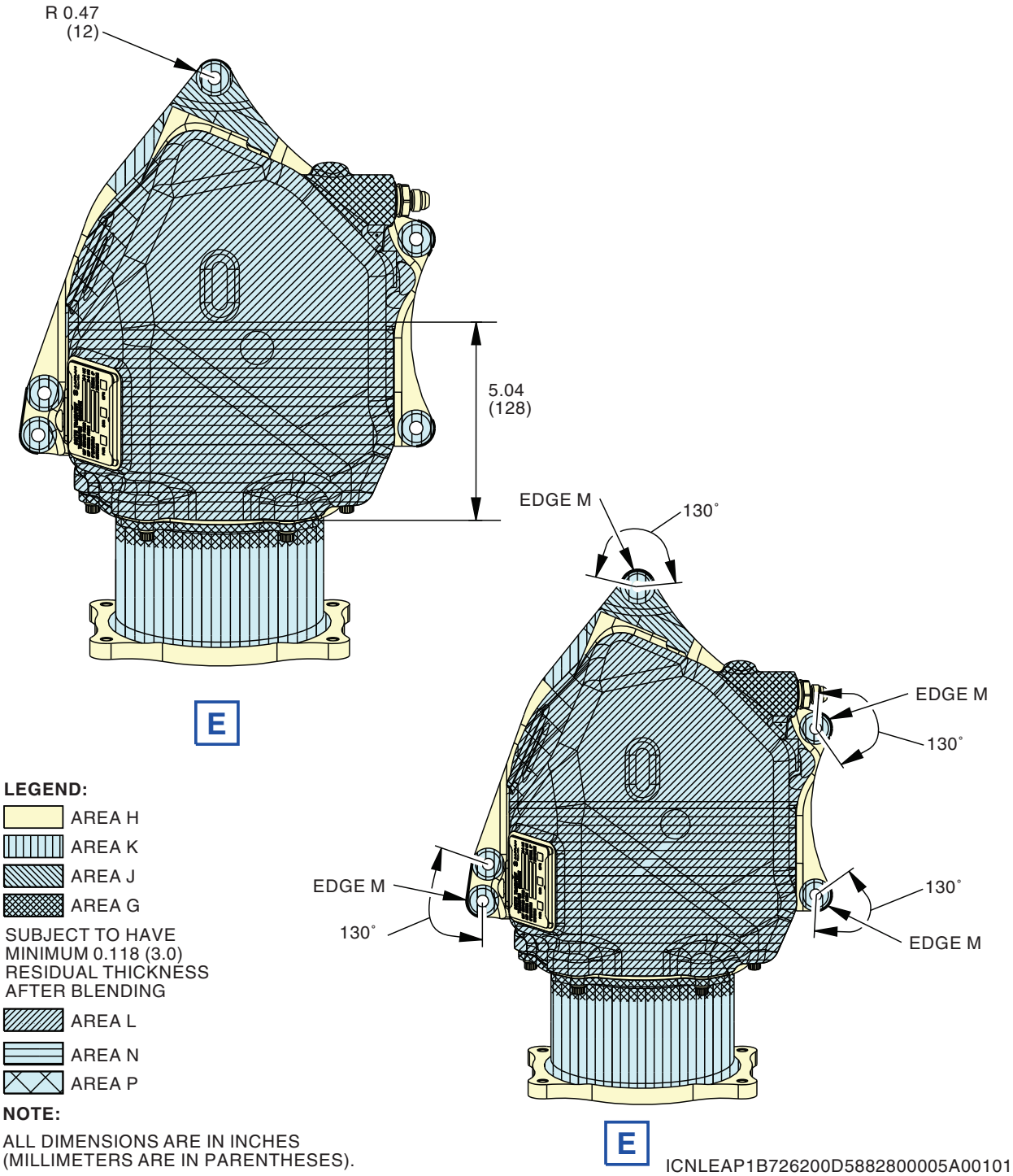
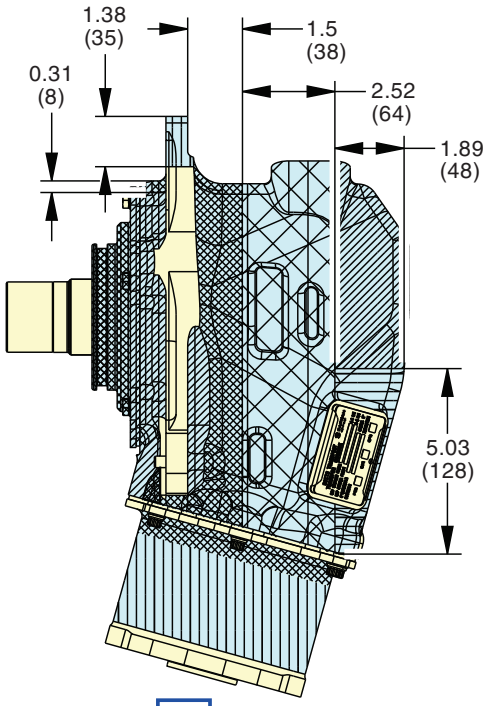
Blending and Spot Touch-Up of the Transfer Gearbox Module
Figure 801/72-62-00-990-804-G00 (Sheet 4 of 6)

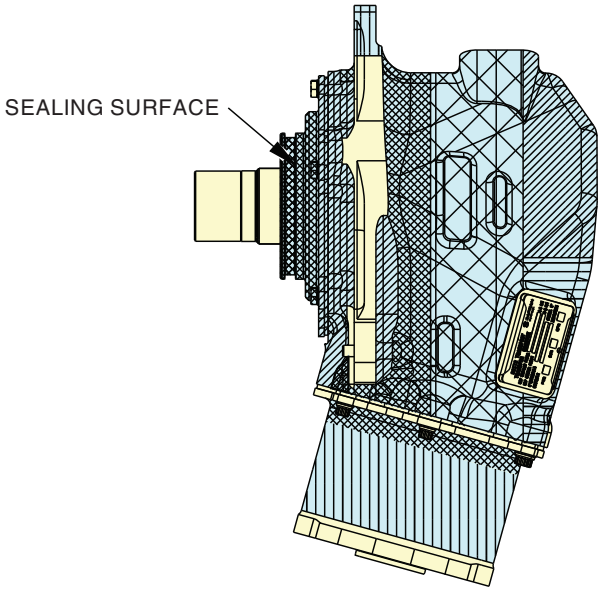
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Blending and Spot Touch-Up of the Transfer Gearbox Module
Figure 801/72-62-00-990-804-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

SUBJECT TO HAVE MINIMUM 0.118 (3.0) RESIDUAL THICKNESS AFTER BLENDING

- AREA L
- AREA N
- AREA P

NOTE:

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES).

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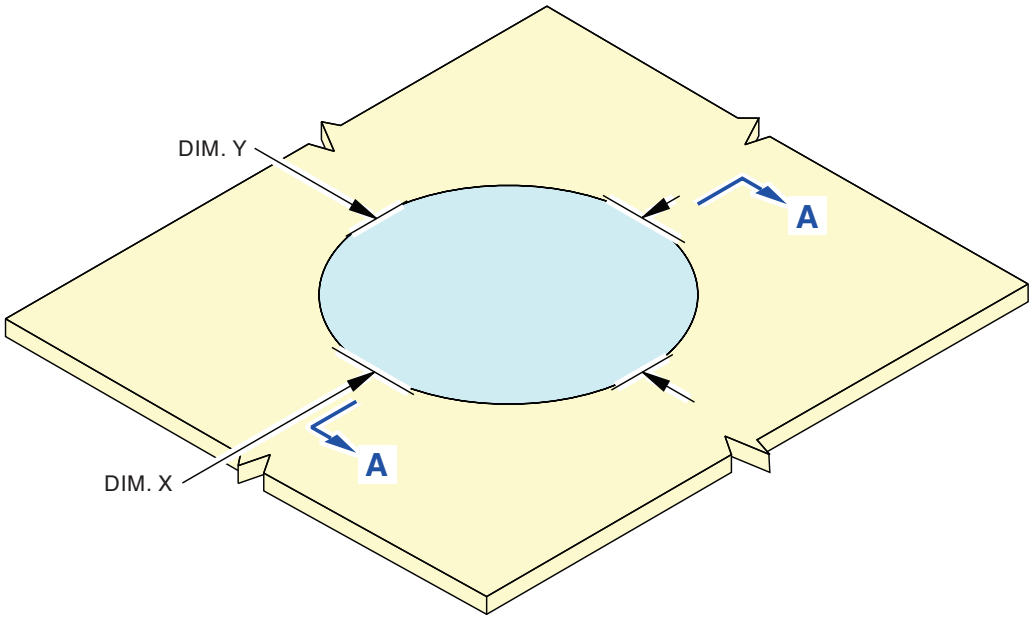
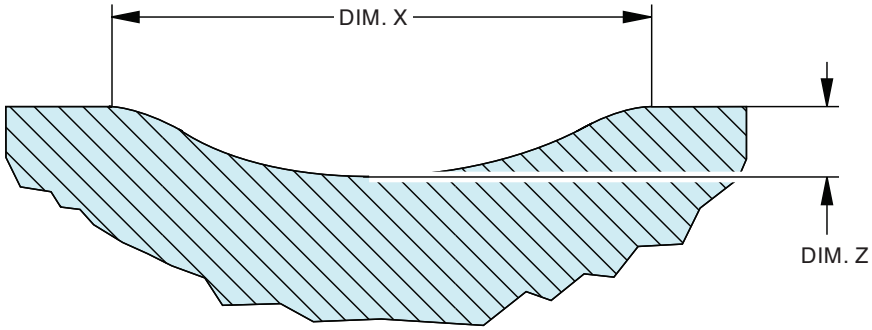
Blending and Spot Touch-Up of the Transfer Gearbox Module
Figure 801/72-62-00-990-804-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLENDING DIMENSIONS OF PLANE AREAS (EXAMPLE)



A-A

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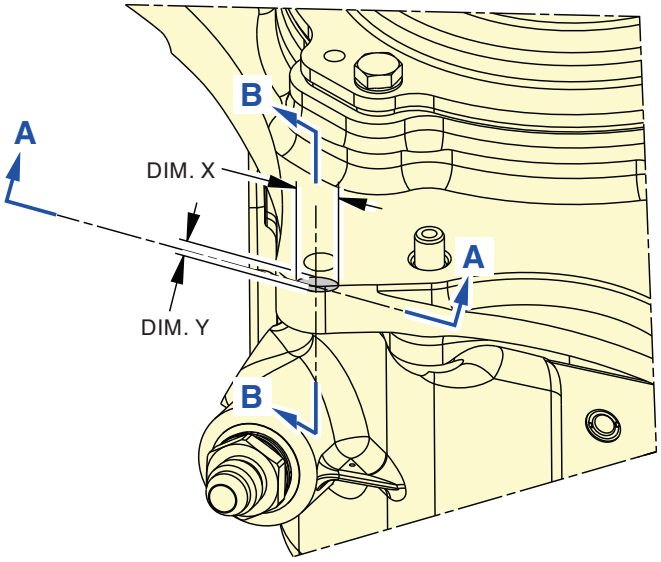
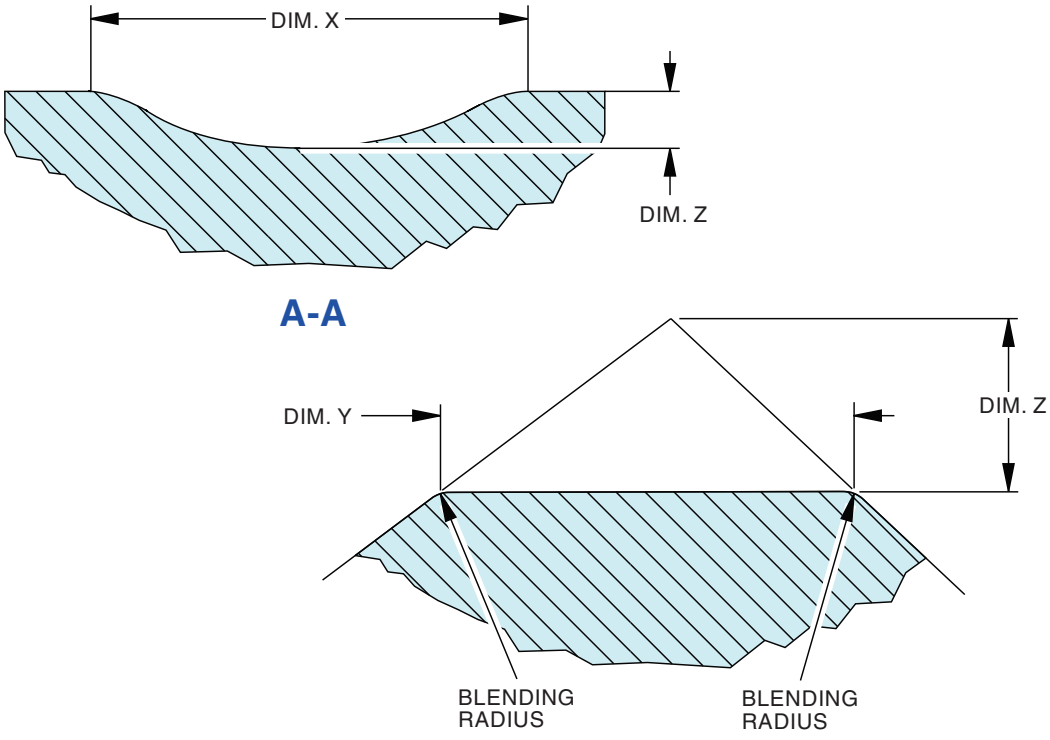
Blending Specific Dimension of Plane Areas of the Transfer Gearbox Module
Figure 802/72-62-00-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLENDING DIMENSIONS OF LOCATION EDGE M (EXAMPLE)



DIM. Y = BLENDING EDGE TO EDGE

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Blending Specific Dimension of Location Edge M of the Transfer Gearbox Module
Figure 803/72-62-00-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-62-00-300-803-G00

3. Blending and Spot Touch Up on Specific Area of the Transfer Gearbox Assembly

(Figure 804 and Figure 805)

A. General

(1) This task gives the instructions to do the blending and spot touch up on a specific area of the Transfer Gearbox (TGB) assembly.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-35-00-270-801-G00	Test for Cracks and Other Defects with Ultrasonic (P/B 601)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50295 [CP2010]	Solvent - cleaning, White Spirit	
C50203 [CP2640]	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	
C50208 [CP2659]	Coating - Chemical Conversion - Bonderite M-CR 1132 Aero Pen (Formerly Alodine 1132)	
G51481 [CP2241]	Tape - Adhesive, Cloth - Permacel 201	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Blending and Spot Touch Up

SUBTASK 72-62-00-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-62-00-860-008-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-00-860-011-G00

(3) Do not remove more parent material than it is necessary.

SUBTASK 72-62-00-300-001-G00

(4) Repair indications without steps and sharp edges.

SUBTASK 72-62-00-300-002-G00

(5) Cover all the damaged area with the hand blending.

G. Blending and Spot Touch Up on Specific Area of the Transfer Gearbox Assembly

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE INJURY OR DAMAGE TO EQUIPMENT.

SUBTASK 72-62-00-860-012-G00

(1) Obey the dimension limits of areas below:

(a) The maximum finish dimension Z is equal to 0.0787 in. (2.00 mm).

1) The minimum description blending limit X is equal to 1 x DIM. Z and minimum description blending limit Y is equal to 1 x DIM. Z.

2) The maximum description blending limit X is equal to 5 x DIM. Z and maximum description blending limit Y is equal to 5 x DIM. Z.

SUBTASK 72-62-00-020-002-G00

(2) Remove the indication on the TGB assembly [5] as follows:

(a) Check for dimension and location of the indication.

1) Make sure that the indication is located in the repairable area (Views C and D, Figure 804).

a) If the indication is out of the repairable area, the part is not repairable with this task.

2) Make sure that the indication depth is not more than the 0.0787 in. (2.00 mm).

a) If the indication depth is more than the limits, the part is not repairable with this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Use Permacel 201 tape, G51481 [CP2241], or locally made masks to mask all the holes, bores, and passageways adjacent to the area that you will repair.

(c) Remove the indication by a smooth hand blending or by a smooth power blending (Figure 805) (TASK 70-43-42-300-802-G00 or TASK 70-43-42-300-803-G00).

1) Make sure that the blending width and length are limited to the impacted area.

(d) Clean the TGB assembly [5] with a brush or cloth moist in acetone solvent, B01058 [CP1039], or White Spirit, B50295 [CP2010] (TASK 70-21-23-100-801-G00).

SUBTASK 72-62-00-320-021-G00

(3) Measure the blended area.

(a) Make sure that the blended area is in the dimensional limits.

SUBTASK 72-62-00-320-022-G00

(4) Measure the residual thickness.

(a) Do this task: Test for Cracks and Other Defects with Ultrasonic, TASK 70-35-00-270-801-G00.

(b) Measure the residual thickness of the impacted area.

NOTE: Take into account the tool measurement uncertainty.

(c) If the residual thickness of the impacted area is greater than or equal to 0.118 in. (3.0 mm), the part is considered serviceable.

(d) If the residual thickness of the impacted area is lower than 0.118 in. (3.0 mm), the part is considered non serviceable.

SUBTASK 72-62-00-230-003-G00

(5) Do an Fluorescent Penetrant Inspection (FPI) of the repaired areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

(a) Do an FPI sensitivity level 2.

(b) Indications less than 0.0118 in. (0.30 mm) are non-interpretable.

(c) Indications more than 0.0118 in. (0.30 mm) are not permitted.

(d) Linear indications are not permitted.

NOTE: Linear indications are indications that are at least four times longer than they are wide.

SUBTASK 72-62-00-350-003-G00

(6) Do a spot touch-up of oxide film coating on the repaired areas (TASK 70-43-07-300-801-G00).

(a) Use only Bonderite M-CR 1200S coating, C50203 [CP2640], or Bonderite M-CR 1132 Aero Pen, C50208 [CP2659].

SUBTASK 72-62-00-420-001-G00

CAUTION

MAKE SURE THAT ALL THE HOLES, OPENINGS, BORES AND PASSAGEWAYS DO NOT HAVE MASKING TAPE OR OTHER UNWANTED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(7) Remove all masking tape and/or locally made masks.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-00-860-009-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-00-860-010-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

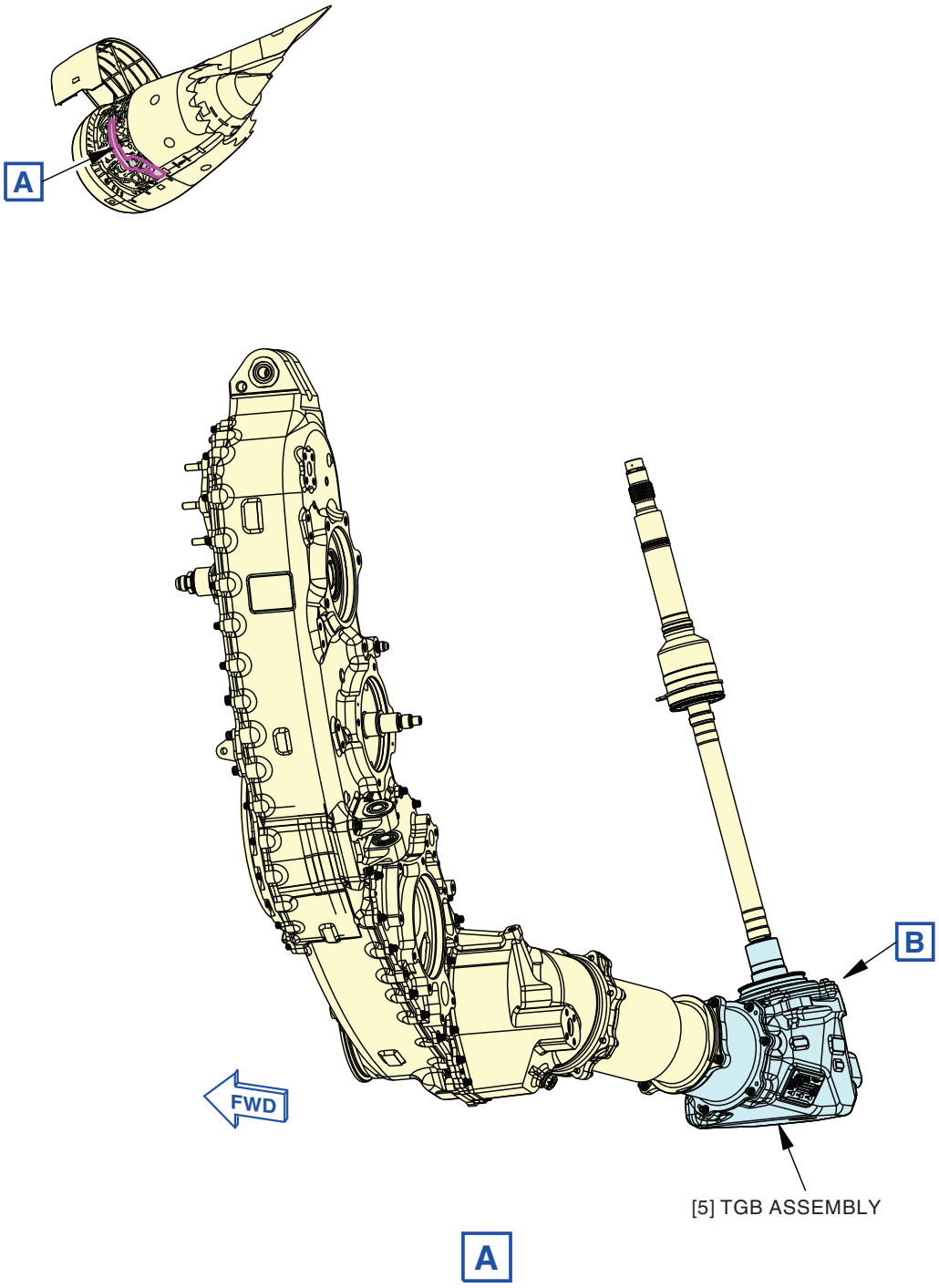
(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-62-00-100-001-G00

(3) Clean the work area and clear the tools and other items.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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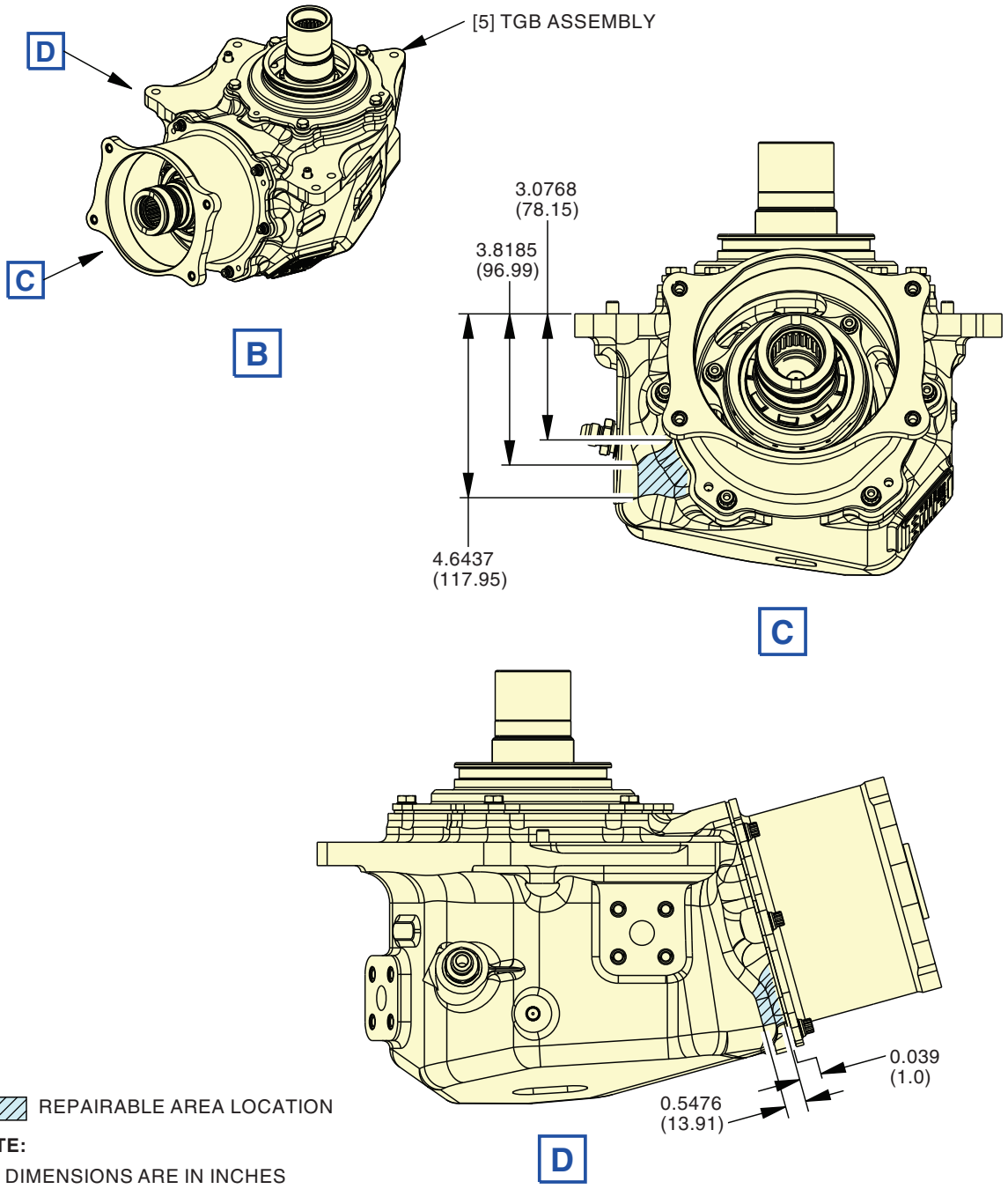
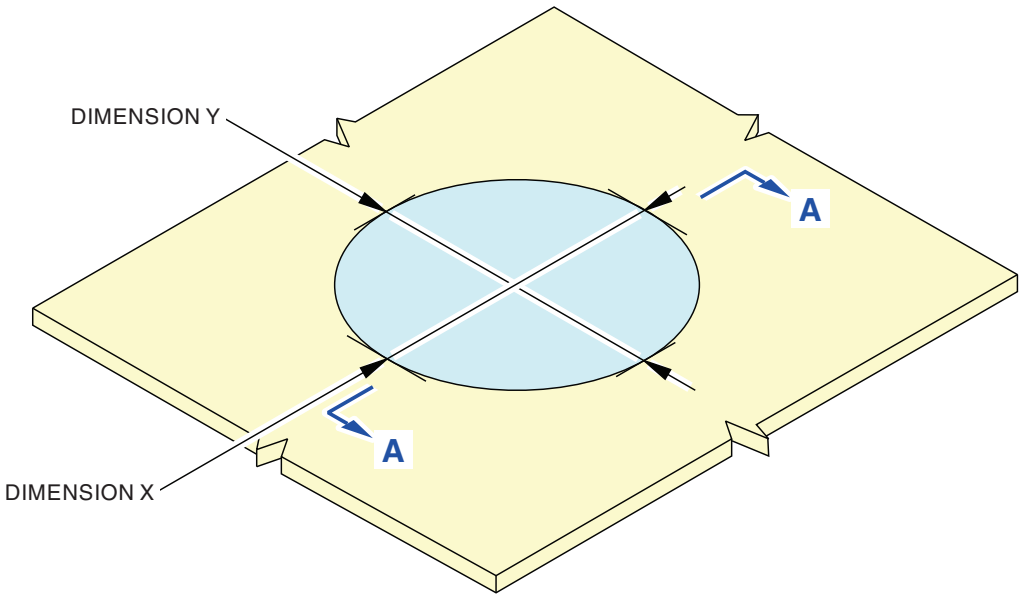
Blending and Spot Touch Up on Specific Area of the Transfer Gearbox Assembly
Figure 804/72-62-00-990-807-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

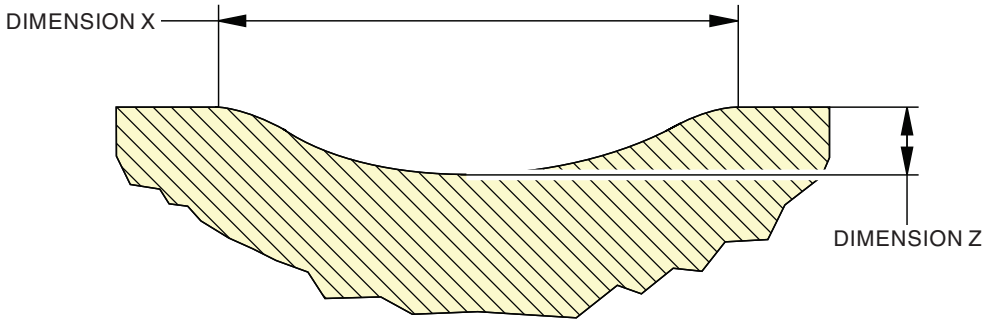


Blending and Spot Touch Up on Specific Area of the Transfer Gearbox Assembly
Figure 804/72-62-00-990-807-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLENDING DIMENSIONS OF PLANE AREAS (EXAMPLE)



A-A

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Blending Dimensions of Plane Areas
Figure 805/72-62-00-990-808-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-62-00-300-804-G00

4. Transfer Gearbox Assembly Corrosion Removal

(Figure 806)

A. General

(1) This task gives the instructions to remove the corrosion on the Transfer Gearbox (TGB) Assembly.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
70-44-50-320-801-G00	Drilling (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50291 [CP2560]	Solvent - cleaning - BONDERITE C-SO IND 79 AERO (TURCO IND79)	
B50295 [CP2010]	Solvent - cleaning, White Spirit	
C50203 [CP2640]	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	
C50208 [CP2659]	Coating - Chemical Conversion - Bonderite M-CR 1132 Aero Pen (Formerly Alodine 1132)	
C50422 [CP2926]	Pen - SurTec 650	
G51481 [CP2241]	Tape - Adhesive, Cloth - Permacel 201	

E. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
423	Engine 2 - Fan Cowl, Left

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Corrosion Removal

SUBTASK 72-62-00-010-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-62-00-020-003-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-00-320-023-G00

(3) Do not remove more parent material than it is necessary.

SUBTASK 72-62-00-320-024-G00

(4) Repair indications without steps and sharp edges.

SUBTASK 72-62-00-320-025-G00

(5) Cover all the damaged area with the hand blending.

H. Transfer Gearbox Assembly Corrosion Removal

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE CORRECT APPLICABLE GLOVES, FACEMASK, EYESHIELDS AND PROTECTIVE CLOTHES AT ALL TIMES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

WARNING ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE AN INJURY OR KILL YOU.

SUBTASK 72-62-00-320-026-G00

(1) Make sure that the corrosion area is not more than the 0.71 in. (18.03 mm) diameter.

(a) If the corrosion area is out of the limits, the part is not repairable with this procedure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-62-00-320-027-G00

(2) Obey the dimension limits as follows:

(a) Dia A must be between 0.3465 in. (8.80 mm) and 0.3582 in. (9.10 mm).

(b) Dim. B must be between 0.583 in. (14.81 mm) and 0.598 in. (15.19 mm).

(c) Maximum finish dimension of Dia C is 0.4035 in. (10.25 mm).

(d) Dim. D must be between 0.0158 in. (0.40 mm) and 0.0236 in. (0.60 mm).

(e) Maximum finish dimension of Dim. E is 0.393 in. (9.98 mm).

(f) Angle F must be between 88 degrees and 92 degrees.

(g) Minimum finish dimension of Radius U is 0.157 in. (3.99 mm).

SUBTASK 72-62-00-320-028-G00

(3) Make sure that the affected area with corrosion is in the repairable area (View B, Figure 806).

SUBTASK 72-62-00-320-029-G00

WARNING

METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

(4) Mask all the holes, bores, and passageways adjacent to the area that you will repair.

(a) Use Permacel 201 tape, G51481 [CP2241], or locally made masks.

(b) If there is corrosion around the screw hole, remove the corrosion by a smooth hand blending (TASK 70-43-42-300-802-G00 or TASK 70-43-42-300-803-G00) and as follows:

1) Clean the TGB assembly with a brush or a cloth moist in solvent, B00682 [CP2011], or White Spirit, B50295 [CP2010], or Ardrox 5503, B50290 [CP2559], or alcohol, B00676 [CP1041], or BONDERITE C-SO IND 79 AERO solvent, B50291 [CP2560].

2) Do a visual examination.

a) Make sure that there is no corrosion.

3) Do a dimensional inspection.

a) Make sure that the residual thickness (Dim. B) is within dimensional limits (View A-A, Figure 806).

(c) If there is corrosion in the screw hole, follow the instructions below:

1) Rebore the hole to remove the corrosion respecting dimensional limits (View A-A, View C, View D, Figure 806), (TASK 70-44-50-320-801-G00).

2) Clean the repaired area with a brush or a cloth moist in solvent, B00682 [CP2011], or White Spirit, B50295 [CP2010], or Ardrox 5503, B50290 [CP2559], or alcohol, B00676 [CP1041], or BONDERITE C-SO IND 79 AERO solvent, B50291 [CP2560].

3) Do a visual examination.

a) Make sure that there is no corrosion.

b) If there is residual corrosion in the hole out of the rebored area, remove the corrosion by a smooth hand blending (TASK 70-43-42-300-802-G00 or TASK 70-43-42-300-803-G00).

4) Do a dimensional inspection (View A-A, View C, View D, Figure 806).

SUBTASK 72-62-00-320-030-G00

(5) Do a fluorescent penetrant inspection of the area repaired area of the TGB assembly (TASK 70-11-06-230-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Sensitivity level 2.

(b) Indications less than 0.0118 in. (0.30 mm) are non-interpretable.

(c) Indications more than 0.0118 in. (0.30 mm) are not permitted.

(d) Linear indications are not permitted.

NOTE: Linear indications are indications that are at least four times longer than they are wide.

(e) Do a spot touch-up of oxide film coating on the repaired area (TASK 70-43-07-300-801-G00).

1) Use only Bonderite M-CR 1200S coating, C50203 [CP2640], Bonderite M-CR 1132 Aero Pen, C50208 [CP2659], or oxide film pen, C50422 [CP2926].

SUBTASK 72-62-00-320-031-G00

CAUTION

MAKE SURE THAT ALL THE HOLES, OPENINGS, BORES AND PASSAGEWAYS DO NOT HAVE MASKING TAPE OR OTHER UNWANTED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE PARTS CAN OCCUR.

(6) Remove the masking tape and/or locally made masks.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-00-320-032-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-00-320-033-G00

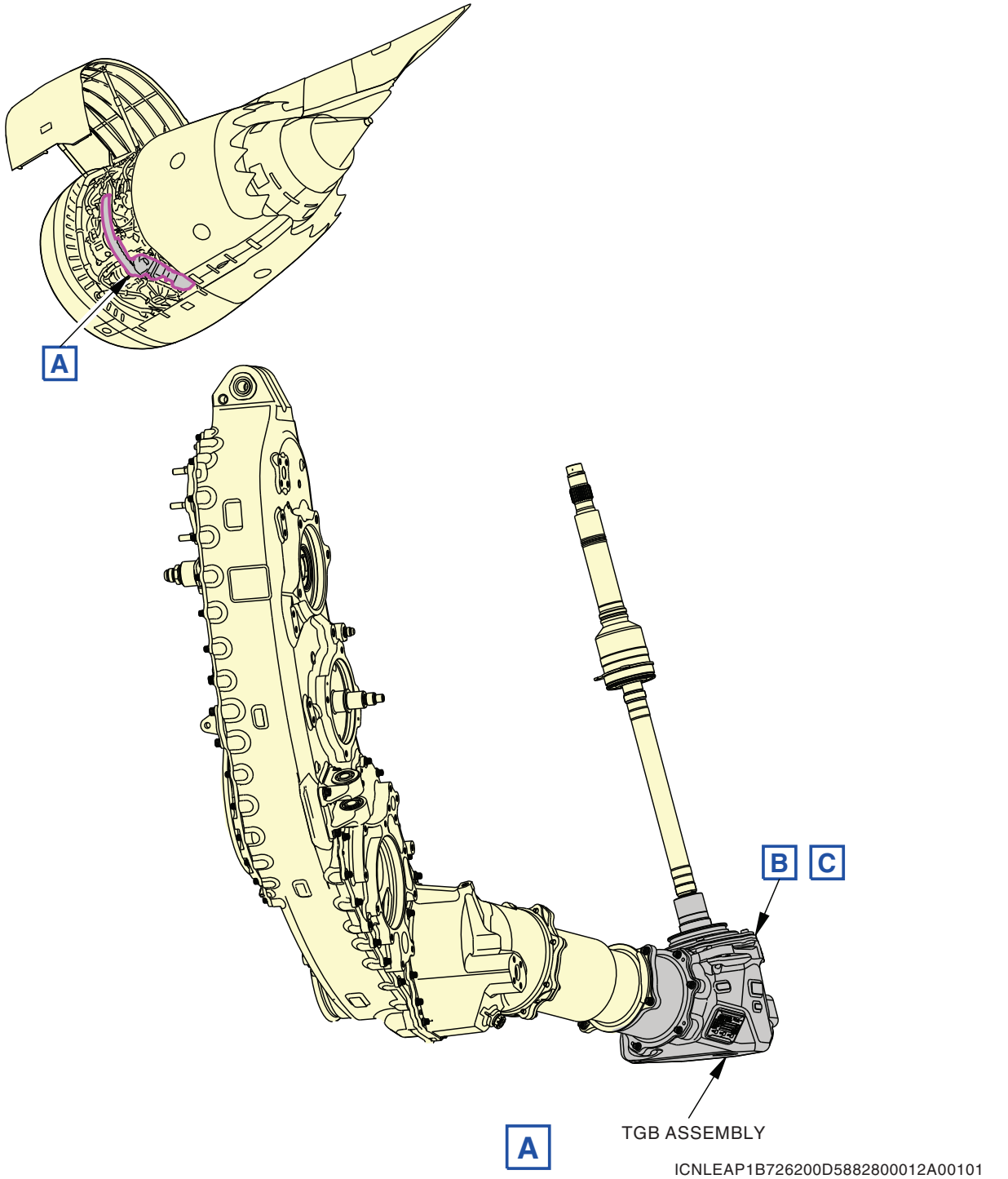
(2) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

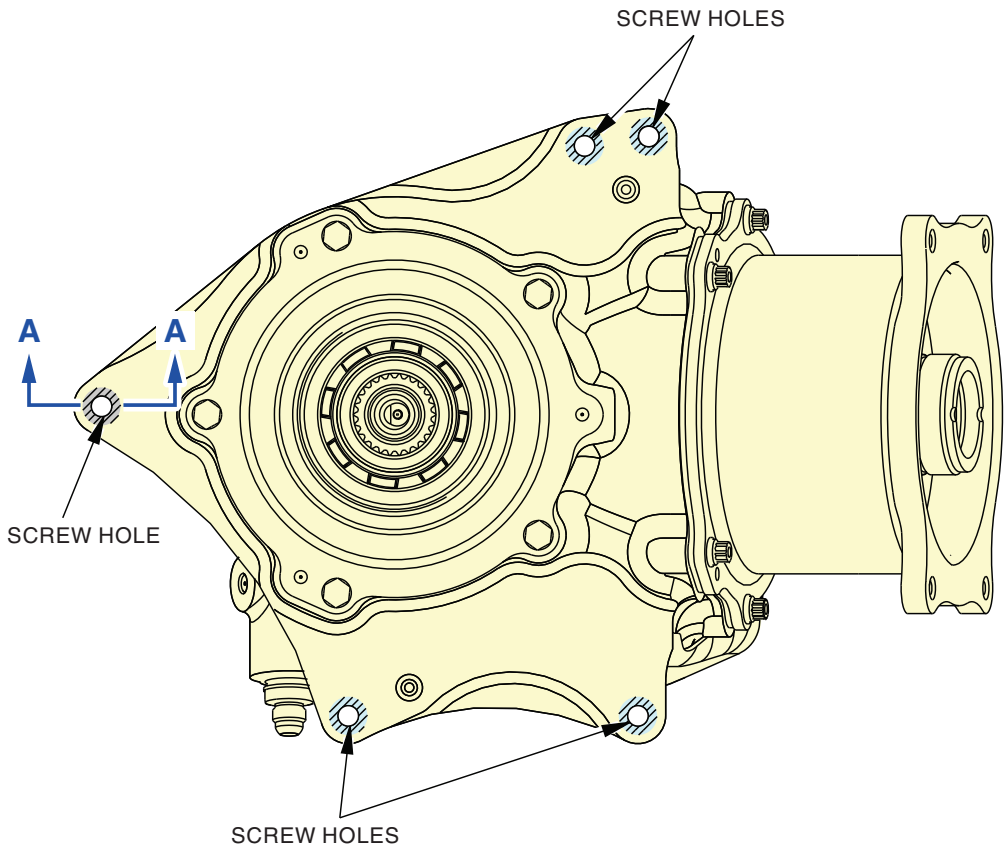
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Assembly Corrosion Removal
Figure 806/72-62-00-990-809-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

LEGEND:

REPAIRABLE AREA DIA 0.78 (20)

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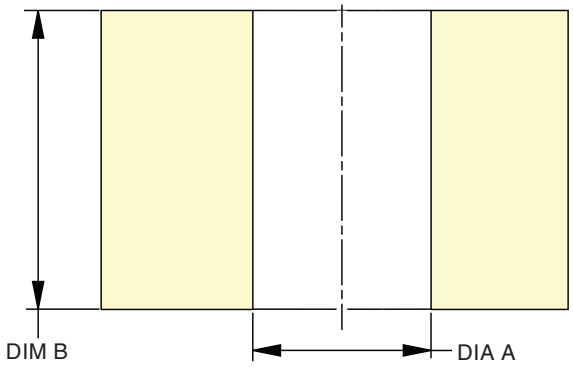
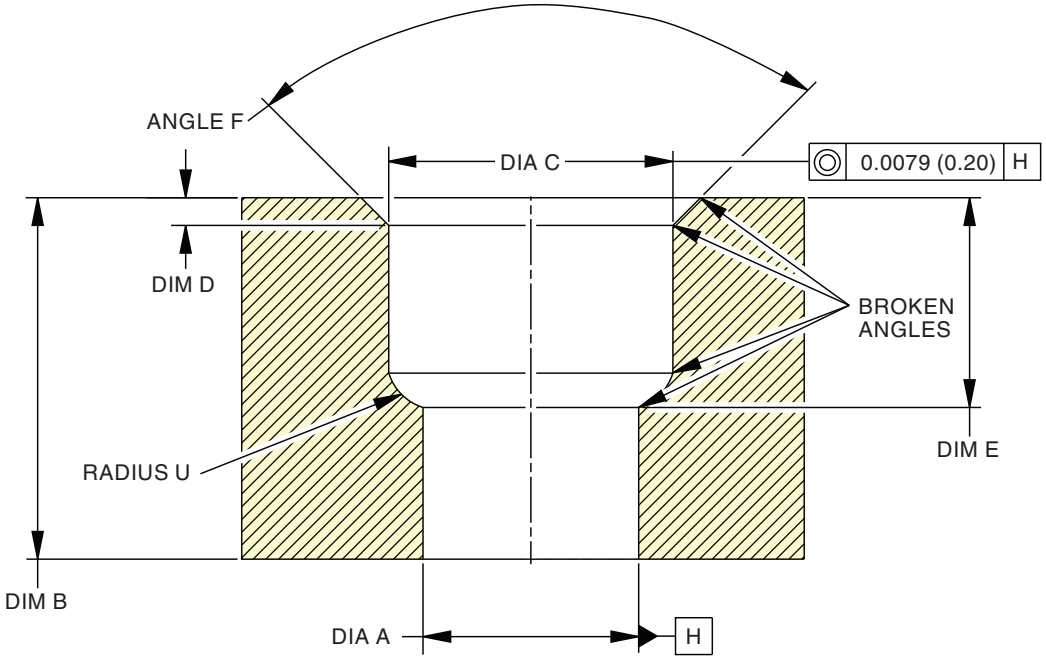
Transfer Gearbox Assembly Corrosion Removal
Figure 806/72-62-00-990-809-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BEFORE REPAIR
TYPICAL VIEW
A-A



AFTER REPAIR
TYPICAL VIEW
A-A

NOTE:

ROUGHNESS ARE IN MICROMETERS WITH MICROINCHES IN PARENTHESES.

ALL DIMENSIONS ARE IN INCHES (mm).

MAX. SURFACE ROUGHNESS: 3.2 (126).

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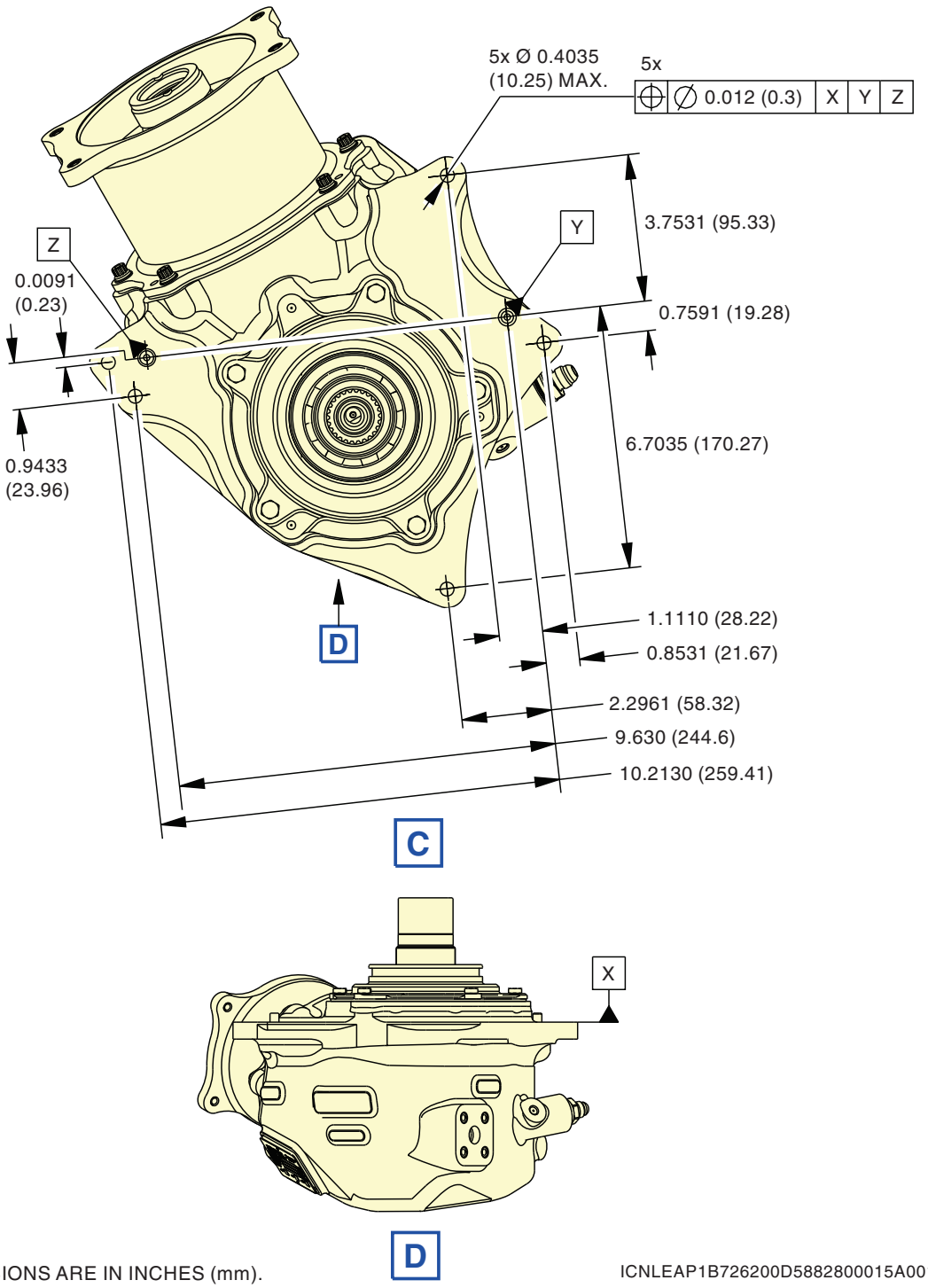
3052838 S0000817637_V1

Transfer Gearbox Assembly Corrosion Removal
Figure 806/72-62-00-990-809-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:
ALL DIMENSIONS ARE IN INCHES (mm).

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Transfer Gearbox Assembly Corrosion Removal
Figure 806/72-62-00-990-809-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRANSFER GEARBOX ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the transfer gearbox assembly

(2) An installation of the transfer gearbox assembly.

TASK 72-62-03-000-801-G00

2. Transfer Gearbox Assembly Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the transfer gearbox assembly.

(2) The figures has been updated to introduce POST SB 72-0021 configuration without the air tube and POST-SB 72-0181 configuration of the Fan Frame Shroud. The instructions are not changed.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-62-06-000-801-G00	Radial Shaft Removal (P/B 401)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14917	Fixture - TGB Removal Part #: 956A8076G02 Supplier: 58828
SPL-14918	Wrench - TGB Shaft Nut Part #: 956A8684G01 Supplier: 58828
SPL-14919	Tool - Sliding Tube, Assembly/Disassembly Part #: 956A8650G01 Supplier: 58828 Opt Part #: 956A8685G03 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 72-62-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-62-03-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-03-010-001-G00

(3) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

G. Transfer Gearbox Assembly Removal

SUBTASK 72-62-03-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the transfer gearbox assembly [1] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the oil resistant container (5 gal)(19 Liter), STD-1055, below the transfer gearbox assembly [1].

(b) Disconnect the oil tube [5] from the transfer gearbox assembly [1] as follows:

1) Loosen the coupling nut [6] of the oil tube [5] from the transfer gearbox assembly [1].

2) Disconnect the oil tube [5] from the transfer gearbox assembly [1].

3) Let the oil drain into the oil resistant container (5 gal)(19 Liter), STD-1055.

(c) Disconnect the oil tube [8] from the transfer gearbox assembly [1] as follows:

1) Remove the four bolts [9] from the oil tube [8].

2) Disconnect the oil tube [8] from the transfer gearbox assembly [1].

3) Let the oil drain into the oil resistant container (5 gal)(19 Liter), STD-1055.

4) Remove and discard the gasket C seal [10] from the oil tube [8].

(d) Disconnect the oil tube [11] from the transfer gearbox assembly [1] as follows:

1) Remove the four bolts [12] from the oil tube [11].

2) Disconnect the oil tube [11] from the transfer gearbox assembly [1].

3) Let the oil drain into the oil resistant container (5 gal)(19 Liter), STD-1055.

4) Remove and discard the gasket C seal [13] from the oil tube [11].

(e) Install the Engine core lock adapter, SPL-14920, on the AGB manual control pad to lock the N2 rotor.

(f) Disconnect the transfer gearbox transfer shaft [17] from the transfer gearbox assembly [1] as follows:

1) Remove the four bolts [15] that attach the transfer gearbox slider [4] to the transfer gearbox assembly [1].

2) Remove the two retainer segments [14].

3) Install the assembly/disassembly sliding tube tool, SPL-14919, on the transfer gearbox slider [4].

4) Move the transfer gearbox slider [4] towards the AGB transfer tube [3] with the assembly/disassembly sliding tube tool, SPL-14919.

5) Loosen the locknut [18] that attaches the transfer gearbox transfer shaft [17] to the transfer gearbox assembly [1] with the shaft nut wrench, SPL-14918.

6) Move the locknut [18] and transfer gearbox transfer shaft [17] towards the AGB transfer tube [3].

7) Remove and discard the slider seal [16] from the groove of the transfer gearbox slider [4].

(g) Unlock the N2 rotor through the accessory gearbox cover.

(h) Remove the transfer gearbox assembly [1] from the fan frame [23] as follows:

1) Install the TGB fixture, SPL-14917, assembly fixture on the transfer gearbox assembly [1].

2) Remove the five bolts [19] and five washers [20] that attach the transfer gearbox assembly [1] to the fan frame [23].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING A MINIMUM OF TWO PERSONS ARE NECESSARY TO HANDLE THE TGB DURING THE REMOVAL PROCEDURE. THE WEIGHT OF THE TGB IS APPROXIMATELY 36.4 LB (16.5 KG). INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR

WARNING MOVE THE TRANSFER GEARBOX ASSEMBLY SLOWLY. THE TRANSFER GEARBOX ASSEMBLY IS HEAVY, INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR

3) Move the transfer gearbox assembly [1] partially out of the fan frame [23].

   NOTE: The transfer gearbox assembly weighs 34.1 lb (15.5 kg).

   NOTE: When you remove the transfer gearbox assembly, make sure that you hold the radial shaft in its place with one hand.

4) Disconnect the transfer gearbox radial shaft [22] from the transfer gearbox assembly [1].

5) Move the transfer gearbox radial shaft [22] into the fan frame [23].

6) Remove the transfer gearbox assembly [1] fully out of the fan frame [23].

7) Do this task: Radial Shaft Removal, TASK 72-62-06-000-801-G00.

8) Remove and discard the preformed packing [21] from the groove of the transfer gearbox assembly [1].

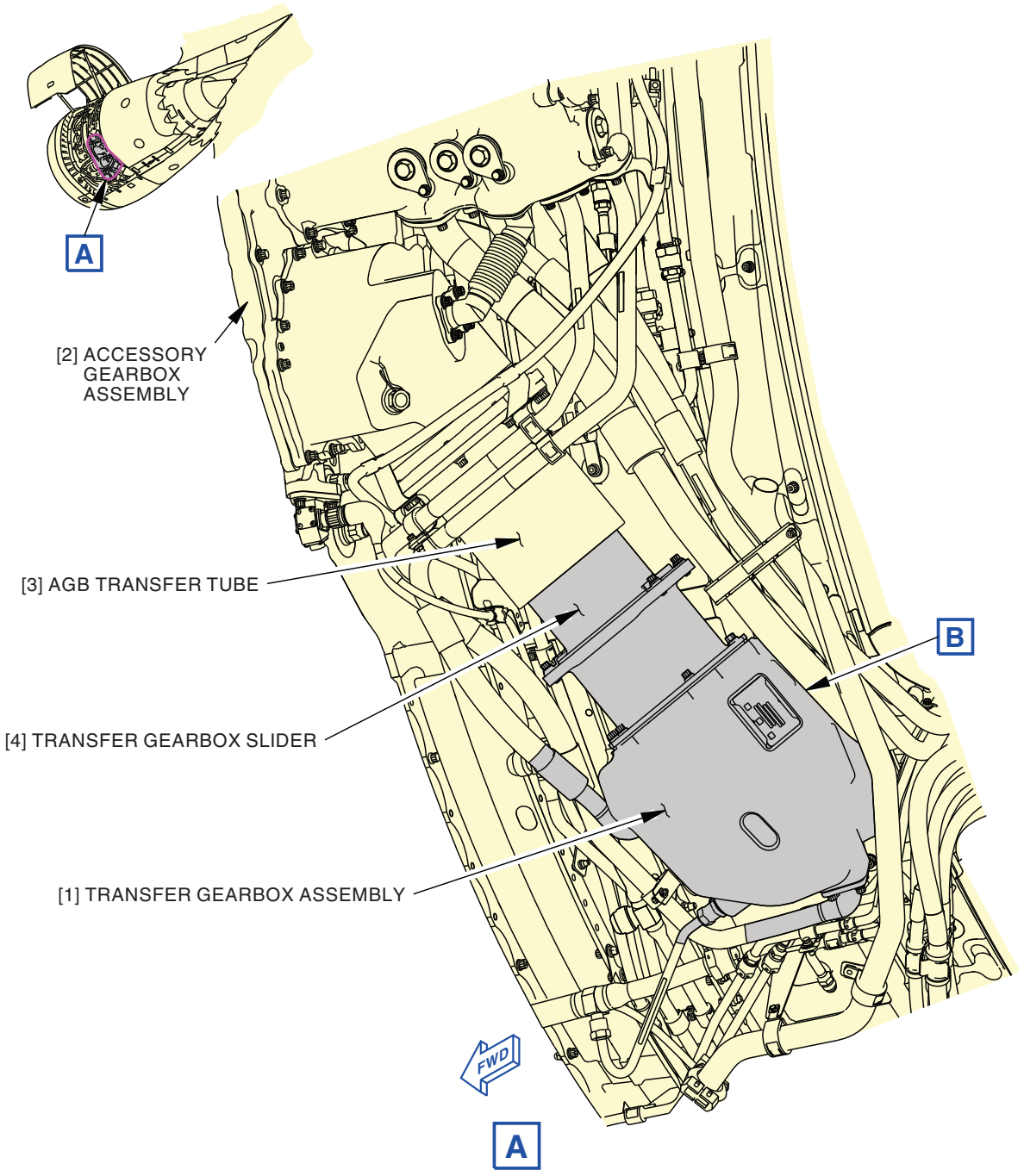
9) Remove the retainer ring [24] from the groove of the transfer gearbox assembly [1].

10) Remove the transfer gearbox assembly [1] from the TGB fixture, SPL-14917.

(i) Install the protective covers and caps on the transfer gearbox assembly [1] and fan frame [23].

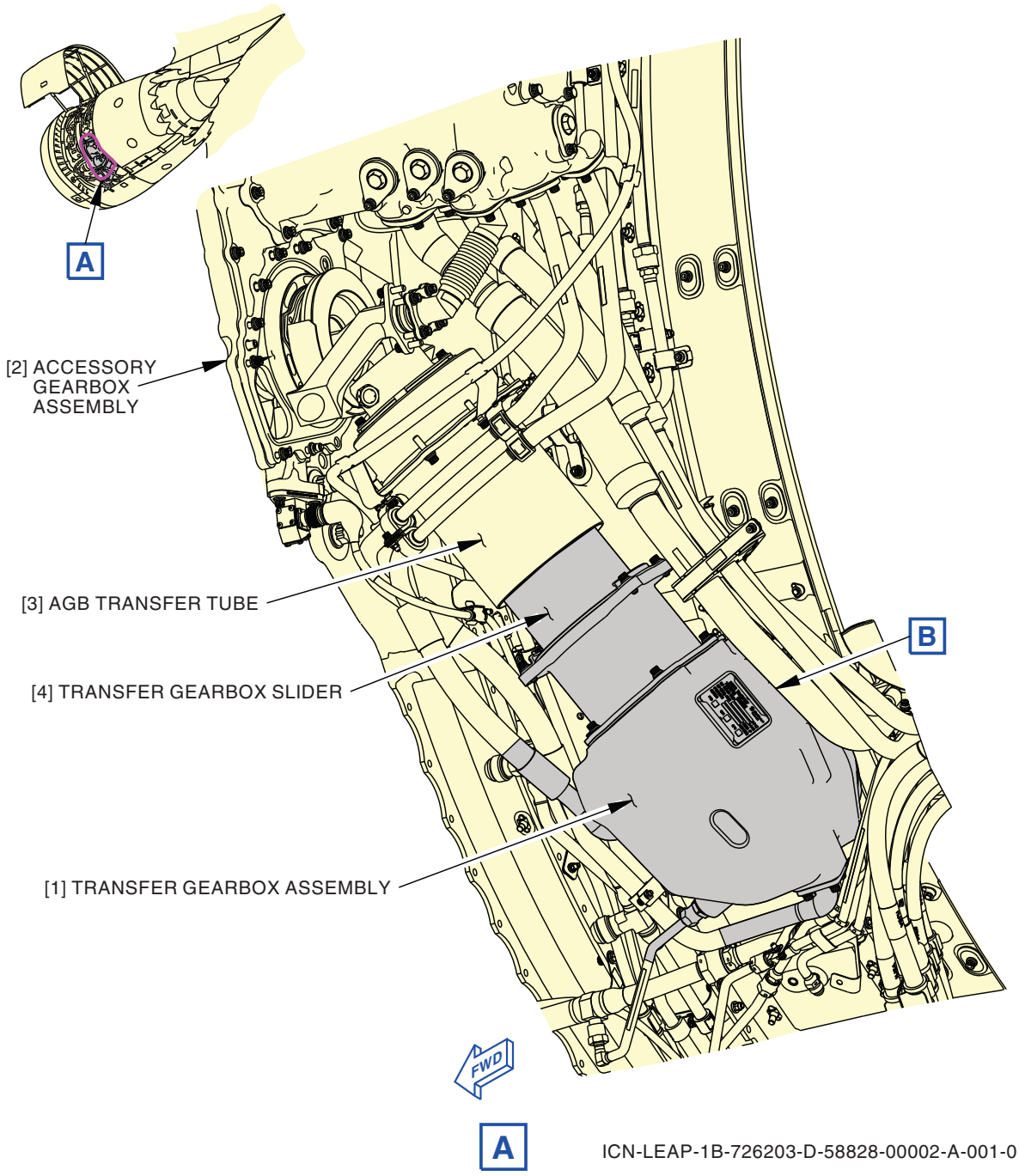
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



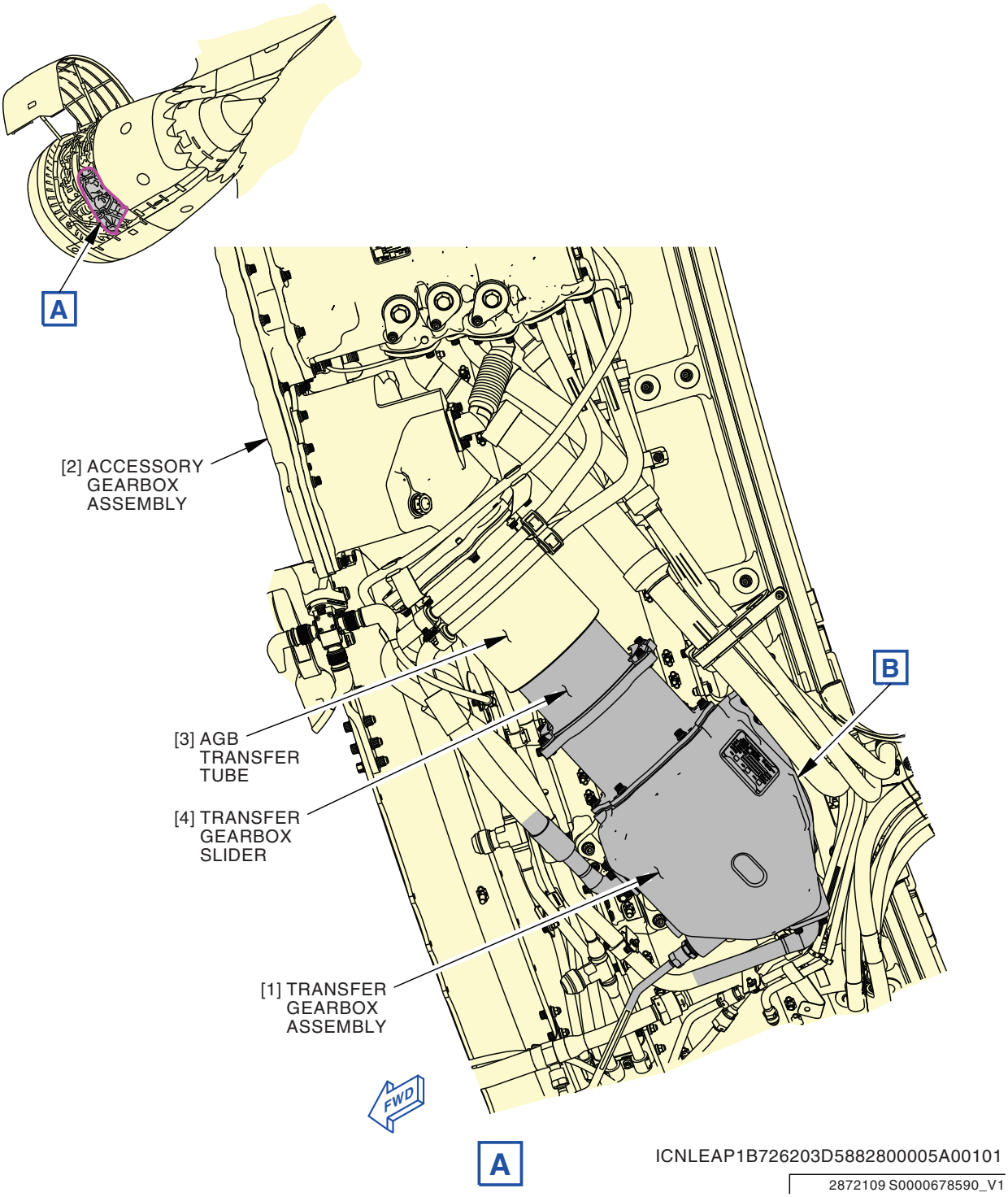
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 1 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



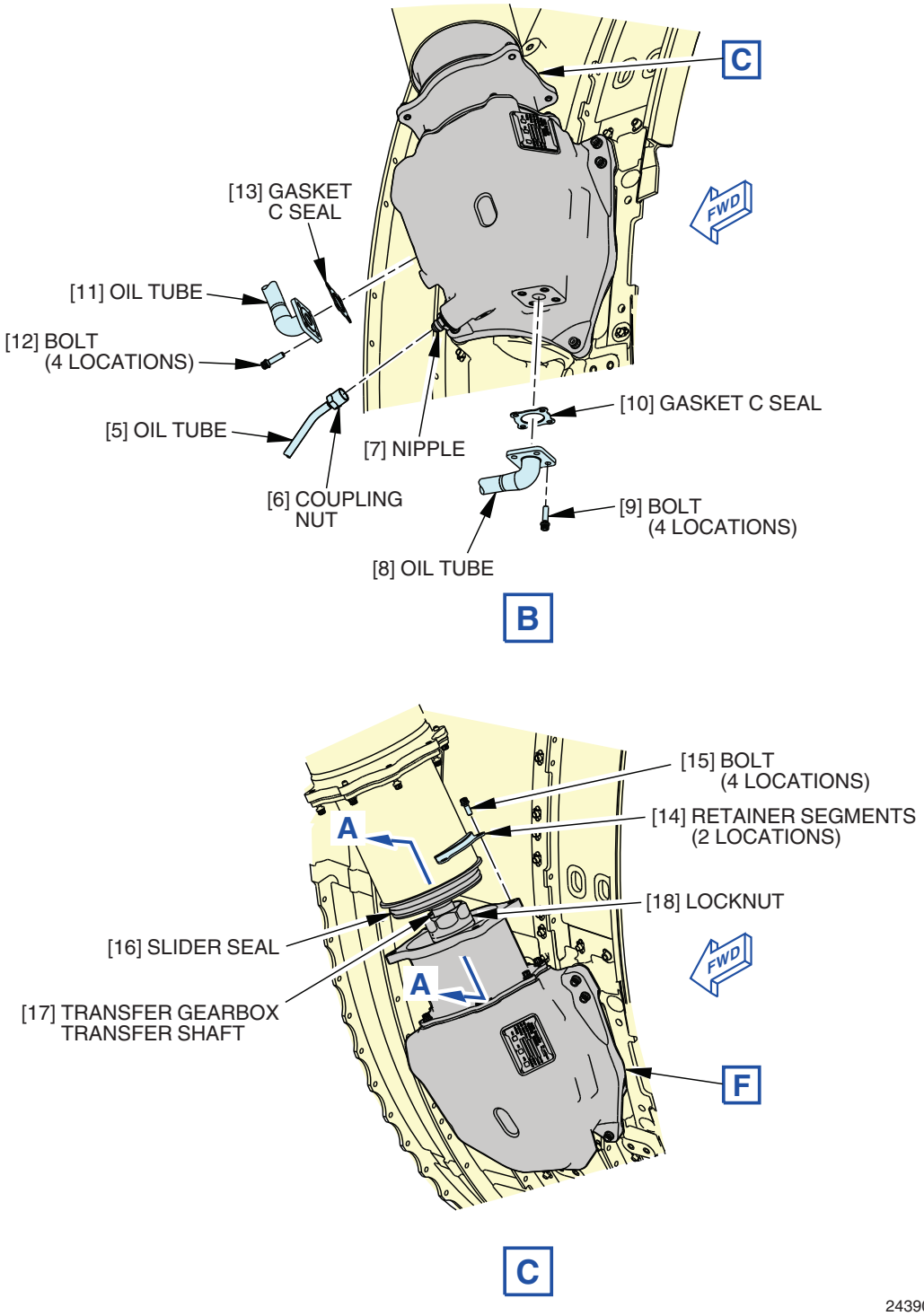
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 2 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 3 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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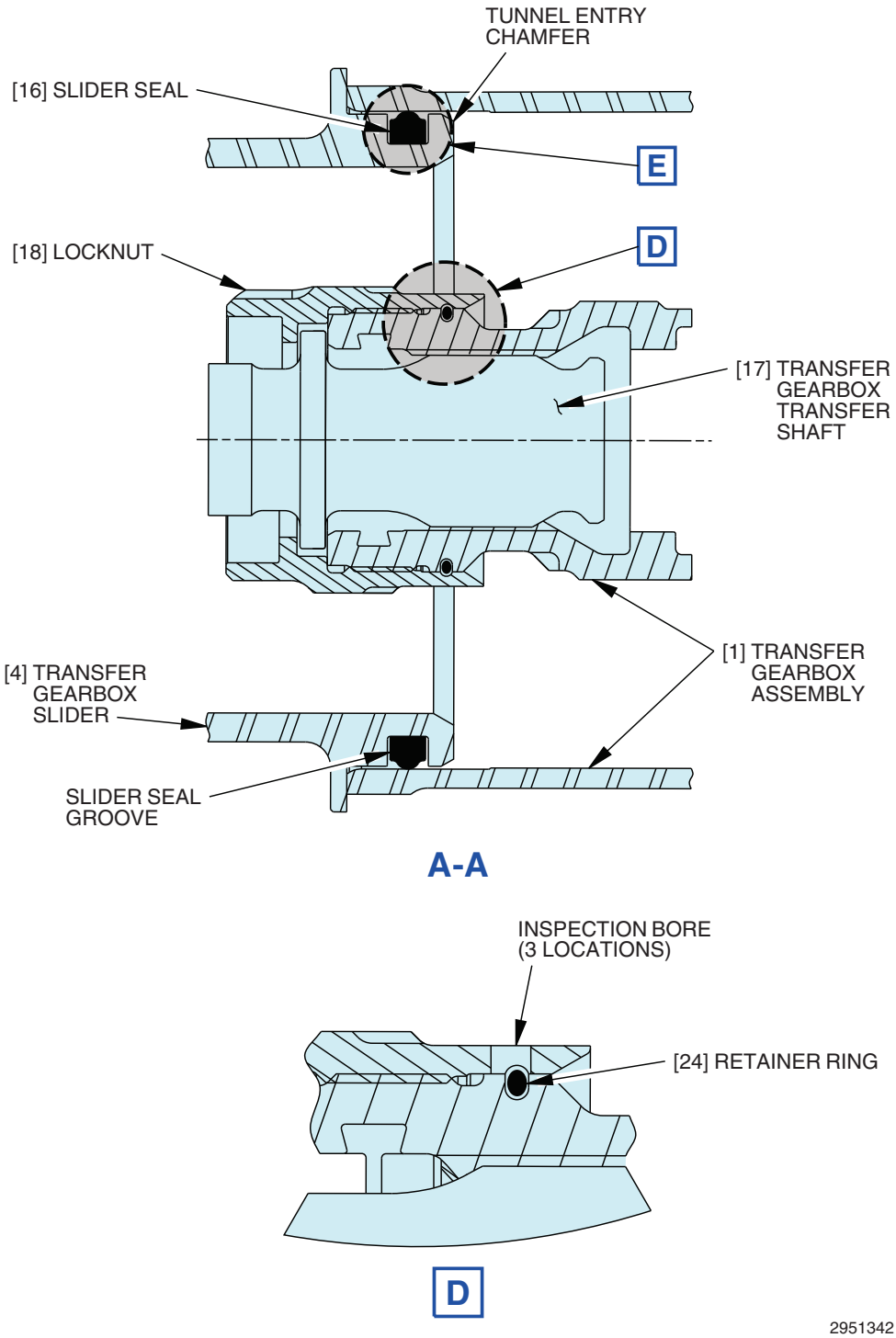
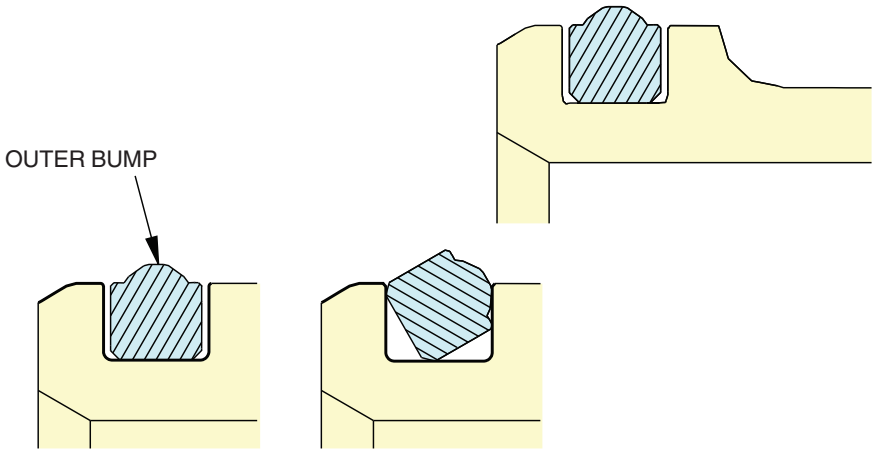
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 5 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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VASELINE OR ENGINE OIL

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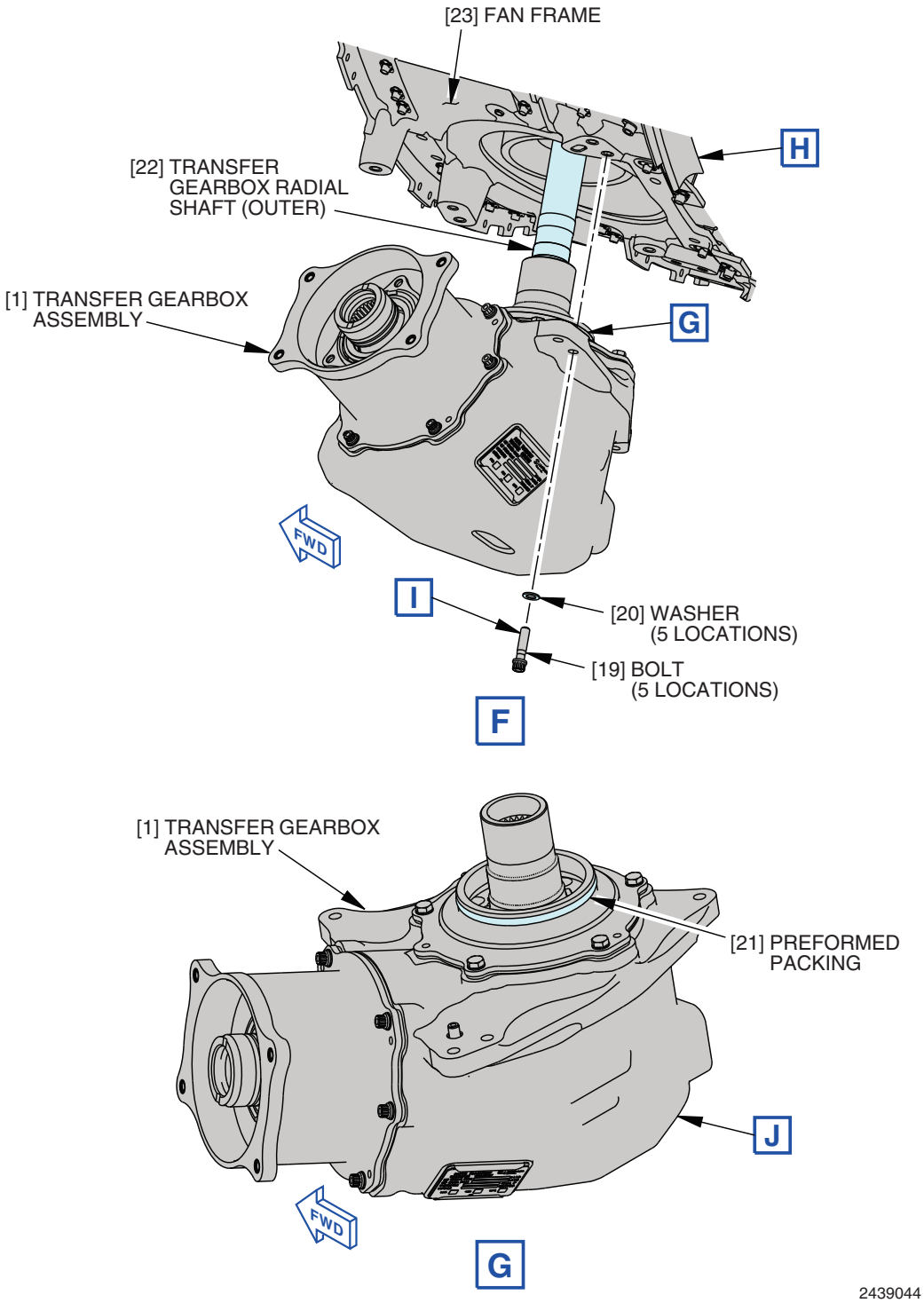
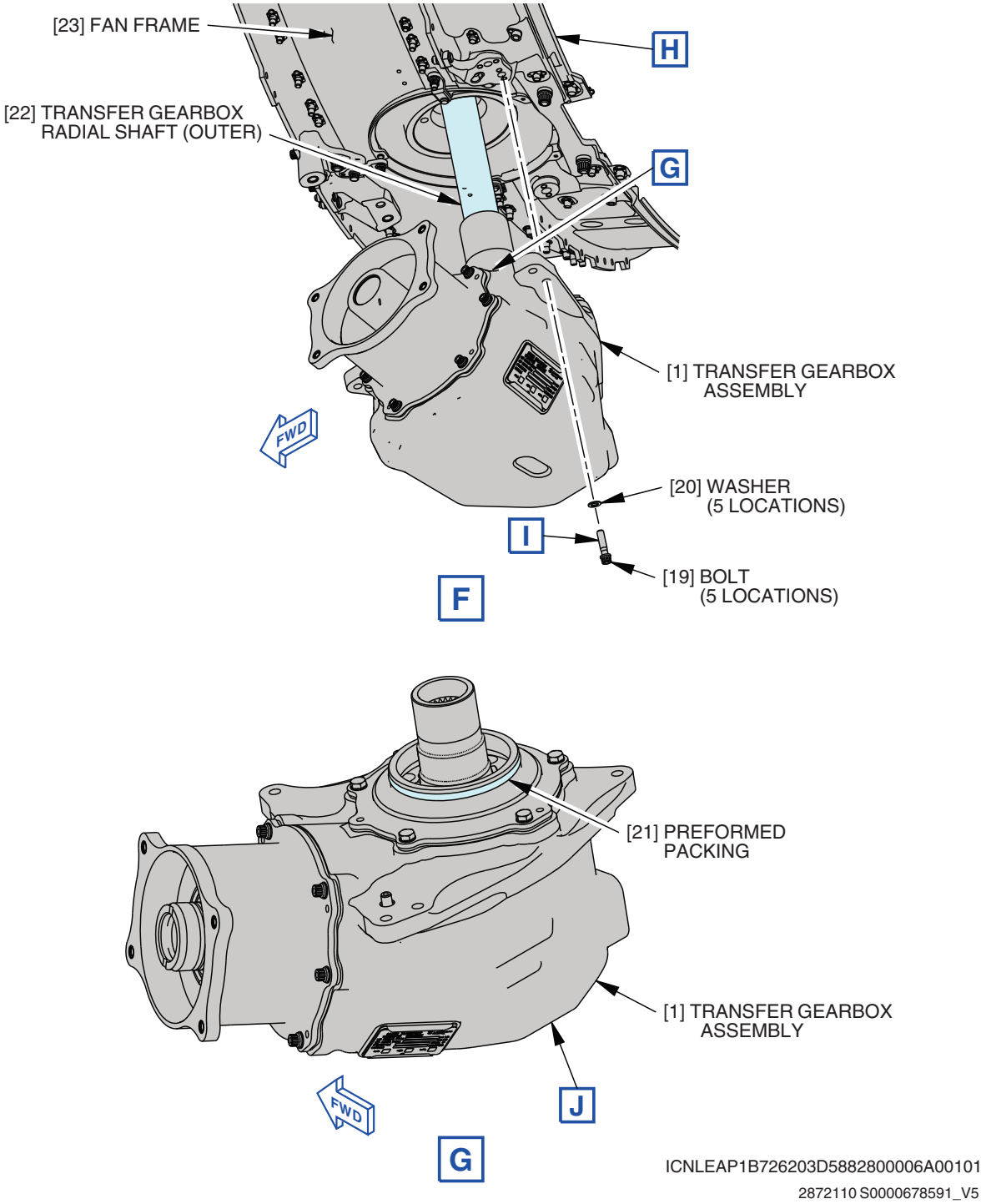
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 6 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



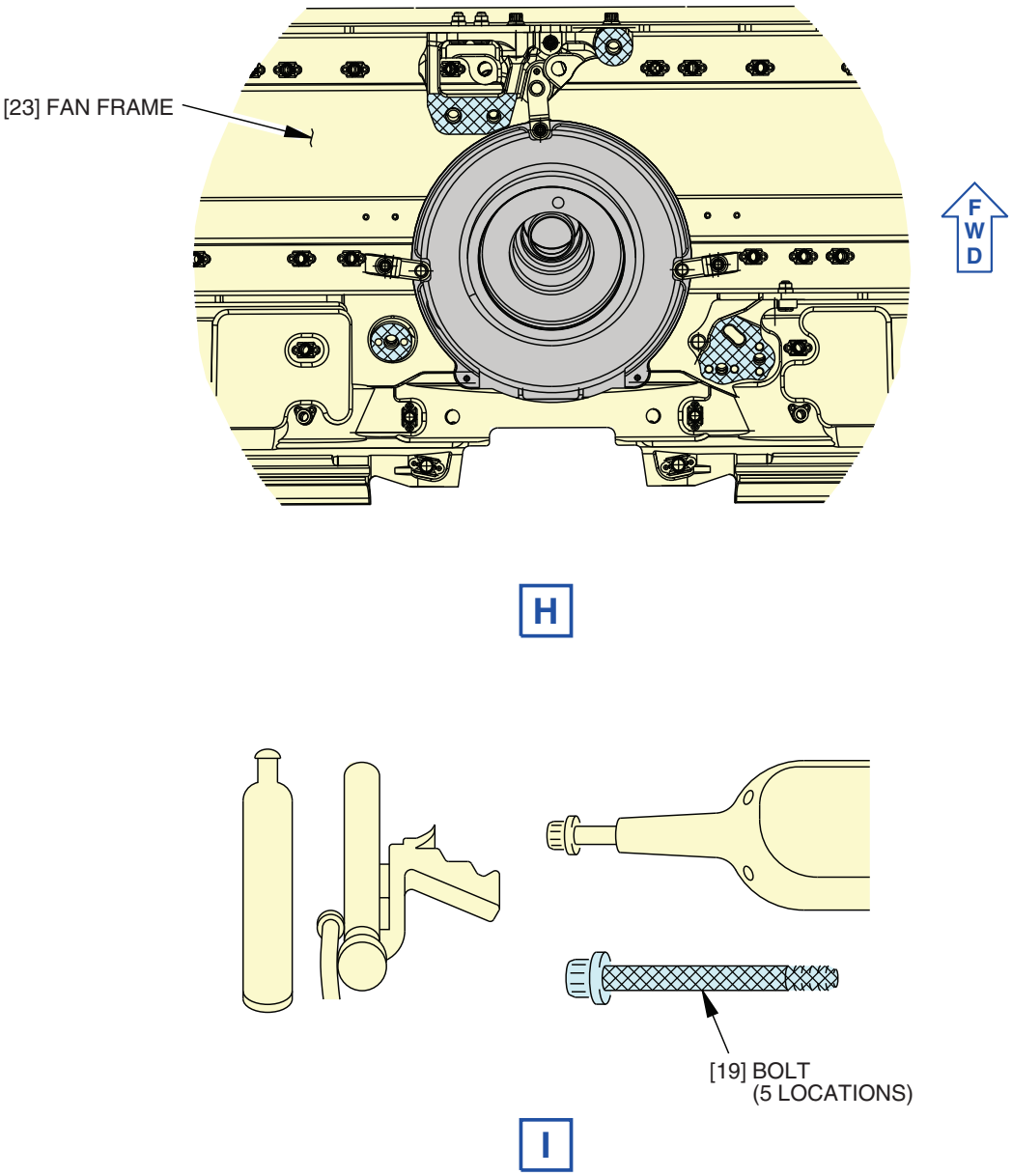
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 7 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



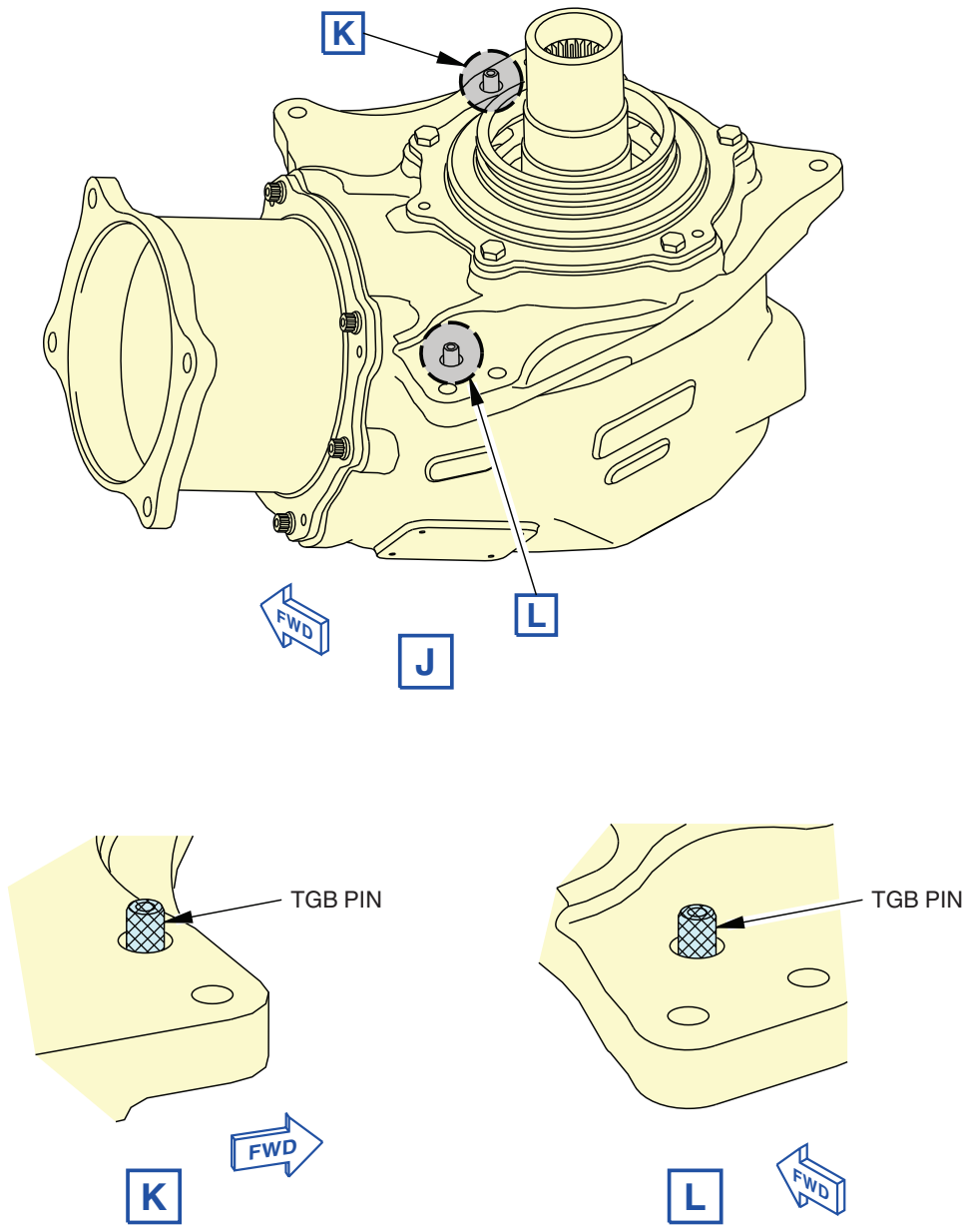
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 8 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



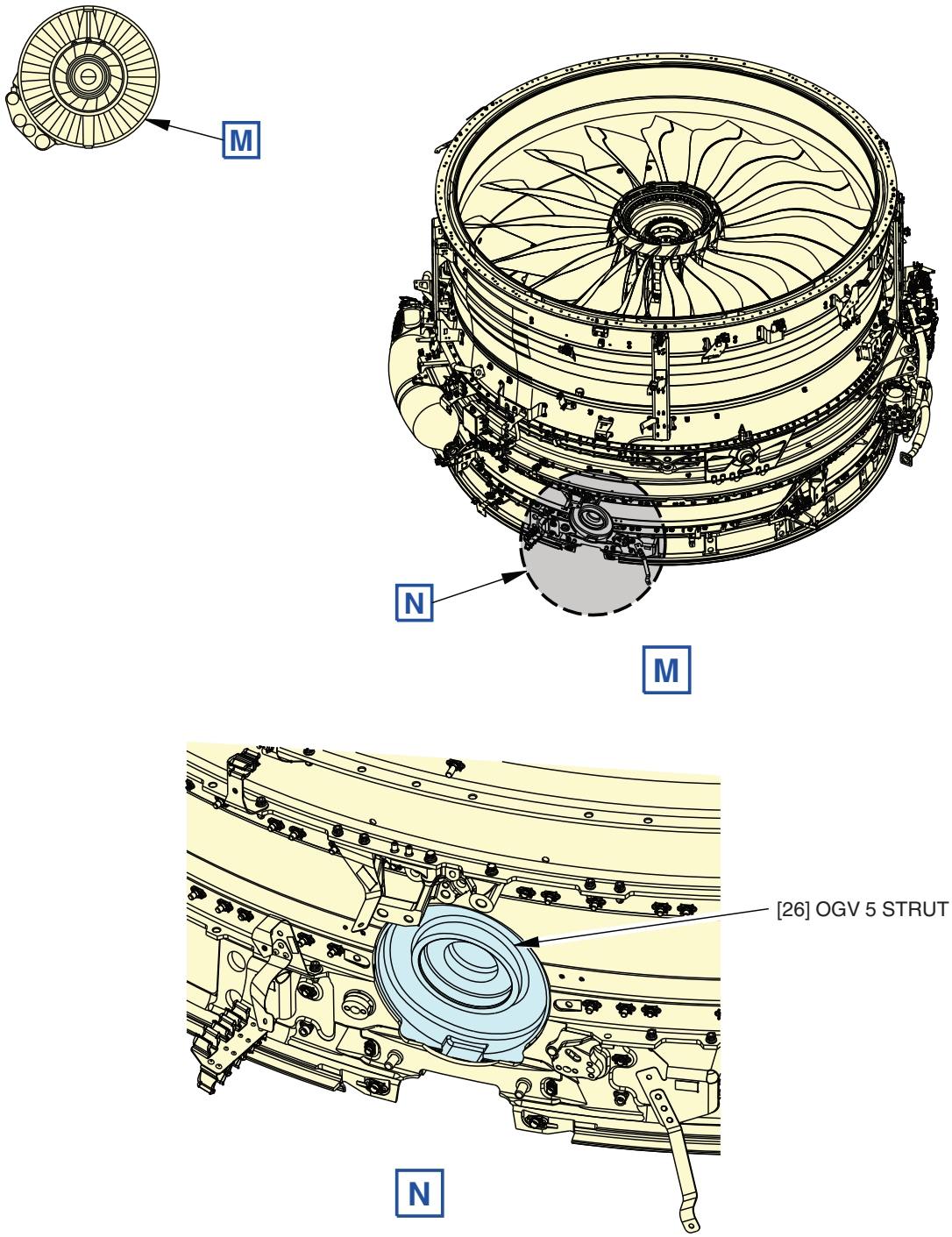
Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 9 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 10 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Transfer Gearbox Assembly Installation
Figure 401/72-62-03-990-801-G00 (Sheet 11 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-62-03-400-801-G00

3. Transfer Gearbox Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the transfer gearbox assembly.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-62-06-400-801-G00	Radial Shaft Installation (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14917	Fixture - TGB Removal Part #: 956A8076G02 Supplier: 58828
SPL-14918	Wrench - TGB Shaft Nut Part #: 956A8684G01 Supplier: 58828
SPL-14919	Tool - Sliding Tube, Assembly/Disassembly Part #: 956A8650G01 Supplier: 58828 Opt Part #: 956A8685G03 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
SPL-15088	Fixture, Install - Transfer Shaft Nut Retainer Part #: 956A8686G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50447	Compound - Jointing [CP2773], Corrosion Inhibiting, Chromate Free, Mastinox CA1010	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Transfer gearbox assembly	72-62-03-01-060	PTF ALL
10	Gasket C seal	72-00-00-12-090	PTF ALL
13	Gasket C seal	72-00-00-12-170	PTF ALL
16	Slider seal	72-00-00-14-320	PTF ALL
21	Preformed packing	72-62-06-01-050	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Transfer Gearbox Assembly Installation

SUBTASK 72-62-03-410-001-G00

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the transfer gearbox assembly [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

WARNING A MINIMUM OF TWO PERSONS ARE NECESSARY TO HANDLE THE TGB DURING THE REMOVAL PROCEDURE. THE WEIGHT OF THE TGB IS APPROXIMATELY 36.4 LB (16.5 KG). INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR

WARNING MOVE THE TRANSFER GEARBOX ASSEMBLY SLOWLY. THE TRANSFER GEARBOX ASSEMBLY IS HEAVY, INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR

(b) Install the transfer gearbox assembly [1] on the TGB fixture, SPL-14917.

NOTE: The transfer gearbox assembly weighs 34.1 lb (15.5 kg).

(c) Remove the protective covers and caps from the transfer gearbox assembly [1] and fan frame [23].

(d) Install the retainer ring [24] on the transfer gearbox assembly [1] with the nut retainer, SPL-15088.

(e) Install the transfer gearbox assembly [1] on the fan frame [23] as follows:

1) Lubricate the threads and under the head of the five bolts [19], two TGB pins, and surface of the fan frame [23] in contact with the transfer gearbox assembly [1], with jointing compound, A50447.

2) Lubricate the new preformed packing [21] with vaseline, D00672 [CP5070].

3) Lubricate the bore and the chamfer of the Outlet Guide Vane (OGV) 5 strut [26] with vaseline, D00672 [CP5070].

4) Install the new preformed packing [21] onto the groove of the transfer gearbox assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If it is necessary, untwist the preformed packings with a spatula.

5) Do this task: Radial Shaft Installation, TASK 72-62-06-400-801-G00.

6) Attach the transfer gearbox assembly [1] to the transfer gearbox radial shaft [22].

a) Make sure that you hold the transfer gearbox radial shaft [22] in place into the fan frame [23] with one hand.

b) Hold the transfer gearbox radial shaft [22] with one hand when you put it into the transfer gearbox assembly [1].

7) Install the transfer gearbox assembly [1] in a correct position on the fan frame [23].

8) Install the five bolts [19] and five washer [20] that attach the transfer gearbox assembly [1] to the fan frame [23].

CAUTION

MAKE SURE THAT YOU USE A GRADUAL CRISS-CROSS PATTERN WHEN YOU TORQUE THE BOLTS. IF NOT, THERE IS A RISK OF LEAKAGE AND IT COULD CAUSE DAMAGE.

9) Tighten the bolts [19] in a gradual criss-cross pattern to 244.15 in-lb (27.59 N·m) - 269.85 in-lb (30.49 N·m).

10) Remove the TGB fixture, SPL-14917, from the transfer gearbox assembly [1].

(f) Attach the transfer gearbox transfer shaft [17] to the transfer gearbox assembly [1] as follows:

1) Lubricate the locknut [18] with engine oil, D00599 [CP2442].

2) Lubricate the slider seal [16] groove and tunnel entry chamfer of the transfer gearbox slider [4] with oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

CAUTION

MAKE SURE THAT THE SLIDER SEAL IS NOT TWISTED. IF YOU DO NOT CORRECTLY INSTALL THE SLIDER SEAL IN THE SLIDER GROOVE, THERE IS A RISK OF OIL LEAKAGE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

3) Install the new slider seal [16] onto the groove of the transfer gearbox slider [4].

a) If it is necessary, untwist the slider seal [16] with a spatula.

4) Lubricate the outward part of the slider seal [16] with oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

5) Install the Engine core lock adapter, SPL-14920, on the AGB manual control pad.

NOTE: This will lock the N2 rotor.

6) Move the transfer gearbox transfer shaft [17] towards the transfer gearbox assembly [1].

7) Use the shaft nut wrench, SPL-14918, to tighten the locknut [18] to 1387.4 in-lb (156.75 N·m) - 1533.4 in-lb (173.25 N·m).

NOTE: The locknut attaches the transfer shaft to the TGB.

8) Make sure that the retainer ring [24] is still engaged into the groove of the shaft of the transfer gearbox assembly [1] after the locknut [18] tightening.

a) Inspect one of the three inspection bores of the locknut [18] to confirm the presence of the retainer ring [24] (View D, Figure 401).

NOTE: You can use a mirror to make the inspection easier.

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<1> If the retainer ring [24] is not in the groove of the shaft of the transfer gearbox assembly [1], do these steps:

<a> Remove the locknut [18].

<b> Re-install the retainer ring [24].

<c> Refer to the steps above and install the locknut [18] again.

b) Make sure that the retainer ring [24] is well expanded in the groove of locknut [18].

<1> Push through one of the three inspection bores of the locknut [18] on the retainer ring [24] with a pin-drift diameter 0.125 in. (3.175 mm) to check that there is a spring effect.

NOTE: You can use any equivalent local tool.

<a> If the retainer ring [24] has no spring effect, replace the transfer gearbox assembly [1].

NOTE: The retainer ring should be trapped in the groove of the shaft of the transfer gearbox.

NOTE: You should send the removed transfer gearbox to the MRO for gearbox bevel gear shaft T02 change.

<b> If the retainer ring [24] has spring effect, no additional action is required.

9) Lubricate the outward part of the slider seal [16] and inner diameter of the transfer gearbox assembly [1] in interface with oil, D00599 [CP2442], or vaseline, D00672 [CP5070], just before installation.

10) Move the transfer gearbox slider [4] towards the transfer gearbox assembly [1] with the assembly/disassembly sliding tube tool, SPL-14919.

11) Lubricate the threads of the four bolts [15] with graphite mineral vaseline, D00601 [CP2101].

12) Install the two retainer segment [14] in a correct position.

13) Install the four bolts [15] that attach the transfer gearbox slider [4] to the transfer gearbox assembly [1].

14) Tighten the bolts [15] to 100.7 in-lb (11.4 N·m) - 111.3 in-lb (12.6 N·m).

(g) Connect the oil tube [11] to the transfer gearbox assembly [1] as follows:

1) Install a new gasket C seal [13] and oil tube [11] in a correct position on the transfer gearbox assembly [1].

2) Install the four bolts [12] that attach the oil tube [11] to the transfer gearbox assembly [1].

3) Tighten the four bolts [12] in a criss cross pattern to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

(h) Connect the oil tube [8] to the transfer gearbox assembly [1] as follows:

1) Install a new gasket C seal [10] and oil tube [8] in a correct position on the transfer gearbox assembly [1].

2) Install the four bolts [9] that attach the oil tube [8] to the transfer gearbox assembly [1].

3) Tighten the four bolts [9] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

(i) Connect the oil tube [5] to the transfer gearbox assembly [1] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Lubricate the coupling nut [6] with engine oil, D00599 [CP2442].

2) Connect the coupling nut [6] to the nipple [7] of the oil tube [5].

3) Tighten the coupling nut [6] to 266 in-lb (30 N·m) - 310 in-lb (35 N·m).

(j) Remove the Engine core lock adapter, SPL-14920, from AGB manual control pad.

(k) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(l) Clean the work area and remove the tools used to do the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-03-410-002-G00

(1) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-62-03-860-003-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-03-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Transfer Gearbox Assembly Installation Test

SUBTASK 72-62-03-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRANSFER SHAFT - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the transfer shaft

(2) An installation of the transfer shaft.

TASK 72-62-04-000-801-G00

2. Transfer Shaft Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the transfer shaft.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-000-801-G00	Transfer Gearbox Assembly Removal (P/B 401)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14918	Wrench - TGB Shaft Nut Part #: 956A8684G01 Supplier: 58828
SPL-14919	Tool - Sliding Tube, Assembly/Disassembly Part #: 956A8650G01 Supplier: 58828 Opt Part #: 956A8685G03 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-62-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

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1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-62-04-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-04-010-001-G00

(3) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

G. Transfer Shaft Removal

SUBTASK 72-62-04-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the transfer shaft [1] as follows:

(a) Lock the N2 rotor with the Engine core lock adapter, SPL-14920, through the accessory gearbox cover (handcranking cover).

(b) Remove the slider [5] as follows:

1) Remove the bolts [3].

2) Remove the two haft segments [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION DO NOT USE METALLIC PARTS TO REMOVE THE SLIDER FROM THE TRANSFER TUBE TGB. THEY CAN CAUSE DAMAGE TO THE PARTS.

3) Remove the slider [5] from the Transfer Gearbox (TGB) with the assembly/disassembly sliding tube tool, SPL-14919.

4) Push the slider [5] entirely into the Accessory Gear Box (AGB) transfer tube.

(c) Disconnect the transfer shaft [1] as follows:

1) Loosen the locking nut [2] with the shaft nut wrench, SPL-14918, and slide it off the transfer shaft [1].

2) Pull the transfer shaft [1] out of the TGB horizontal bevel gear.

3) Push the transfer shaft [1] entirely into the AGB.

(d) Remove the slider seal [7] from the slider groove and discard it.

(e) Remove the transfer gearbox assembly (Transfer Gearbox Assembly Removal, TASK 72-62-03-000-801-G00).

CAUTION DO NOT APPLY SIDE LOADS WHICH COULD BEND THE SPLINED DRIVESHAFT DURING REMOVAL, OR INSTALLATION OF THE COMPONENT. DAMAGE CAN OCCUR.

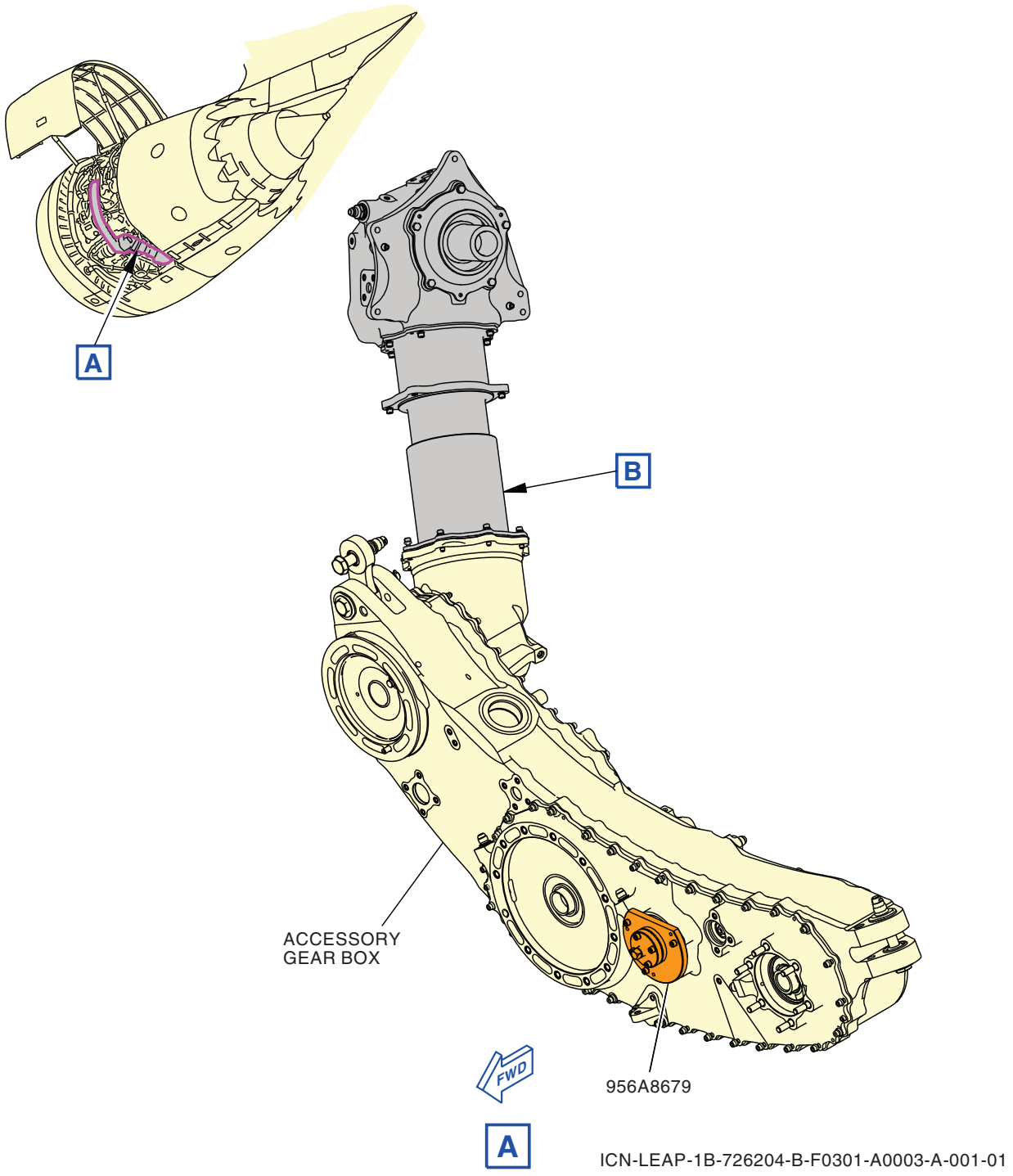
(f) Remove the transfer shaft [1] from the AGB.

1) Hold the locking nut [2] with one hand and gently pull out the transfer shaft [1] from the AGB with the other hand.

2) Carefully pull out the transfer shaft [1] through the locking nut [2].

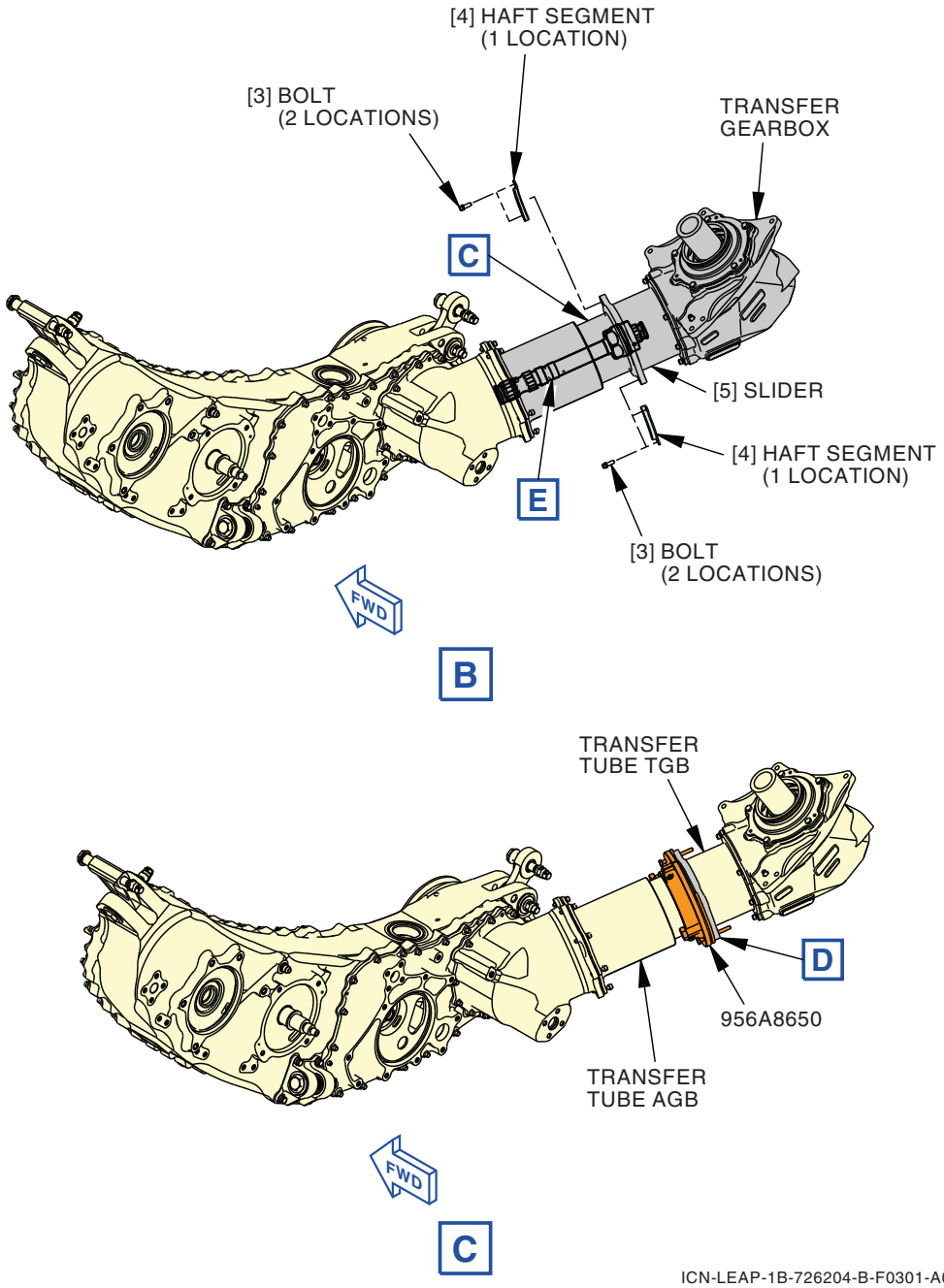
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



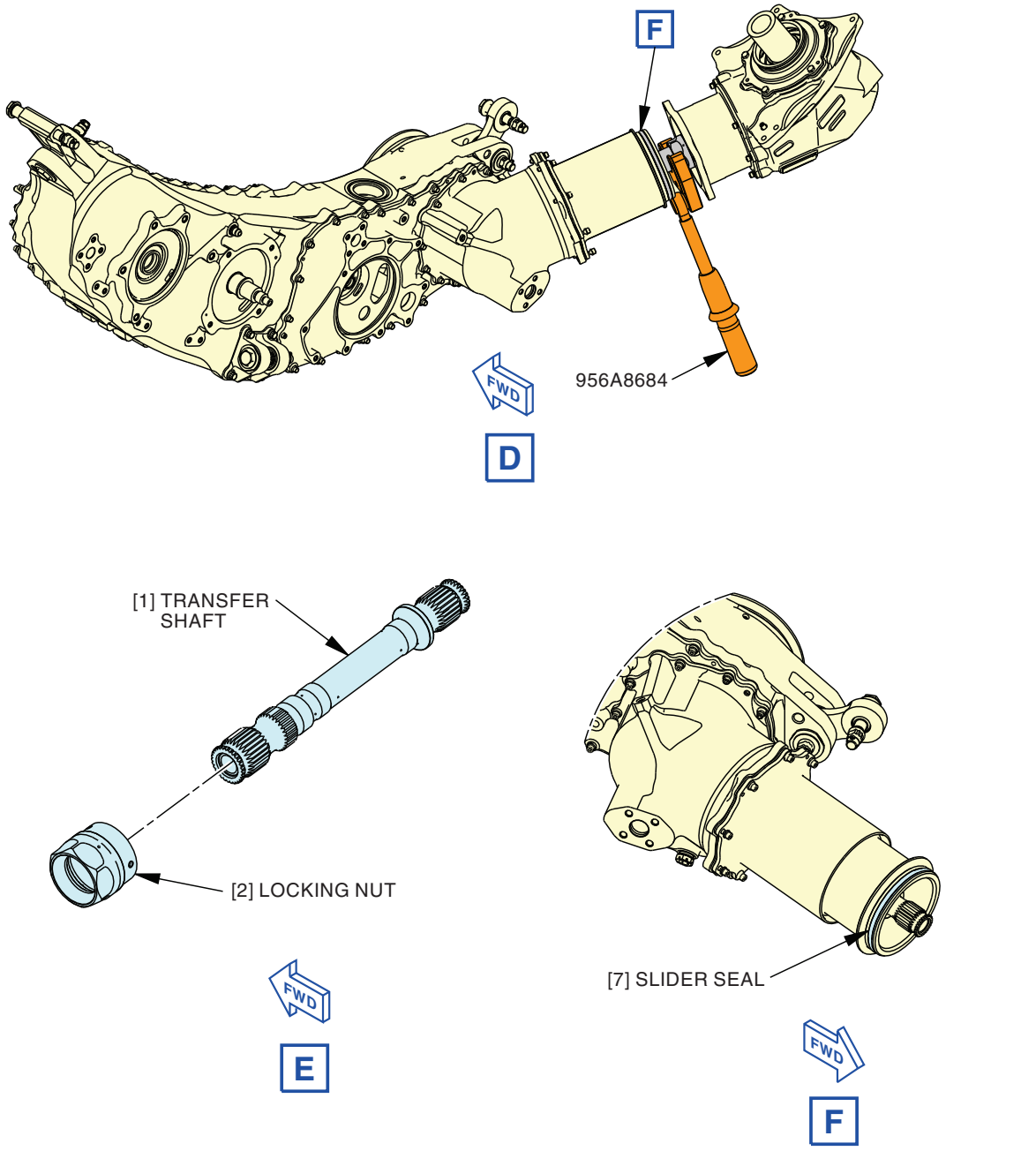
Transfer Shaft Installation
Figure 401/72-62-04-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Shaft Installation
Figure 401/72-62-04-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Transfer Shaft Installation
Figure 401/72-62-04-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-62-04-400-801-G00

3. Transfer Shaft Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the transfer shaft.

(2) You must do the tests that are listed in the power plant test reference table after you install the transfer shaft.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-400-801-G00	Transfer Gearbox Assembly Installation (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14918	Wrench - TGB Shaft Nut Part #: 956A8684G01 Supplier: 58828
SPL-14919	Tool - Sliding Tube, Assembly/Disassembly Part #: 956A8650G01 Supplier: 58828 Opt Part #: 956A8685G03 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Transfer shaft	72-00-00-14-280	PTF ALL
7	Slider seal	72-00-00-14-310	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Transfer Shaft Installation

SUBTASK 72-62-04-410-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT APPLY FORCE TO THE TRANSFER SHAFT. DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Install the transfer shaft [1] as follows:

(a) Lubricate the locking nut [2] with oil, D00599 [CP2442].

(b) Carefully slide the transfer shaft [1] through the locking nut [2] and push it onto the transfer shaft [1] flange.

(c) Hold the locking nut [2] with one hand and engage the transfer shaft [1] into the Accessory Gear Box (AGB) with the other hand.

(d) Slide the transfer shaft [1] into the AGB.

(e) Gently engage the splines shaft of the transfer shaft [1] into the AGB L1 bevel gear.

CAUTION MAKE SURE THAT THE SLIDER SEAL IS NOT TWISTED. IF YOU DO NOT CORRECTLY INSTALL THE SLIDER SEAL IN THE SLIDER GROOVE, THERE IS A RISK OF OIL LEAKAGE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(f) Install the new slider seal [7] as follows:

1) Lubricate the groove of the slider seal [7] with the oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

2) Install the slider seal [7] onto the slider groove.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure that the seal is correctly engaged into the slider groove.

4) Lubricate the outward part of the slider seal [7] with oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

(g) Install the Transfer Gearbox (TGB) assembly (Transfer Gearbox Assembly Installation, TASK 72-62-03-400-801-G00).

(h) Examine the AGB and TGB interface areas for cleanliness and damage.

1) Transfer shaft splines

2) Locking nut

3) Slider cavity

4) Transfer tube TGB cavity.

(i) Connect the transfer shaft [1] as follows:

1) Do a check on the alignment between AGB transfer shaft and TGB port.

2) Carefully slide the transfer shaft [1] through the TGB.

3) Gently engage the splines shaft of the transfer shaft [1] into the TGB horizontal bevel gear.

4) Slide the locking nut [2] onto the transfer shaft [1].

5) Tighten the locking nut [2] to 1328 in-lb (150 N·m) - 1593 in-lb (180 N·m) with the shaft nut wrench, SPL-14918.

(j) Install the slider [5] as follows:

1) Slide the slider [5] and install it on the TGB flange with the assembly/disassembly sliding tube tool, SPL-14919.

2) Install the two haft segments [4].

3) Install the bolts [3].

4) Tighten the bolts [3] to 97.4 in-lb (11.0 N·m) - 106.2 in-lb (12.0 N·m).

(k) Unlock the N2 rotor with the Engine core lock adapter, SPL-14920, through the accessory gearbox cover (handcranking cover).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-04-410-002-G00

(1) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-62-04-860-003-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-04-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Transfer Shaft Installation Test

SUBTASK 72-62-04-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRANSFER GEARBOX SLIDER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks.

(1) Transfer Gearbox Slider removal.

(2) Transfer Gearbox Slider installation.

TASK 72-62-05-000-801-G00

2. Transfer Gearbox Slider Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Transfer Gearbox Slider.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-000-801-G00	Transfer Gearbox Assembly Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-62-05-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-62-05-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-62-05-010-001-G00

(3) Do this task: Transfer Gearbox Assembly Removal, TASK 72-62-03-000-801-G00.

G. Transfer Gearbox Slider Removal

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

SUBTASK 72-62-05-010-002-G00

(1) Remove the Transfer Gearbox (TGB) slider [1] as follows:

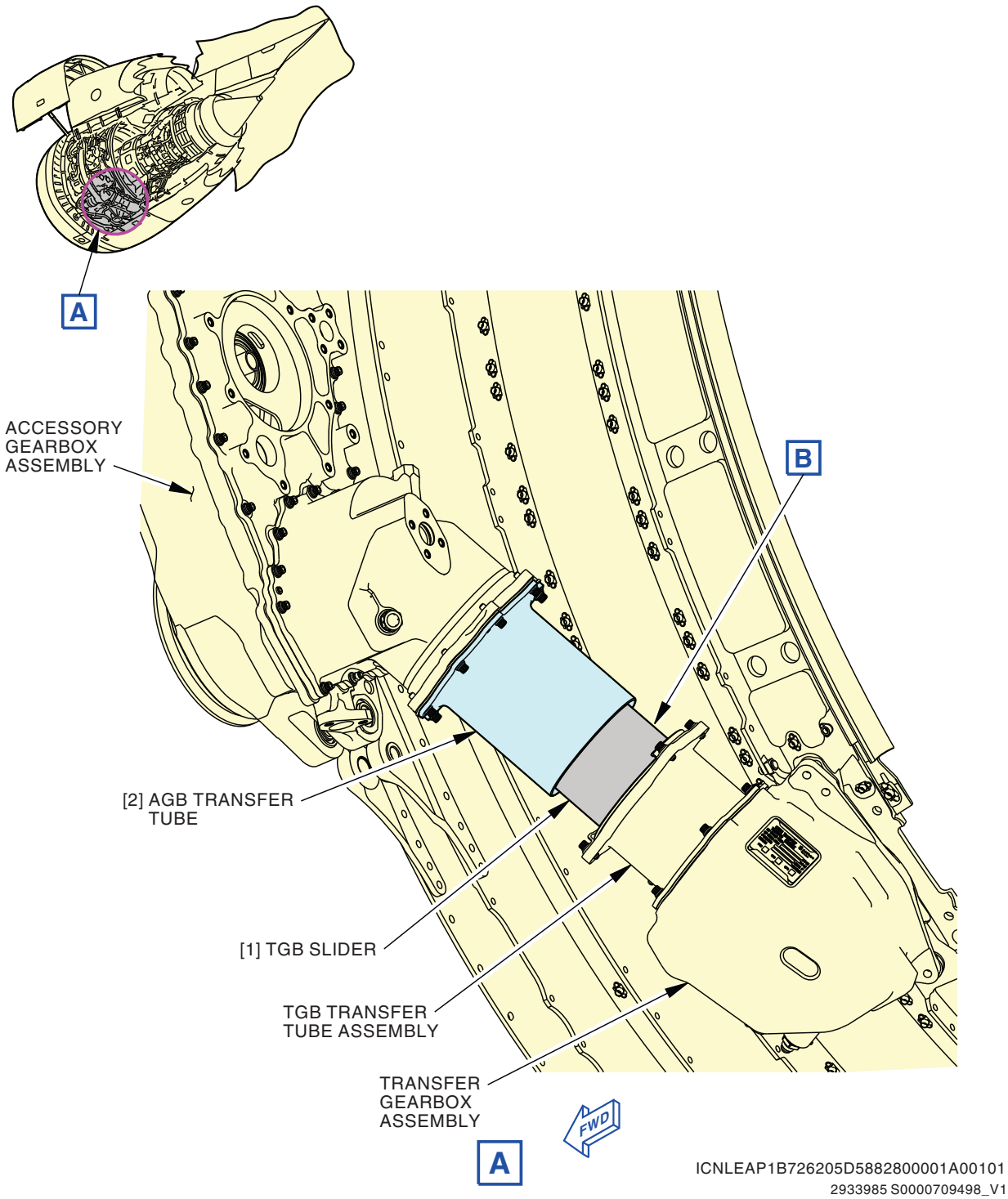
(a) Put the oil resistant container (5 gal)(19 Liter), STD-1055 below the TGB slider [1].

(b) Pull and remove the TGB slider [1] out of the AGB transfer tube [2].

(c) Remove and discard the TBG slider [1] ball joint seal [3] and seal [4].

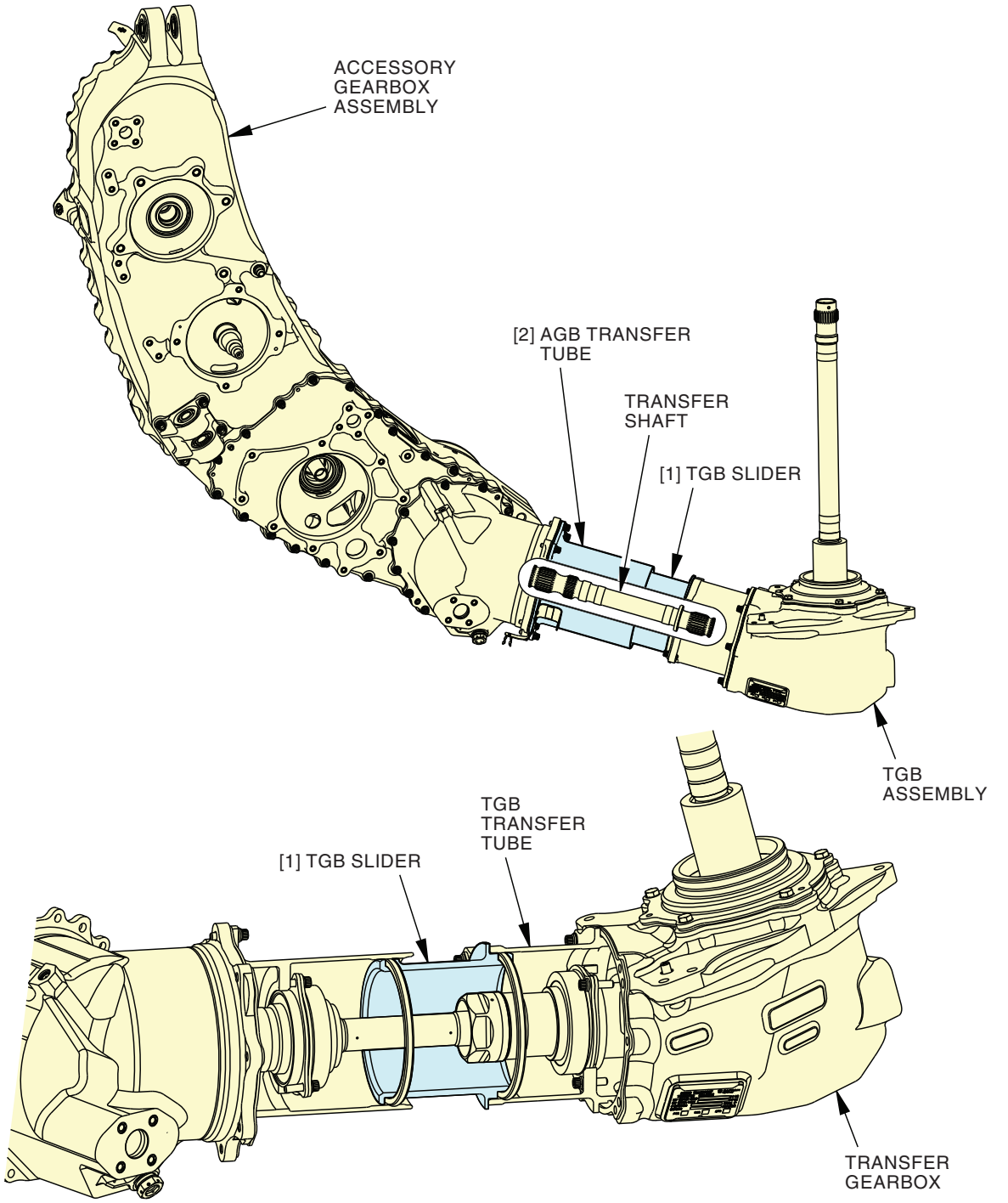
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



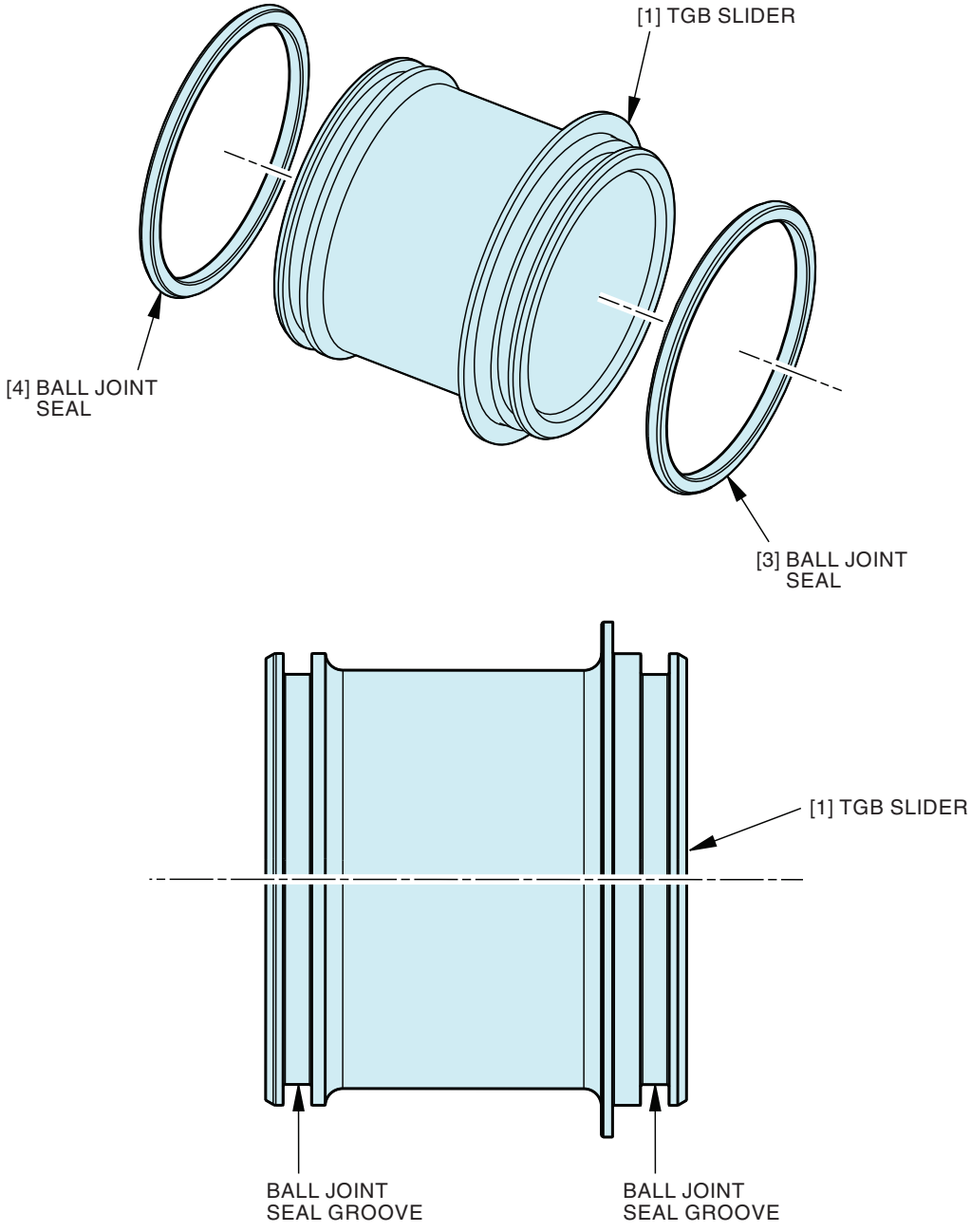
Transfer Gearbox Slider Installation
Figure 401/72-62-05-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Slider Installation
Figure 401/72-62-05-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transfer Gearbox Slider Installation
Figure 401/72-62-05-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-62-05-400-801-G00

3. Transfer Gearbox Slider Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the transfer gearbox slider.

(2) You must do the tests that are listed in the power plant test reference table after you install the transfer shaft.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-400-801-G00	Transfer Gearbox Assembly Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-809	Spatula
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Transfer Gearbox Slider Installation

SUBTASK 72-62-05-410-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the Transfer Gearbox (TGB) slider [1] as follows:

(a) Put the oil resistant container (5 gal)(19 Liter), STD-1055, below the TGB slider [1].

(b) Lubricate the new TGB Slider Seal Grooves seal [3] and seal [4] with engine oil, D00599 [CP2442] or vaseline, D00672 [CP5070]. (Figure 401 (Sheet 2)

CAUTION MAKE SURE THAT YOU INSTALL THE TGB SLIDER-BALL JOINT-SEALS CORRECTLY AND ARE NOT TWISTED. IF THE TGB SLIDER-BALL JOINT-SEALS ARE TWISTED IN THE SLIDER GROOVES, THERE IS A RISK OF OIL LEAKAGE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Install the two new TGB Ball Joint Seals seal [3] and seal [4] into the TGB slider [1] grooves.

NOTE: The outer bump of the TGB slider Ball Joint Seals must be visible all around, it's perimeter on 360°F (182°C).

(d) Make sure that the new TGB Slider Ball Joint Seals seal [3] and seal [4] are correctly engaged into the TGB slider [1] grooves..

(e) If necessary, use a spatula, STD-809 to untwist the TGB Slider Ball Joints seal [3] and seal [4] after installation in the TGB slider [1] grooves.

(f) Lubricate the outward part of the TGB Slider Ball Joint seal [3] and seal [4] with engine oil, D00599 [CP2442].

(g) Install and push the TGB slider [1] in the AGB transfer tube [2].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-05-410-001-G00

(1) Do this task: Transfer Gearbox Assembly Installation, TASK 72-62-03-400-801-G00.

SUBTASK 72-62-05-860-003-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-62-05-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Transfer Gearbox Slider Installation Test

SUBTASK 72-62-05-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RADIAL SHAFT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Radial Shaft Removal

(2) Radial Shaft Installation.

TASK 72-62-06-000-801-G00

2. Radial Shaft Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the radial shaft.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-000-801-G00	Transfer Gearbox Assembly Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14881	Radial Shaft Push/Pull Tool Part #: 956A8043G01 Supplier: 58828
STD-199	Container - 2 U.S. Gallon (7.6 l) (oil resistant)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 72-62-06-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-62-06-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-62-06-010-001-G00

(3) Remove the transfer gearbox assembly [2] (TASK 72-62-03-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Radial Shaft Removal

SUBTASK 72-62-06-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE FORCE ON THE RADIAL SHAFT. DAMAGE TO THE RADIAL SHAFT CAN OCCUR.

(1) Remove the radial shaft [1] as follows:

(a) Put a container, STD-199 below the radial shaft.

CAUTION DO NOT USE FORCE ON THE RADIAL SHAFT. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE RADIAL SHAFT.

(b) Install the push/pull tool, SPL-14881 to the end of the radial shaft [1].

(c) Carefully remove the radial shaft [1] fully out of the fan frame shroud [3].

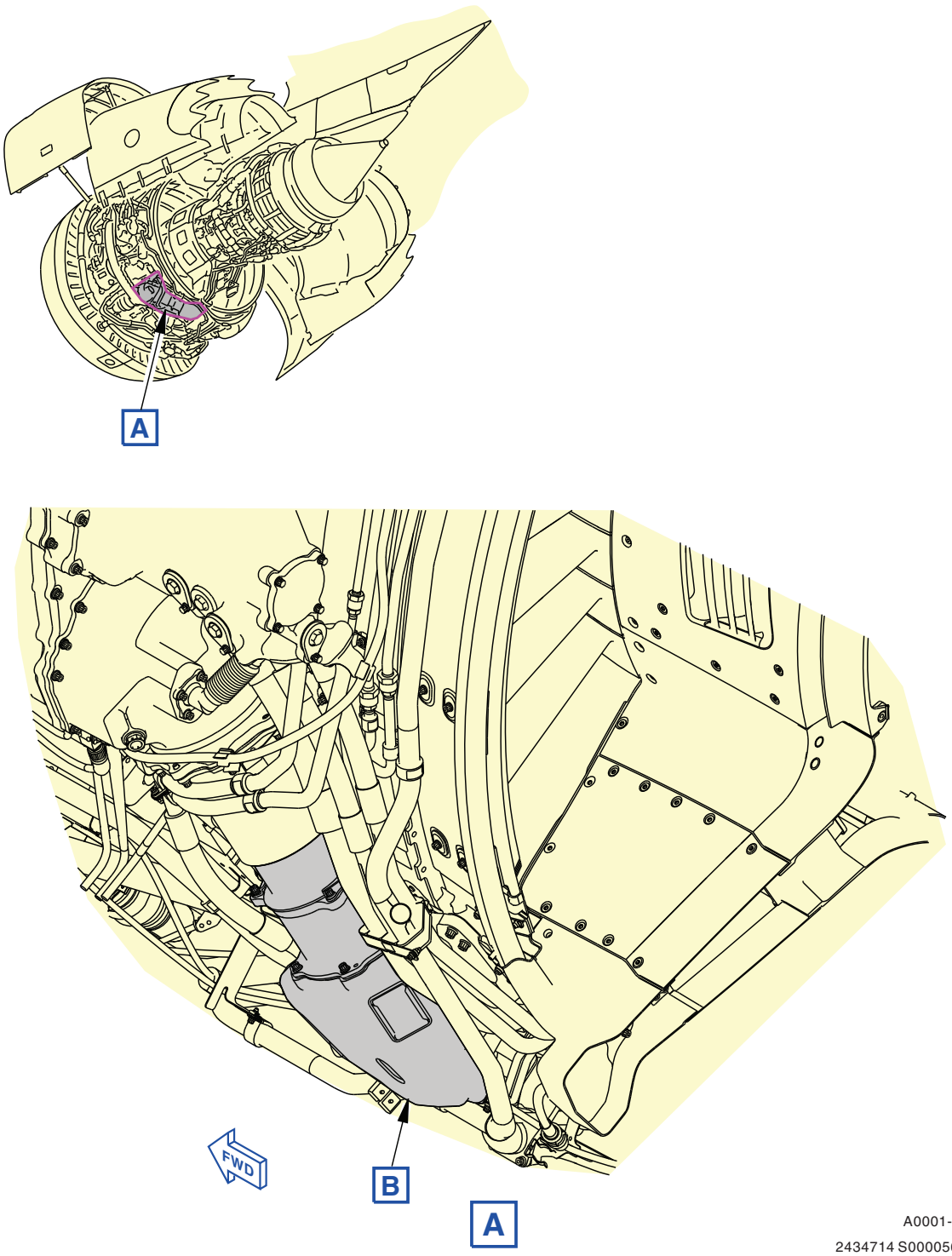
(d) Remove the radial shaft [1] from the engine.

(e) Remove the push/pull tool, SPL-14881 from the radial shaft [1].

(f) Install the protective covers on the fan frame shroud [3].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Radial Shaft Installation
Figure 401/72-62-06-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

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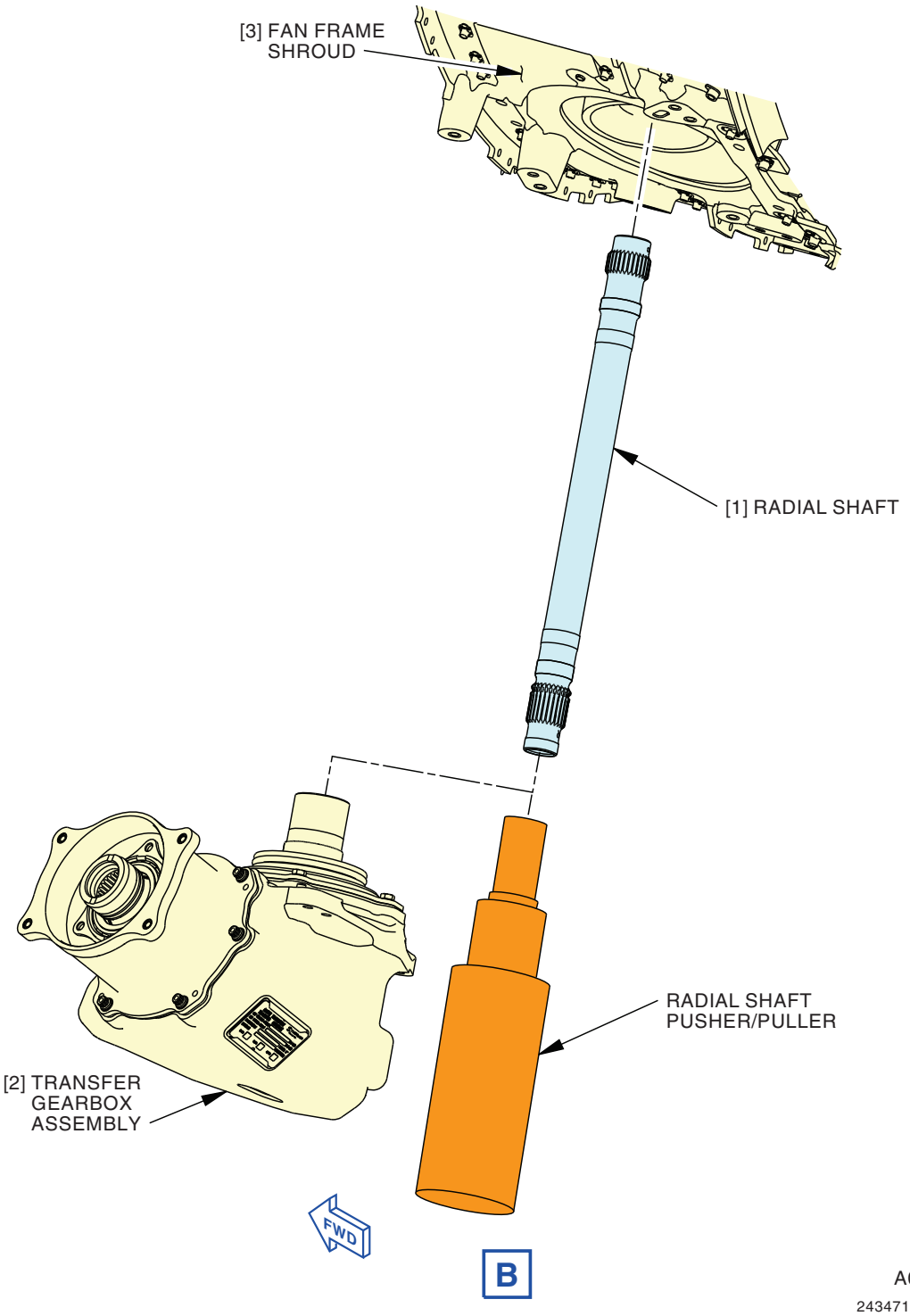
Radial Shaft Installation
Figure 401/72-62-06-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Radial Shaft Installation
Figure 401/72-62-06-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-62-06-400-801-G00

3. Radial Shaft Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the radial shaft.

(2) You must do the tests that are listed in the power plant test reference table after you install the radial shaft.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-400-801-G00	Transfer Gearbox Assembly Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14881	Radial Shaft Push/Pull Tool Part #: 956A8043G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Radial shaft	72-62-06-01-060	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Radial Shaft Installation

SUBTASK 72-62-06-410-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

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WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE FORCE ON THE RADIAL SHAFT. DAMAGE TO THE RADIAL SHAFT CAN OCCUR.

(1) Install the radial shaft [1] as follows:

(a) Remove the protective covers from the fan frame shroud [3].

(b) Make sure that the mating faces are clean and clear of unwanted materials.

CAUTION DO NOT USE FORCE ON THE RADIAL SHAFT. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE RADIAL SHAFT.

(c) Install the push/pull tool, SPL-14881 to the end of the radial shaft [1].

(d) Install the radial shaft [1] in a correct position on the engine.

(e) Move the radial shaft [1] into the fan frame shroud [3].

(f) Remove the push/pull tool, SPL-14881 from the radial shaft [1].

(g) Clean the work area and clear the tools and other items.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-62-06-410-002-G00

(1) Install the transfer gearbox assembly [2] (TASK 72-62-03-400-801-G00).

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SUBTASK 72-62-06-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-62-06-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Radial Shaft Installation Test

SUBTASK 72-62-06-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX MODULE - REPAIR

1. General

A. This procedure has this task:

(1) Blending and Spot Touch-Up of the Accessory Gearbox Module.

TASK 72-63-00-300-802-G00

2. Blending and Spot Touch-Up of the Accessory Gearbox Module

(Figure 801, Figure 802, Figure 803, Figure 804)

A. General

(1) This task gives the instructions to do the blending and spot touch-up of the Accessory Gearbox (AGB) Module, made up of the AGB assembly and the AGB slider [7].

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
71-11-04-000-801-G00	Fan Cowl Panels Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-400-801-G00	Fan Cowl Panels Installation (P/B 401)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50295 [CP2010]	Solvent - cleaning, White Spirit	
C50005	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
C50208 [CP2659]	Coating - Chemical Conversion - Bonderite M-CR 1132 Aero Pen (Formerly Alodine 1132)	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2

G. Prepare for the Blending and Spot Touch-Up

SUBTASK 72-63-00-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-00-860-006-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-00-020-002-G00

(3) On the applicable engine, remove the left fan cowl panel (TASK 71-11-04-000-801-G00).

(a) Remove these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Dimensional Information

SUBTASK 72-63-00-320-009-G00

WARNING MAKE SURE THAT YOU READ THE MATERIAL SAFETY DATA SHEET (MSDS) FOR THE MATERIALS YOU WILL USE, AND WEAR THE PROTECTIVE CLOTHING AND EQUIPMENT IF IT IS SPECIFIED. REFER TO THE APPLICABLE MANUFACTURER'S OR SUPPLIER'S MSDS FOR TALK WITH THE LOCAL SAFETY DEPARTMENT OR AUTHORITIES FOR THE PROCEDURES TO DISCARD THIS HAZARDOUS AGENT.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE AN INJURY OR KILL YOU.

(1) Blend the area to repair as follow:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-63-00-320-010-G00

(2) Do not remove more parent material than necessary.

(a) Indications must be repaired without steps and sharp edges.

(b) The hand blending must cover all the damage area.

SUBTASK 72-63-00-320-011-G00

(3) Make sure that the quantity of blend repairs on:

(a) The AGB housing [2] is less or equal than 10.

(b) The front cover [3] is less or equal than 10.

(c) The rear cover [4] is less or equal than 10.

(d) The AGB transfer housing [5] is less or equal than 10.

(e) The AGB transfer tube [6] is less or equal than 10.

(f) The AGB slider [7] is less or equal than 20.

(g) The manual control cover [8] is less or equal than 6.

SUBTASK 72-63-00-320-012-G00

(4) Make sure that the distance between two blends must be equal to the width of maximum indication observed on repaired area, if not the indications must be included on one blend.

SUBTASK 72-63-00-320-013-G00

(5) Blending is not permitted on an area already blended.

SUBTASK 72-63-00-320-014-G00

(6) Obey the dimension limits of areas below:

(a) On area H the maximum finish dimension Z is equal to 0.0354 in. (0.90 mm).

1) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

2) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

(b) On area K the maximum finish dimension Z is equal to 0.0196 in. (0.50 mm).

1) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

2) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

(c) On area J the maximum finish dimension Z is equal to 0.0039 in. (0.10 mm).

1) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

2) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

(d) On area P the maximum finish dimension Z in depth is equal to 0.0137 in. (0.35 mm).

NOTE: The total blended area must not exceed an angular portion of 30° of the bore diameter.

1) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

2) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) On area R the maximum finish dimension Z in depth is equal to 0.0196 in. (0.50 mm).

NOTE: The total blended area must not exceed 1.644 in^2 (1061 mm^2) (10% of the total starter interface flat surface).

1) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

2) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

(f) On locations 2 and 3 on lugs the maximum finish dimension Z is equal to 0.0250 in. (0.64 mm).

1) The minimum description blending limit X on location 2 and 3 is equal to 40 x dim.Z.

2) The maximum description blending limit X on location 2 and 3 is equal to 45 x dim.Z.

3) The description blending limit Y on location 2 and 3 is edge to edge.

(g) Blending is not allowed on the area G.

CAUTION

MAKE SURE THAT AREA L HAS A MINIMUM REMAINING THICKNESS OF 0.118 INCH (3.0 MM) AFTER YOU BLEND IT. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE ENGINE FAILURE.

(h) Areas L is subject to have minimum 0.118 in. (3.0 mm) residual thickness after blending.

1) On area L the maximum finish dimension Z is equal to 0.0354 in. (0.90 mm), if only after blending the measured residual thickness of material is minimum 0.118 in. (3.0 mm).

2) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

3) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

(i) On locations edge M the maximum finish dimension Z is equal to 0.0118 in. (0.30 mm).

1) The minimum description blending limit X on edge M is equal to 40 x dim.Z.

2) The maximum description blending limit X on edge M is equal to 45 x dim.Z.

3) The description blending limit Y on edge M is edge to edge.

(j) On locations edge N the maximum finish dimension Z is equal to 0.0629 in. (1.60 mm).

NOTE: The total blended area must not exceed an angular portion of 18° of the bore diameter.

1) The minimum description blending limit X on edge N is equal to 4 x dim.Z.

2) The maximum description blending limit X on edge N is equal to 5 x dim.Z.

3) The description blending limit Y on edge N is edge to edge.

(k) On locations edge O the maximum finish dimension Z is equal to 0.0748 in. (1.90 mm).

NOTE: The total blended area must not exceed an angular portion of 18° of the bore diameter.

1) The minimum description blending limit X on edge O is equal to 4 x dim.Z.

2) The maximum description blending limit X on edge O is equal to 5 x dim.Z.

3) The description blending limit Y on edge O is edge to edge.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Blending and Spot Touch-Up of the AGB Module

SUBTASK 72-63-00-320-015-G00

(1) Check for dimensions, locations and quantities of indications on areas located on Figure 802 and Figure 803 and compare with repair limits.

CAUTION

YOU MUST NOT BLEND THE PART IN NOT PERMITTED AREA. THIS CAN CAUSE DAMAGE TO THE PART.

(a) If the indications or repair area is more than the limits, the part is not repairable by this procedure.

(b) Blending is not allowed on the area G.

(c) Blending is not permitted on an area already blended.

(d) Make sure that the distance between two blends must be equal to the width of maximum indication observed on repaired area, if not the indications must be included on one blend.

SUBTASK 72-63-00-320-016-G00

(2) Remove the indications from the repair area by a smooth hand blending.

(a) For general hand blending:

1) Sharp edges can be blended out, using abrasive tools. Coarse grades of abrasives or files may be used for fast metal removal, but the parts must then be given a smooth surface finish with fine grades of abrasives or crocus cloth.

2) Avoid removing metal from leading and trailing edges of part sections in such a way that the edges become thin or sharp; blend so as to maintain approximately the original contour.

3) Use typical abrasive tools:

- Abrasive sheets
- Abrasive belt
- Abrasive disks
- Abrasive stones
- Hand files
- Crocus cloth.

(b) For dimensional information: (Figure 802, Figure 803 and Figure 804)

NOTE: If one blending impacts several areas, consider the lowest depth criteria.

SUBTASK 72-63-00-120-002-G00

(3) Clean the repair area of the AGB module [1] with a brush or a cloth using Ardrox 5503, B50290 [CP2559], solvent, B00682 [CP2011], or White Spirit, B50295 [CP2010].

SUBTASK 72-63-00-230-002-G00

(4) Do the applicable fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

(a) No cracks are permitted.

SUBTASK 72-63-00-350-002-G00

(5) Do a spot touch-up of oxide film coating on the repaired areas (TASK 70-43-07-300-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use only Bonderite M-CR 1200S Aero coating, C50005, and Bonderite M-CR 1132 Aero Pen, C50208 [CP2659].

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-00-420-002-G00

(1) On the applicable engine, install the left fan cowl panel (TASK 71-11-04-400-801-G00).

(a) Install these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-00-860-007-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-00-640-002-G00

(3) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 72-63-00-860-008-G00

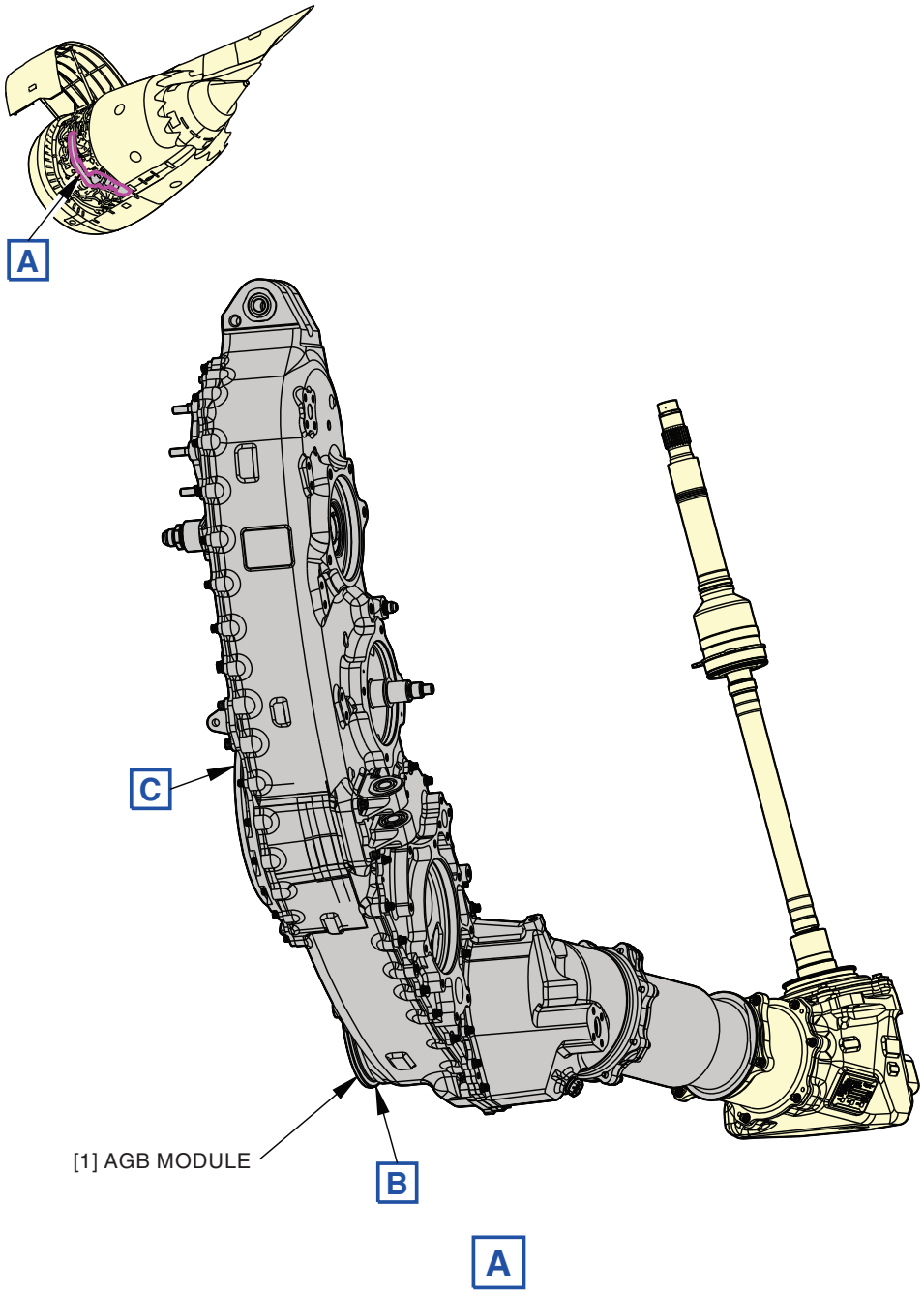
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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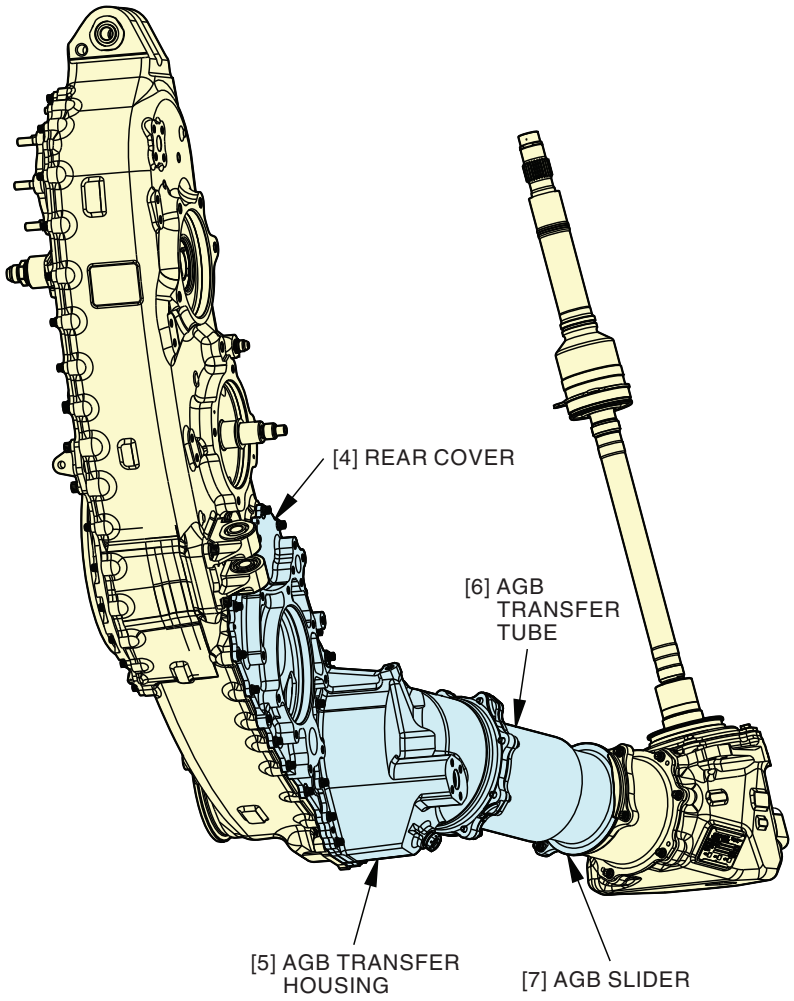
2925997 S0000705833_V1

Accessory Gearbox (AGB) Module
Figure 801/72-63-00-990-805-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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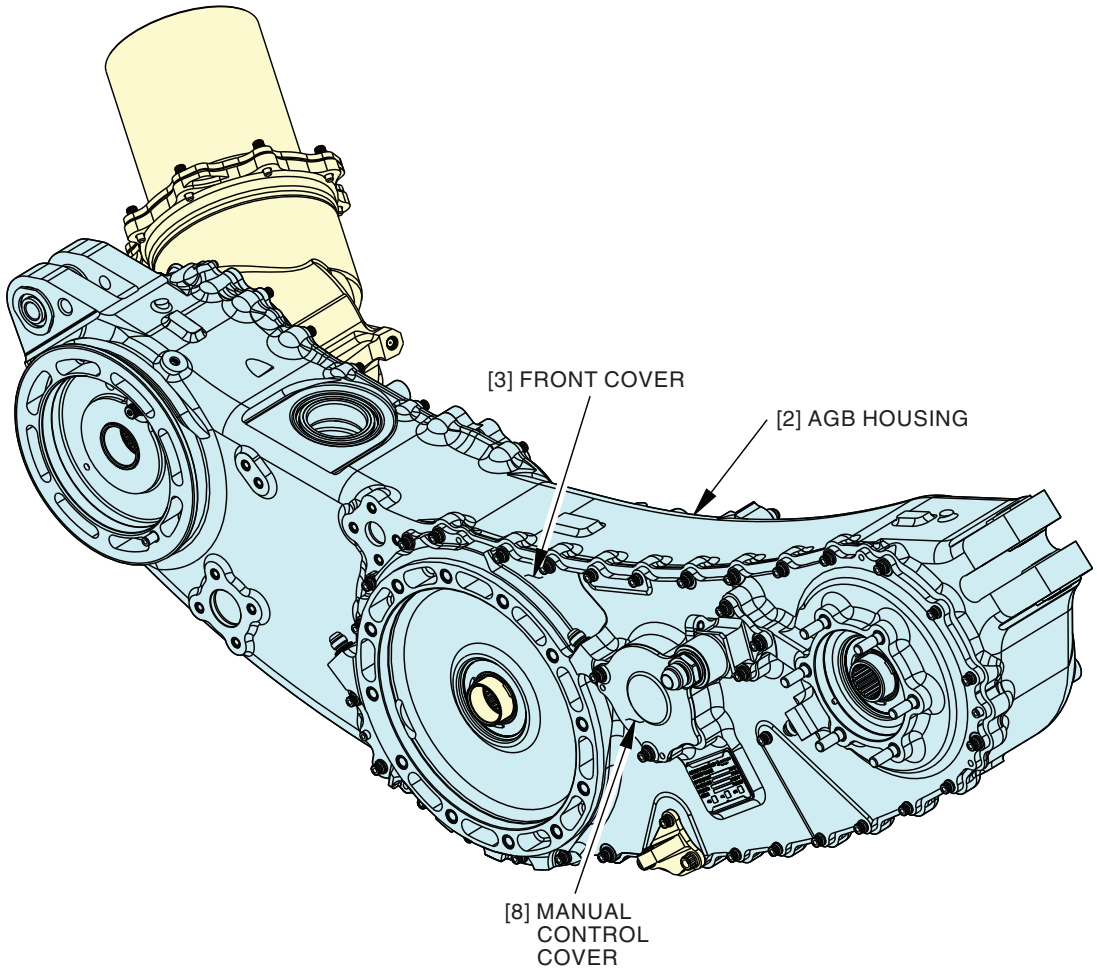
2926004 S0000705834_V1

Accessory Gearbox (AGB) Module
Figure 801/72-63-00-990-805-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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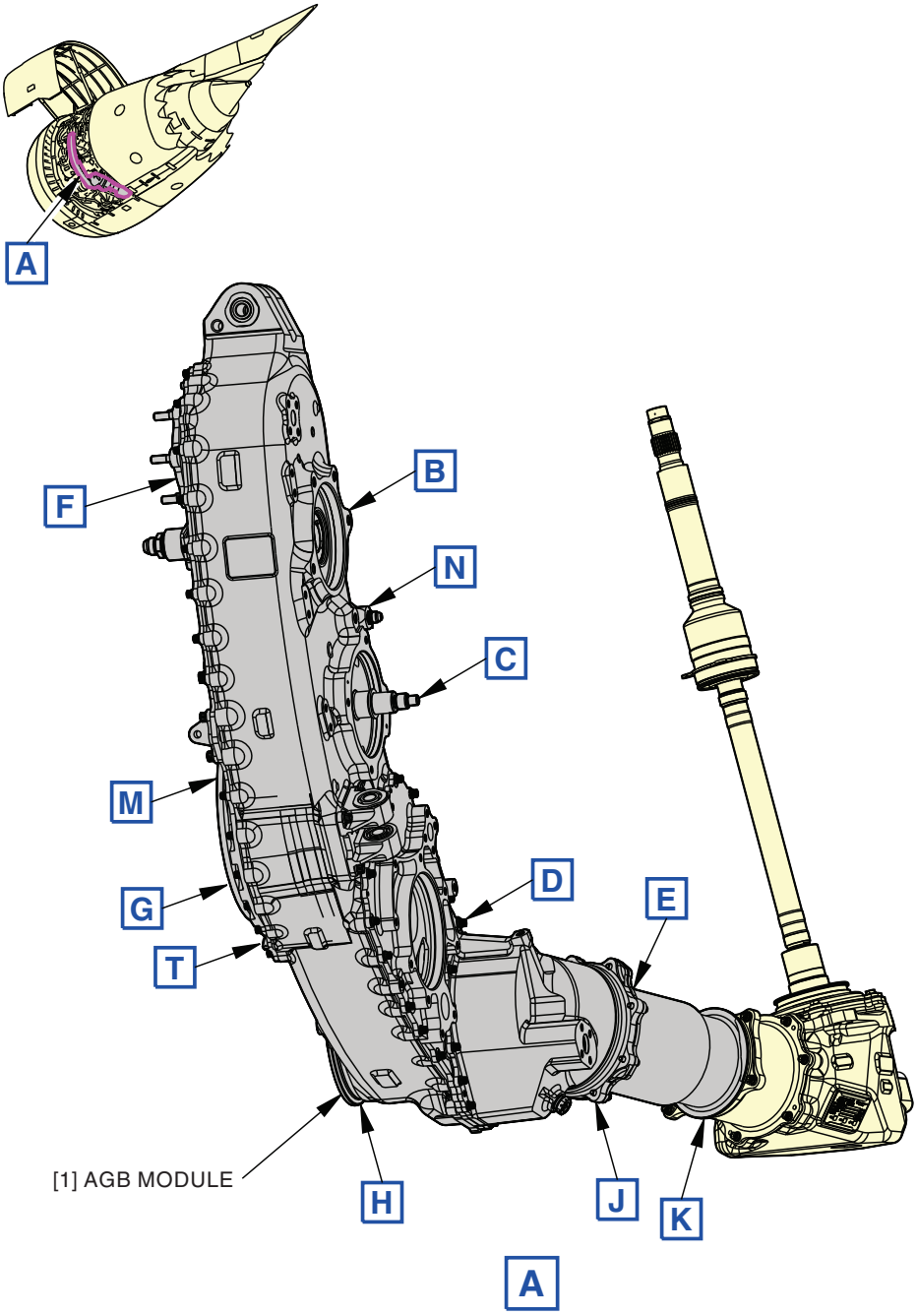
2926005 S0000705836_V1

Accessory Gearbox (AGB) Module
Figure 801/72-63-00-990-805-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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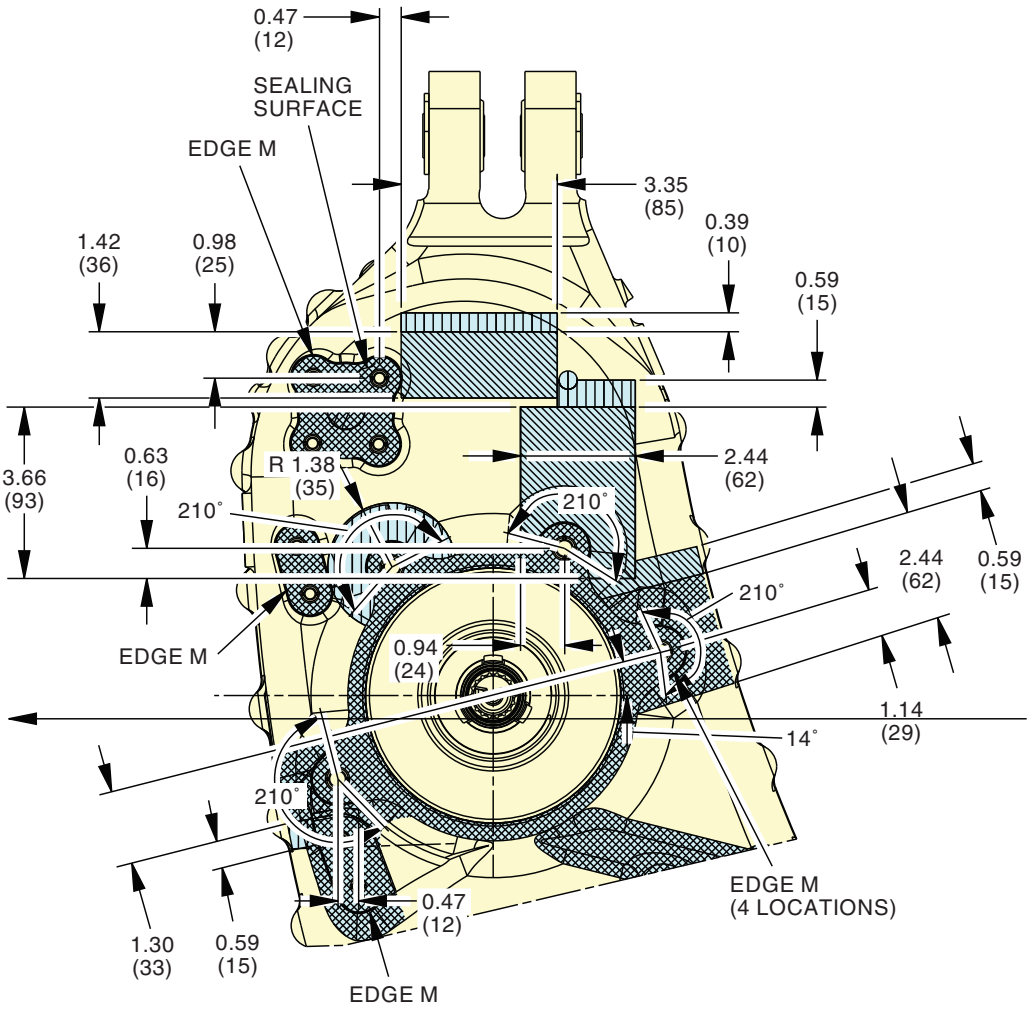
2926008 S0000705837_V1

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 1 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

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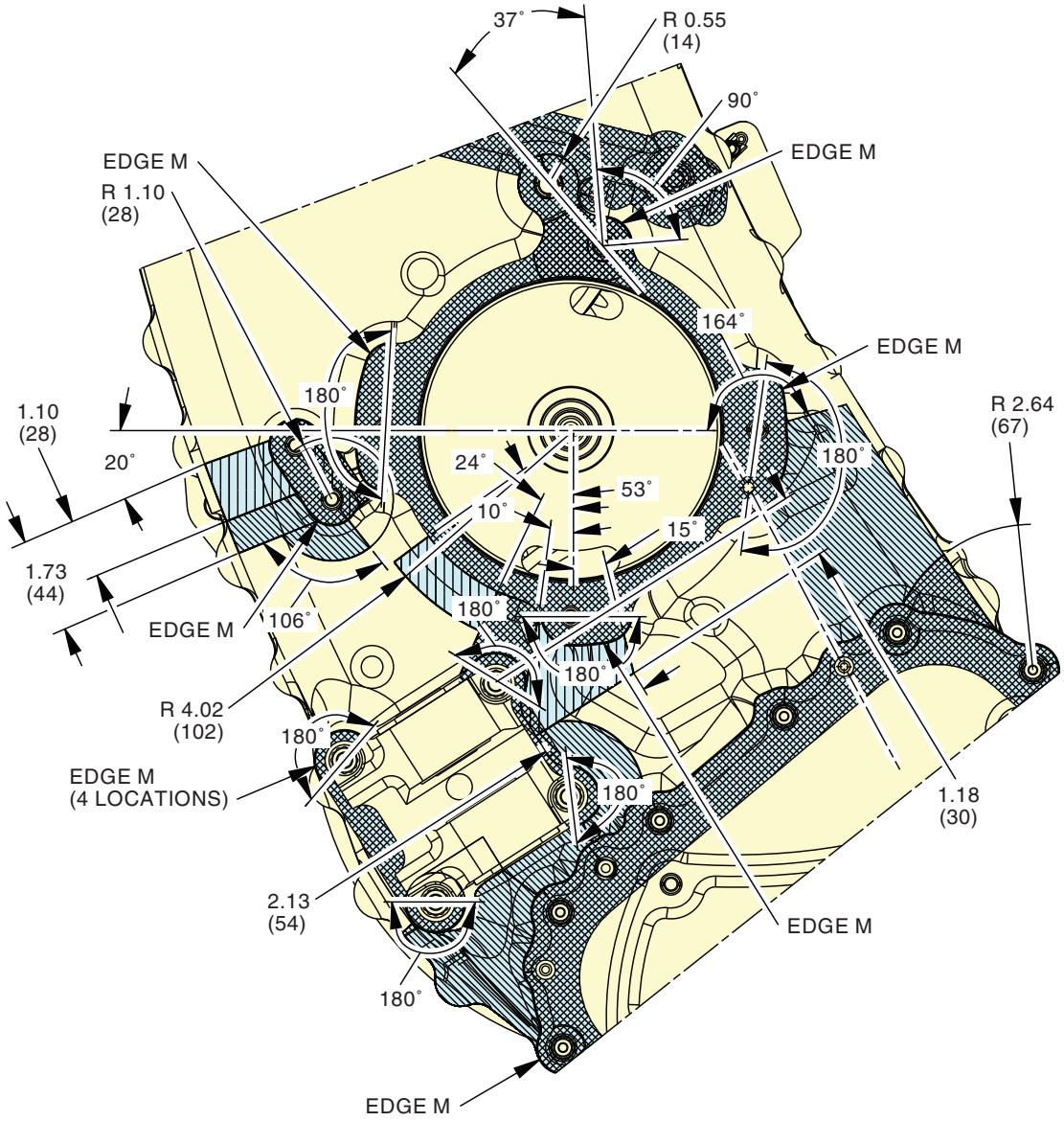
2926010 S0000705838_V1

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 2 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

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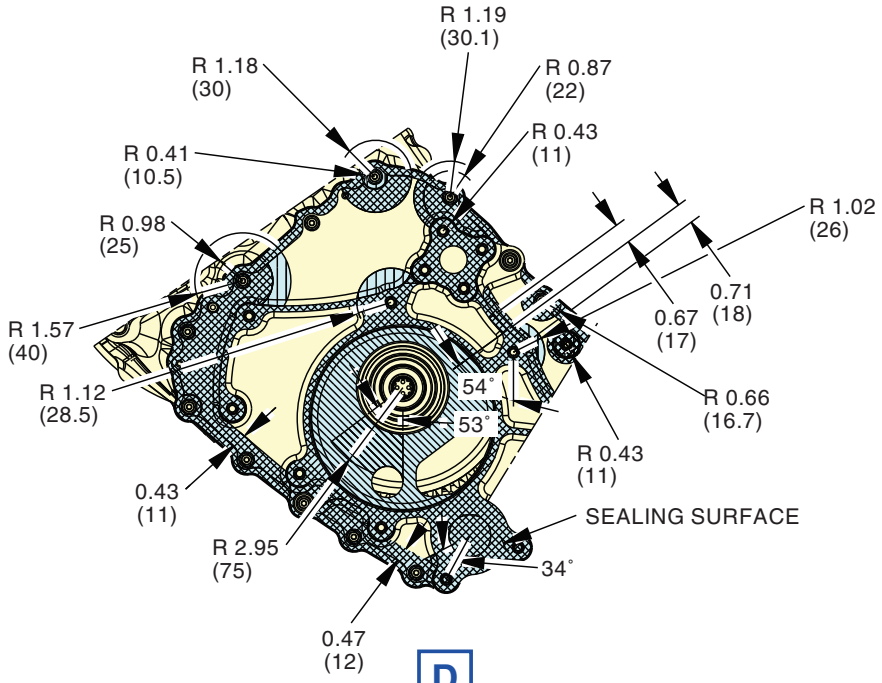
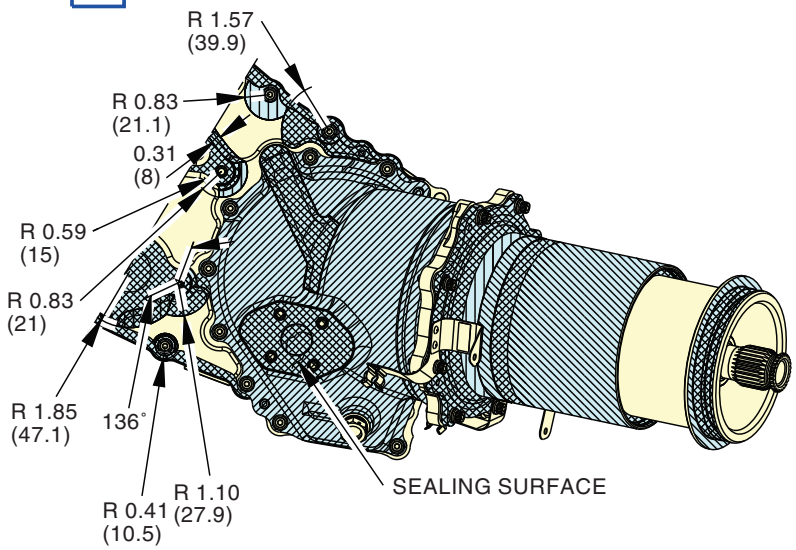
2926012 S0000705839_V1

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 3 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

SUBJECT TO HAVE MINIMUM 0.118 (3.0) RESIDUAL THICKNESS AFTER BLENDING

- AREA L

NOTE:

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES).

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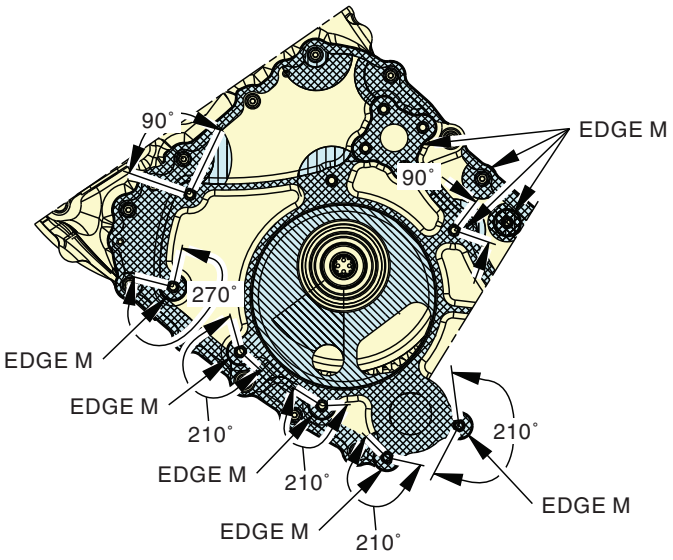
2926013 S0000705840_V1

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 4 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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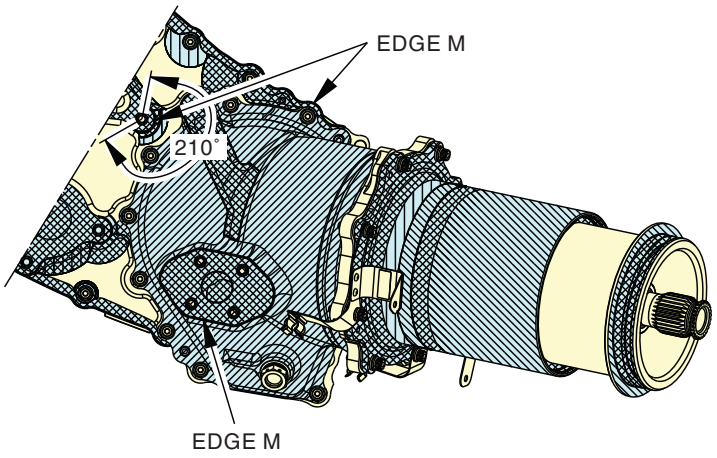

E

LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

SUBJECT TO HAVE MINIMUM 0.118 (3.0) RESIDUAL THICKNESS AFTER BLENDING

- AREA L

NOTE:

ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES).

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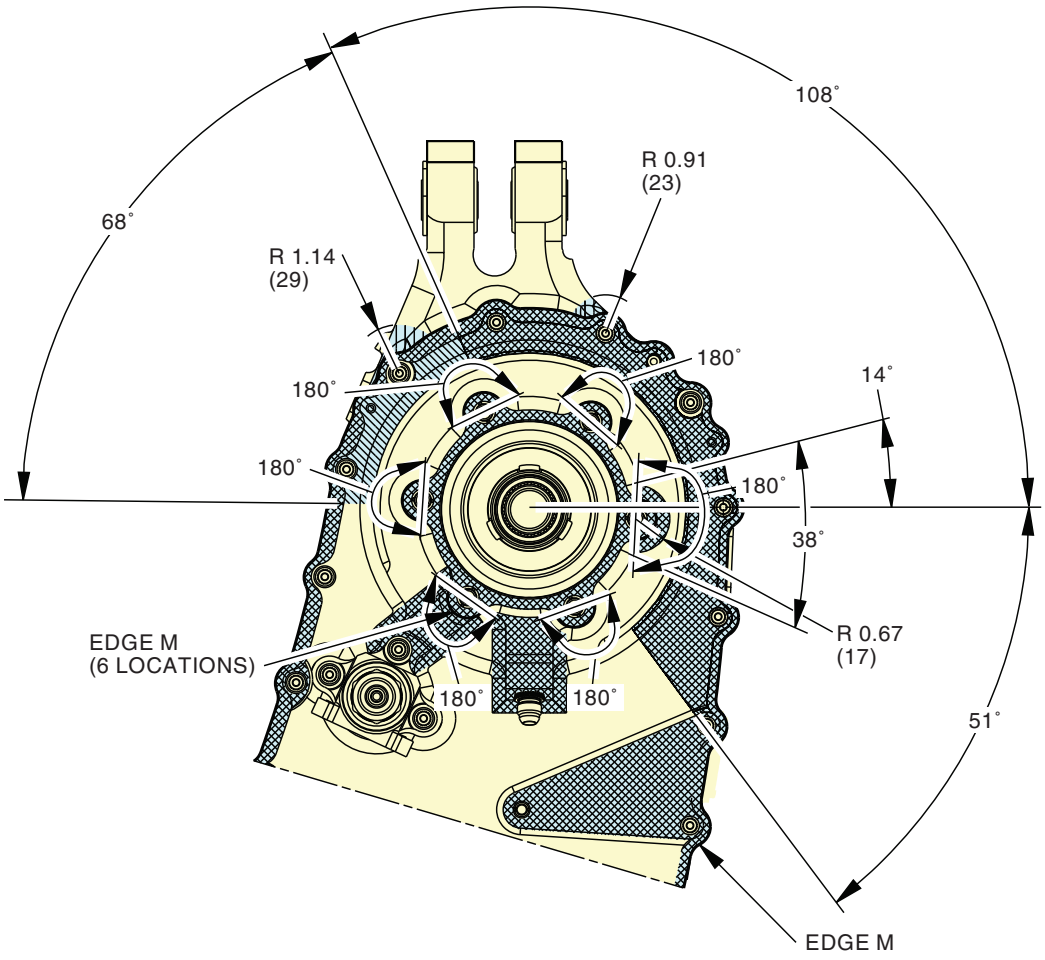
2926019 S0000705841_V1

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 5 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



F

LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

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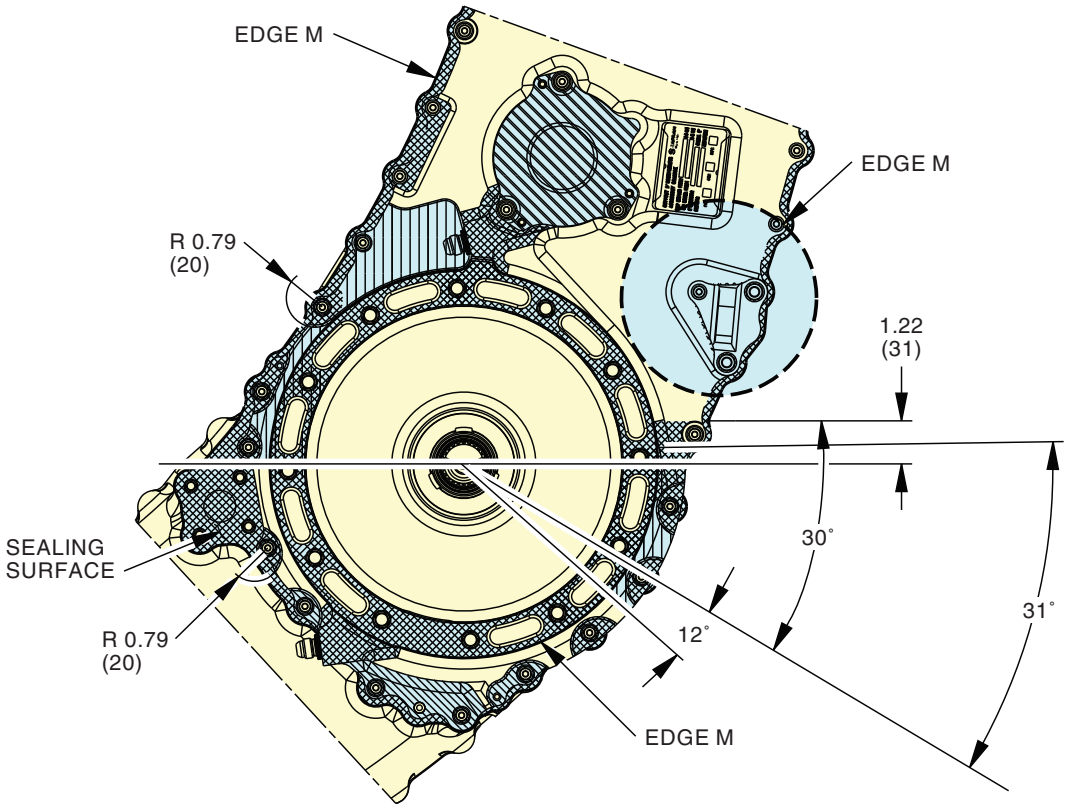
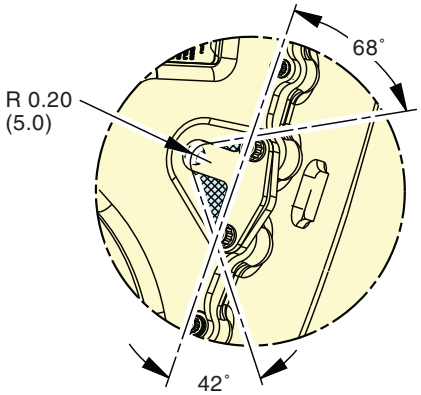
2926021 S0000705842_V1

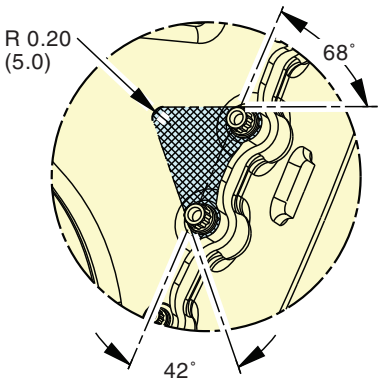
Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





CONFIGURATION 2



CONFIGURATION 3

LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G
- AREA Q

NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

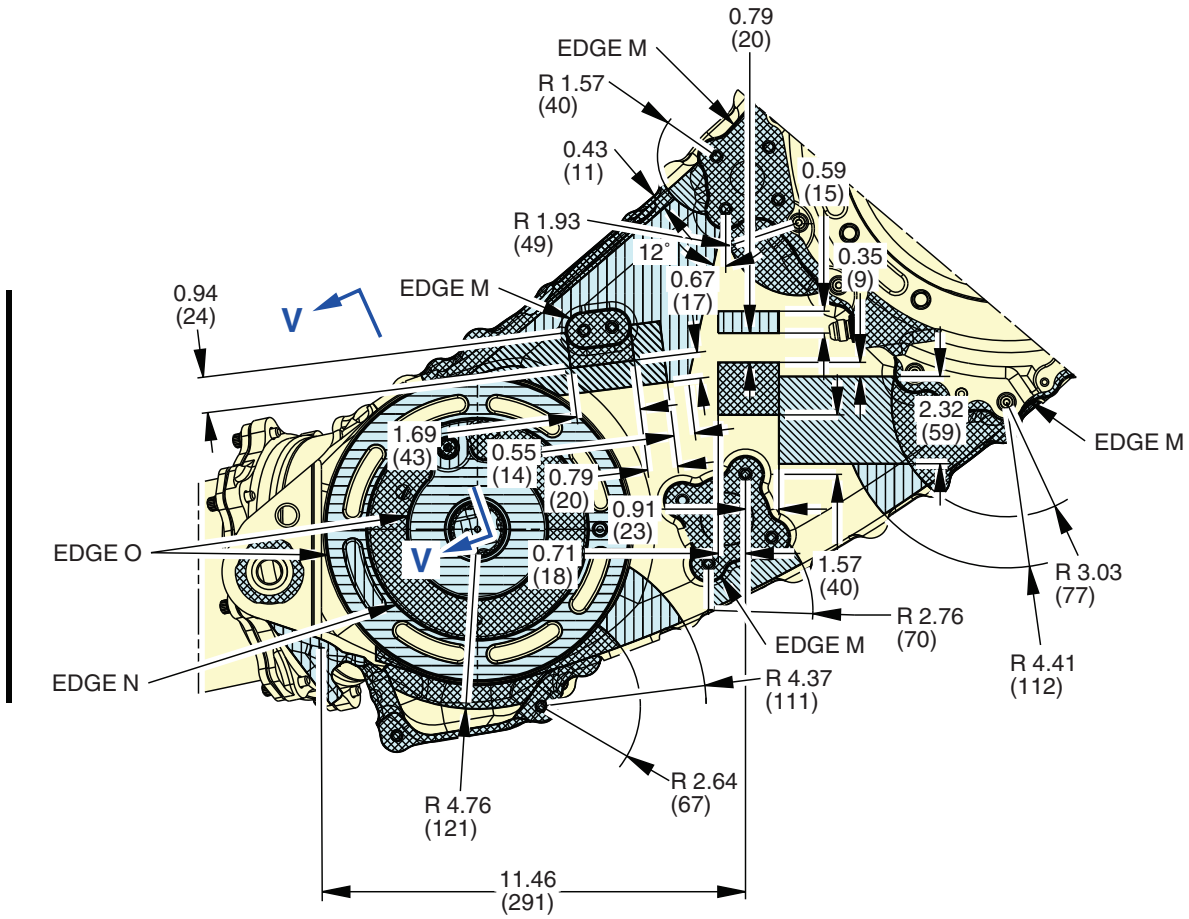
G

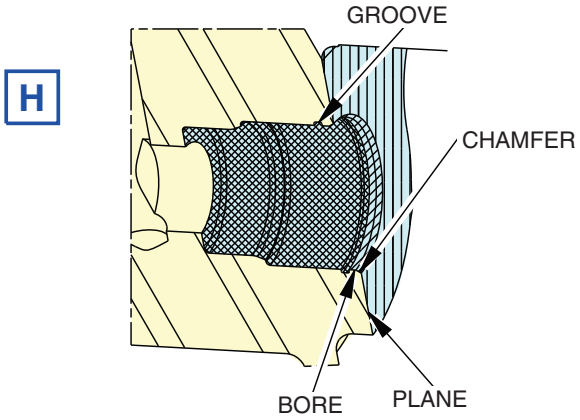
ICNLEAP1B726300D5882800009A00101

2926022 S0000705843_V1

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 7 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G
- AREA P
- AREA R

NOTE:

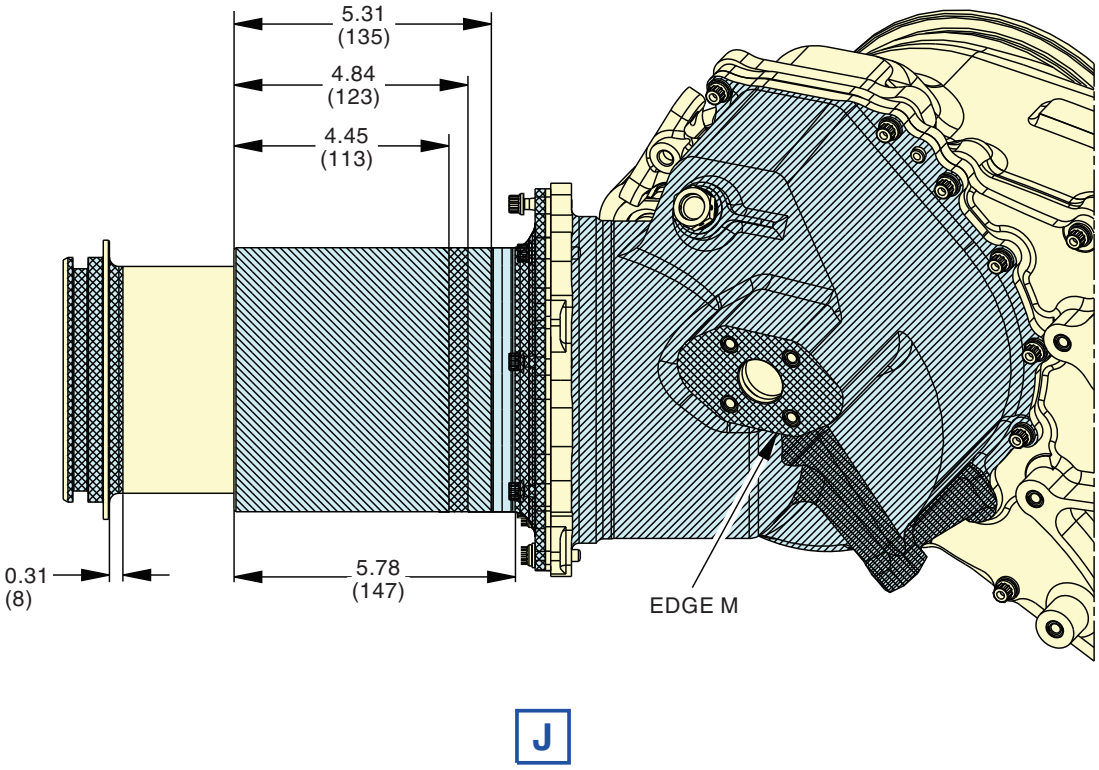
ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

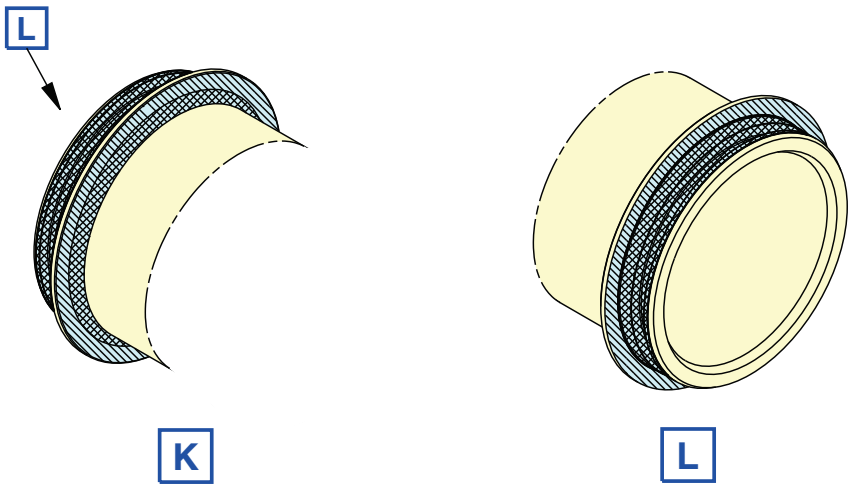
V-V

ICNLEAP1B726300D5882800010A00101
2926023 S0000705844_V2

Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 8 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

- AREA H
- AREA K
- AREA J
- AREA G

SUBJECT TO HAVE MINIMUM 0.118 (3.0) RESIDUAL THICKNESS AFTER BLENDING

- AREA L

NOTE:

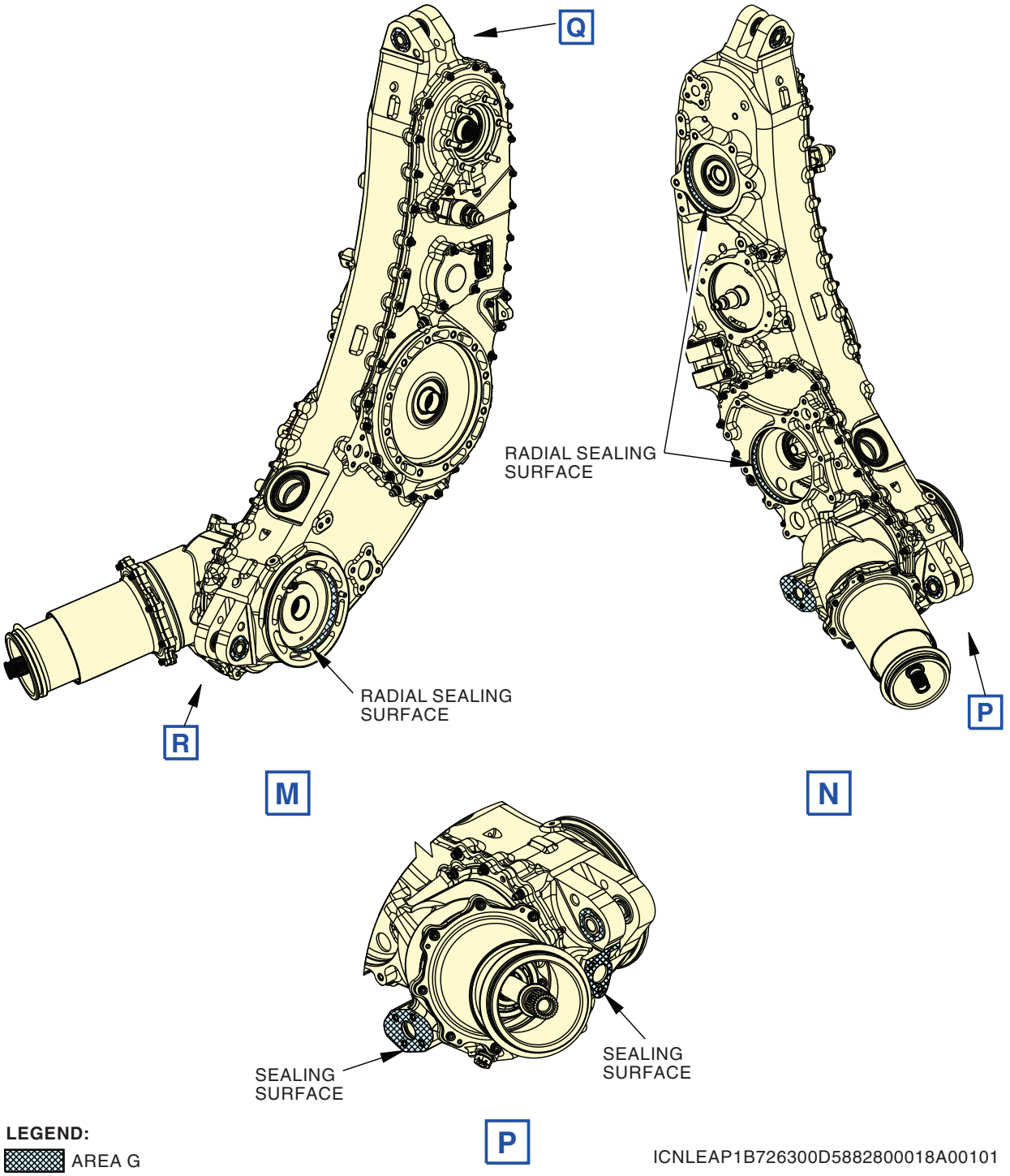
ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES).

ICNLEAP1B726300D5882800017A00101

2926024 S0000705845_V1

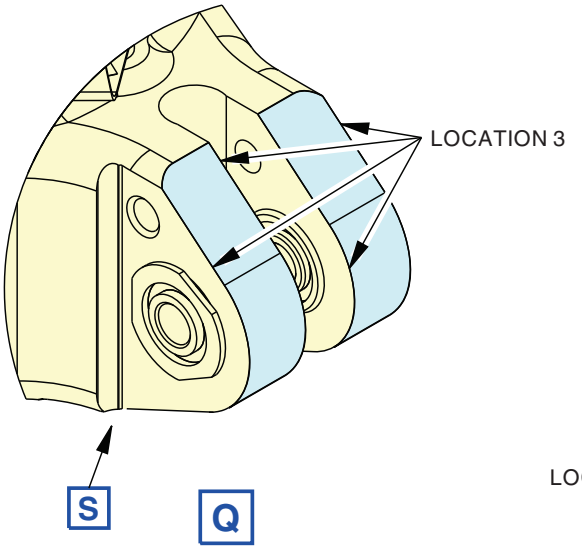
Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 9 of 10)

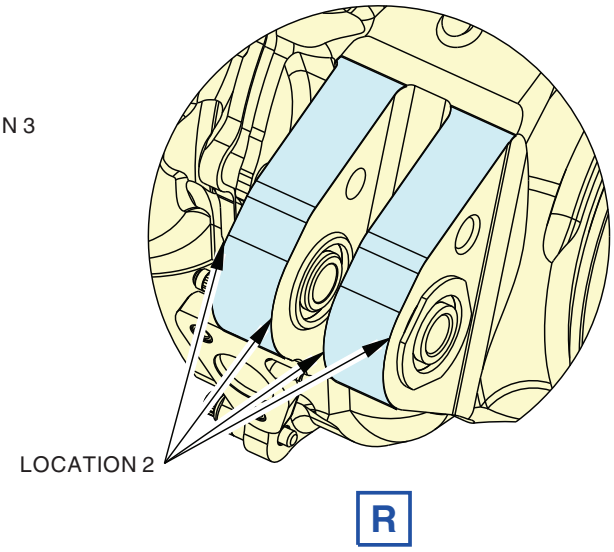
ECCN 9E991 BOEING PROPRIETARY - See title page for details

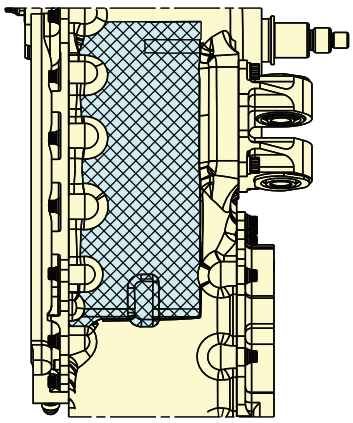


Blending and Spot Touch-Up of the Accessory Gearbox (AGB) Module
Figure 802/72-63-00-990-806-G00 (Sheet 10 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details







LEGEND:

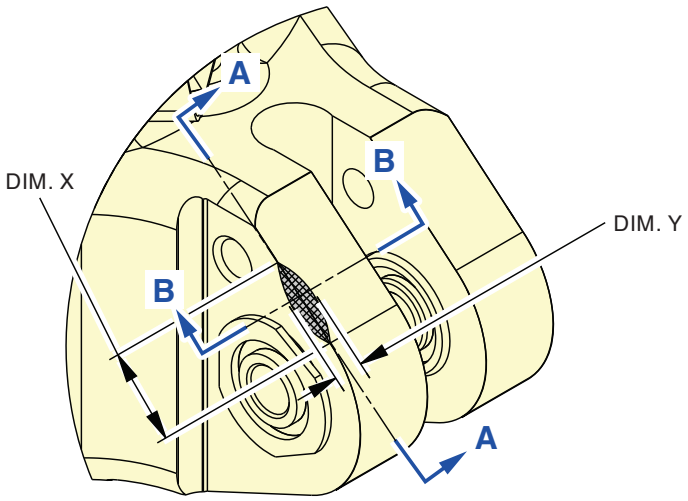
AREA G

ICNLEAP1B726300D5882800006A00101

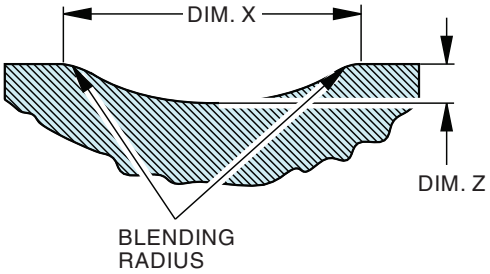
2926033 S0000705849_V1

Blending Specific Dimensions of the Lug and Edge M of the Accessory Gearbox (AGB) Module
Figure 803/72-63-00-990-807-G00 (Sheet 1 of 2)

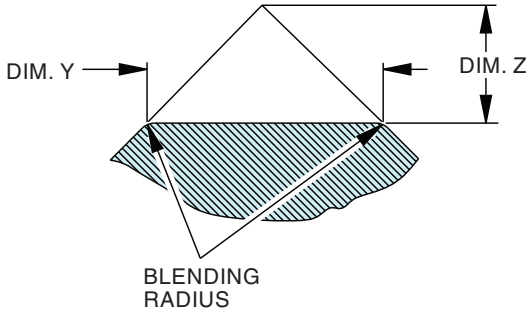
ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLENDING DIMENSIONS OF LOCATION 2, LOCATION 3 AND EDGE M (EXAMPLE)



A-A



B-B

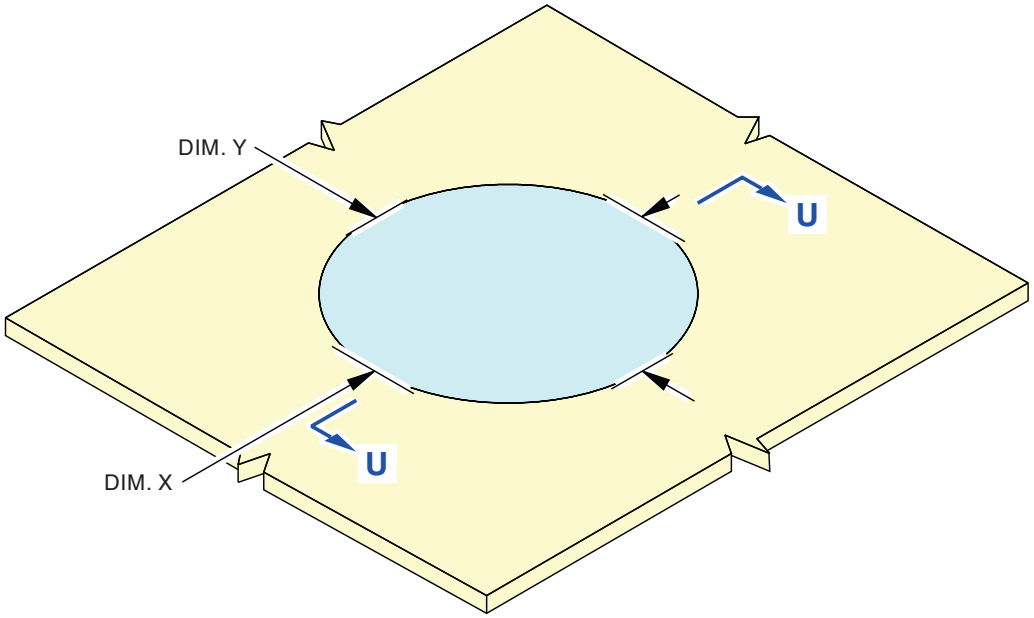
DIM. Y = BLENDING EDGE TO EDGE

ICNLEAP1B726300D5882800011A00101

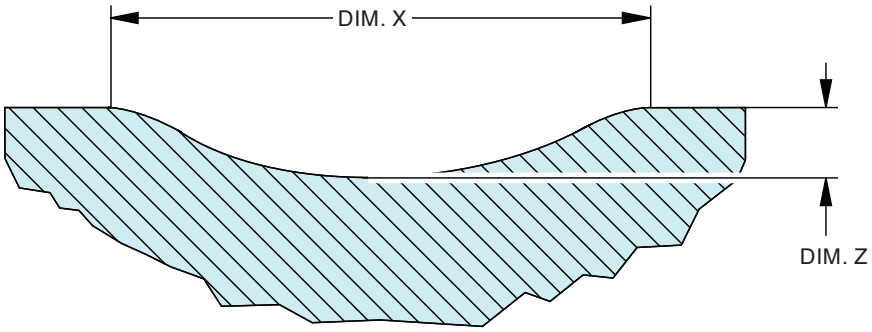
2926035 S0000705850_V1

Blending Specific Dimensions of the Lug and Edge M of the Accessory Gearbox (AGB) Module
Figure 803/72-63-00-990-807-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLENDING DIMENSIONS OF PLANE AREAS (EXAMPLE)



U-U

ICNLEAP1B726300D5882800005A00101

2926038 S0000705853_V1

Blending Specific Dimensions of plane areas of the Accessory Gearbox (AGB) Module
Figure 804/72-63-00-990-808-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX ASSEMBLY - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Accessory gearbox assembly oil replenishing.

TASK 72-63-01-610-802-G00

2. Accessory Gearbox Assembly Oil Replenishing

(Figure 201)

A. General

(1) This task gives the instructions to replenish the accessory gearbox assembly with oil.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-790-804-G00	Test No. 4 - Idle Leak Test (P/B 501)
71-11-04-000-801-G00	Fan Cowl Panels Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-400-801-G00	Fan Cowl Panels Installation (P/B 401)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Gasket seal	72-00-00-13-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Replenishing

SUBTASK 72-63-01-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-01-010-003-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-01-020-003-G00

(3) On the applicable engine, remove the left fan cowl panel (TASK 71-11-04-000-801-G00).

(a) Remove these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. Accessory Gearbox Assembly Oil Replenishing

SUBTASK 72-63-01-020-004-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Disconnect the oil tube [2] from the accessory gearbox assembly [1] as follows:

(a) Remove the four bolts [3] from the oil tube [2].

(b) Disconnect the oil tube [2] from the accessory gearbox assembly [1].

(c) Remove and discard the gasket seal [4] from the oil tube [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-63-01-610-003-G00

CAUTION

DO NOT PUT MORE THAN THE MAXIMUM CAPACITY OF APPROVED ENGINE OIL IN THE ACCESSORY GEARBOX ASSEMBLY. IF YOU ADD TO MUCH ENGINE OIL IN THE ACCESSORY GEARBOX ASSEMBLY, TO MUCH SERVICING CAN OCCUR.

CAUTION

USE ONLY APPROVED ENGINE OIL WHEN SERVICING THE ENGINE OIL SYSTEMS. DO NOT INTERMIX WITH OTHER OIL TYPES.

(2) Replenish the accessory gearbox assembly [1] with 2 gal (6 l) of approved engine oil through the vent line input bore of the oil tube [2].

NOTE: Use the same approved engine oil currently used on the engine to replenish the accessory gearbox assembly [1].

NOTE: Use local manufactured tool (plastic tube and funnel) to replenish the accessory gearbox assembly [1] with engine oil through the vent line input bore of the oil tube [2].

NOTE: Diameter of the vent line input bore of the oil tube [2] is 0.787 in. (20 mm).

SUBTASK 72-63-01-420-003-G00

(3) Connect the oil tube [2] to the accessory gearbox assembly [1] as follows:

(a) Lubricate the new gasket seal [4] with engine oil, D00599 [CP2442].

(b) Install the gasket seal [4] and oil tube [2] on the accessory gearbox assembly [1].

(c) Install the four bolts [3] that attach the oil tube [2] to the accessory gearbox assembly [1].

(d) Tighten the four bolts [3] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-01-420-004-G00

(1) On the applicable engine, install the left fan cowl panel (TASK 71-11-04-400-801-G00).

(a) Install these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-01-410-003-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-01-610-004-G00

(3) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 72-63-01-860-008-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

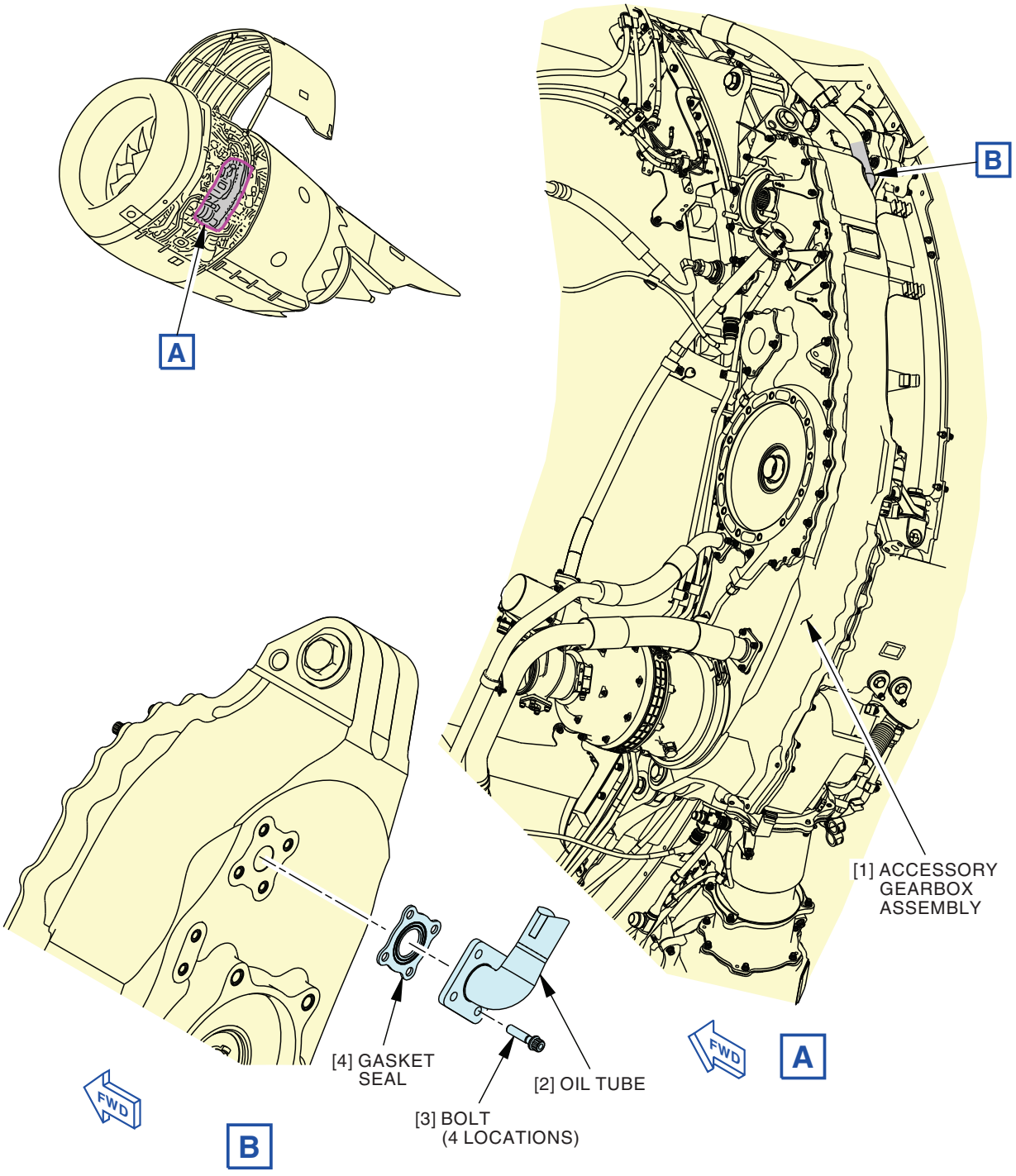
(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-63-01-790-002-G00

(5) Do this task: Test No. 4 - Idle Leak Test, TASK 71-00-00-790-804-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2784328 S0000631217_V1

Accessory Gearbox Assembly Oil Replenishing
Figure 201/72-63-01-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the accessory gearbox assembly

(2) An installation of the accessory gearbox assembly.

TASK 72-63-01-000-801-G00

2. Accessory Gearbox Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the accessory gearbox assembly.

NOTE: The figures has been updated to introduce POST-SB 72-0181 configuration of the Fan Frame Shroud. The instructions are not changed.

B. References

Reference	Title
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
24-11-11-000-801	Integrated Drive Generator (IDG) Removal (P/B 401)
29-11-11-000-801	Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal (P/B 401)
71-11-04-000-801-G00	Fan Cowl Panels Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-000-801-G00	Transfer Gearbox Assembly Removal (P/B 401)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
73-11-01-000-801-G00	Main Fuel Pump Removal (P/B 401)
73-21-08-000-801-G00	Permanent Magnet Alternator Removal (P/B 401)
77-12-02-000-801-G00	N2 Speed Sensor Removal (P/B 401)
79-21-03-000-801-G00	Lubrication Unit Removal (P/B 401)
79-32-01-000-801-G00	Oil Pressure and Temperature Sensor Removal (P/B 401)
80-11-01-000-801-G00	Engine Starter Removal (P/B 401)
80-11-07-000-801-G00	Starter Air Valve Manual Control Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14690	Fixture - Accessory Gearbox (AGB) Remove/Install Part #: 956A7018G02 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
SPL-25234	Pallet - Storage, Accessory Gearbox (AGB) Part #: 956A6650G01 Supplier: 58828
STD-193	Container
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Preformed packing	72-63-01-03-380	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 72-63-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-01-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-01-020-001-G00

(3) On the applicable engine, remove the left fan cowl panel (TASK 71-11-04-000-801-G00).

(a) Remove these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Accessory Gearbox Assembly Removal

SUBTASK 72-63-01-010-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

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WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the accessory gearbox assembly [1] as follows:

(a) Open the oil tank filler cap (TASK 12-13-11-600-803).

(b) Drain the oil tank (TASK 12-13-11-600-803).

(c) Put an 8 gal (30 l) oil resistant container, STD-193, below the accessory gearbox assembly [1].

(d) Drain the accessory gearbox assembly [1] as follows:

1) Cut the safety cable [4] or safety wire [4] from the drain plug [2].

2) Remove the drain plug [2] from the accessory gearbox assembly [1].

3) Remove and discard the preformed packing [3] from the drain plug [2].

4) Let the oil drain into the container.

(e) Install the drain plug [2] on the accessory gearbox assembly [1] as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [2] with engine oil, D00599 [CP2442].

3) Lubricate the new preformed packing [3] with pure mineral vaseline, D00672 [CP5070].

4) Install the new preformed packing [3] on the groove of the drain plug [2].

5) Install the drain plug [2] on the accessory gearbox assembly [1].

6) Tighten the drain plug [2] to 186 in-lb (21 N·m) - 210 in-lb (24 N·m).

7) Install the safety wire [4] (safety wire, G02345 [CP8001]) or safety cable [4] (safety cable, G50065 [CP8006]) on the drain plug [2].

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(f) Remove the accessory gearbox manual control cover (TASK 72-63-04-000-801-G00).

(g) Lock the N2 rotor with the Engine core lock adapter, SPL-14920, through the accessory gearbox cover (handcranking cover).

(h) Remove the starter air valve manual control (TASK 80-11-07-000-801-G00).

(i) Remove the N2 speed sensor (TASK 77-12-02-000-801-G00).

(j) Remove the engine starter (TASK 80-11-01-000-801-G00).

(k) Remove the integrated drive generator (TASK 24-11-11-000-801).

(l) Remove the engine driven hydraulic pump (TASK 29-11-11-000-801).

(m) Remove the main fuel pump (TASK 73-11-01-000-801-G00).

(n) Remove the permanent magnet alternator (TASK 73-21-08-000-801-G00).

(o) Remove the lubrication unit lube and scavenge pump (TASK 79-21-03-000-801-G00).

(p) Disconnect the transfer gearbox assembly (TASK 72-62-03-000-801-G00).

(q) Remove the oil pressure and temperature sensor (TASK 79-32-01-000-801-G00).

(r) Remove the drain tubes from the accessory gearbox assembly [1] as follows:

1) Disconnect the drain tube [24] from the accessory gearbox assembly [1] as follows:

a) Loosen the coupling nut [25] from drain tube [24].

b) Disconnect the drain tube [24] from the accessory gearbox assembly [1].

c) Remove the bolt [26] that attaches the clamp [27] and drain tube [24] to the bracket [29].

2) Disconnect the drain tube [30] from the accessory gearbox assembly [1] as follows:

a) Loosen the coupling nut [31] from drain tube [30].

b) Disconnect the drain tube [30] from the accessory gearbox assembly [1].

c) Remove the bolt [32] that attaches the clamp [33] and drain tube [30] to the bracket [35].

3) Disconnect the drain tube [36] from the accessory gearbox assembly [1] as follows:

a) Loosen the coupling nut [37] from drain tube [36].

b) Disconnect the drain tube [36] from the accessory gearbox assembly [1].

4) Disconnect the drain tube [38] from the accessory gearbox assembly [1] as follows:

a) Loosen the coupling nut [39] from drain tube [38].

b) Disconnect the drain tube [38] from the accessory gearbox assembly [1].

5) Remove the engine brackets from the accessory gearbox assembly [1] as follows:

a) Remove the two bolts [5] that attach the bracket [6] to the accessory gearbox assembly [1].

b) Remove the bracket [6] from the accessory gearbox assembly [1].

c) Remove the two bolts [7] that attach the bracket [8] to the accessory gearbox assembly [1].

d) Remove the bracket [8] from the accessory gearbox assembly [1].

e) Remove the two bolts [9] that attach the bracket [10] to the accessory gearbox assembly [1].

f) Remove the bracket [10] from the accessory gearbox assembly [1].

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g) Remove the two bolts [20] that attach the bracket [21] to the accessory gearbox assembly [1].

h) Remove the bracket [21] from the accessory gearbox assembly [1].

i) Remove the two bolts [22] that attach the bracket [23] to the accessory gearbox assembly [1].

j) Remove the bracket [23] from the accessory gearbox assembly [1].

6) Remove the bonding strap [49] from the accessory gearbox assembly [1] as follows:

a) Remove the bolt [50] that attaches the bonding strap [49] to the accessory gearbox assembly [1].

b) Remove the bonding strap [49] from the accessory gearbox assembly [1].

7) Disconnect the oil tube [40] from the accessory gearbox assembly [1] as follows:

a) Remove the four bolts [41] from the oil tube [40].

b) Disconnect the oil tube [40] from the accessory gearbox assembly [1].

c) Remove and discard the gasket seal [42] from the oil tube [40].

8) Disconnect the oil tube [43] from the accessory gearbox assembly [1] as follows:

a) Remove the four bolts [44] from the oil tube [43].

b) Disconnect the oil tube [43] from the accessory gearbox assembly [1].

c) Remove and discard the gasket seal [45] from the oil tube [43].

9) Disconnect the oil tube [46] from the accessory gearbox assembly [1] as follows:

a) Remove the four bolts [47] from the oil tube [46].

b) Disconnect the oil tube [46] from the accessory gearbox assembly [1].

c) Remove and discard the gasket seal [48] from the oil tube [46].

10) Disconnect the oil tube [51] from the accessory gearbox assembly [1] as follows:

a) Remove the four bolts [52] from the oil tube [51].

b) Disconnect the oil tube [51] from the accessory gearbox assembly [1].

c) Remove and discard the gasket seal [53] from the oil tube [51].

11) Disconnect the oil tube [54] from the accessory gearbox assembly [1] as follows:

a) Loosen the coupling nut [55].

b) Disconnect the oil tube [54] from the accessory gearbox assembly [1].

12) Disconnect the oil tube [56] from the accessory gearbox assembly [1] as follows:

a) Remove the four bolts [57] from the oil tube [56].

b) Disconnect the oil tube [56] from the accessory gearbox assembly [1].

c) Remove and discard the gasket seal [58] from the oil tube [56].

PTF ALL POST SB 737-78-1095; With MDOS-GSE hose support brackets

13) Remove the bracket [81] from the accessory gearbox assembly [1] as follows:

a) Remove the bolt [79], washer [80], washer [83], and nut [84] from the bracket [81] and support [82].

b) Remove the bracket [81] and support [82].

PTF ALL

14) Remove the accessory gearbox assembly [1] as follows:

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WARNING MOVE THE ACCESSORY GEARBOX ASSEMBLY SLOWLY. THE ACCESSORY GEARBOX ASSEMBLY IS HEAVY, INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

a) Remove the accessory gearbox links from the accessory gearbox assembly [1] as follows:

NOTE: The accessory gearbox assembly weighs 143.5 lb (65.1 kg).

<1> Remove the accessory gearbox assembly [1] from axial link assembly [65] as follows:

<a> Remove and discard the cotter pin [66] from the mount axis [69].

<b> Loosen the nut [67] and washer [68] from the mount axis [69].

<c> Remove the mount axis [69] that attaches the axial link assembly [65] to the axial lug [70].

<d> Install the AGB remove/install fixture, SPL-14690, on the accessory gearbox assembly [1].

<2> Remove the accessory gearbox assembly [1] from the stirrup assembly [59] as follows:

<a> Remove and discard the cotter pin [60] from the mount axis [63].

<b> Loosen the nut [61] and washer [62] from the mount axis [63].

<c> Remove the mount axis [63] that attaches the stirrup assembly [59] to the accessory gearbox housing lug [64].

<3> Remove the accessory gearbox assembly [1] from link assembly [71] as follows:

<a> Remove and discard the cotter pin [72] from the mount axis [75].

<b> Loosen the nut [73] and washer [74] from the mount axis [75].

<c> Remove the mount axis [75] that attaches the link assembly [71] to the accessory gearbox mounting lug [76].

<4> Remove the accessory gearbox assembly [1] from the spigot assembly [77] which attaches to the fan frame shroud [78].

15) Remove the accessory gearbox assembly [1] from the engine.

16) You can use the AGB pallet, SPL-25234, to store the accessory gearbox assembly [1].

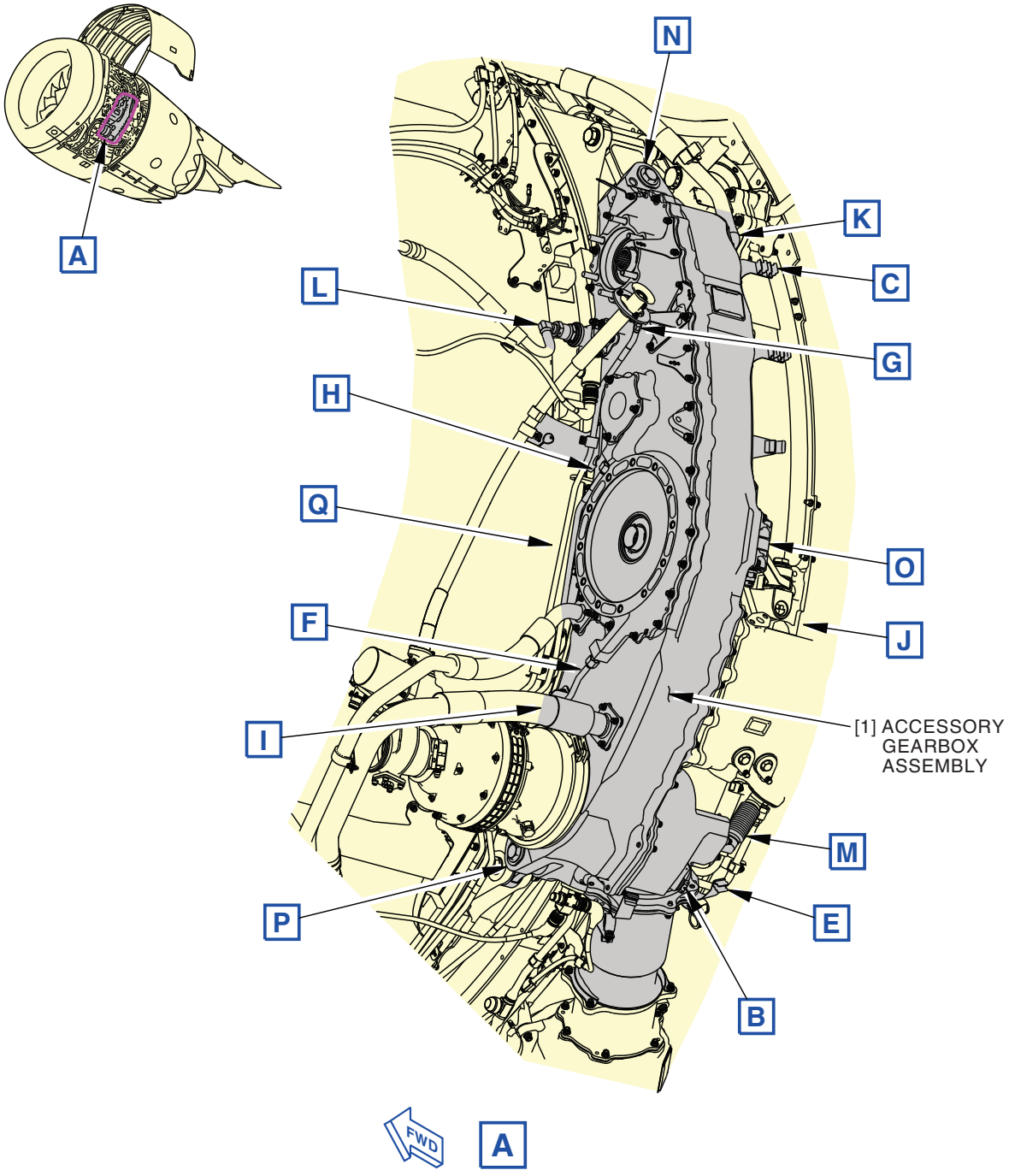
17) Remove the AGB remove/install fixture, SPL-14690, from the accessory gearbox assembly [1].

(s) Install protective covers on all the open ports of the accessory gearbox assembly [1].

(t) Install protective caps on the drain and oil tubes.

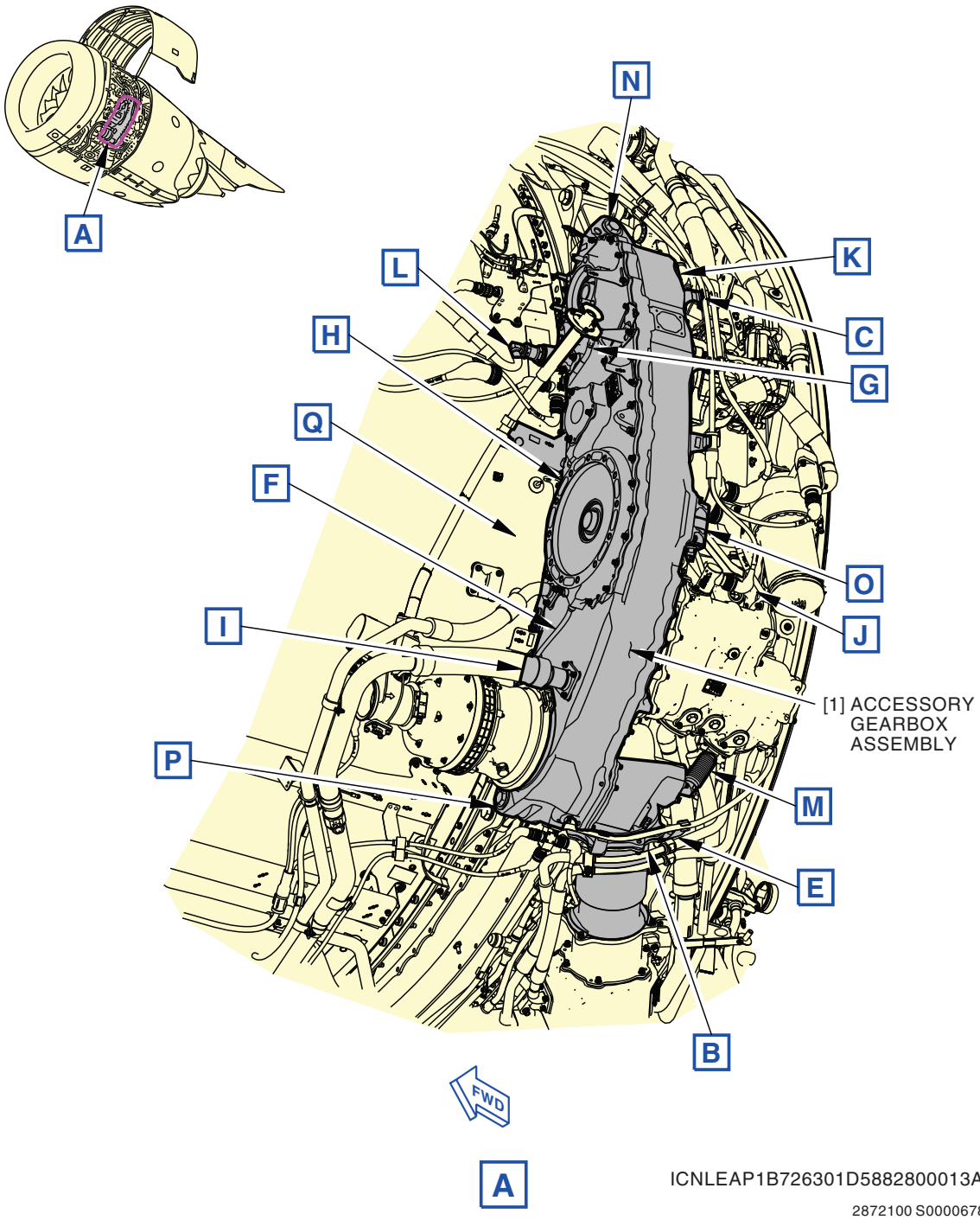
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



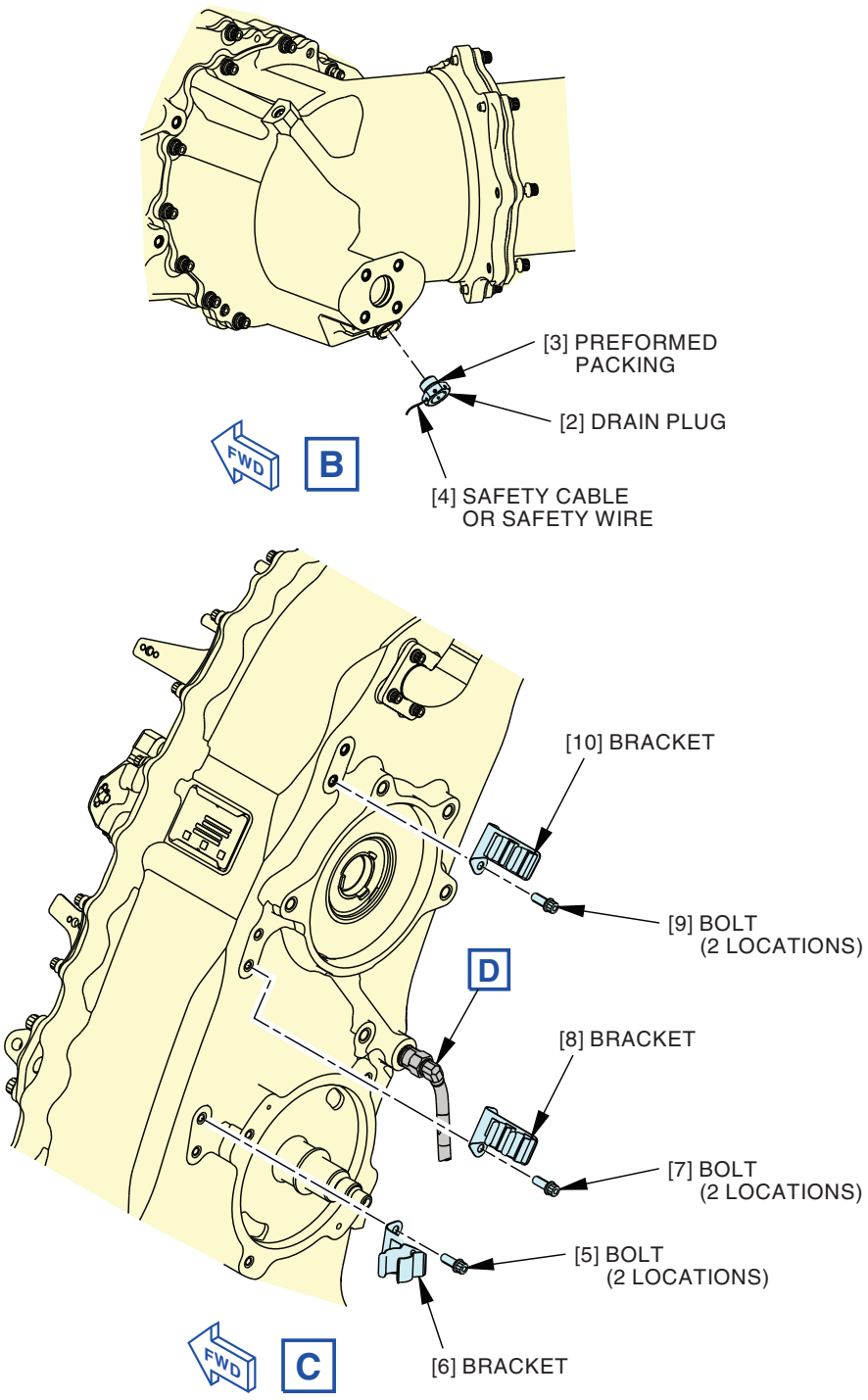
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 1 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



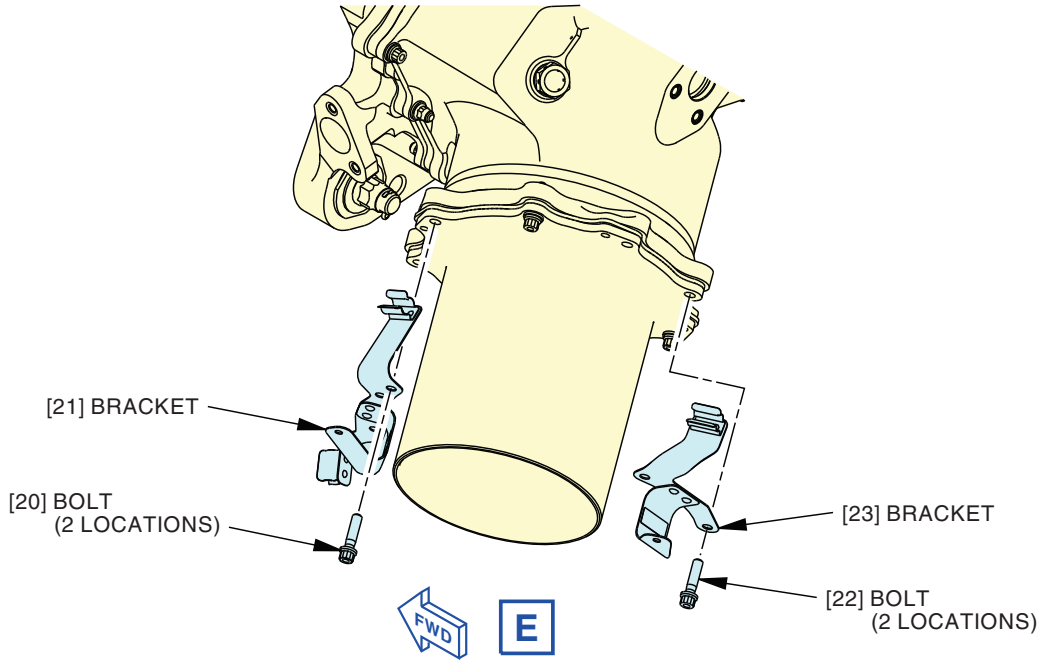
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 2 of 16)

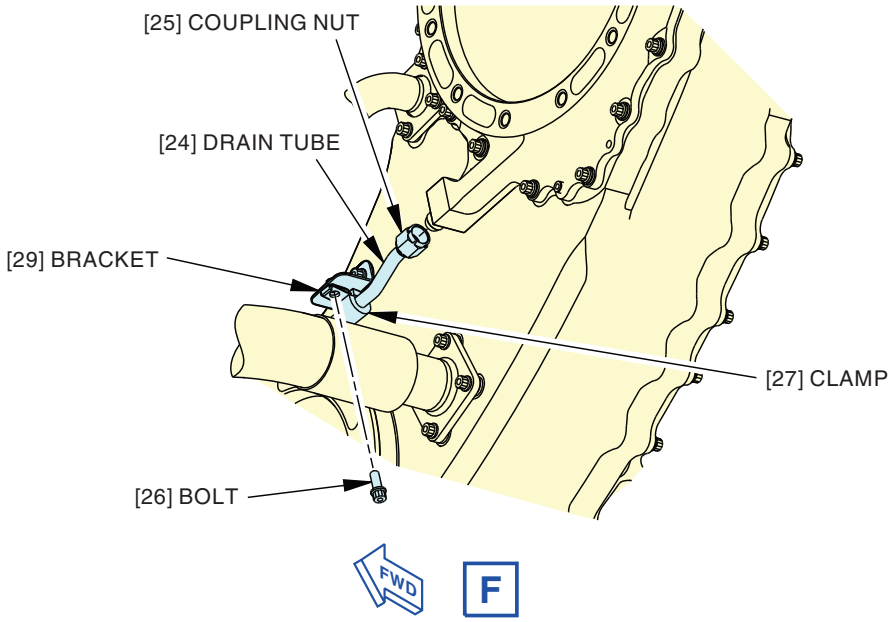
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 3 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

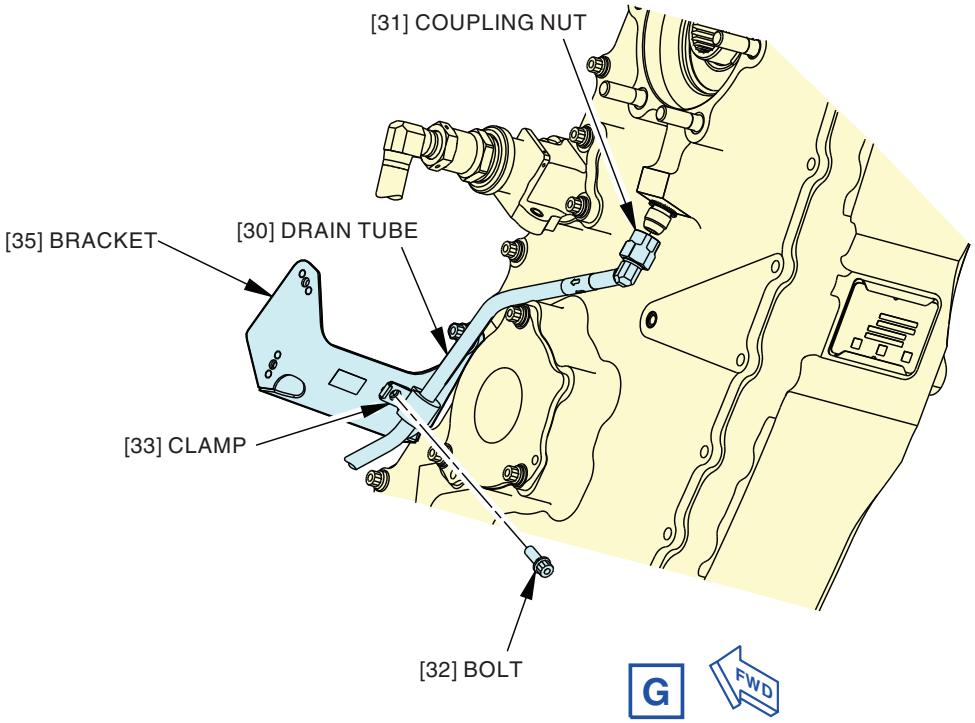




2439530 S0000565668_V3

Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 4 of 16)

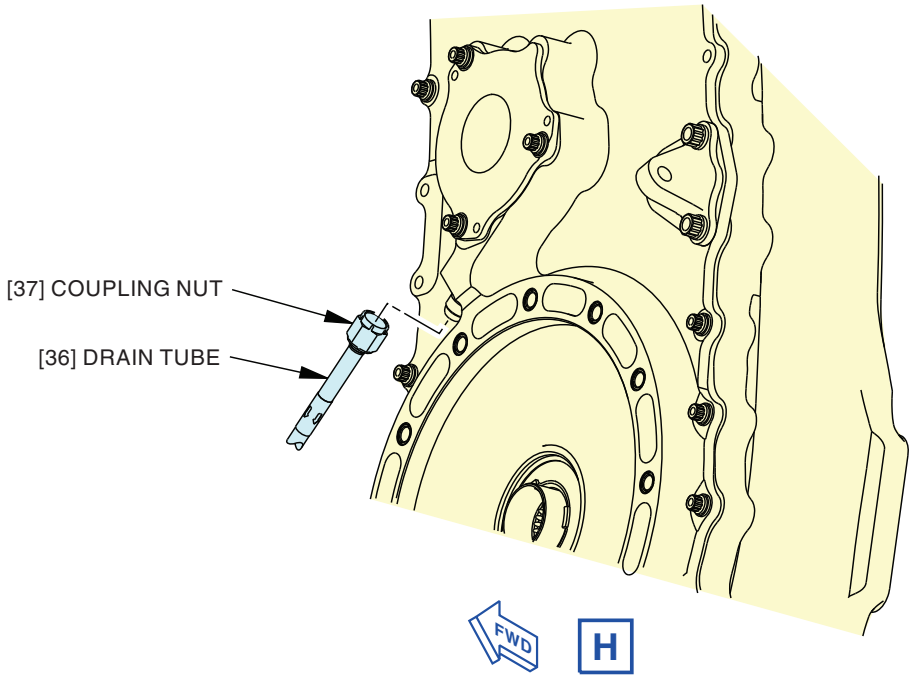
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2439531 S0000565669_V4

Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 5 of 16)

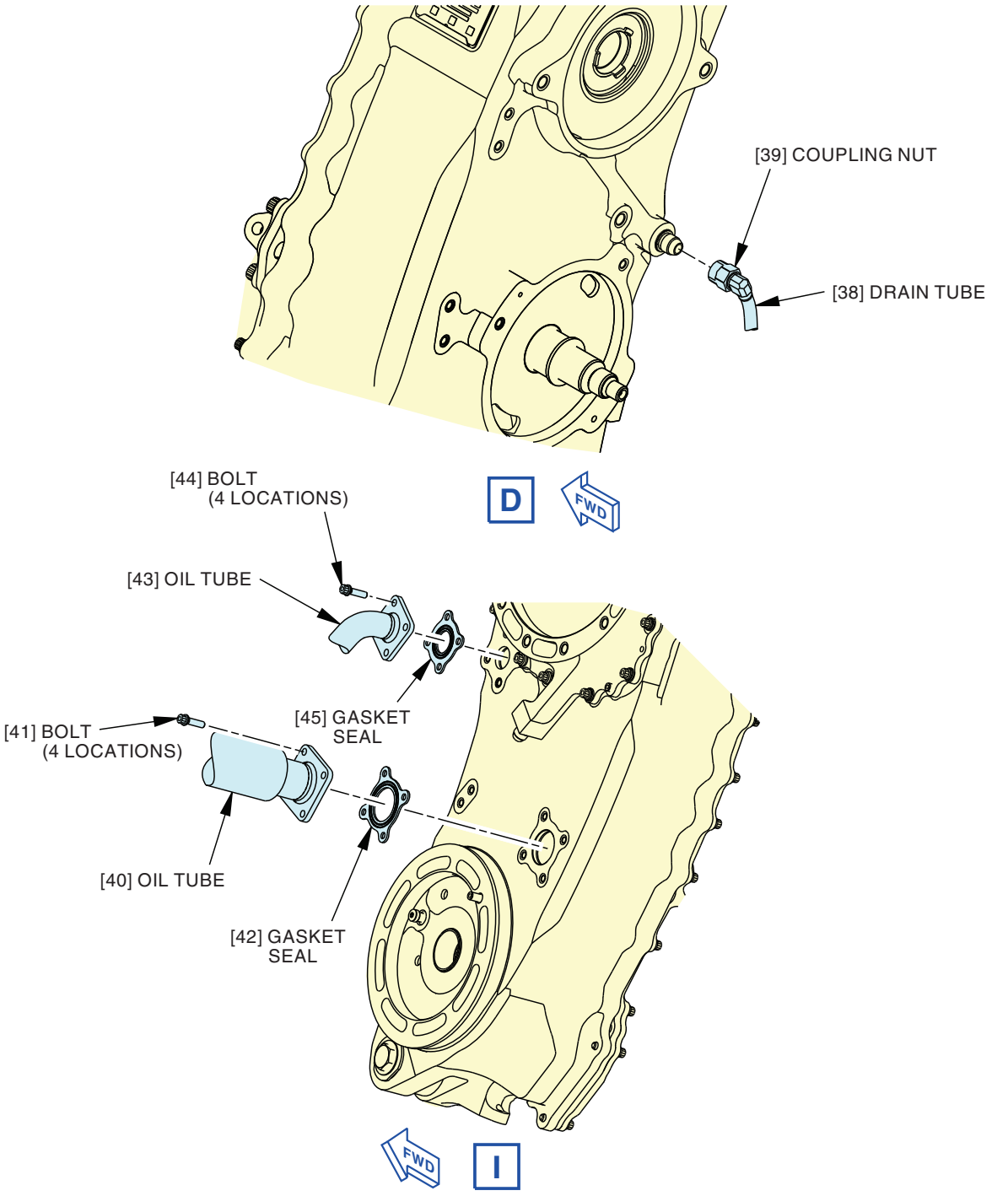
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2439533 S0000565670_V3

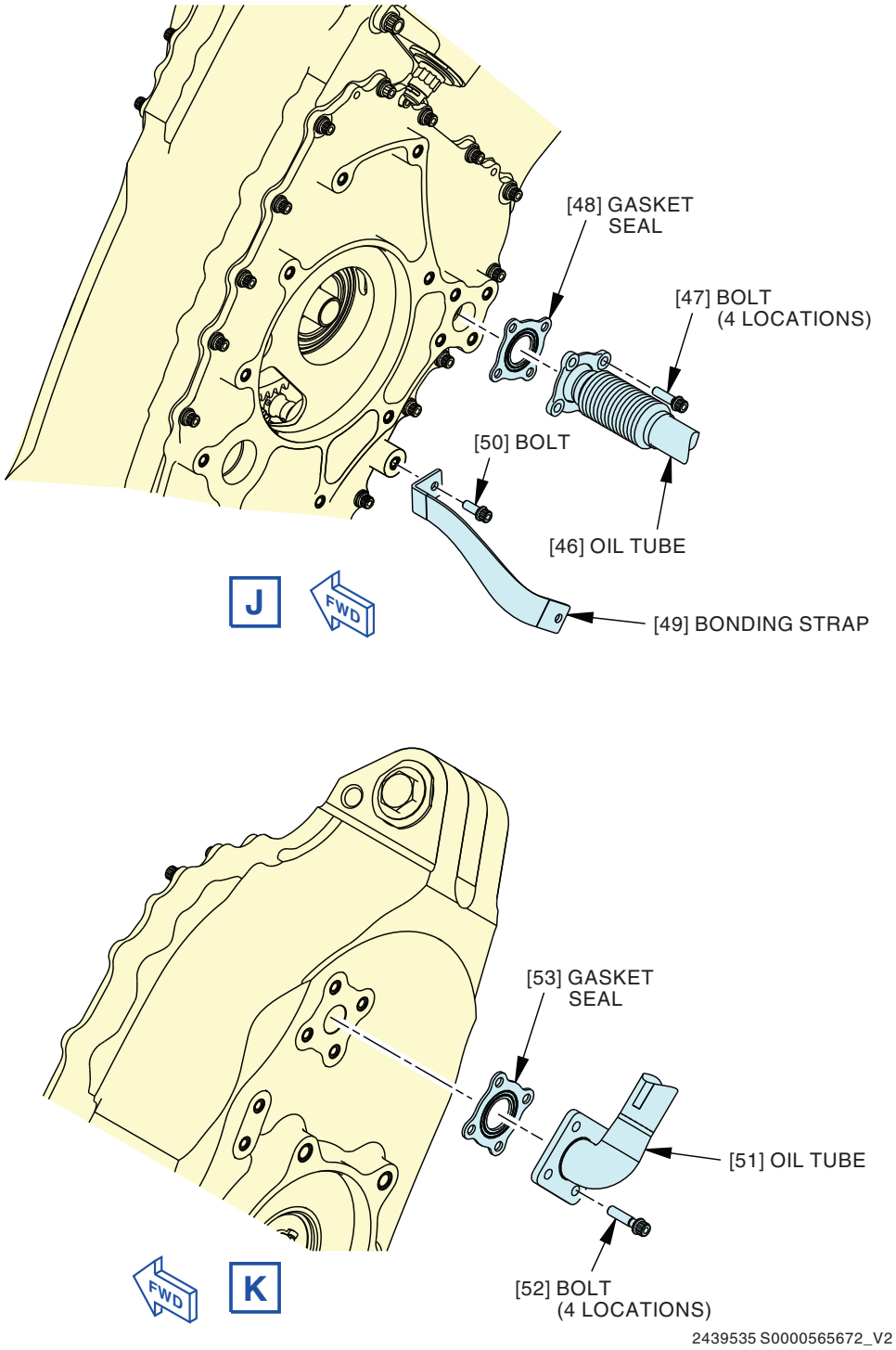
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 6 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



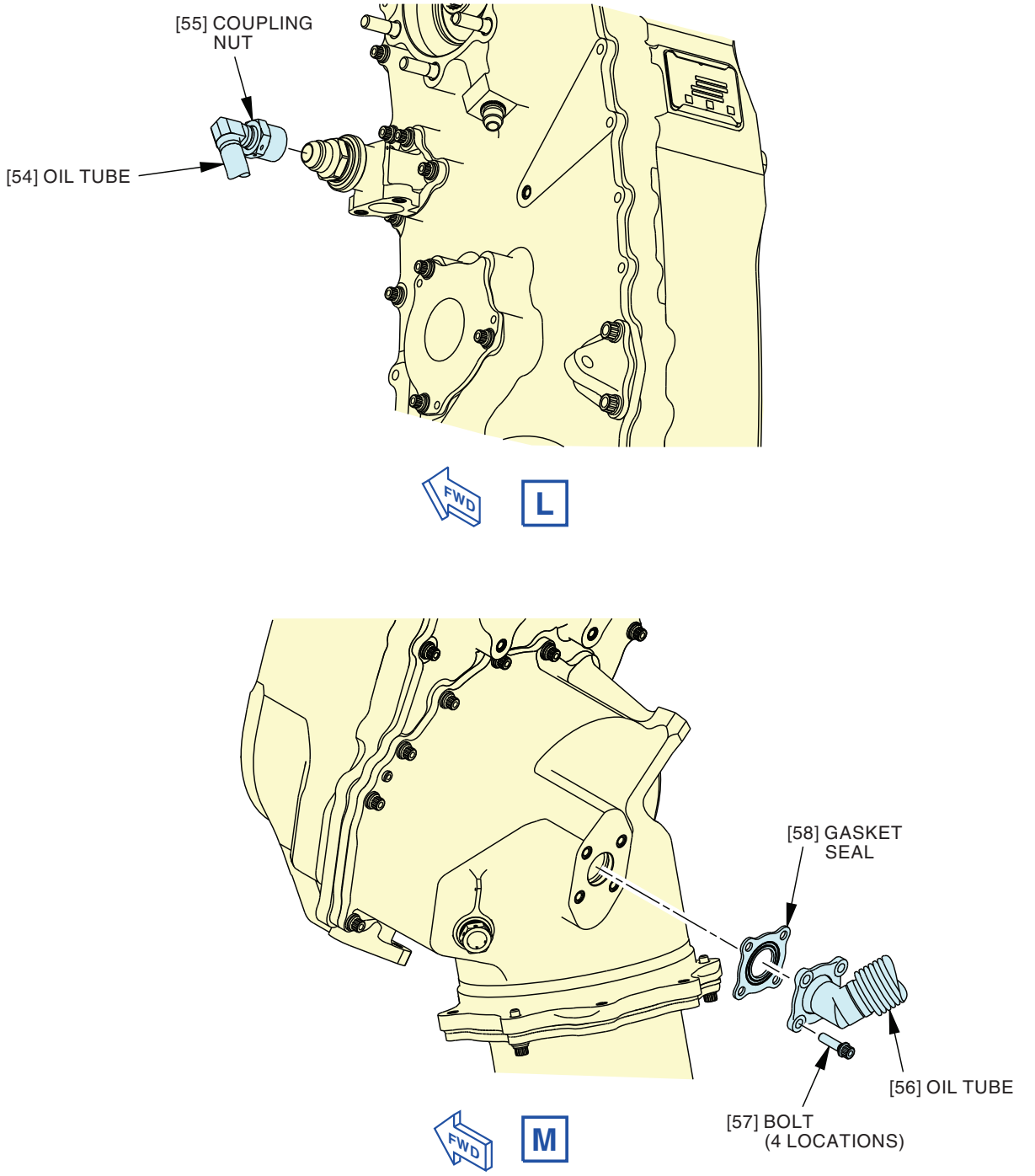
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 7 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



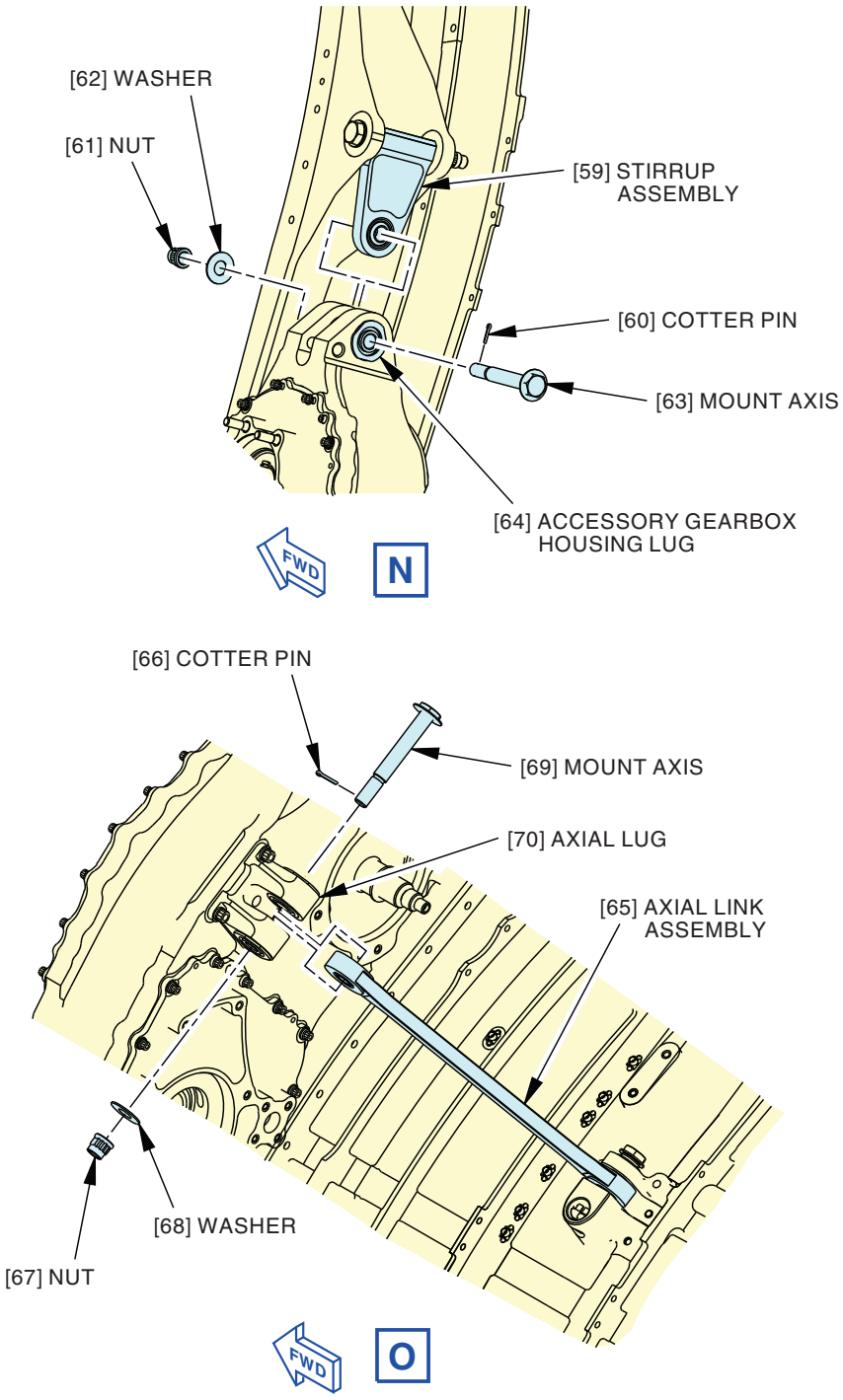
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 8 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 9 of 16)

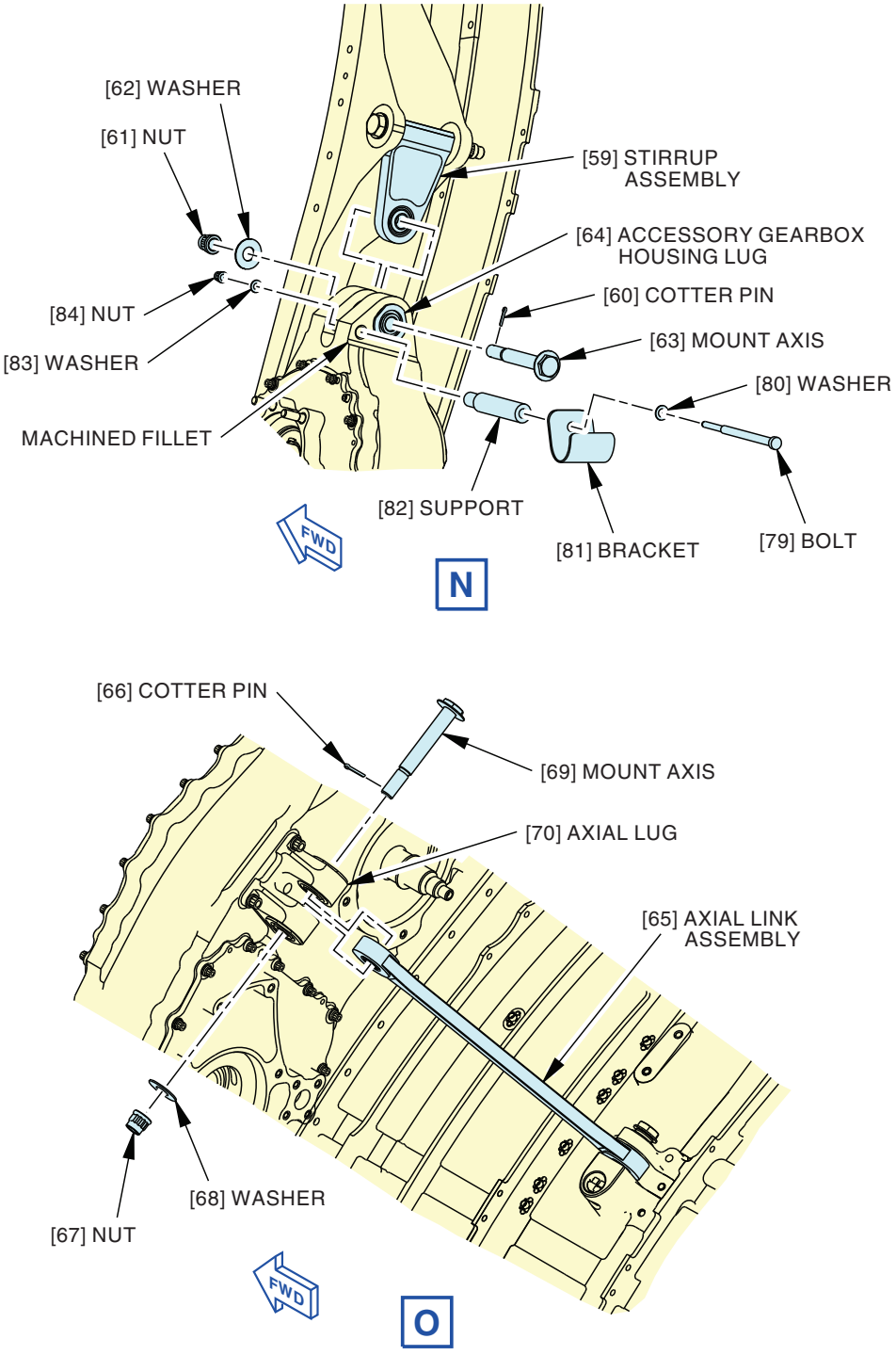
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2439537 S0000565674_V3

Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 10 of 16)

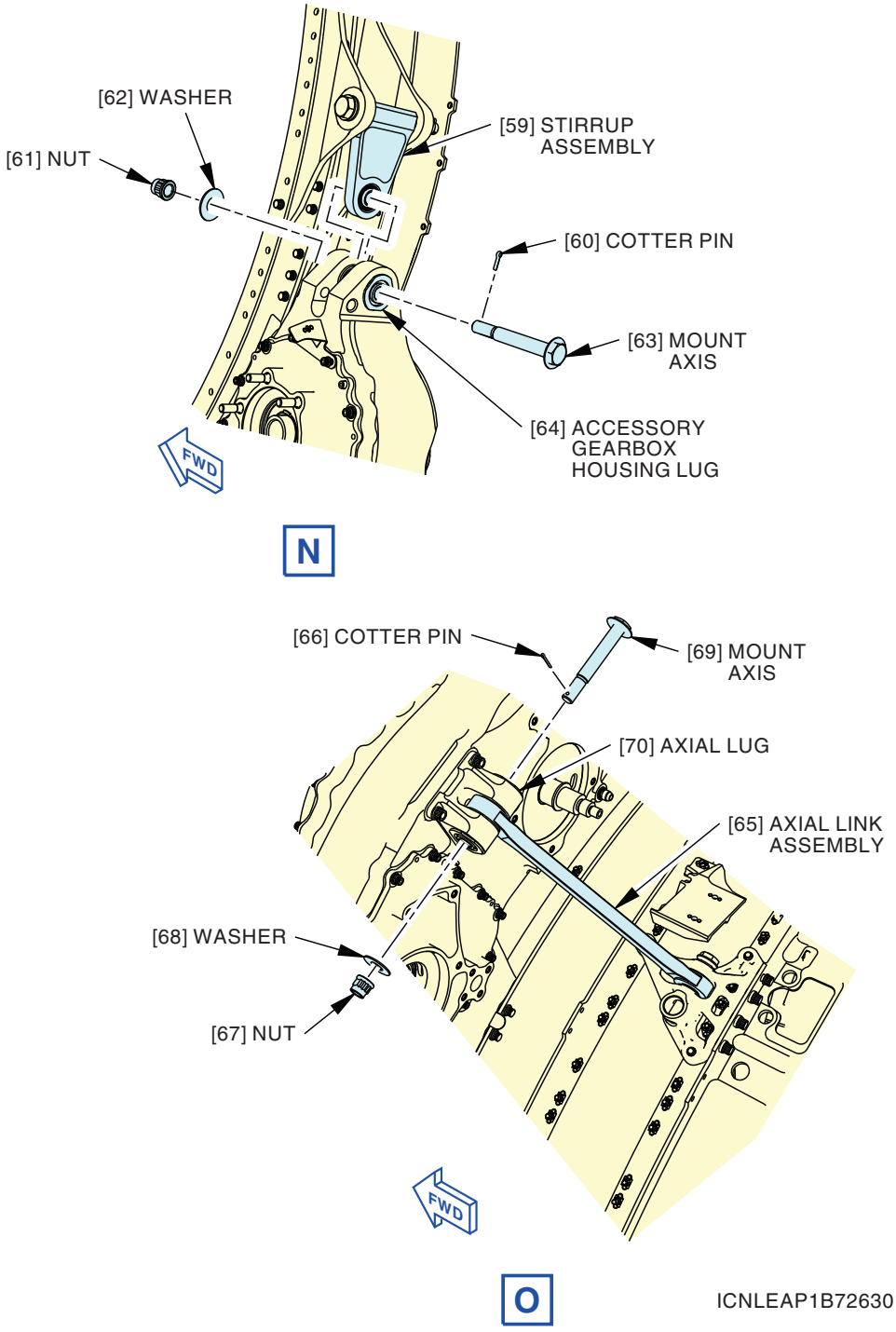
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2970865 S0000749293_V1

Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 11 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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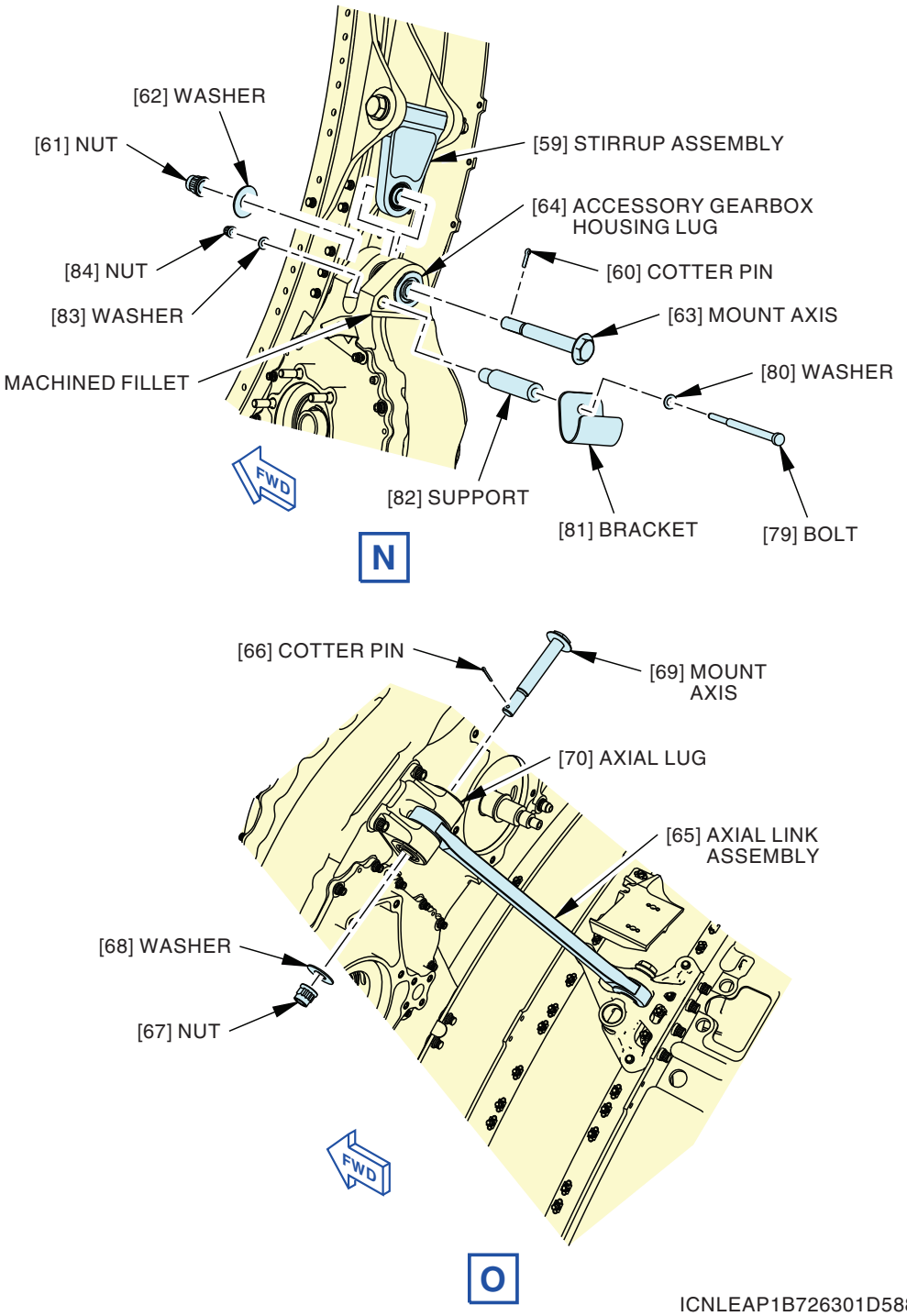
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 12 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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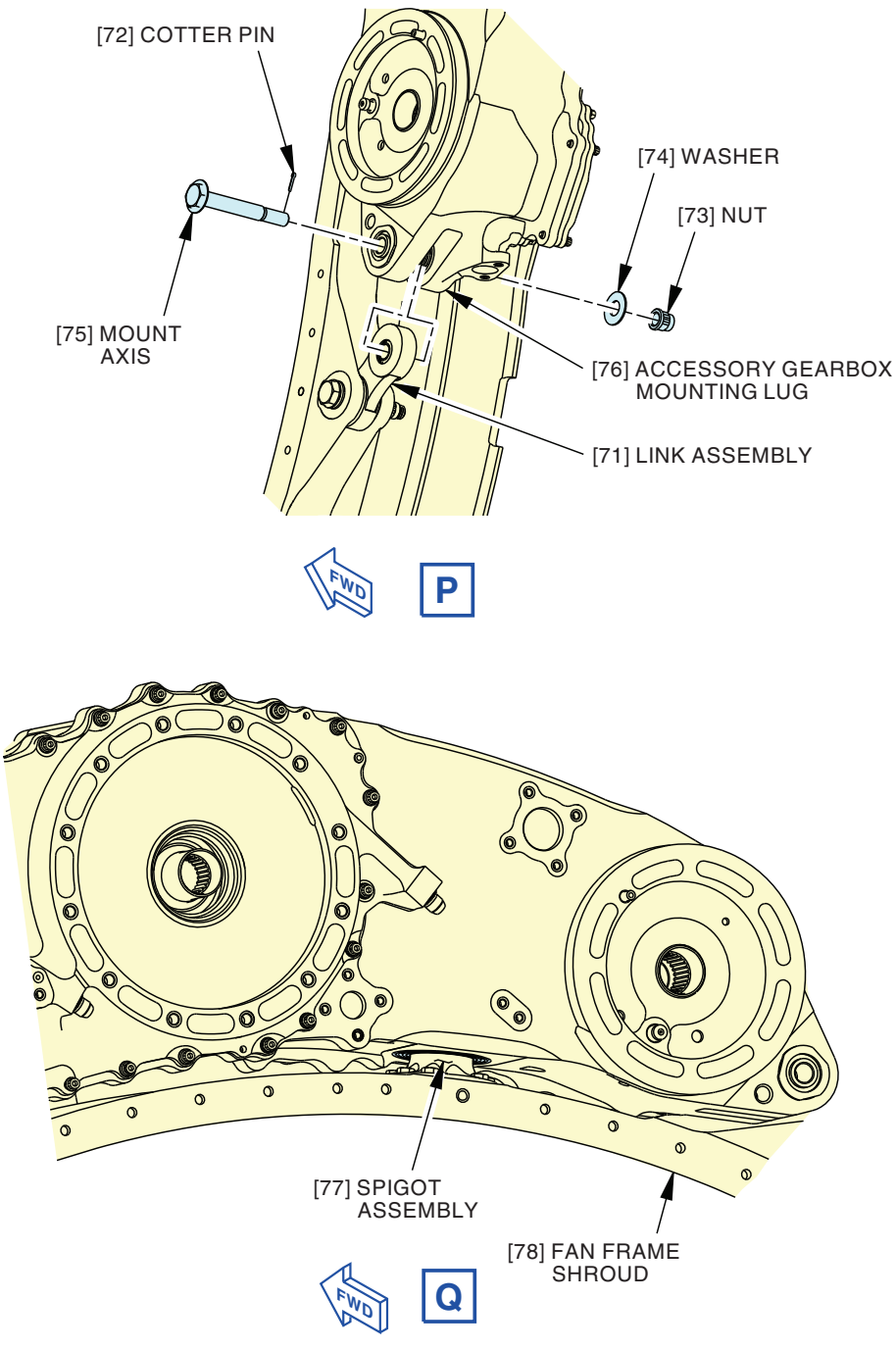
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 13 of 16)

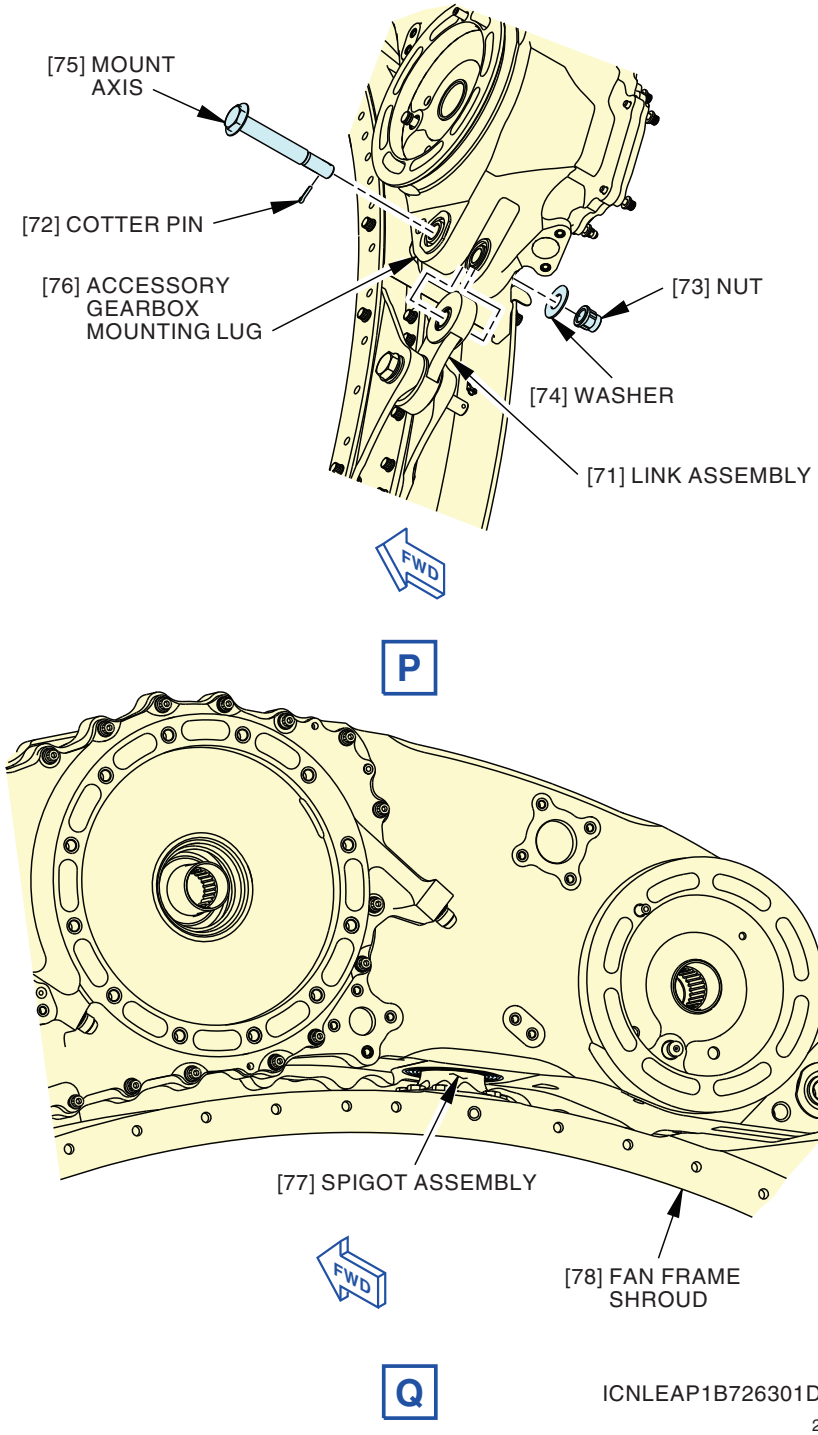
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2439538 S0000565675_V2

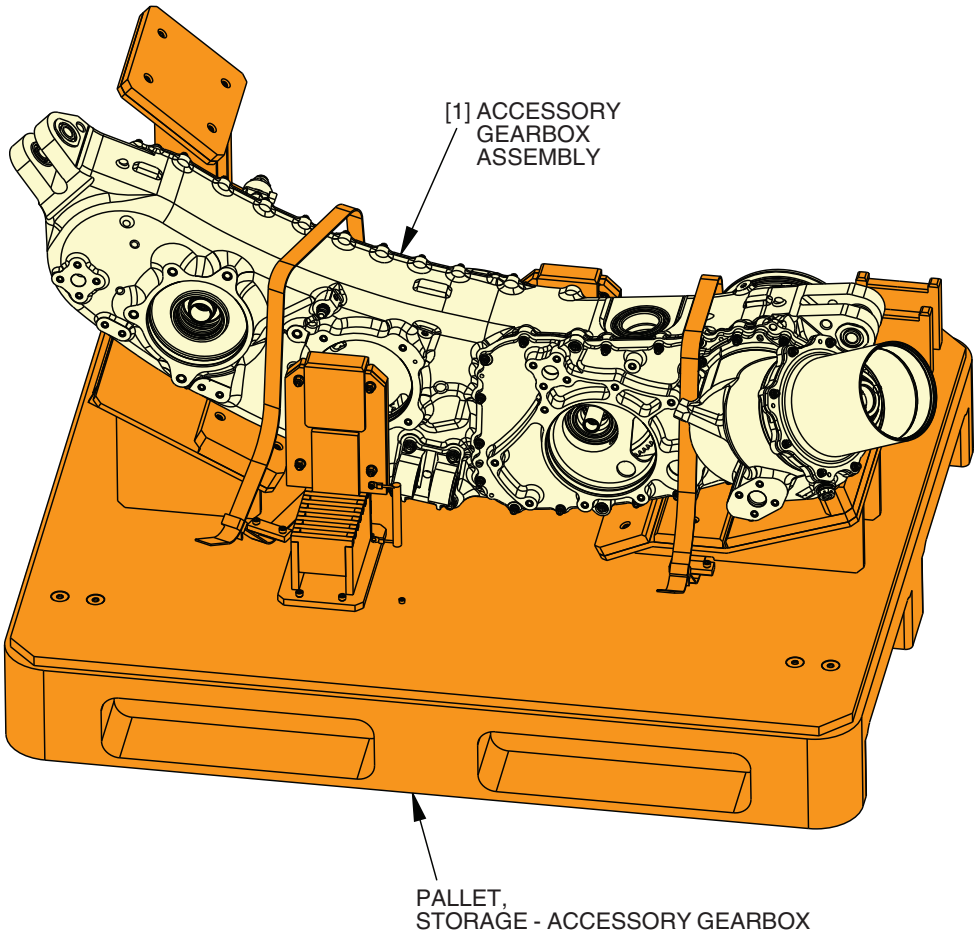
Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 14 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 15 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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3086595 S0000839715_V1

Accessory Gearbox Assembly Installation
Figure 401/72-63-01-990-801-G00 (Sheet 16 of 16)

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TASK 72-63-01-400-801-G00

3. Accessory Gearbox Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the accessory gearbox assembly.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
24-11-11-400-801	Integrated Drive Generator (IDG) Installation (P/B 401)
29-11-11-400-801	Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-400-801-G00	Fan Cowl Panels Installation (P/B 401)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-62-03-400-801-G00	Transfer Gearbox Assembly Installation (P/B 401)
72-63-01-610-802-G00	Accessory Gearbox Assembly Oil Replenishing (P/B 201)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)
73-11-01-400-801-G00	Main Fuel Pump Installation (P/B 401)
73-21-08-400-801-G00	Permanent Magnet Alternator Installation (P/B 401)
77-12-02-400-801-G00	N2 Speed Sensor Installation (P/B 401)
79-21-03-400-801-G00	Lubrication Unit Installation (P/B 401)
79-32-01-400-801-G00	Oil Pressure and Temperature Sensor Installation (P/B 401)
80-11-01-400-801-G00	Engine Starter Installation (P/B 401)
80-11-07-400-801-G00	Starter Air Valve Manual Control Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14690	Fixture - Accessory Gearbox (AGB) Remove/Install Part #: 956A7018G02 Supplier: 58828
SPL-14920	Adapter - Core Engine Lock Part #: 956A8679G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50469	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63, Type II, Class B-4
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
42	Gasket seal	72-00-00-13-370	PTF ALL
45	Gasket seal	72-00-00-13-510	PTF ALL
48	Gasket seal	72-00-00-12-230	PTF ALL
53	Gasket seal	72-00-00-13-070	PTF ALL
60	Cotter pin	72-00-00-14-330	PTF ALL
66	Cotter pin	72-00-00-14-160	PTF ALL
72	Cotter pin	72-00-00-14-240	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Accessory Gearbox Assembly Installation

SUBTASK 72-63-01-410-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the accessory gearbox assembly [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers from all open ports of the accessory gearbox assembly [1].

(c) Remove the protective caps from the drain and oil tubes that you will install for the accessory gearbox assembly [1].

(d) Install the AGB remove/install fixture, SPL-14690, on the accessory gearbox assembly [1].

WARNING

MOVE THE ACCESSORY GEARBOX ASSEMBLY SLOWLY. THE ACCESSORY GEARBOX ASSEMBLY IS HEAVY, INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

(e) Install the accessory gearbox assembly [1] in a correct position on the fan frame shroud [78].

NOTE: The accessory gearbox assembly weighs 143.5 lb (65.1 kg).

(f) Install the accessory gearbox assembly [1] on the accessory gearbox links as follows:

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1) Install the accessory gearbox assembly [1] on the spigot assembly [77] that attaches to the fan frame shroud [78].
2) Install the accessory gearbox assembly [1] on the link assembly [71] as follows:
   a) Install the link assembly [71] on the accessory gearbox mounting lug [76].
   b) Install the mount axis [75] that attaches the link assembly [71] to the accessory gearbox mounting lug [76].
   c) Lubricate the threads of the nut [73] with pure mineral vaseline, D00672 [CP5070].
   d) Install the washer [74] and nut [73] that attach the mount axis [75] to the link assembly [71].
   e) Tighten the nut [73] to 420.4 in-lb (47.5 N·m) - 464.6 in-lb (52.5 N·m).
   f) Install the new cotter pin [72] on the mount axis [75].
3) Install the stirrup assembly [59] on the accessory gearbox assembly [1] as follows:
   a) Install the stirrup assembly [59] in a correct position on the accessory gearbox housing lug [64].
   b) Install the mount axis [63] that attaches the stirrup assembly [59] to the accessory gearbox housing lug [64].
   c) Lubricate the threads of the nut [61] with pure mineral vaseline, D00672 [CP5070].
   d) Install the washer [62] and nut [61] that attaches the mount axis [63] to the stirrup assembly [59].
   e) Tighten the nut [61] to 420.4 in-lb (47.5 N·m) - 464.6 in-lb (52.5 N·m).
   f) Install the new cotter pin [60] on the mount axis [63].
4) Install the accessory gearbox assembly [1] on the axial link assembly [65] as follows:
   a) Remove the AGB remove/install fixture, SPL-14690, from the accessory gearbox assembly [1].
   b) Install the axial link assembly [65] in a correct position on the axial lug [70].
   c) Install the mount axis [69] that attaches the axial link assembly [65] to the axial lug [70].
   d) Lubricate the threads of the nut [67] with pure mineral vaseline, D00672 [CP5070].
   e) Install the washer [68] and nut [67] that attach the mount axis [69] to the axial link assembly [65].
   f) Tighten the nut [67] to 420.4 in-lb (47.5 N·m) - 464.6 in-lb (52.5 N·m).
   g) Install the new cotter pin [66] on the mount axis [69].

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5) Install the bracket [81] on the accessory gearbox assembly [1] as follows:
   a) Insert the smaller cylinder of the support [82] into the hole in the accessory gearbox assembly [1].
      <1> To prevent interference with the accessory gearbox assembly [1], make sure that the flat surface on the larger cylinder of the support [82] is parallel with the machined fillet of the accessory gearbox assembly [1].

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PTF ALL POST SB 737-78-1095; With MDOS-GSE hose support brackets (Continued)

b) Put the bracket [81] into its position.

<1> Make sure that the bottom side of the bracket [81] is parallel with the machined fillet of the accessory gearbox assembly [1].

c) Turn the bracket [81] 30 ± 10 degrees counterclockwise from its parallel position with machined fillet.

d) Install the washer [80] and bolt [79].

<1> Apply one layer of Never Seez, D50294, to the threads of the bolt [79].

<2> Apply sealant, A50469, to the shank of the bolt [79] for a wet installation.

e) Install the washer [83] and nut [84].

<1> Tighten the nut [84] to 70 in-lb (7.91 N·m) - 74 in-lb (8.36 N·m).

PTF ALL

6) Connect the oil tube [56] to the accessory gearbox assembly [1] as follows:

a) Install the new gasket seal [58] and oil tube [56] on the accessory gearbox assembly [1].

b) Install the four bolts [57] that attach the oil tube [56] to the accessory gearbox assembly [1].

c) Tighten the four bolts [57] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

7) Connect the oil tube [54] to the accessory gearbox assembly [1] as follows:

a) Lubricate the threads of the coupling nut [55] with engine oil, D00599 [CP2442].

b) Connect the coupling nut [55] to the accessory gearbox assembly [1].

c) Tighten the coupling nut [55] to 664 in-lb (75 N·m) - 752 in-lb (85 N·m).

8) Connect the oil tube [51] to the accessory gearbox assembly [1] as follows:

a) Lubricate the new gasket seal [53] with engine oil, D00599 [CP2442].

b) Install the gasket seal [53] and oil tube [51] on the accessory gearbox assembly [1].

c) Install the four bolts [52] that attach the oil tube [51] to the accessory gearbox assembly [1].

d) Tighten the four bolts [52] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

9) Connect the oil tube [46] to the accessory gearbox assembly [1] as follows:

a) Lubricate the new gasket seal [48] with engine oil, D00599 [CP2442].

b) Install the gasket seal [48] and oil tube [46] in the correct position on the accessory gearbox assembly [1].

c) Install the four bolts [47] that attach the oil tube [46] to the accessory gearbox assembly [1].

d) Tighten the four bolts [47] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

10) Connect the oil tube [43] to the accessory gearbox assembly [1] as follows:

a) Lubricate the new gasket seal [45] with engine oil, D00599 [CP2442].

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b) Lubricate the threads of the four bolts [44] with graphite mineral vaseline, D00601 [CP2101].

c) Install the gasket seal [45] and oil tube [43] on the accessory gearbox assembly [1].

d) Install the four bolts [44] that attach the oil tube [43] to the accessory gearbox assembly [1].

e) Tighten the four bolts [44] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

11) Connect the oil tube [40] to the accessory gearbox assembly [1] as follows:

a) Lubricate the new gasket seal [42] with engine oil, D00599 [CP2442].

b) Install the gasket seal [42] and oil tube [40] on the accessory gearbox assembly [1].

c) Install the four bolts [41] that attach the oil tube [40] to the accessory gearbox assembly [1].

d) Tighten the four bolts [41] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

12) Install the bonding strap [49] on the accessory gearbox assembly [1] as follows:

a) Install the bonding strap [49] on the accessory gearbox assembly [1].

b) Install the bolt [50] that attaches the bonding strap [49] to the accessory gearbox assembly [1].

c) Tighten the bolt [50] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

13) Install all the engine brackets on the accessory gearbox assembly [1] as follows:

a) Install the bracket [23] on the accessory gearbox assembly [1] as follows:

<1> Lubricate the threads of the two bolts [22] with engine oil, D00599 [CP2442].

<2> Install the two bolts [22] that attach the bracket [23] to the accessory gearbox assembly [1].

<3> Tighten the two bolts [22] to 97 in-lb (11.0 N·m) - 106 in-lb (12.0 N·m).

b) Install the bracket [21] on the accessory gearbox assembly [1] as follows:

<1> Lubricate the threads of the two bolts [20] with engine oil, D00599 [CP2442].

<2> Install the two bolts [20] that attach the bracket [21] to the accessory gearbox assembly [1].

<3> Tighten the two bolts [20] to 97 in-lb (11.0 N·m) - 106 in-lb (12.0 N·m).

c) Install the bracket [10] on the accessory gearbox assembly [1] as follows:

<1> Install the two bolts [9] that attach the bracket [10] to the accessory gearbox assembly [1].

<2> Tighten the two bolts [9] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

d) Install the bracket [8] on the accessory gearbox assembly [1] as follows:

<1> Install the two bolts [7] that attach the bracket [8] to the accessory gearbox assembly [1].

<2> Tighten the two bolts [7] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

e) Install the bracket [6] on the accessory gearbox assembly [1] as follows:

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<1> Install the two bolts [5] that attach the bracket [6] to the accessory gearbox assembly [1].

<2> Tighten the two bolts [5] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

14) Install the drain tubes on the accessory gearbox assembly [1] as follows:

a) Connect the drain tube [38] to the accessory gearbox assembly [1] as follows:

<1> Lubricate the threads of the coupling nut [39] with engine oil, D00599 [CP2442].

<2> Connect the coupling nut [39] of the drain tube [38] to the accessory gearbox assembly [1].

<3> Tighten the coupling nut [39] to 267.0 in-lb (30.2 N·m) - 283.0 in-lb (32.0 N·m).

b) Connect the drain tube [36] to the accessory gearbox assembly [1] as follows:

<1> Lubricate the threads of the coupling nut [37] with engine oil, D00599 [CP2442].

<2> Connect the coupling nut [37] of the drain tube [36] to the accessory gearbox assembly [1].

<3> Tighten the coupling nut [37] to 267.0 in-lb (30.2 N·m) - 283.0 in-lb (32.0 N·m).

c) Connect the drain tube [30] to the accessory gearbox assembly [1] as follows:

<1> Lubricate the threads of the coupling nut [31] and the bolt [32] with engine oil, D00599 [CP2442].

<2> Connect the coupling nut [31] of the drain tube [30] to the accessory gearbox assembly [1].

<3> Tighten the coupling nut [31] to 267.0 in-lb (30.2 N·m) - 283.0 in-lb (32.0 N·m).

<4> Install the bolt [32] that attaches the clamp [33] and drain tube [30] to the bracket [35].

<5> Tighten the bolt [32] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

d) Connect the drain tube [24] to the accessory gearbox assembly [1] as follows:

<1> Lubricate the threads of the coupling nut [25] and the bolt [26] with engine oil, D00599 [CP2442].

<2> Connect the coupling nut [25] of the drain tube [30] to the accessory gearbox assembly [1].

<3> Tighten the coupling nut [25] to 267.0 in-lb (30.2 N·m) - 283.0 in-lb (32.0 N·m).

<4> Install the bolt [26] that attaches the clamp [27] and drain tube [24] to the bracket [29].

<5> Tighten the bolt [26] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

15) Install the oil pressure and temperature sensor (TASK 79-32-01-400-801-G00).

16) Connect the transfer gearbox assembly (TASK 72-62-03-400-801-G00).

17) Install the lubrication unit lube and scavenge pump (TASK 79-21-03-400-801-G00).

18) Install the permanent magnet alternator (TASK 73-21-08-400-801-G00).

19) Install the main fuel pump (TASK 73-11-01-400-801-G00).

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20) Install the engine driven hydraulic pump (TASK 29-11-11-400-801).

21) Install the integrated drive generator (TASK 24-11-11-400-801).

22) Install the engine starter (TASK 80-11-01-400-801-G00).

23) Install the starter air valve manual control (TASK 80-11-07-400-801-G00).

24) Install the N2 speed sensor (TASK 77-12-02-400-801-G00).

25) Unlock the N2 rotor with the Engine core lock adapter, SPL-14920, through the accessory gearbox cover (handcranking cover).

26) Install the accessory gearbox manual control cover (TASK 72-63-04-400-801-G00).

SUBTASK 72-63-01-160-001-G00

(2) Clean the work area and clear the tools and all items used for the task.

SUBTASK 72-63-01-610-002-G00

(3) Do this task: Accessory Gearbox Assembly Oil Replenishing, TASK 72-63-01-610-802-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-01-420-001-G00

(1) On the applicable engine, install the left fan cowl panel (TASK 71-11-04-400-801-G00).

(a) Install these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-01-860-003-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-01-600-001-G00

(3) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 72-63-01-860-004-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Accessory Gearbox Assembly Installation Test

SUBTASK 72-63-01-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MANUAL CONTROL COVER - REPAIR

1. General

A. This procedure has this task:

(1) Blending and Spot Touch-Up of the Manual Control Cover.

TASK 72-63-03-300-802-G00

2. Blending and Spot Touch-Up of the Manual Control Cover

(Figure 801, Figure 802)

A. General

(1) This task gives the instructions to do the blending and spot touch-up of the manual control cover (MCC) [1] of the Accessory Gearbox (AGB).

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50295 [CP2010]	Solvent - cleaning, White Spirit	
C50005	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
C50208 [CP2659]	Coating - Chemical Conversion - Bonderite M-CR 1132 Aero Pen (Formerly Alodine 1132)	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

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G. Prepare for the Blending and Spot Touch-Up

SUBTASK 72-63-03-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-03-860-006-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Dimensional Information

WARNING MAKE SURE THAT YOU READ THE MATERIAL SAFETY DATA SHEET (MSDS) FOR THE MATERIALS YOU WILL USE, AND WEAR THE PROTECTIVE CLOTHING AND EQUIPMENT IF IT IS SPECIFIED. REFER TO THE APPLICABLE MANUFACTURER'S OR SUPPLIER'S MSDS FOR TALK WITH THE LOCAL SAFETY DEPARTMENT OR AUTHORITIES FOR THE PROCEDURES TO DISCARD THIS HAZARDOUS AGENT.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE AN INJURY OR KILL YOU.

SUBTASK 72-63-03-320-010-G00

(1) Blend the area to repair as follow:

SUBTASK 72-63-03-320-011-G00

(2) Do not remove more parent material than necessary.

(a) Indications must be repaired without steps and sharp edges.

(b) The hand blending must cover all the damage area.

SUBTASK 72-63-03-320-012-G00

(3) Make sure that the quantity of blend repairs on:

(a) The manual control cover (MCC) [1] is less or equal than 6.

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SUBTASK 72-63-03-320-013-G00

(4) Make sure that the distance between two blends must be equal to the width of maximum indication observed on repaired area, if not the indications must be included on one blend.

SUBTASK 72-63-03-320-014-G00

(5) Blending is not permitted on an area already blended.

SUBTASK 72-63-03-320-015-G00

(6) Obey the dimension limits of areas below:

(a) On area Q the maximum finish dimension Z is equal to 0.0098 in. (0.25 mm).

1) The minimum description blending limit X is equal to 40 x dim.Z and the minimum description blending limit Y is equal to 40 x dim.Z.

2) The maximum description blending limit X is equal to 45 x dim.Z and the maximum description blending limit Y is equal to 45 x dim.Z.

(b) Blending is not allowed on the area G.

I. Blending and Spot Touch-Up of the Manual Control Cover

SUBTASK 72-63-03-320-016-G00

(1) Check for dimensions, locations and quantities of indications on areas located on Figure 801 and compare with repair limits.

CAUTION

YOU MUST NOT BLEND THE PART IN NOT PERMITTED AREA. THIS CAN CAUSE DAMAGE TO THE PART.

(a) If the indications or repair area is more than the limits, the part is not repairable by this procedure.

(b) Blending is not allowed on the area G.

(c) Blending is not permitted on an area already blended.

(d) Make sure that the distance between two blends must be equal to the width of maximum indication observed on repaired area, if not the indications must be included on one blend (Figure 802).

SUBTASK 72-63-03-320-017-G00

(2) Remove the indications from the repair area by a smooth hand blending.

(a) For general hand blending:

1) Sharp edges can be blended out, using abrasive tools. Coarse grades of abrasives or files may be used for fast metal removal, but the parts must then be given a smooth surface finish with fine grades of abrasives or crocus cloth.

2) Avoid removing metal from leading and trailing edges of part sections in such a way that the edges become thin or sharp; blend so as to maintain approximately the original contour.

3) Use typical abrasive tools:

- Abrasive sheets
- Abrasive belt
- Abrasive disks
- Abrasive stones
- Hand files

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Crocus cloth.

(b) For dimensional information: (Figure 802).

NOTE: If one blending impacts several areas, consider the lowest depth criteria.

SUBTASK 72-63-03-120-002-G00

(3) Clean the repair area of the manual control cover (MCC) [1] with a brush or a cloth using Ardrox 5503, B50290 [CP2559], solvent, B00682 [CP2011], or White Spirit, B50295 [CP2010].

SUBTASK 72-63-03-230-002-G00

(4) Do the applicable fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

(a) No cracks are permitted.

SUBTASK 72-63-03-350-002-G00

(5) Do a spot touch-up of oxide film coating on the repaired areas (TASK 70-43-07-300-801-G00).

(a) Use only Bonderite M-CR 1200S Aero coating, C50005, and Bonderite M-CR 1132 Aero Pen, C50208 [CP2659].

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-03-860-007-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-03-860-008-G00

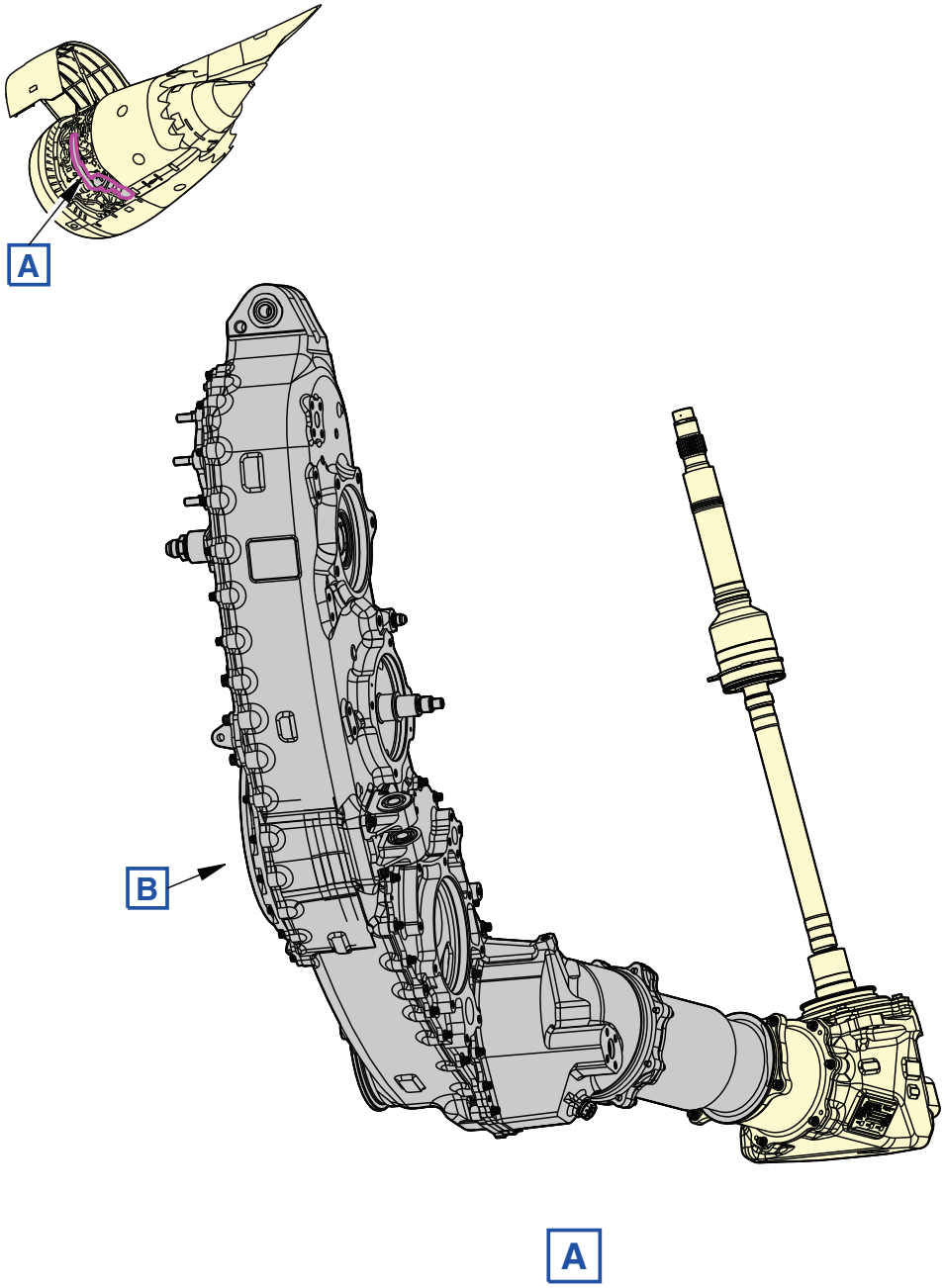
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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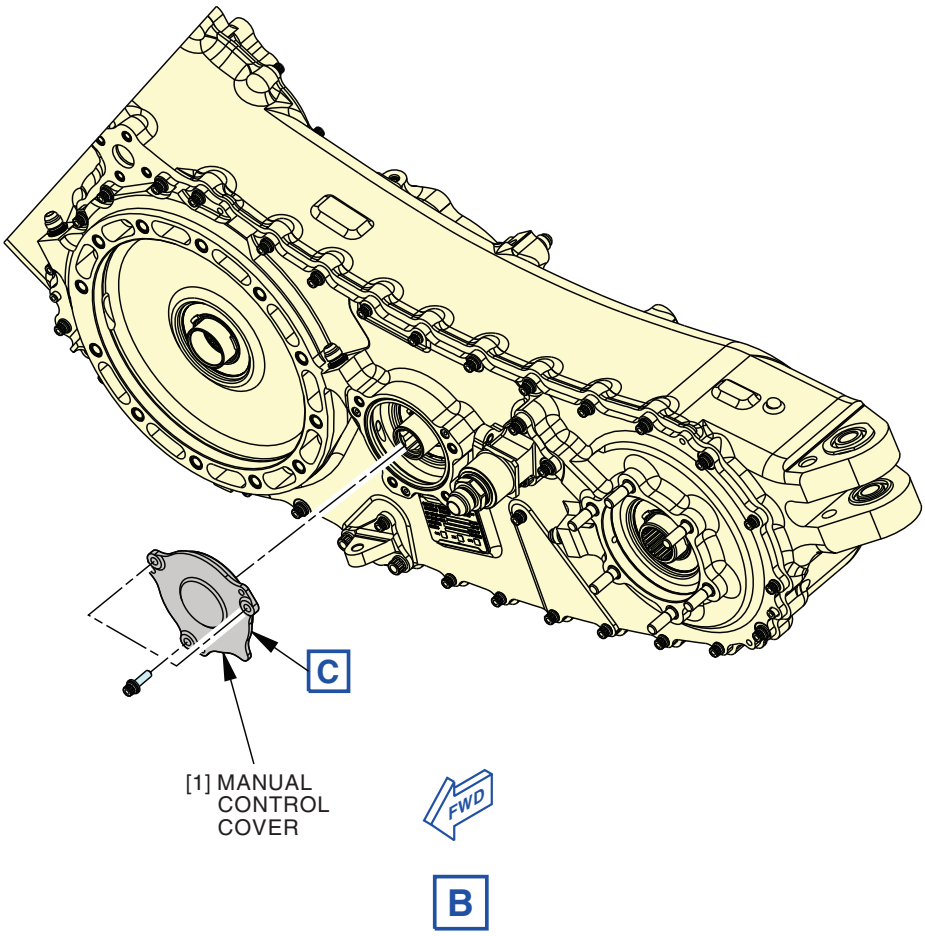
Blending and spot touch-up of the AGB Manual-Control Cover
Figure 801/72-63-03-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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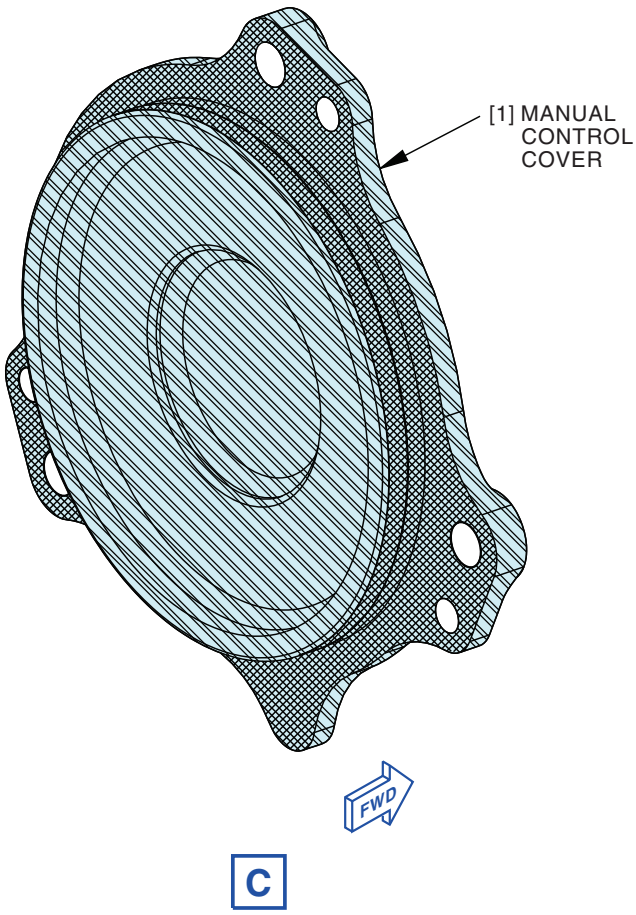
Blending and spot touch-up of the AGB Manual-Control Cover
Figure 801/72-63-03-990-803-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

MATERIAL: 2618A

COATING: ANODIZING

AREA Q

AREA G

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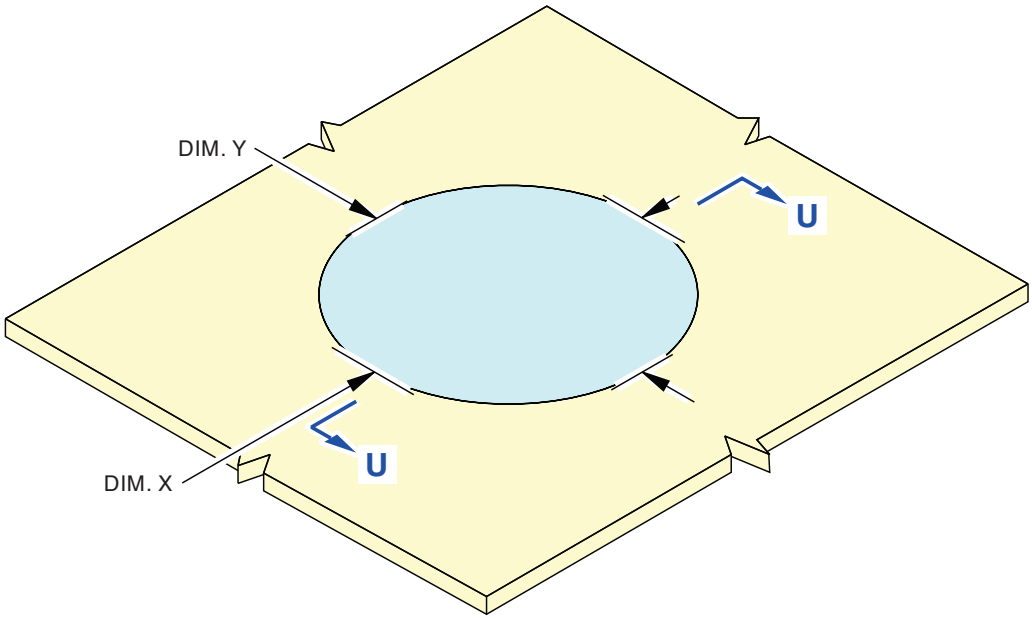
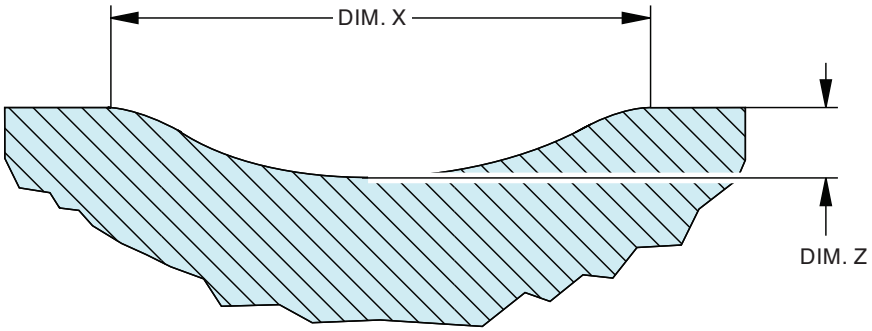
Blending and spot touch-up of the AGB Manual-Control Cover
Figure 801/72-63-03-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLENDING DIMENSIONS OF PLANE AREAS (EXAMPLE)



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Blending specific dimensions of plane areas of the AGB Manual-Control Cover
Figure 802/72-63-03-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX MANUAL CONTROL COVER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Accessory gearbox manual control cover removal

(2) Accessory gearbox manual control cover installation.

TASK 72-63-04-000-801-G00

2. Accessory Gearbox Manual Control Cover Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the manual control cover on the accessory gearbox.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14904	Tool Set - Jackscrew Part #: 956A1609G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-63-04-860-002-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 72-63-04-860-003-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Accessory Gearbox Manual Control Cover Removal

SUBTASK 72-63-04-010-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the manual control cover [1] from the Accessory Gearbox (AGB) assembly as follows:

(a) Remove the three bolts [2] and the three washers [3] that attach the manual control cover [1] to the AGB.

(b) Install the three jackscrews from the Tool Set, SPL-14904 (or the three bolts [2]) in the three extract holes of the manual control cover [1].

1) Turn the three jackscrews until you can remove the manual control cover [1] from the AGB.

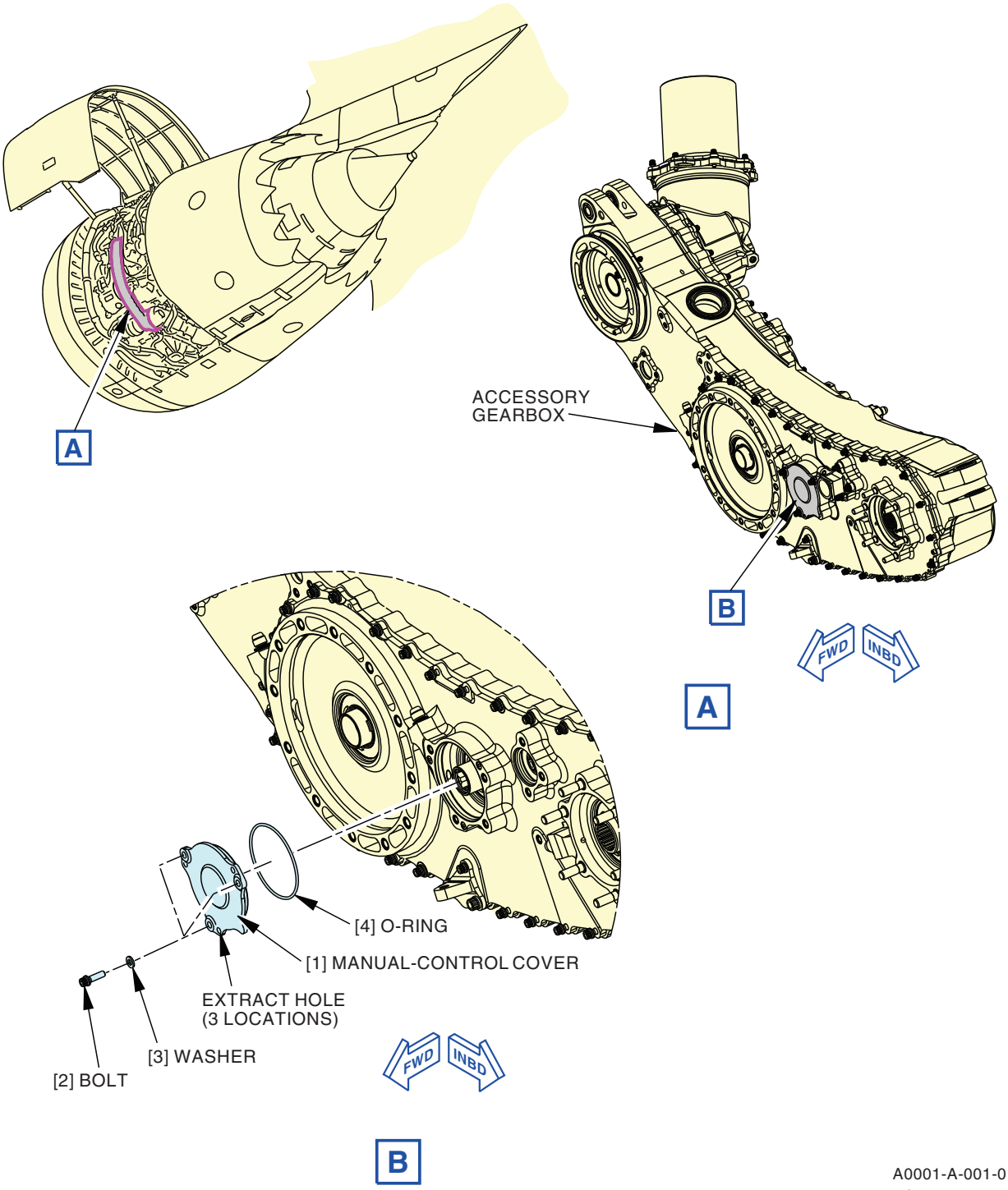
(c) Remove the manual control cover [1].

(d) Remove and discard the O-ring [4].

(e) Remove the three bolts [2] from the extract holes.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Accessory Gearbox Manual Control Cover Installation
Figure 401/72-63-04-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-63-04-400-801-G00

3. Accessory Gearbox Manual Control Cover Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the manual control cover on the accessory gearbox.

(2) You must do the tests that are listed in the power plant test reference table after you install the manual control cover.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Manual control cover	72-63-01-04-170	PTF ALL
4	O-ring	72-63-01-04-180	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Accessory Gearbox Manual Control Cover Installation

SUBTASK 72-63-04-410-001-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE COVER ON THE AGB. OIL LEAKS WILL OCCUR WITHOUT THE COVER. DAMAGE TO THE ENGINE CAN OCCUR.

(1) Install the manual control cover [1] on the accessory gearbox (AGB) assembly:

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU INSTALL NEW O-RINGS ON AGB COVER. USED O-RINGS CAN CAUSE OIL LEAKS AND DAMAGE TO THE ENGINE CAN OCCUR.

(a) Lubricate the new O-ring [4] with engine oil, D00599 [CP2442].

(b) Install the new O-ring [4] on the manual control cover [1].

(c) Install the manual control cover [1] on the AGB assembly.

(d) Lubricate the three bolts [2] with engine oil, D00599 [CP2442].

(e) Install the three bolts [2] with the three washers [3] on the manual control cover [1] manually.

(f) Tighten the bolts [2] to 97.4 in-lb (11.0 N·m) to 106.2 in-lb (12.0 N·m).

(g) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-04-860-004-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-04-860-005-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Accessory Gearbox Manual Control Cover Installation Test

SUBTASK 72-63-04-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX MANUAL CONTROL COVER - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Accessory gearbox manual control inspection

TASK 72-63-04-210-801-G00

2. Accessory Gearbox Manual Control Inspection

(Figure 601)

A. General

(1) This task gives the instructions on how to do the visual inspection of the accessory gearbox manual control cover.

(2) The accessory gearbox manual control cover is referred to as the manual control cover.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-63-03-300-802-G00	Blending and Spot Touch-Up of the Manual Control Cover (P/B 801)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 72-63-04-860-018-G00

(1) For the left engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 72-63-04-860-019-G00

(2) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

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SUBTASK 72-63-04-860-008-G00

(3) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-04-860-009-G00

(4) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-04-010-002-G00

(5) On the applicable engine, do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

G. Accessory Gearbox Manual Control Cover Inspection

SUBTASK 72-63-04-211-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Visual Inspection of the AGB manual control cover [1] as follows:

(a) Make sure that the manual control cover [1] to be examined is clean.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Visual inspection of the adjacent areas.

1) Do a visual inspection of the external surfaces of all the removed components and other areas of the engine that are seen after the removal of the components.

2) Refer to applicable section for the inspection criteria.

(c) Examine all the areas of the manual control cover [1] for:

1) Cracks:

a) Maximum serviceable limit: Not serviceable.

b) Maximum repairable limit: None.

c) Repair method: Not repairable.

(d) Examine the seal groove [2] for:

1) Scratches, nicks, dents, and scores:

a) Maximum serviceable limit: Not serviceable.

b) Maximum repairable limit: None.

c) Repair method: Not repairable.

(e) Examine the non functional surfaces for:

1) Lack of protection

a) Maximum serviceable limit: Any amount, 5 percent of inside or outside protection.

(f) Examine all other surfaces for:

1) Lack of protection:

a) Maximum serviceable limit: Not serviceable.

2) Scratches, nicks, dents, and scores:

a) Not permitted

b) Maximum repairable limit: In the limits of the applicable

c) Repair method: Refer to Blending and Spot Touch-Up of the Manual Control Cover, TASK 72-63-03-300-802-G00.

(g) Clean the work area and clear the tools and other items.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-04-410-002-G00

(1) On the applicable engine, do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00

SUBTASK 72-63-04-860-010-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-04-860-011-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

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(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-63-04-860-020-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 72-63-04-860-021-G00

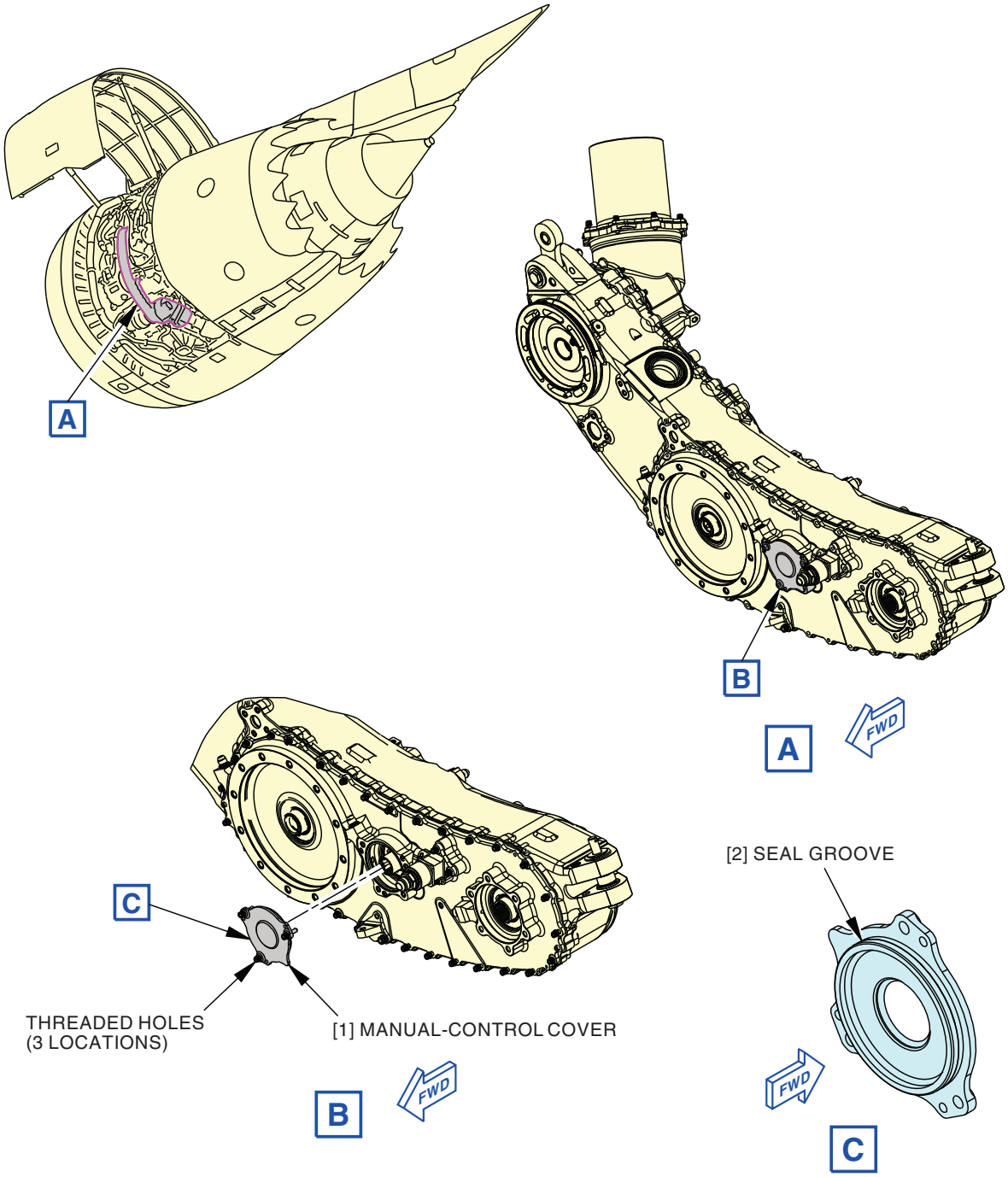
(5) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Accessory Gearbox Manual Control Cover Inspection
Figure 601/72-63-04-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX MANUAL CONTROL COVER - CLEANING/PAINTING

1. General

A. This procedure has one task:

(1) Accessory gearbox manual control cover cleaning.

TASK 72-63-04-100-801-G00

2. Accessory Gearbox Manual Control Cover Cleaning

(Figure 701)

A. General

(1) This task gives the instructions on how to clean the accessory gearbox manual control cover.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50295 [CP2010]	Solvent - cleaning, White Spirit	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Prepare for the Cleaning

SUBTASK 72-63-04-860-013-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-04-860-014-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-04-100-001-G00

(3) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

H. Accessory Gearbox Manual Control Cover Cleaning

SUBTASK 72-63-04-100-002-G00

(1) Clean the manual control cover [1]:

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

(a) Remove oil, grease and loose dirt from the manual control cover [1].

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) Clean the components with a brush or cloth and the solvent, B00682 [CP2011] or the White Spirit, B50295 [CP2010] or the Ardrox 5503, B50290 [CP2559].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-04-860-015-G00

(1) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-63-04-860-016-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-04-860-017-G00

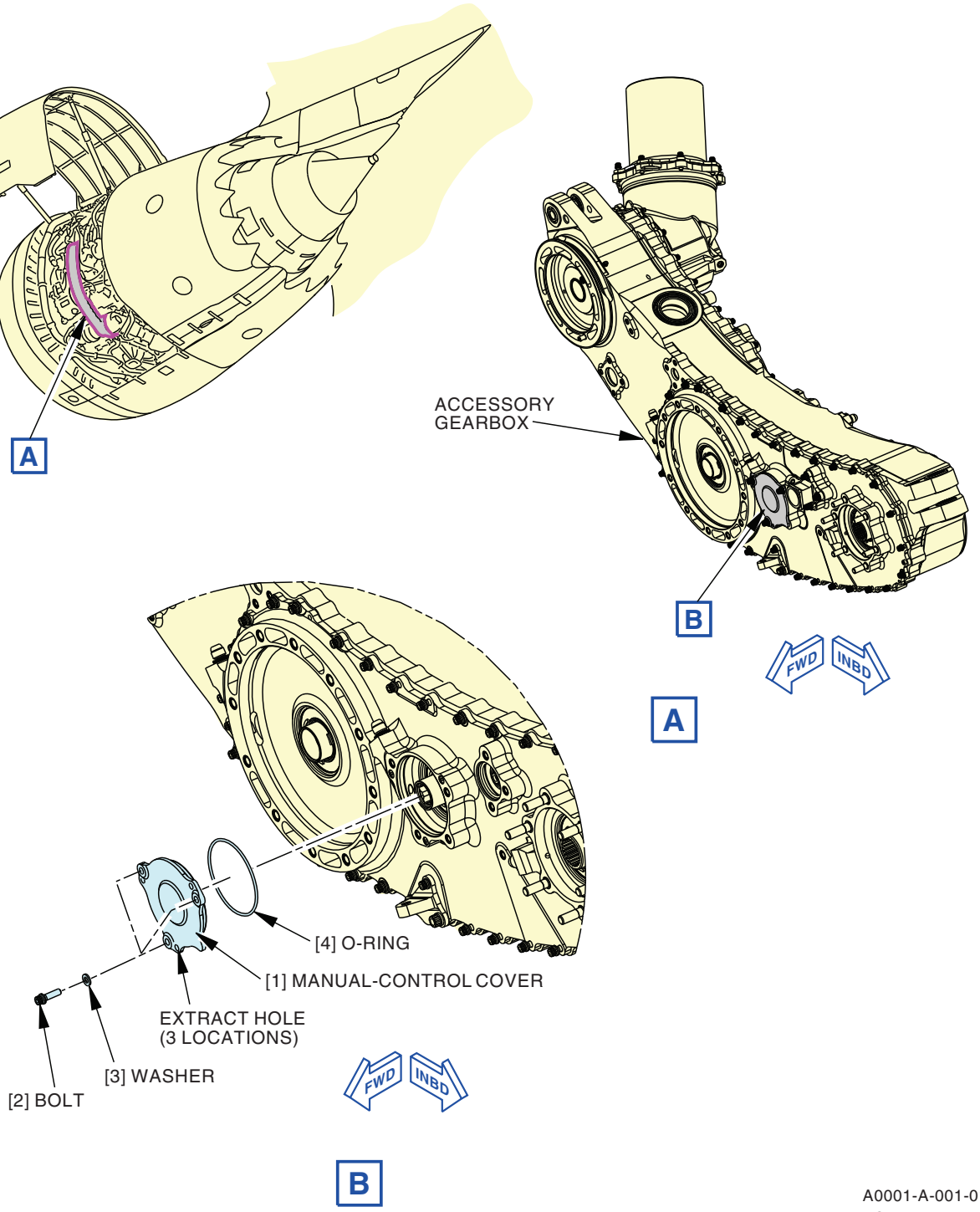
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Accessory Gearbox Manual Control Cover Cleaning
Figure 701/72-63-04-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX LINK - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Accessory gearbox link removal

(2) Accessory gearbox link installation.

TASK 72-63-05-000-801-G00

2. Accessory Gearbox Link Removal

(Figure 401 or Figure 402)

A. General

(1) This task provides the instructions on how to remove the accessory gearbox links.

(2) The accessory gearbox links included these parts: the stirrup assembly, the axial link assembly, and the link assembly.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14690	Fixture - Accessory Gearbox (AGB) Remove/Install Part #: 956A7018G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-63-05-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-63-05-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Accessory Gearbox Link Removal

SUBTASK 72-63-05-020-001-G00

(1) Remove the accessory gearbox links [33] as follows:

(a) Remove the stirrup assembly [2] as follows:

1) Install the AGB remove/install fixture, SPL-14690 on the accessory gearbox assembly [1].
2) Remove and discard the cotter pin [3] from the mount axis [6].
3) Loosen the nut [4] and the washer [5] from the mount axis [6].
4) Remove the mount axis [6] that attaches the stirrup assembly [2] to the accessory gearbox assembly housing lug [7].
5) Remove and discard the cotter pin [8] from the mount axis [10].
6) Loosen the nut [9] from the mount axis [10].
7) Remove the mount axis [10] that attaches the stirrup assembly [2] to the fan frame shroud [32].
8) Remove the spacer [11] from the mount axis [10].

(b) Remove the axial link assembly [12] as follows:

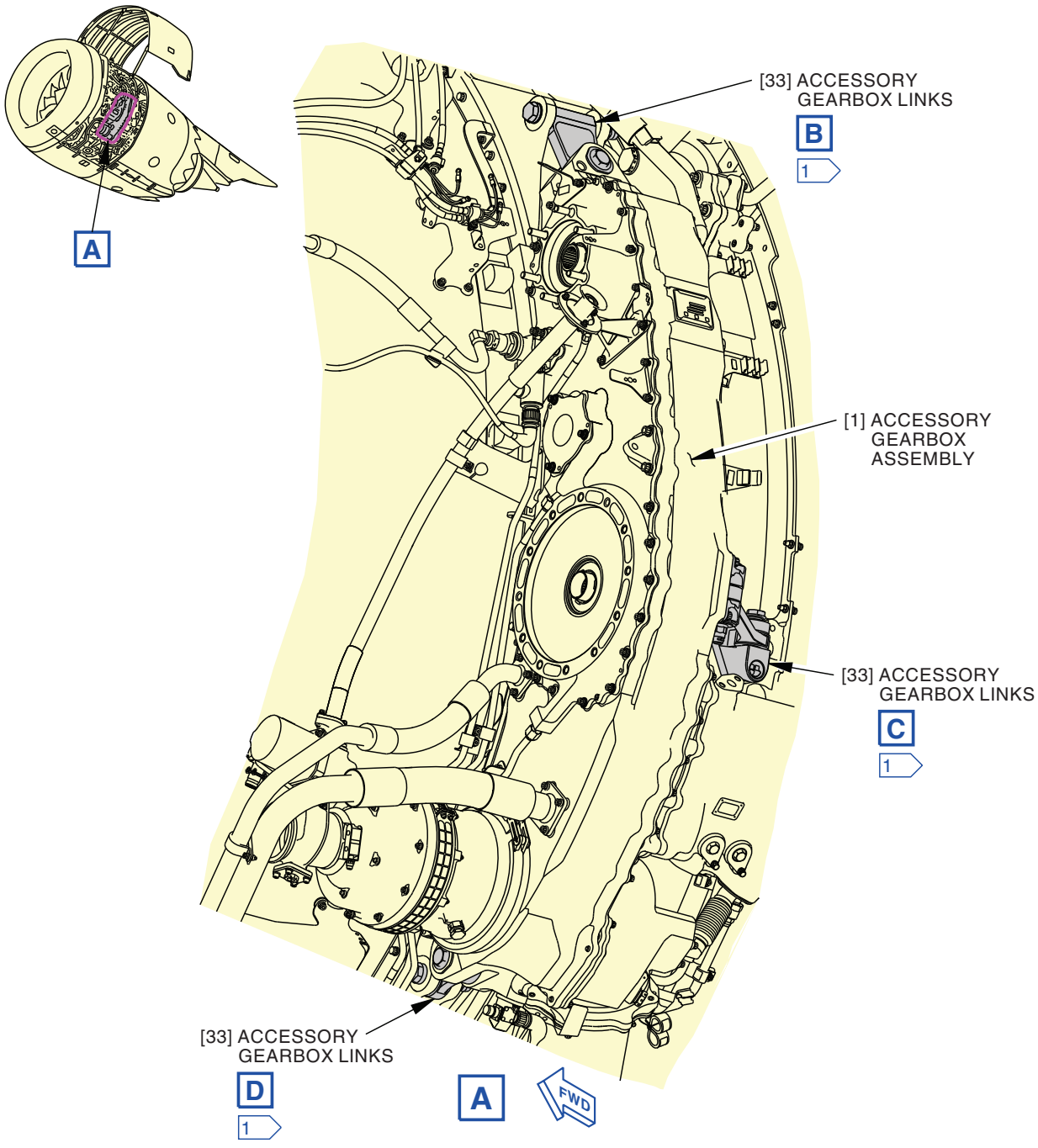
1) Remove and discard the cotter pin [13] from the mount axis [16].
2) Loosen the nut [14] and the washer [15] from the mount axis [16].
3) Remove the mount axis [16] that attaches the axial link assembly [12] to the axial lug [17].
4) Loosen the mount axis [18] that attaches the axial link assembly [12] to the fan frame shroud [32].
5) Remove the spacer [19] from the mount axis [18].
6) Remove the barrel nut retainer [21] and the barrel nut [20].

(c) Remove the link assembly [22] as follows:

1) Remove and discard the cotter pin [23] from the mount axis [26].
2) Remove the nut [24] and the washer [25] from the mount axis [26].
3) Remove the mount axis [26] that attaches the link assembly [22] to the accessory gearbox assembly mounting lug [27].
4) Remove and discard the cotter pin [28] from the mount axis [30].
5) Loosen the nut [29] from the mount axis [30].
6) Remove the mount axis [30] that attaches the link assembly [22] to the fan frame shroud [32].
7) Remove the spacer [31] from the mount axis [30].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 ACCESSORY GEARBOX LINKS INCLUDED THESE PARTS: THE STIRRUP ASSEMBLY, THE AXIAL LINK ASSEMBLY, AND THE LINK ASSEMBLY

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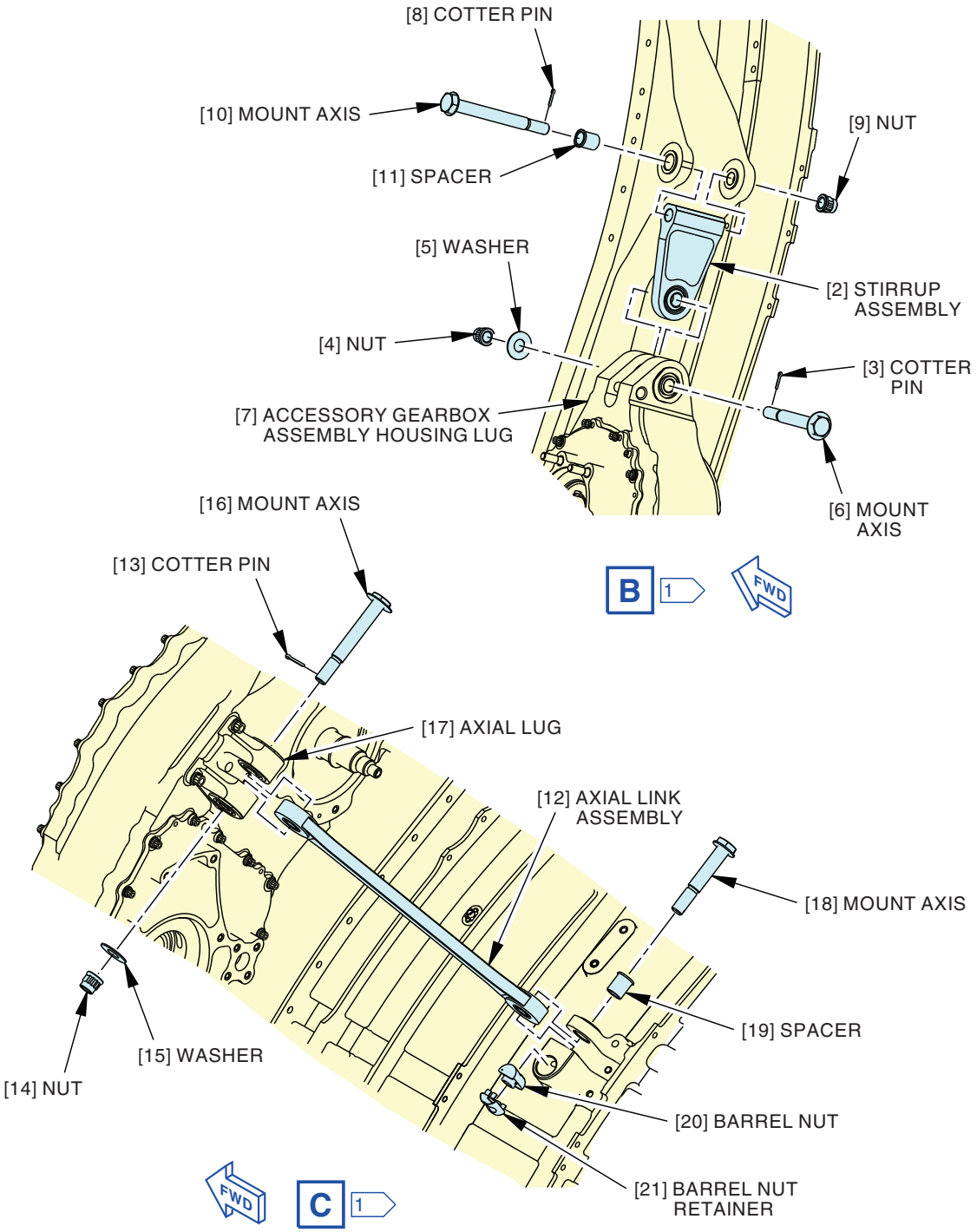
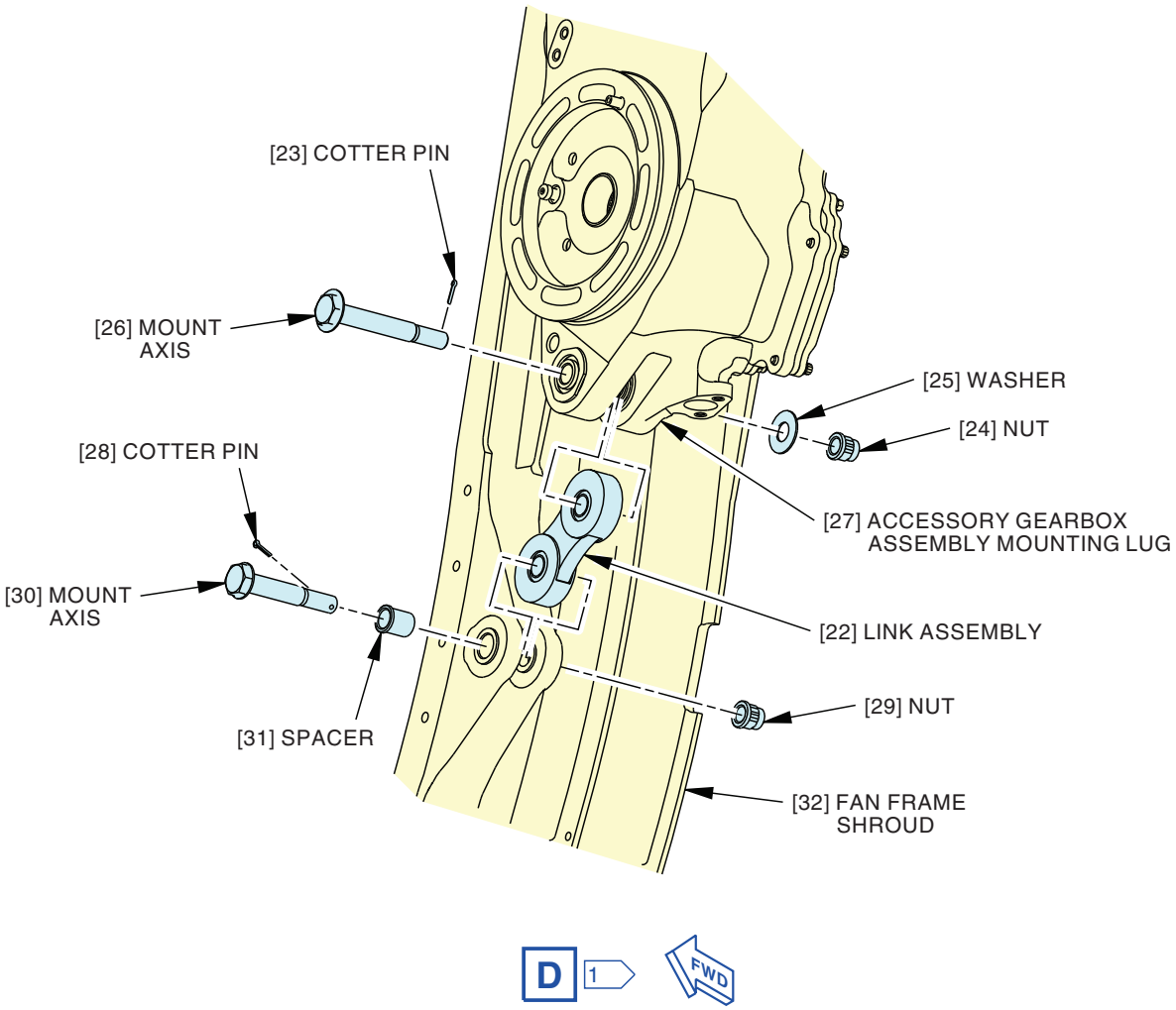
Accessory Gearbox Link Installation
Figure 401/72-63-05-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Link Installation
Figure 401/72-63-05-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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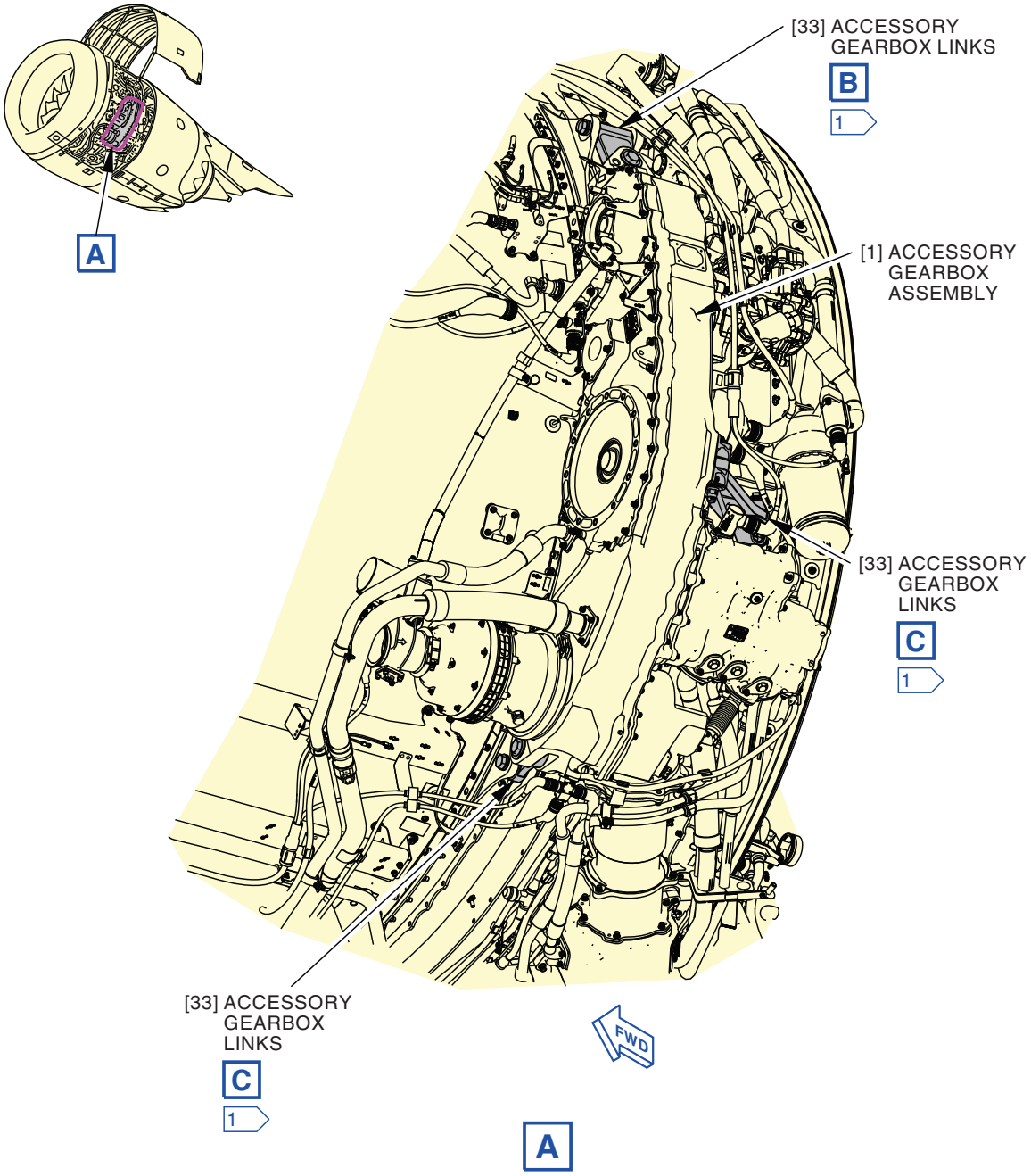
Accessory Gearbox Link Installation
Figure 401/72-63-05-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 ACCESSORY GEARBOX LINKS INCLUDED THESE PARTS: THE STIRRUP ASSEMBLY, THE AXIAL LINK ASSEMBLY, AND THE LINK ASSEMBLY.

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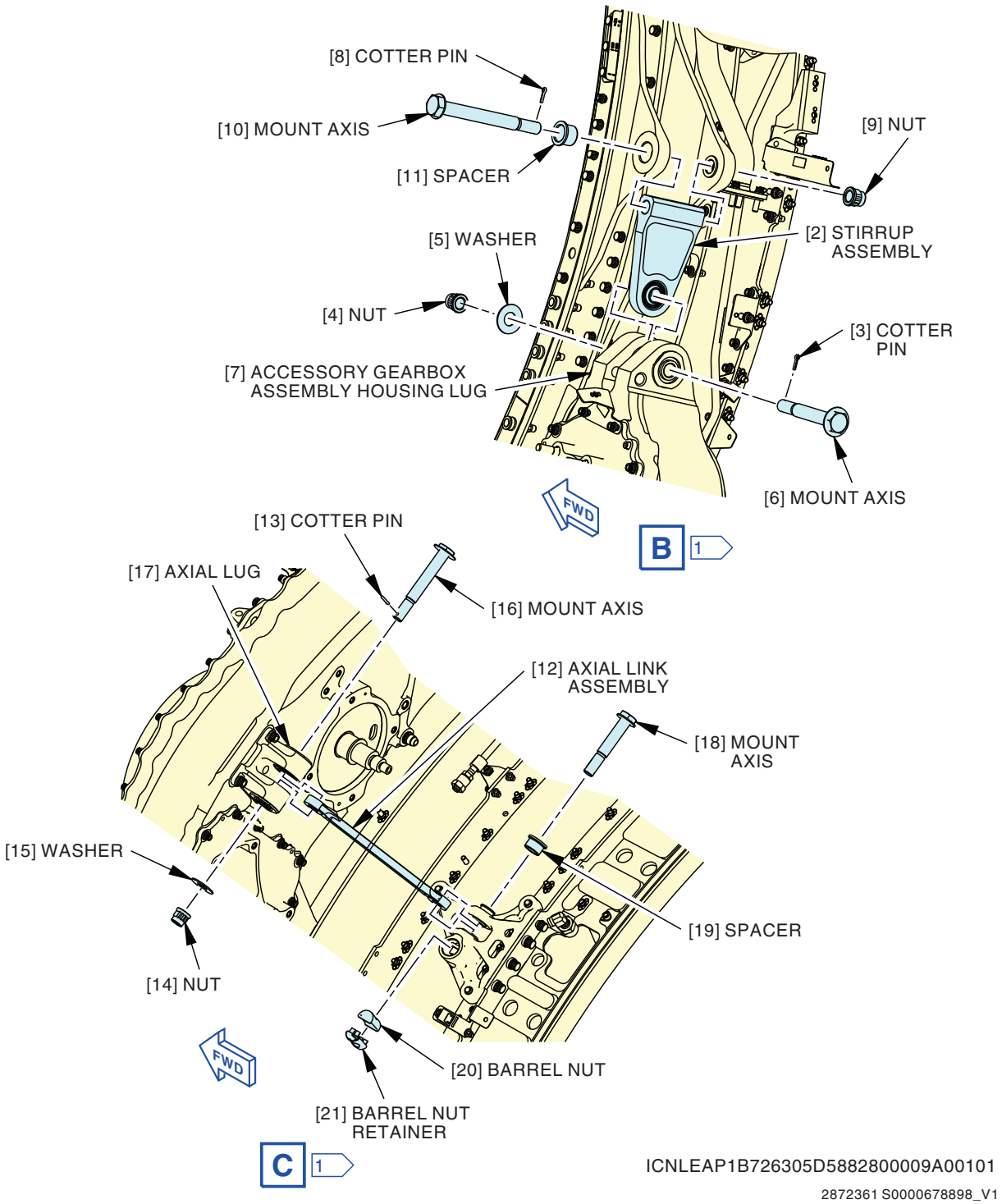
2872360 S0000678897_V1

Accessory Gearbox Link Installation
Figure 402/72-63-05-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Link Installation
Figure 402/72-63-05-990-803-G00 (Sheet 2 of 3)

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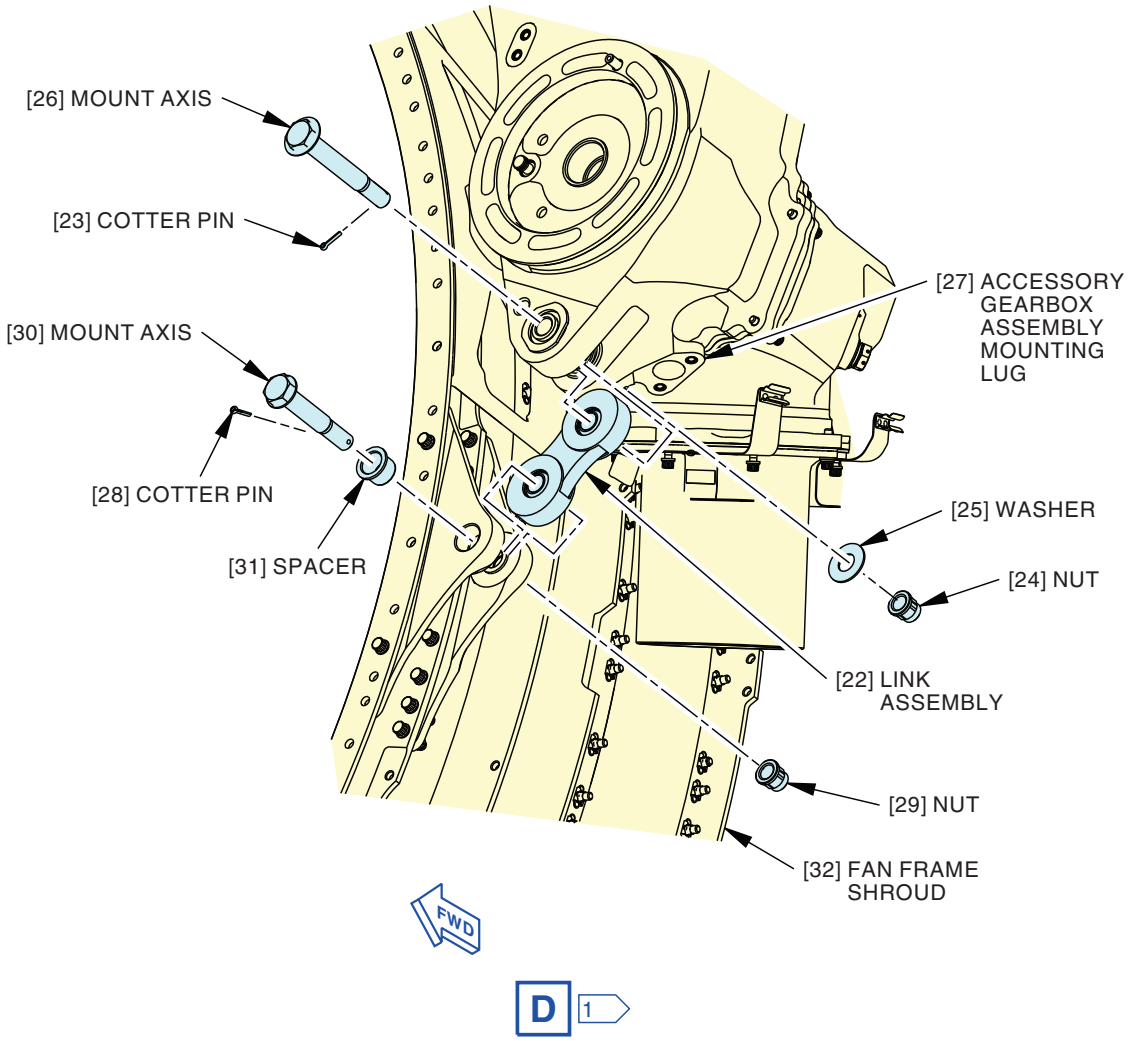

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Accessory Gearbox Link Installation
Figure 402/72-63-05-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-63-05-400-801-G00

3. Accessory Gearbox Link Installation

(Figure 401 or Figure 402)

A. General

(1) This task provides the instructions on how to install the accessory gearbox links.

(2) The accessory gearbox links included these parts: the stirrup assembly, the axial link assembly, and the link assembly.

(3) You must do the tests that are listed in the power plant test reference table after you install the accessory gearbox links.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14660	Positioner - Fan Disk Barrel Nut Part #: 956A6071G01 Supplier: 58828
SPL-14690	Fixture - Accessory Gearbox (AGB) Remove/Install Part #: 956A7018G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Stirrup assembly	72-00-00-14-370	PTF ALL
3	Cotter pin	72-00-00-14-330	PTF ALL
8	Cotter pin	72-00-00-14-200	PTF ALL
12	Axial link assembly	72-00-00-14-460	PTF ALL
13	Cotter pin	72-00-00-14-160	PTF ALL
20	Barrel nut	72-00-00-14-480	PTF ALL
22	Link assembly	72-00-00-14-420	PTF ALL
23	Cotter pin	72-00-00-14-240	PTF ALL
28	Cotter pin	72-00-00-14-380	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Accessory Gearbox Link Installation

SUBTASK 72-63-05-420-001-G00

(1) Install the accessory gearbox links [33] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the link assembly [22] as follows:

1) Install the link assembly [22] in a correct position on the fan frame shroud [32].

2) Install the spacer [31] on the mount axis [30].

3) Install the mount axis [30] that attaches the link assembly [22] to the fan frame shroud [32].

4) Lubricate the threads of the nut [29] with the graphite mineral vaseline, D00601 [CP2101].

5) Install the nut [29] that attaches the mount axis [30] to the link assembly [22].

6) Tighten the nut [29] to 420.4 in-lb (47.5 N·m) to 464.6 in-lb (52.5 N·m).

7) Install the new cotter pin [28] on the mount axis [30].

8) Install the link assembly [22] in a correct position on the accessory gearbox assembly mounting lug [27].

9) Install the mount axis [26] that attaches the link assembly [22] to the accessory gearbox assembly mounting lug [27].

10) Lubricate the threads of the nut [24] with the graphite mineral vaseline, D00601 [CP2101].

11) Install the nut [24] and the washer [25] that attach the mount axis [26] to the link assembly [22].

12) Tighten the nut [24] to 420.4 in-lb (47.5 N·m) to 464.6 in-lb (52.5 N·m).

13) Install the new cotter pin [23] on the mount axis [26].

(c) Install the axial link assembly [12] as follows:

NOTE: Due to self-locking feature limited to 5 cycles, barrel nut [20] must be discarded every 5 axial link assembly [12] removals.

1) Use the positioner, SPL-14660 to install the barrel nut retainer [21] and the barrel nut [20] in a correct position.

2) Install the axial link assembly [12] in a correct position on the fan frame shroud [32].

3) Install the spacer [19] on the mount axis [18].

4) Lubricate the threads of the mount axis [18] with graphite mineral vaseline, D00601 [CP2101].

5) Install the mount axis [18] that attaches the axial link assembly [12] to the fan frame shroud [32]

6) Tighten the mount axis [18] to 420.4 in-lb (47.5 N·m) to 464.6 in-lb (52.5 N·m).

7) Install the axial link assembly [12] in a correct position on the axial lug [17].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

8) Install the mount axis [16] that attaches the axial link assembly [12] to the axial lug [17].

9) Lubricate the threads of the nut [14] with the graphite mineral vaseline, D00601 [CP2101].

10) Install the nut [14] and the washer [15] that attach the mount axis [16] to the axial link assembly [12].

11) Tighten the nut [14] to the 420.4 in-lb (47.5 N·m) to 464.6 in-lb (52.5 N·m).

12) Install the new cotter pin [13] on the mount axis [16].

(d) Install the stirrup assembly [2] as follows:

1) Install the stirrup assembly [2] in a correct position on the fan frame shroud [32].

2) Install the spacer [11] on the mount axis [10].

3) Install the mount axis [10] that attaches the stirrup assembly [2] to the fan frame shroud [32].

4) Lubricate the threads of the nut [9] with the graphite mineral vaseline, D00601 [CP2101].

5) Install the nut [9] that attaches the mount axis [10] to the stirrup assembly [2].

6) Tighten the nut [9] to 420.4 in-lb (47.5 N·m) to 464.6 in-lb (52.5 N·m).

7) Install the new cotter pin [8] on the mount axis [10].

8) Install the stirrup assembly [2] in a correct position on the accessory gearbox assembly housing lug [7].

9) Install the mount axis [6] that attaches the stirrup assembly [2] to the accessory gearbox assembly housing lug [7].

10) Lubricate the threads of the nut [4] with the graphite mineral vaseline, D00601 [CP2101].

11) Install the nut [4] and washer [5] that attach the mount axis [6] to the stirrup assembly [2].

12) Tighten the nut [4] to 420.4 in-lb (47.5 N·m) to 464.6 in-lb (52.5 N·m).

13) Install the new cotter pin [3] on the mount axis [6].

(e) Remove the AGB remove/install fixture, SPL-14690 from the accessory gearbox assembly [1].

(f) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-05-860-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-05-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Accessory Gearbox Link Installation Test

SUBTASK 72-63-05-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX LINK - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Accessory gearbox links visual inspection.

TASK 72-63-05-200-801-G00

2. Accessory Gearbox Links Visual Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the visual inspection of the accessory gearbox links.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-63-05-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-05-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Accessory Gearbox Links Visual Inspection

SUBTASK 72-63-05-211-001-G00

(1) Do the inspection of the interface between coupling rod and spherical bearing of the axial link assembly [2] and the AGB link assembly [3] as follows:

(a) Axial and radial play

1) Not permitted.

2) Replace the link assembly.

(b) Seized-up

1) Not permitted.

2) Replace the link assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-05-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-63-05-860-006-G00

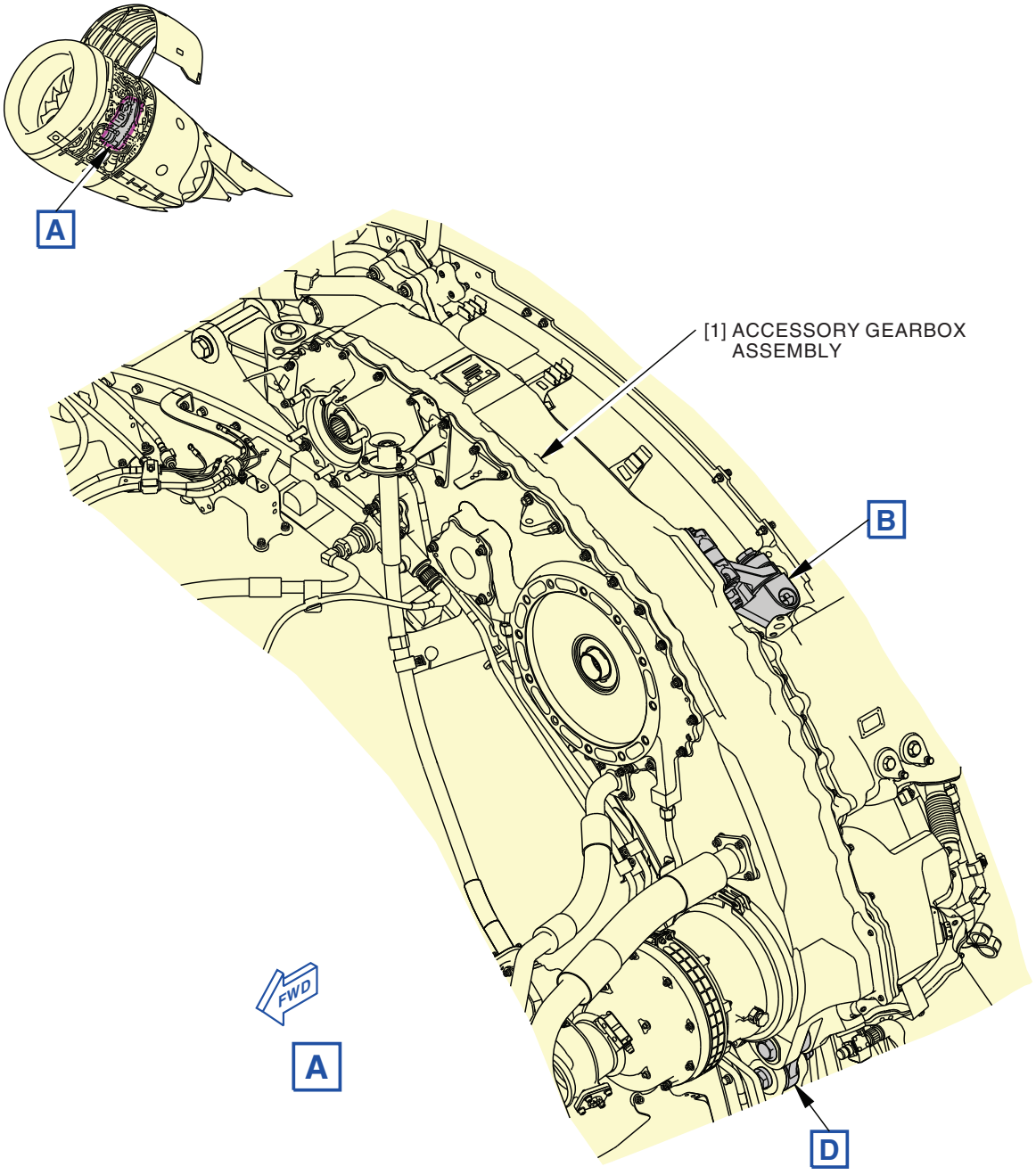
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

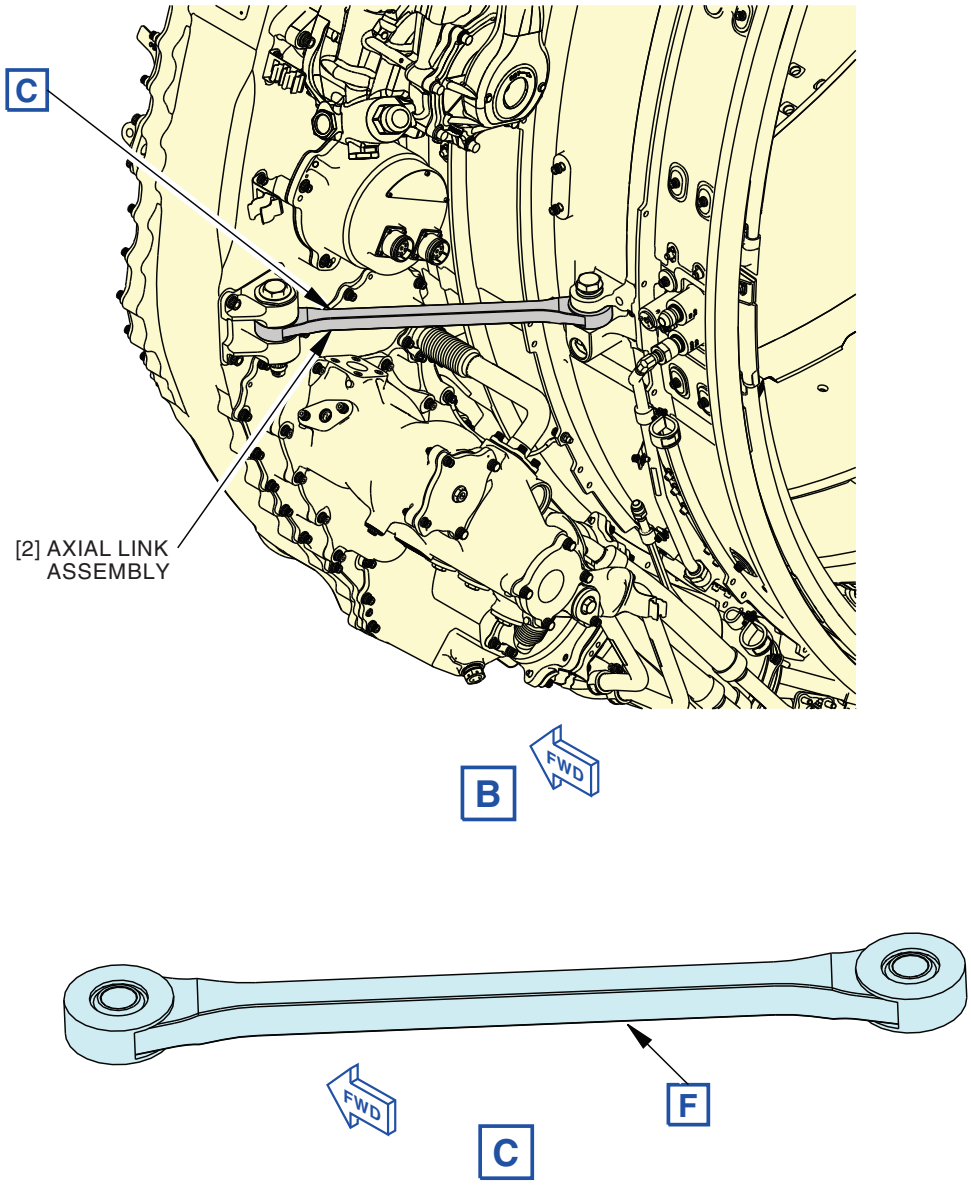
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Links Visual Inspection
Figure 601/72-63-05-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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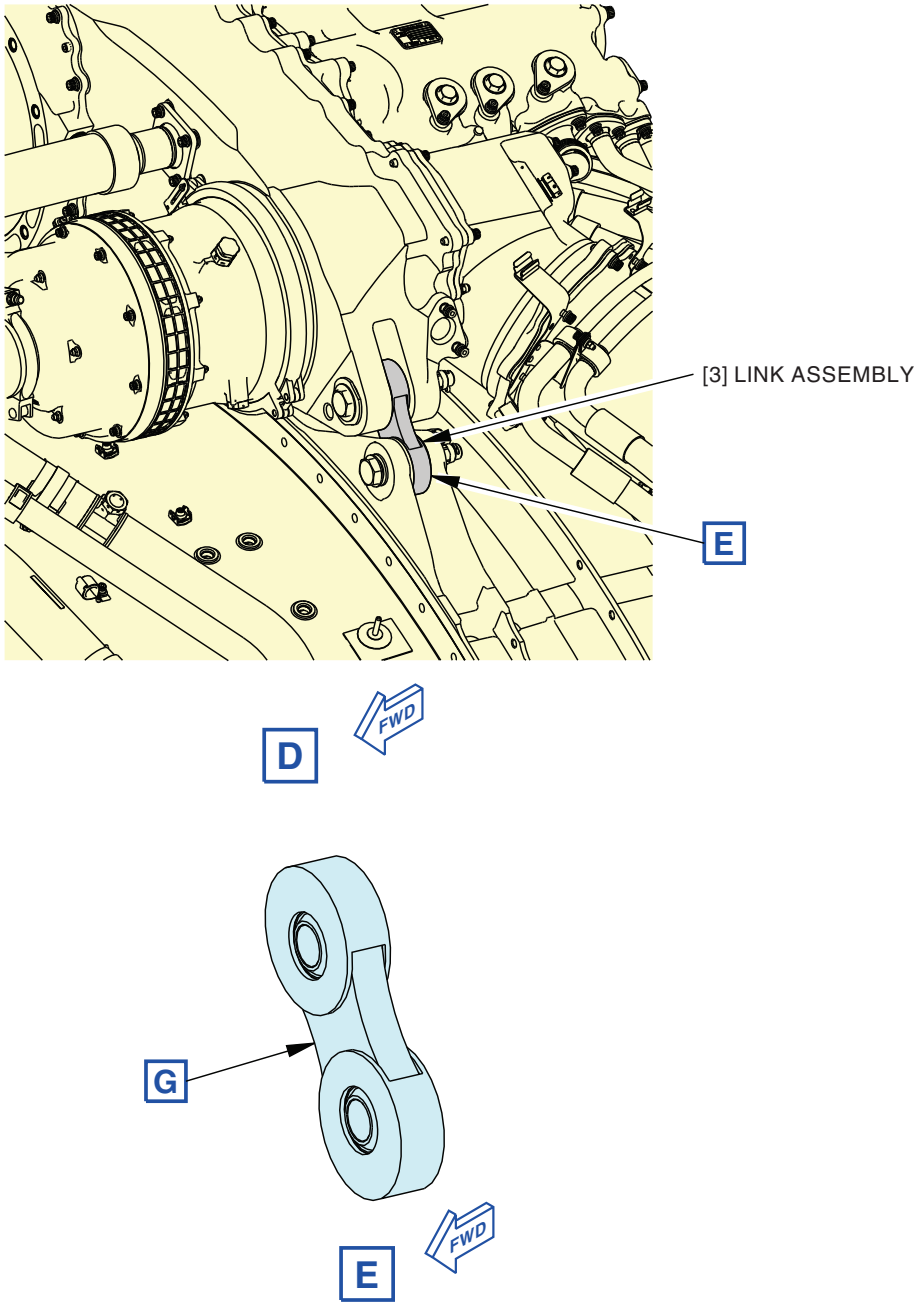
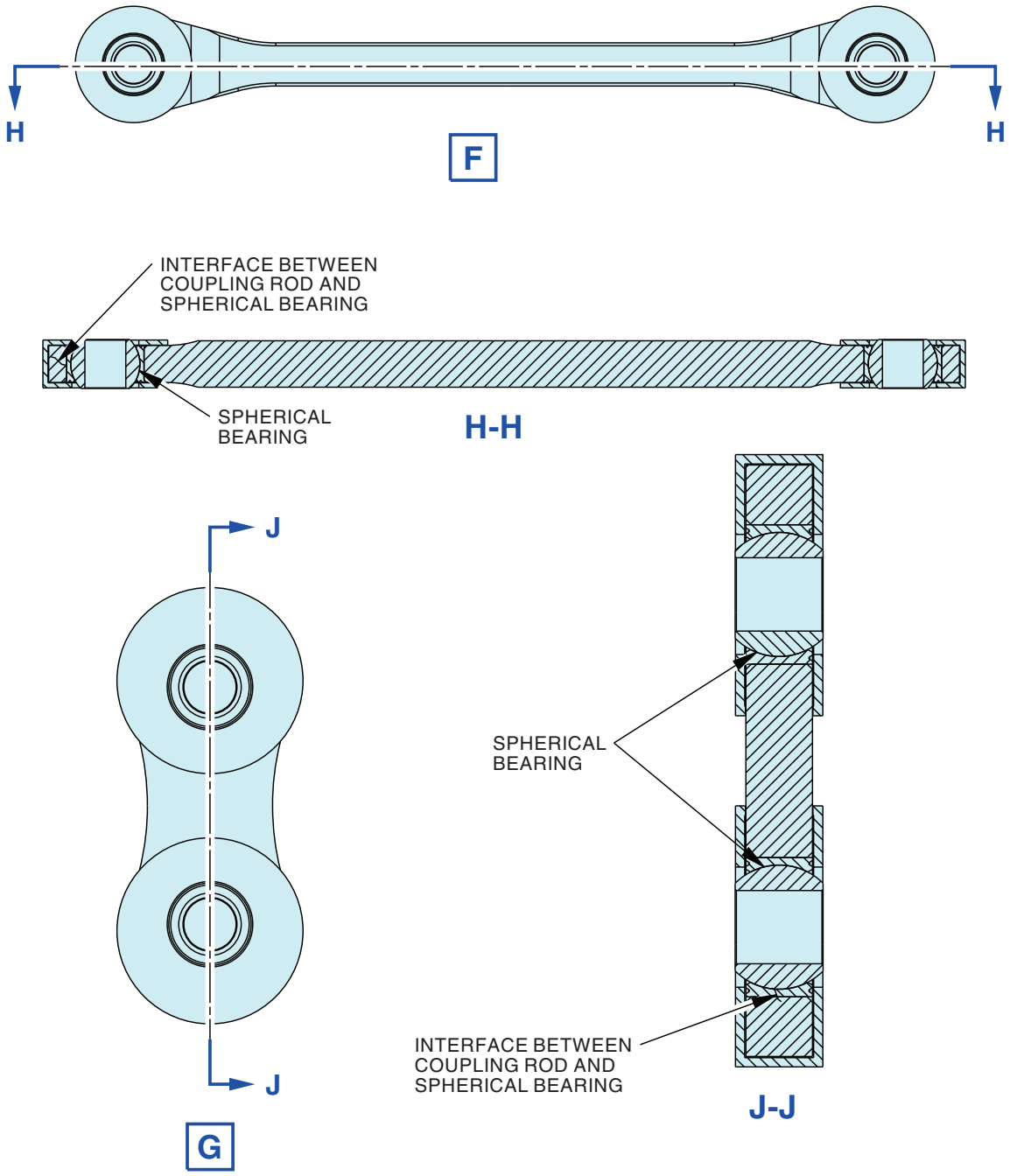
Accessory Gearbox Links Visual Inspection
Figure 601/72-63-05-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Links Visual Inspection
Figure 601/72-63-05-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Accessory Gearbox Links Visual Inspection
Figure 601/72-63-05-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX OIL PRESSURE TEMPERATURE SENSOR SUPPORT ASSEMBLY - REMOVAL/ INSTALLATION

1. General

A. This procedure has two tasks:

(1) Accessory gearbox oil pressure temperature sensor support assembly removal

(2) Accessory gearbox oil pressure temperature sensor support assembly installation.

TASK 72-63-07-000-801-G00

2. Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the accessory gearbox oil pressure temperature sensor support assembly.

(2) The accessory gearbox oil pressure temperature sensor support assembly is referred to as the AGB OPT sensor support assembly.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-63-07-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-63-07-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Removal

SUBTASK 72-63-07-020-001-G00

WARNING ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE AN INJURY OR KILL YOU.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove AGB OPT sensor support assembly [3] as follows:

(a) Remove the three screws [1] and the three washers [2].

(b) Remove and discard the AGB OPT sensor support assembly [3] from the AGB front cover [8].

(c) Remove and discard the O-ring [4] from the AGB OPT sensor support assembly [3].

(d) Remove and discard the O-ring [5] from the AGB OPT sensor support assembly [3].

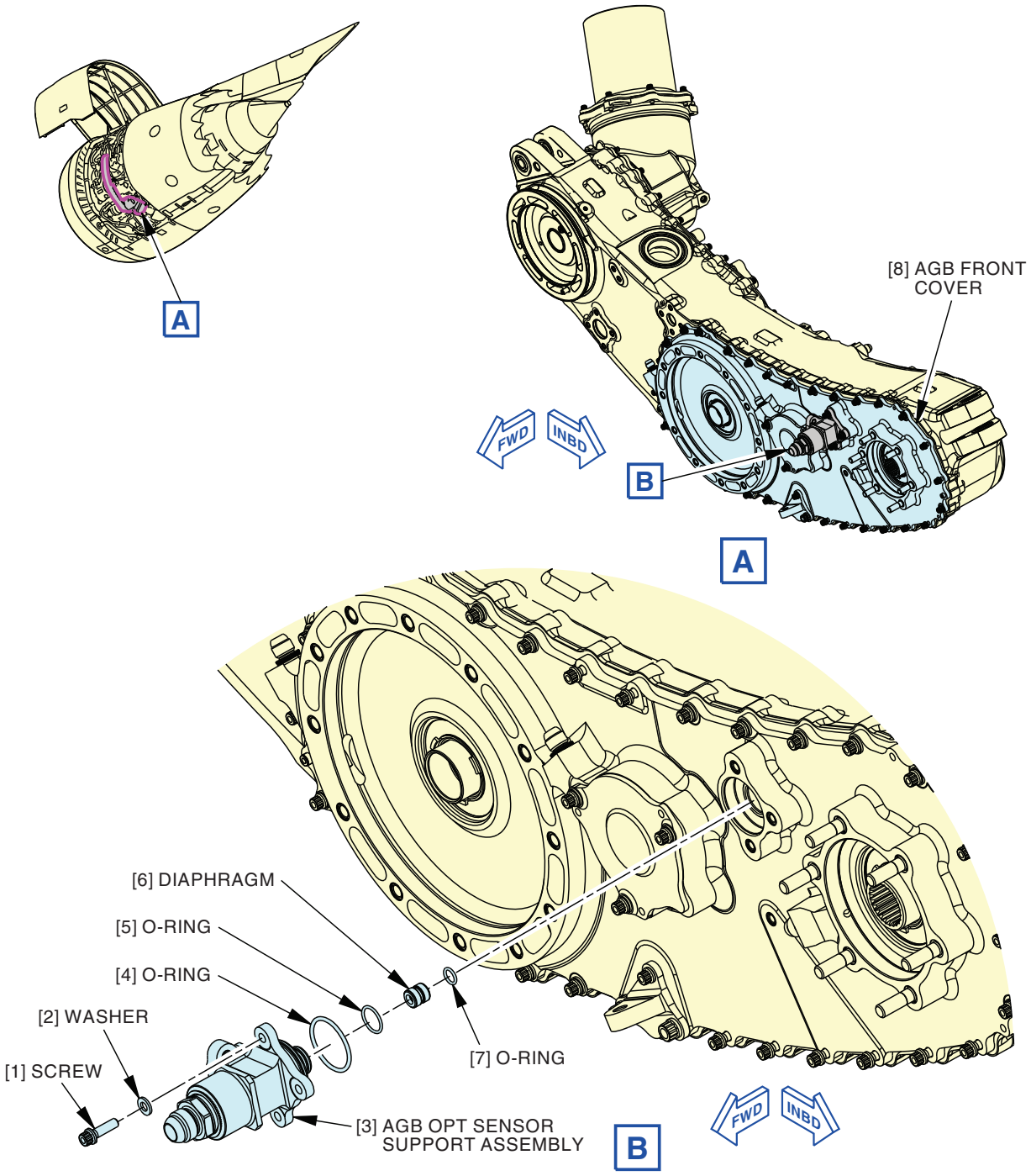
(e) Remove the diaphragm [6] and the O-ring [7] from the AGB OPT sensor support assembly [3].

1) Discard the O-ring [7].

(f) Install the protective covers on the AGB OPT sensor support assembly [3] and the open port on the AGB front cover [8].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Installation
Figure 401/72-63-07-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-63-07-400-801-G00

3. Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the accessory gearbox oil pressure temperature sensor support assembly.

(2) The accessory gearbox oil pressure temperature sensor support assembly is referred to as the AGB OPT sensor support assembly.

(3) You must do the tests that are listed in the power plant test reference table after you install the AGB OPT sensor support assembly.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
D50327 [CP5076]	Grease - General Purpose, Synthetic	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	AGB OPT sensor support assembly	72-63-01-04-070	PTF ALL
4	O-ring	72-63-01-04-080	PTF ALL
5	O-ring	72-63-01-04-090	PTF ALL
7	O-ring	72-63-01-04-110	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Installation

SUBTASK 72-63-07-420-001-G00

WARNING

ALWAYS USE APPROVED EQUIPMENT AND PROCEDURES TO LIFT PARTS. STAY OUT FROM BELOW LOADS. IF THE PARTS ACCIDENTALLY MOVE OR FALL, THEY CAN CAUSE AN INJURY OR KILL YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the AGB OPT sensor support assembly [3]:

(a) Remove the protective covers from the AGB OPT sensor support assembly [3] and the open port on the AGB front cover [8].

(b) Lubricate the new O-ring [7], the new O-ring [5], and the new O-ring [4] with the vaseline, D00672 [CP5070].

(c) Install the new O-ring [7] and the diaphragm [6] on the AGB OPT sensor support assembly [3].

(d) Install the new O-ring [5] on the AGB OPT sensor support assembly [3].

(e) Install the new O-ring [4] on the AGB OPT sensor support assembly [3].

(f) Apply the general purpose grease, D50327 [CP5076] on the three holes of the AGB OPT sensor support assembly [3] and the AGB front cover [8]

(g) Install the AGB OPT sensor support assembly [3] on the AGB front cover [8].

(h) Install the three screws [1] with the three washers [2] on the AGB OPT sensor support assembly [3] manually.

(i) Tighten the screws [1] to 97.35 in-lb (11.00 N·m) - 106.20 in-lb (12.00 N·m).

(j) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-07-860-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-63-07-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Installation Test

SUBTASK 72-63-07-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX COIL ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the accessory gearbox coil assembly

(2) An installation of the accessory gearbox coil assembly.

TASK 72-63-08-000-801-G00

2. Accessory Gearbox Coil Assembly Removal

(Figure 401)

A. General

(1) This task gives instructions to remove AGB (Accessory Gearbox) coil assembly.

B. References

Reference	Title
80-11-01-000-801-G00	Engine Starter Removal (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 72-63-08-000-001-G00

(1) Do this task: Engine Starter Removal, TASK 80-11-01-000-801-G00.

E. Coil Assembly Removal

SUBTASK 72-63-08-000-002-G00

(1) Remove the coil assembly [2] from the AGB housing as follows:

CAUTION

DO NOT USE A SHARP TOOL WHEN YOU REMOVE THE RETAINING RING FROM THE ACCESSORY GEARBOX. IF YOU DO NOT OBEY, DAMAGE TO THE ACCESSORY GEARBOX CAN OCCUR

(a) Remove the retaining ring [1] from the AGB housing.

(b) Remove the coil assembly [2] from the AGB housing.

(c) Remove the O-ring [3] from the coil assembly [2].

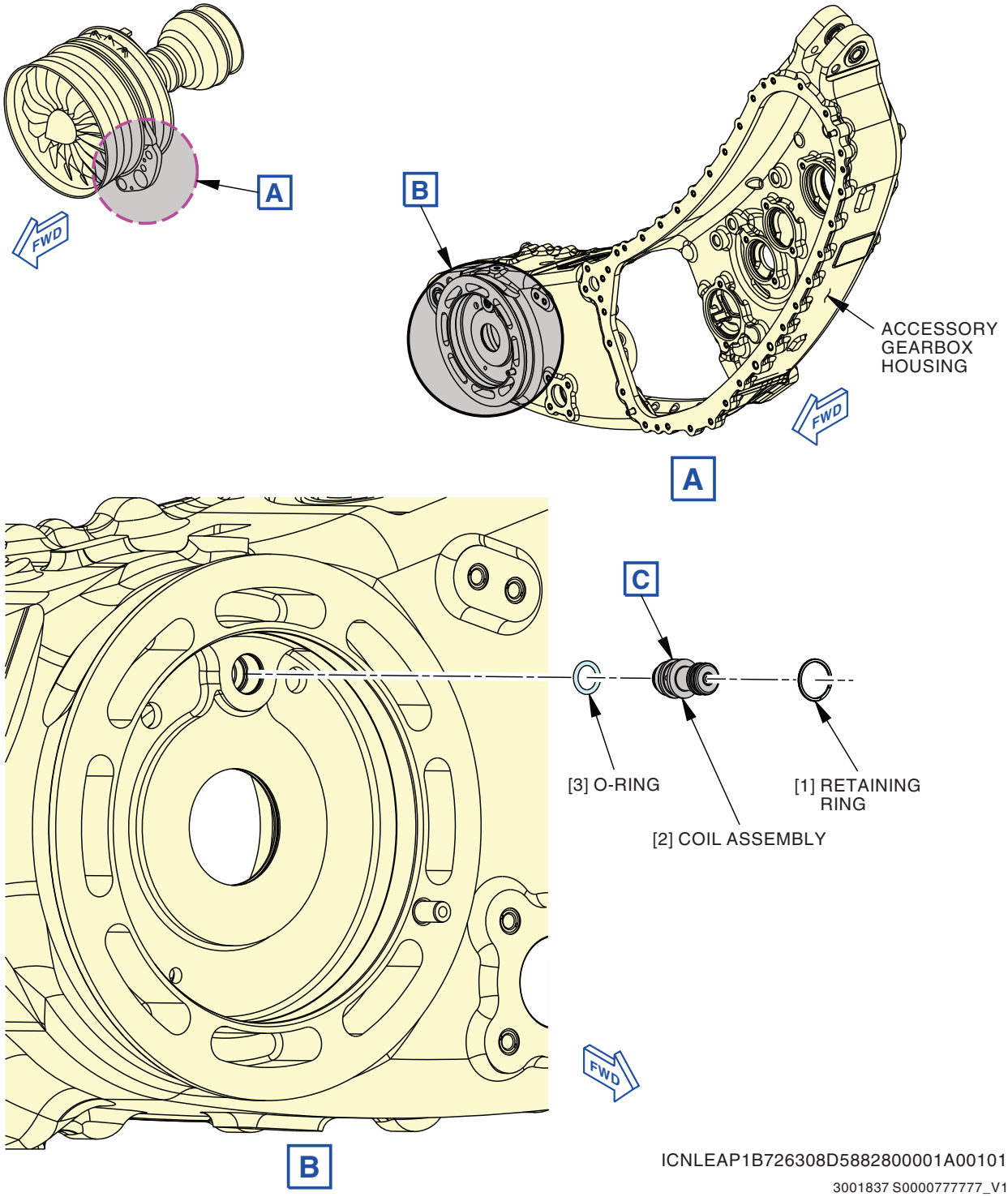
(d) Discard the coil assembly [2] and O-ring [3].

SUBTASK 72-63-08-480-001-G00

(2) Install the protective covers on all openings.

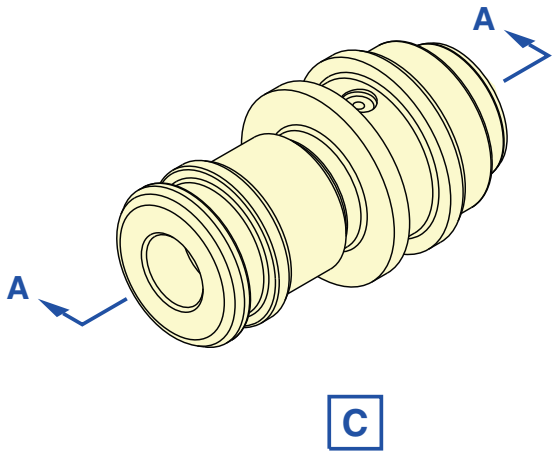
END OF TASK

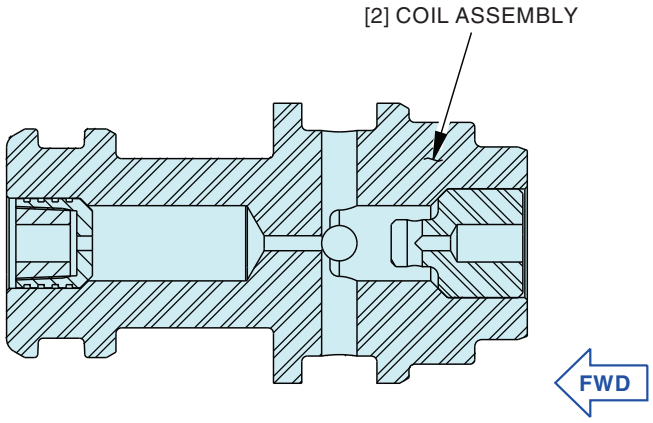
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Coil Assembly Installation
Figure 401/72-63-08-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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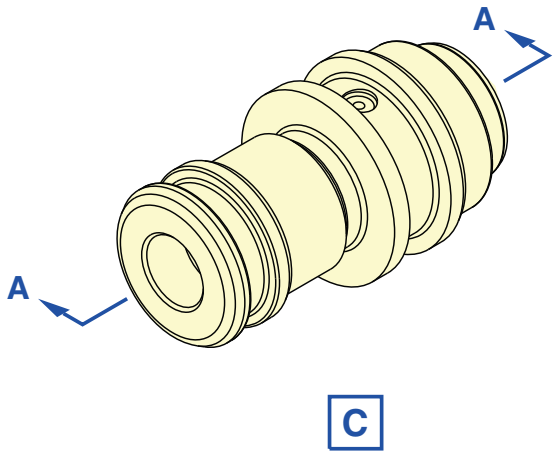
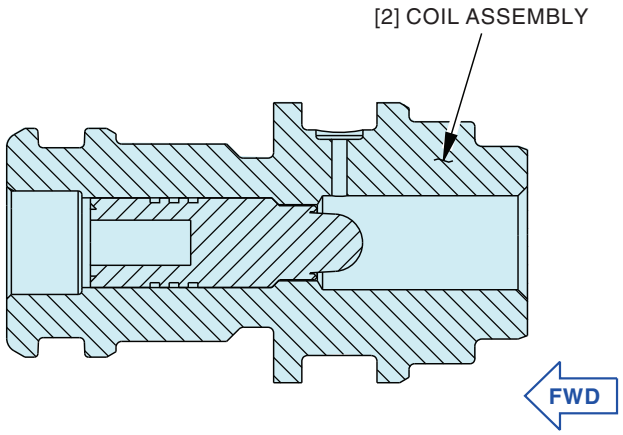
Accessory Gearbox Coil Assembly Installation
Figure 401/72-63-08-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





A-A

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Accessory Gearbox Coil Assembly Installation
Figure 401/72-63-08-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-63-08-400-801-G00

3. Accessory Gearbox Coil Assembly Installation

(Figure 401)

A. General

(1) This task gives instructions to install AGB (Accessory Gearbox) coil assembly.

B. References

Reference	Title
80-11-01-400-801-G00	Engine Starter Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Coil assembly	72-63-01-04-200	PTF ALL
3	O-ring	72-63-01-04-210	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Coil Assembly Installation

SUBTASK 72-63-08-080-001-G00

(1) Remove the protective covers from all openings.

SUBTASK 72-63-08-210-001-G00

(2) Make sure that the mating surfaces of the AGB housing are clean and clear of unwanted materials.

SUBTASK 72-63-08-400-001-G00

(3) Install the new coil assembly [2] in the AGB housing as follows:

(a) Lubricate the new O-ring [3] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(b) Install the O-ring [3] on the new coil assembly [2].

(c) Make sure that the recess of the coil assembly [2] in the AGB housing is clear of unwanted materials.

(d) Install the coil assembly [2] it its recess of the AGB housing.

(e) Install the retaining ring [1] to secure the coil assembly [2] in its position.

1) If the groove for the retaining ring [1] is not visible, axially push on the coil assembly [2].

(f) Make sure that the retaining ring [1] in fully engaged in the groove of the AGB housing on all its perimeter.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-63-08-410-001-G00

(1) Do this task: Engine Starter Installation, TASK 80-11-01-400-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY GEARBOX ROTATING SEALS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Accessory gearbox rotating seals removal

(2) Accessory gearbox rotating seals installation.

TASK 72-64-01-000-801-G00

2. Accessory Gearbox Rotating Seals Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the Accessory Gearbox (AGB) rotating seals.

(2) The rotating seals include these parts: the O-ring, the mating ring, the O-ring, the rotating seal housing and the retaining ring.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14666	Kit - Tool, AGB Rotating Seal Part #: 956A8900G01 Supplier: 58828
STD-809	Spatula
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-64-01-860-001-G00

(1) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

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SUBTASK 72-64-01-860-002-G00

(2) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-64-01-860-003-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Accessory Gearbox Seals Removal

SUBTASK 72-64-01-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the rotating seal [25] from the AGB manual control pad [1] as follows:

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CAUTION

DO NOT USE A SCREWDRIVER OR EQUIVALENT TOOLS TO REMOVE THE ROTATING-SEALS. DAMAGE TO THE ROTATING-SEALS OR THE ACCESSORY GEARBOX COVER CAN OCCUR.

(a) Install a spatula, STD-809 into the removal notch of the retaining ring [5].

(b) Pull out the first end of the retaining ring [5] with the spatula, STD-809.

(c) Turn the retaining ring [5] until it is free from the groove on the AGB manual control pad [1].

(d) Remove the retaining ring [5].

(e) Insert the three locking pins of the rotating seal housing [6] into the three slots on the housing removal/installation tool from the seal replacement tool, SPL-14666 kit.

(f) Rotate the removal/installation tool into a position that will allow you to pull the rotating seal housing [6].

(g) Pull the housing removal/installation tool to remove the rotating seal housing [6] from the AGB manual control pad [1].

(h) Remove and discard the O-ring [7] from the rotating seal housing [6].

(i) Remove the rotating seal housing [6] from the removal and installation tool.

(j) Insert the two locking pins of the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit into the slots of the mating ring [8].

(k) Rotate the removal/installation tool 90 degrees.

(l) Pull the removal/installation tool to remove the mating ring [8] from the AGB manual control pad [1].

(m) Remove and discard the O-ring [9] from the mating ring [8].

SUBTASK 72-64-01-020-002-G00

(2) Remove the rotating seal [25] from the IDG pad [2] as follows:

CAUTION

DO NOT USE A SCREWDRIVER OR EQUIVALENT TOOLS TO REMOVE THE ROTATING-SEALS. DAMAGE TO THE ROTATING-SEALS OR THE ACCESSORY GEARBOX COVER CAN OCCUR.

(a) Install a spatula, STD-809 into the removal notch of the retaining ring [10].

(b) Pull out the first end of the retaining ring [10] with the spatula, STD-809.

(c) Turn the retaining ring [10] until it is free from the groove on the IDG pad [2].

(d) Remove the retaining ring [10].

(e) Insert the three locking pins of the rotating-seal housing [11] into the three slots on the housing removal/installation tool from the seal replacement tool, SPL-14666 kit.

(f) Rotate the housing removal/installation tool into a position that will allow you to pull the rotating-seal housing [11].

(g) Pull the housing removal/installation tool to remove the rotating-seal housing [11].

(h) Remove and discard the O-ring [12] from the rotating-seal housing [11].

(i) Insert the two locking pins of the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit into the two slots of the mating ring [13].

(j) Rotate the mating ring removal/installation tool 90 degrees.

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(k) Pull the removal/installation tool to remove the mating ring [13] from the IDG pad [2].

(l) Remove and discard the O-ring [14] from the mating ring [13].

SUBTASK 72-64-01-020-003-G00

(3) Remove the rotating seal [25] from the hydraulic pump pad [3] as follows:

CAUTION

DO NOT USE A SCREWDRIVER OR EQUIVALENT TOOLS TO REMOVE THE ROTATING-SEALS. DAMAGE TO THE ROTATING-SEALS OR THE ACCESSORY GEARBOX COVER CAN OCCUR.

(a) Install a spatula, STD-809 into the removal notch of the retaining ring [15].

(b) Pull out the first end of the retaining ring [15] with the spatula, STD-809.

(c) Turn the retaining ring [15] until it is free from the groove on the hydraulic pump pad [3].

(d) Remove the retaining ring [15].

(e) Insert the three locking pins of the rotating-seal housing [16] into the three slots on the housing removal/installation tool from the seal replacement tool, SPL-14666 kit.

(f) Rotate the housing removal/installation tool into a position that will allow you to pull on the rotating-seal housing [16].

(g) Pull the removal/installation tool to remove the rotating-seal housing [16] from the hydraulic pump pad [3].

(h) Remove and discard the O-ring [17] from the rotating-seal housing [16].

(i) Insert the two locking pins of the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit into the two slots of the mating ring [18].

(j) Rotate the removal/installation tool 90 degrees.

(k) Pull the removal/installation tool to remove the mating ring [18] from the hydraulic pump pad [3].

(l) Remove and discard the O-ring [19] from the mating ring [18].

SUBTASK 72-64-01-020-004-G00

(4) Remove the rotating seal [25] from the main fuel pump pad [4] as follows:

CAUTION

DO NOT USE A SCREWDRIVER OR EQUIVALENT TOOLS TO REMOVE THE ROTATING-SEALS. DAMAGE TO THE ROTATING-SEALS OR THE ACCESSORY GEARBOX COVER CAN OCCUR.

(a) Install a spatula, STD-809 into the removal notch of the retaining ring [20].

(b) Pull out the first end of the retaining ring [20] with the spatula, STD-809.

(c) Turn the retaining ring [20] until it is free from the groove on the main fuel pump pad [4].

(d) Remove the retaining ring [20].

(e) Insert the three locking pins of the rotating seal housing [21] into the three slots of the housing removal/installation tool from the seal replacement tool, SPL-14666 kit.

(f) Rotate the removal/installation tool into a position that will allow you to pull on the rotating seal housing [21].

(g) Pull the removal/installation tool to remove the rotating seal housing [21] from the main fuel pump pad [4].

(h) Remove and discard the O-ring [22] from the rotating seal housing [21].

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(i) Insert the two locking pins of the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit into the two slots of the mating ring [23].

(j) Rotate the removal/installation tool 90 degrees.

(k) Pull the removal/installation tool to remove the mating ring [23] from the main fuel pump pad [4].

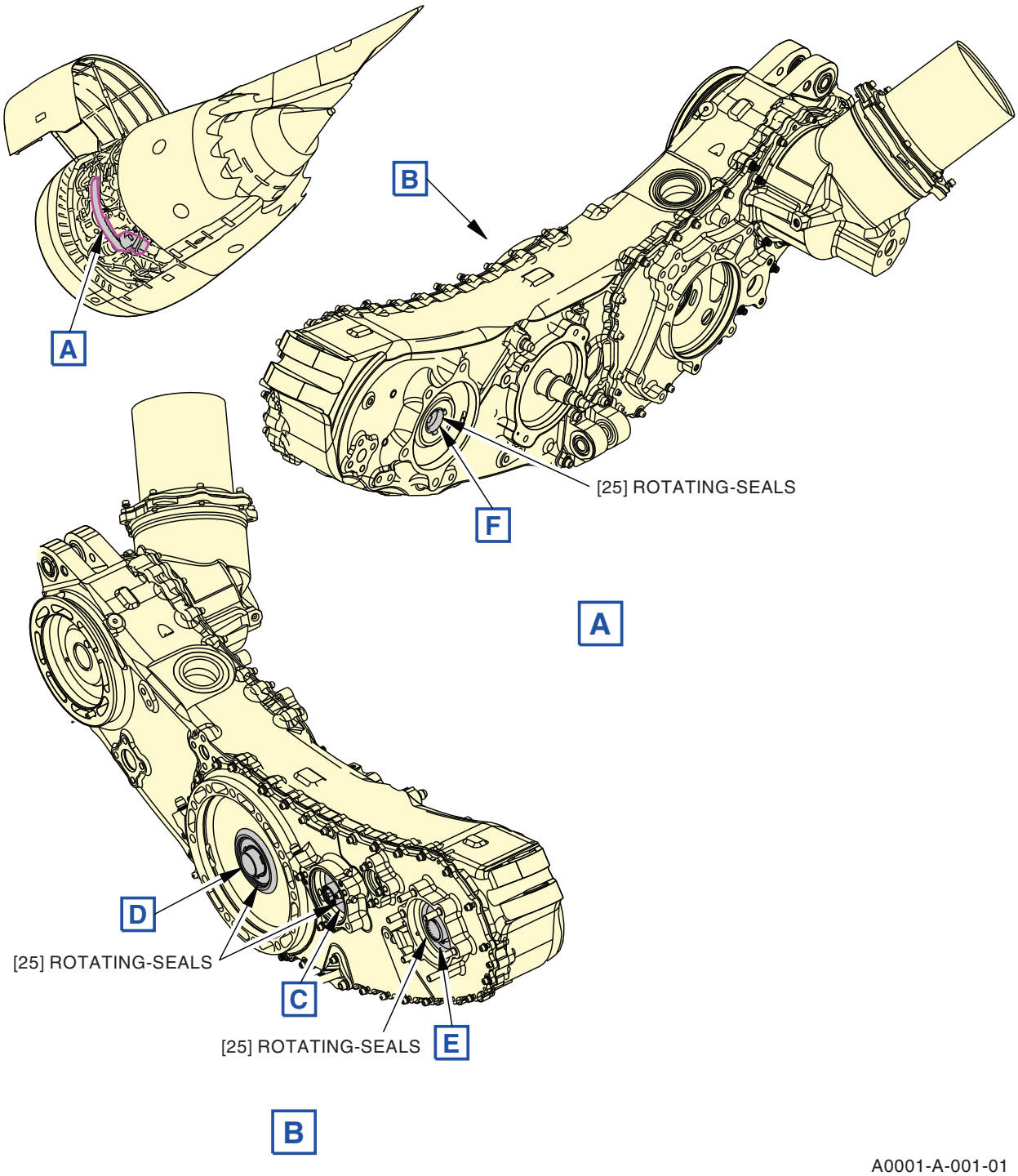
(l) Remove and discard the O-ring [24] from the mating ring [23].

SUBTASK 72-64-01-080-001-G00

(5) Install protective plugs to the AGB openings where the seals have been removed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

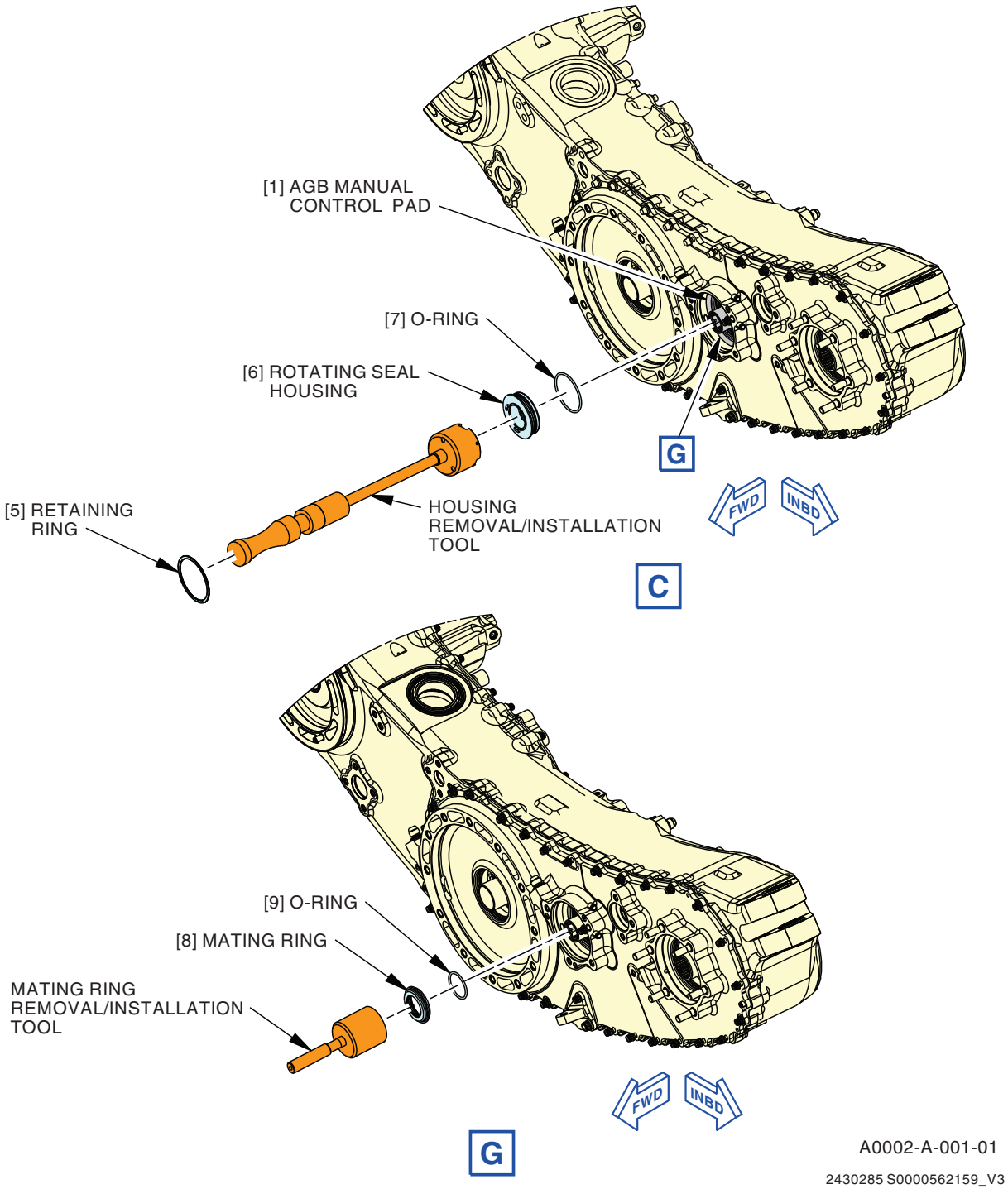


A0001-A-001-01

2429483 S0000562158_V2

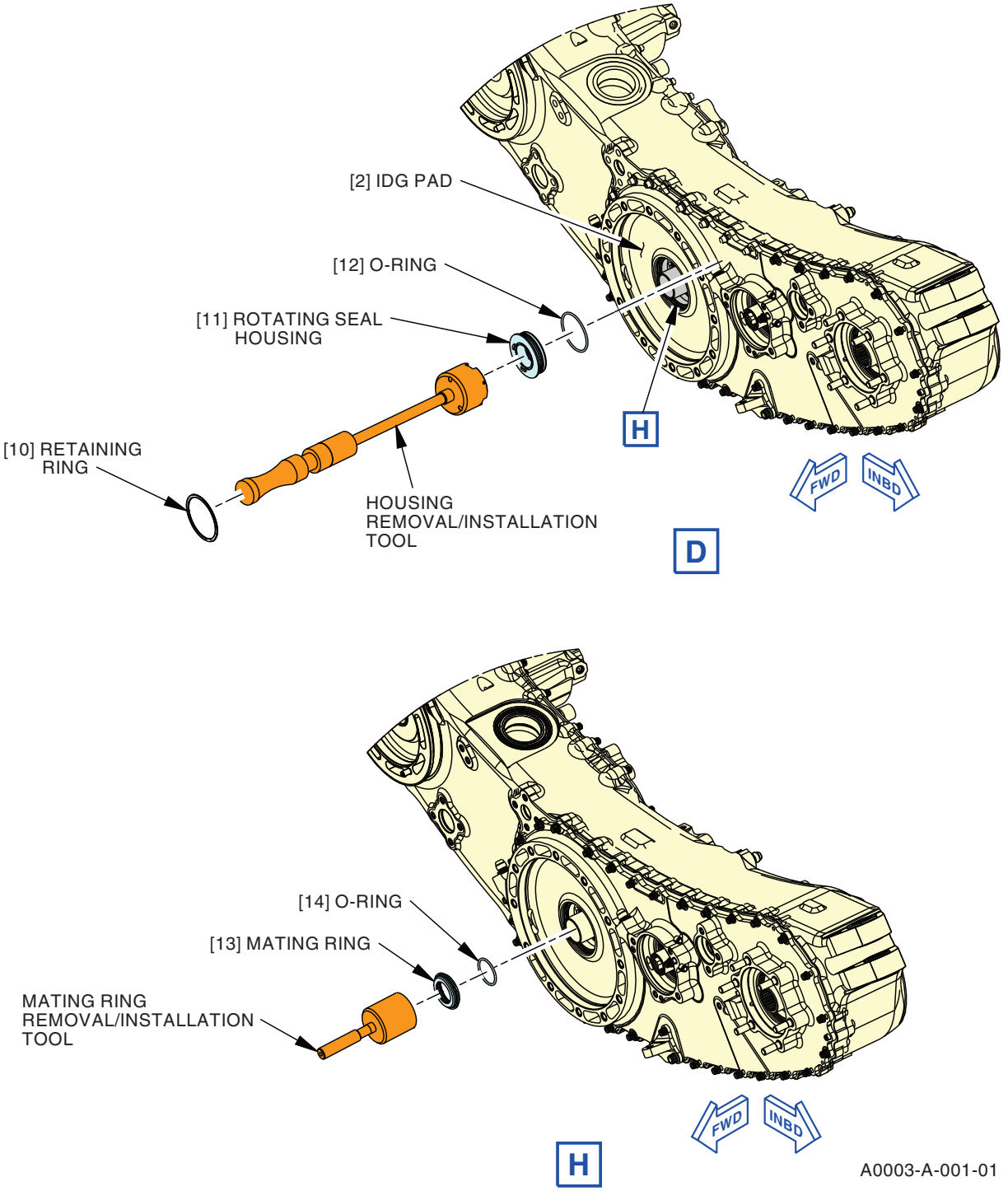
Accessory Gearbox Rotating Seals Installation
Figure 401/72-64-01-990-802-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



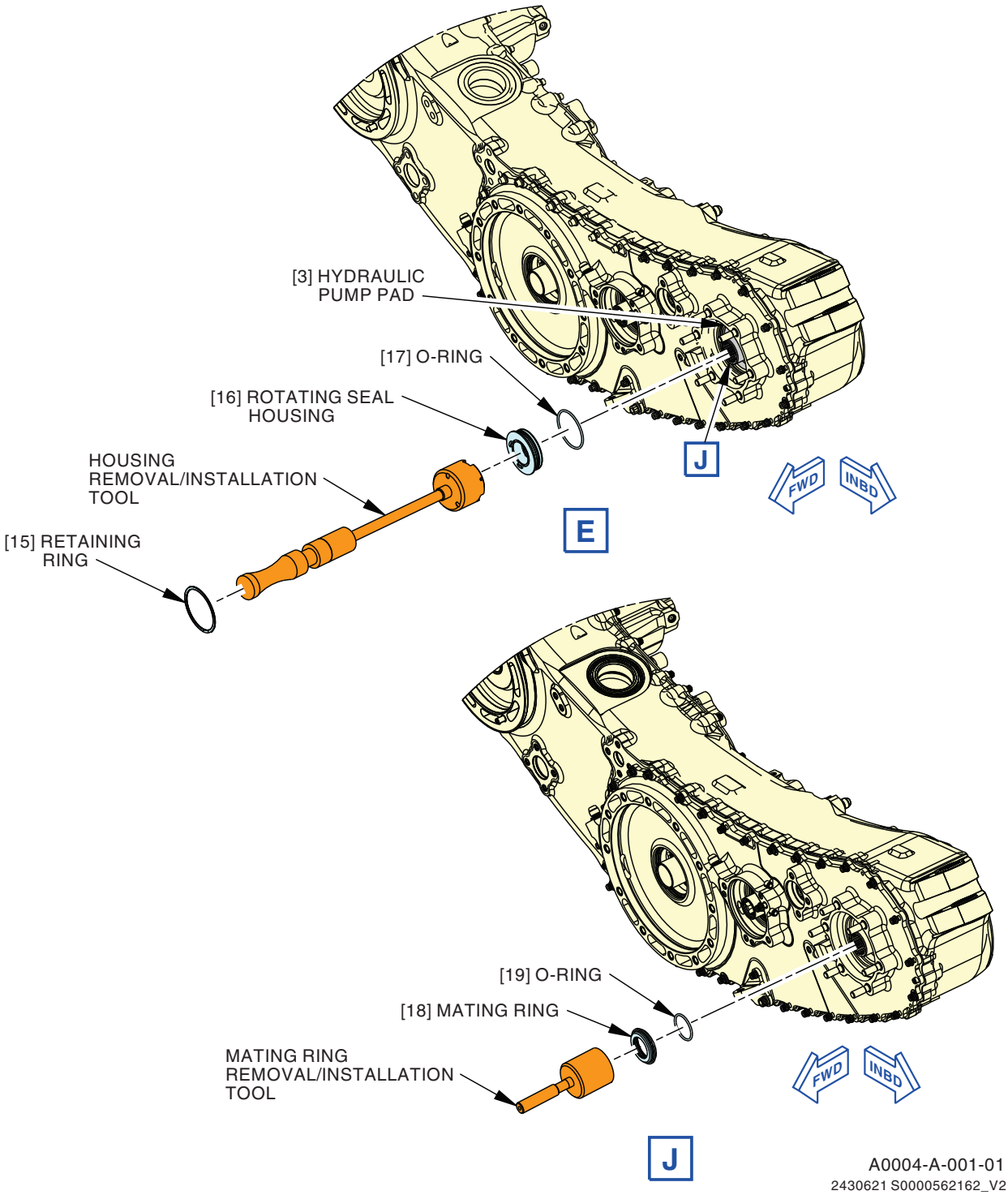
Accessory Gearbox Rotating Seals Installation
Figure 401/72-64-01-990-802-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



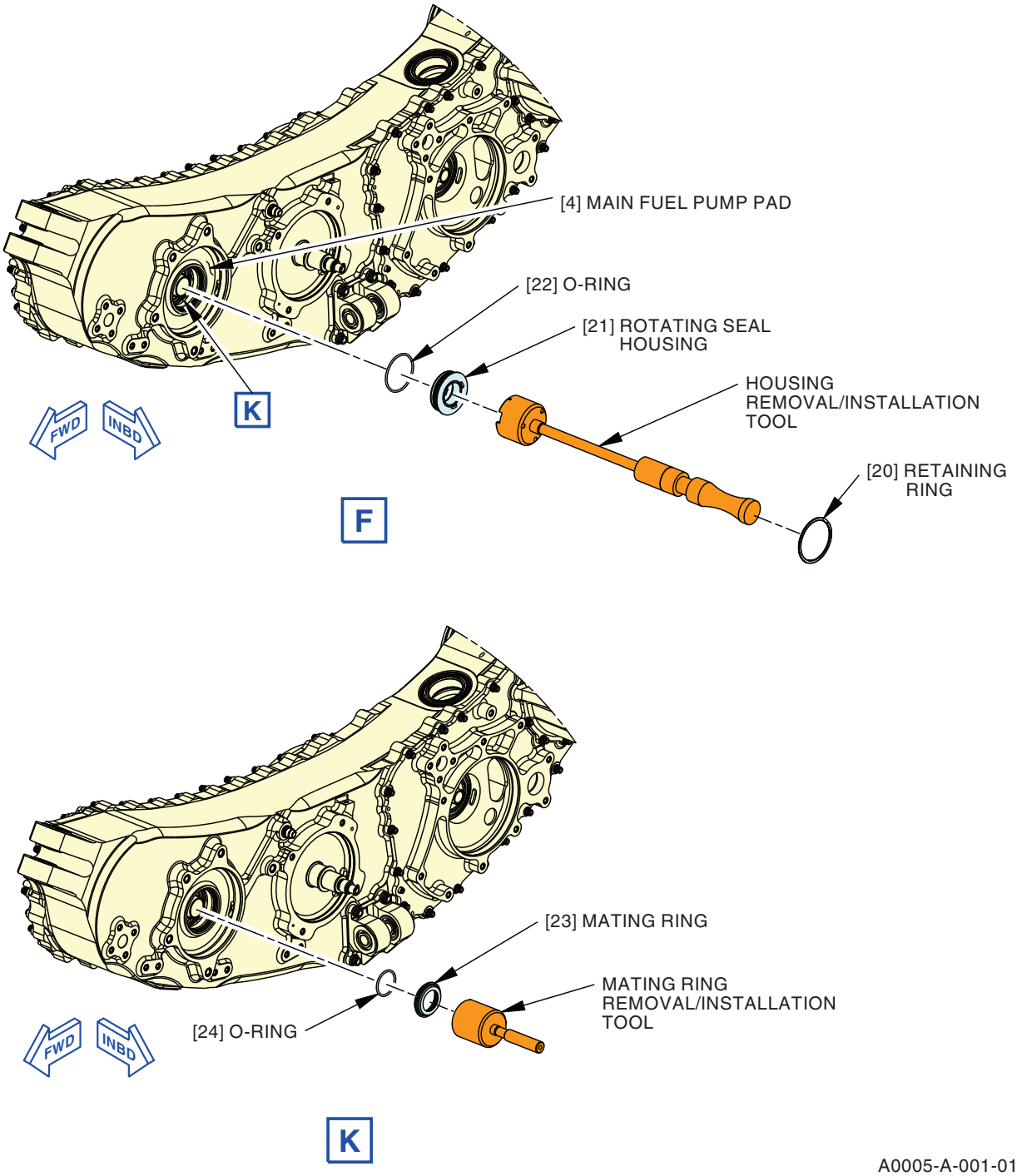
Accessory Gearbox Rotating Seals Installation
Figure 401/72-64-01-990-802-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Gearbox Rotating Seals Installation
Figure 401/72-64-01-990-802-G00 (Sheet 4 of 5)

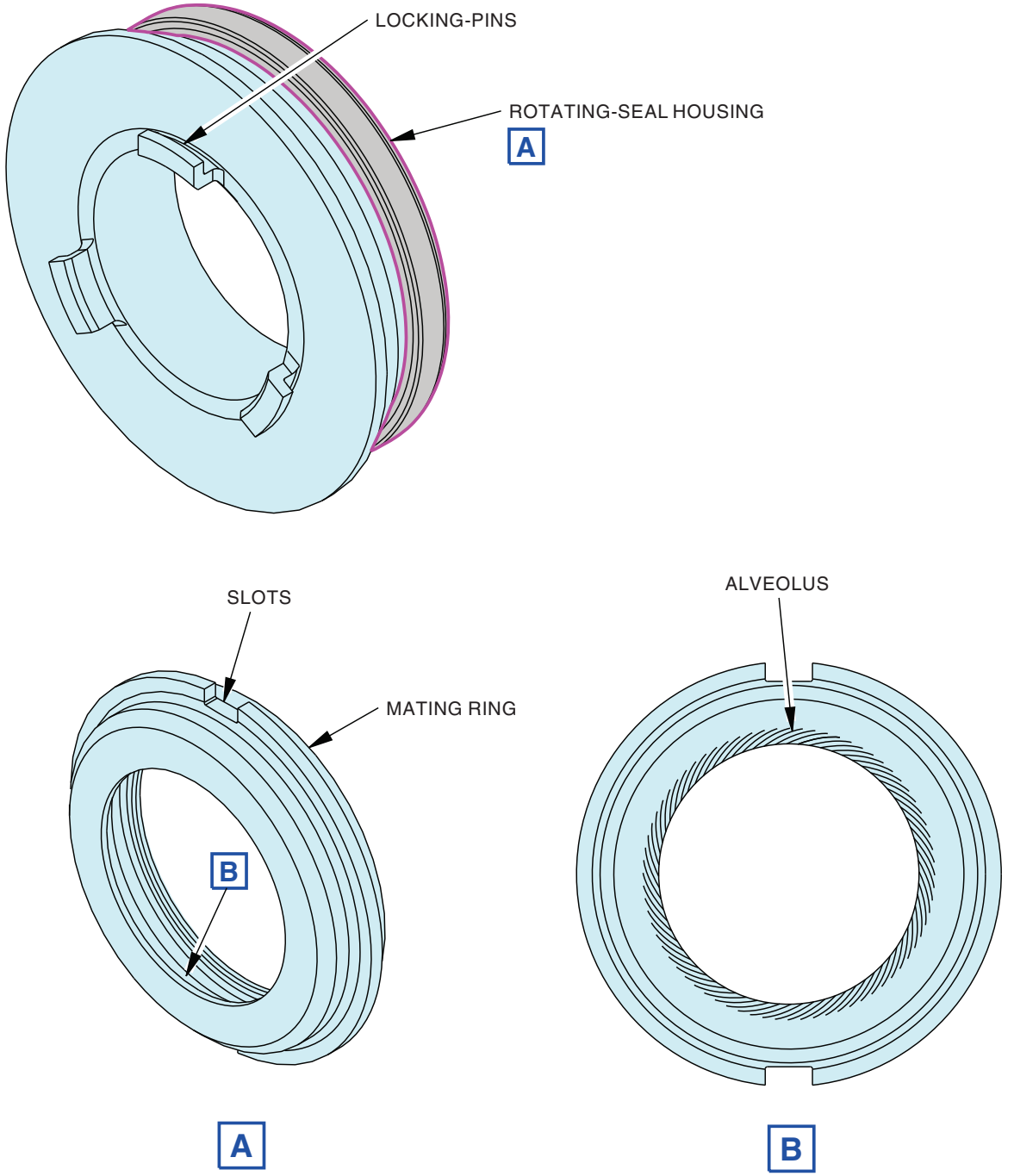
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0005-A-001-01
2430636 S0000562163_V2

Accessory Gearbox Rotating Seals Installation
Figure 401/72-64-01-990-802-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0006-A-001-01
2429099 S0000562157_V1

Accessory Gearbox Rotating Seal Types
Figure 402/72-64-01-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-64-01-400-801-G00

3. Accessory Gearbox Rotating-Seals Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the Accessory Gearbox (AGB) rotating-seals.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14666	Kit - Tool, AGB Rotating Seal Part #: 956A8900G01 Supplier: 58828
STD-809	Spatula
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50337 [CP2766]	Lubricant - Assembly Fluid 1	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	O-ring	72-63-01-04-310	PTF ALL
9	O-ring	72-63-01-04-300	PTF ALL
12	O-ring	72-63-01-04-450	PTF ALL
14	O-ring	72-63-01-04-250	PTF ALL
17	O-ring	72-63-01-04-370	PTF ALL
19	O-ring	72-63-01-04-360	PTF ALL
22	O-ring	72-63-01-04-440	PTF ALL
24	O-ring	72-63-01-04-430	PTF ALL
25	Rotating seal	72-63-01-04-220	PTF ALL
		72-63-01-04-270	PTF ALL
		72-63-01-04-330	PTF ALL
		72-63-01-04-400	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Accessory Gearbox Rotating Seals Installation

SUBTASK 72-64-01-080-002-G00

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the protective plugs from the AGB openings where you will install a seal.

SUBTASK 72-64-01-420-001-G00

(2) Install the rotating seal [25] on the AGB manual control pad [1] as follows:

(a) Lubricate the new O-ring [9] with engine oil, D00599 [CP2442].

(b) Install the new O-ring [9] into the groove of the mating ring [8].

NOTE: Make sure that the alveolus is not visible when you install the removal/installation tool on the mating ring.

(c) Install the mating ring [8] on the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit.

(d) Rotate the mating ring removal/installation tool 90 degrees.

(e) Install the mating ring [8] into the groove of the AGB manual control pad [1] and push it to the thrust of the spur gear.

(f) Make sure that the alveolus is visible after you install the mating ring [8]

(g) Remove the mating ring removal/installation tool.

(h) Make sure that the mating ring face is completely free of damage and dirt mark.

(i) Lubricate the new O-ring [7] with Assembly Fluid 1 lubricant, D50337 [CP2766].

(j) Install the new O-ring [7] into the groove of rotating seal housing [6].

(k) Install the rotating seal housing [6] on the housing removal/installation tool from the seal replacement tool, SPL-14666 kit.

(l) Insert the three pins of the rotating seal housing [6] into the three slots of the housing removal/installation tool.

(m) Rotate the seal replacement tool, SPL-14666 to a position that will hold the rotating seal housing [6].

(n) Install the rotating seal housing [6] into the groove of the AGB manual control pad [1] until the thrust of the mating ring [8].

(o) Remove the housing removal/installation tool.

(p) Make sure that the groove for the retaining ring [5] is clean.

(q) Turn the retaining ring [5] until it is installed in its groove.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(r) Push the first end of the retaining ring [5] with a spatula, STD-809.

(s) Use the spatula, STD-809 to complete the installation of the retaining ring [5].

(t) Make sure that the retaining ring [5] is installed in the correct position.

SUBTASK 72-64-01-420-002-G00

(3) Install the rotating seal [25] on the IDG pad [2] as follows:

(a) Lubricate the new O-ring [14] with engine oil, D00599 [CP2442].

(b) Install the new O-ring [14] into the groove of the mating ring [13].

(c) Install the mating ring [13] on the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit.

(d) Make sure that the alveolus is not visible after you install the mating ring [13] on the removal/installation tool.

(e) Insert the two locking pins of the mating ring removal/installation tool into the two slots of the mating ring [13].

(f) Rotate the mating ring removal/installation tool 90 degrees.

(g) Install the mating ring [13] into the groove of the IDG pad [2] and push it to the thrust of the spur gear.

(h) Remove the mating ring removal/installation tool.

(i) Make sure that the alveolus is visible after you install the mating ring.

(j) Make sure that the mating ring face is completely free of damage and dirt mark

(k) Lubricate the new O-ring [12] with Assembly Fluid 1 lubricant, D50337 [CP2766].

(l) Install the new O-ring [12] into the groove of the rotating-seal housing [11].

(m) Install the rotating-seal housing [11] on the housing removal/installation tool from the seal replacement tool, SPL-14666 kit.

(n) Insert the three pins of the rotating-seal housing [11] into the three slots of the housing removal/installation tool.

(o) Rotate the housing removal/installation tool into position to hold the rotating-seal housing [11].

(p) Install the rotating-seal housing [11] into the groove of the IDG pad [2] until the thrust of the mating ring [13].

(q) Remove the housing removal/installation tool.

(r) Make sure that the groove on the retaining ring [10] is clean.

(s) Turn the retaining ring [10] until it is installed in its groove.

(t) Push the first end of the retaining ring [10] with a spatula, STD-809.

(u) Use the spatula, STD-809 to complete the installation of the retaining ring [10].

(v) Make sure that the retaining ring [10] is installed in the correct position.

SUBTASK 72-64-01-420-003-G00

(4) Install the rotating seal [25] on the hydraulic pump pad [3] port as follows:

(a) Lubricate the new O-ring [19] with engine oil, D00599 [CP2442].

(b) Install the new O-ring [19] into the groove of the mating ring [18].

(c) Put the mating ring [18] on the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Make sure that the alveolus is not visible after you install the mating ring [18] on the removal/installation tool.

(e) Insert the two locking pins of the removal/installation tool into the two slots of the mating ring [18].

(f) Rotate the mating ring removal/installation tool 90 degrees.

(g) Install the mating ring [18] into the groove of the hydraulic pump pad [3] port and push until the thrust of the spur gear.

(h) Make sure that the alveolus is visible after you install the mating ring [18].

(i) Make sure that the mating ring face is completely free of damage and dirt mark

(j) Lubricate the new O-ring [17] with Assembly Fluid 1 lubricant, D50337 [CP2766].

(k) Install the new O-ring [17] into the groove of the rotating-seal housing [16].

(l) Install the rotating-seal housing [16] on the housing removal and installation tool from the seal replacement tool, SPL-14666 kit.

(m) Insert the three pins of the rotating-seal housing [16] into the three slots of the housing removal and installation tool.

(n) Rotate the housing removal and installation tool into position to hold rotating-seal housing [16].

(o) Install the rotating-seal housing [16] into the groove of the hydraulic pump pad [3] port until the thrust of the mating ring [18].

(p) Remove the housing removal and installation tool from the rotating-seal housing [16].

(q) Make sure that the groove for the retaining ring [15] is clean.

(r) Turn the retaining ring [15] until it is installed in its groove.

(s) Push the first end of the retaining ring [15] with a spatula, STD-809.

(t) Use the spatula, STD-809 to complete the installation of the retaining ring [15].

(u) Make sure that the retaining ring [15] is installed in the correct position.

SUBTASK 72-64-01-420-004-G00

(5) Install the rotating seal [25] on the main fuel pump pad [4] as follows:

(a) Lubricate the new new O-ring [24] with engine oil, D00599 [CP2442].

(b) Install the new new O-ring [24] into the groove of the mating ring [23].

(c) Install the mating ring [23] on the mating ring removal/installation tool from the seal replacement tool, SPL-14666 kit.

(d) Make sure that the alveolus is not visible after you install the mating ring [23] on the removal/installation tool.

(e) Insert the two locking pins of the mating ring removal/installation tool into the two slots of the mating ring [23].

(f) Rotate the mating ring removal/installation tool 90 degrees.

(g) Install the mating ring [23] into the groove of the main fuel pump pad [4] and push until the thrust of the spur gear.

(h) Remove the mating ring removal/installation tool.

(i) Make sure that the alveolus is visible after you install the mating ring.

(j) Make sure that the mating ring face is completely free of damage and dirt mark

(k) Lubricate the new O-ring [22] with Assembly Fluid 1 lubricant, D50337 [CP2766].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Install the new O-ring [22] into the groove of the rotating seal housing [21].

(m) Install the rotating seal housing [21] on the housing removal and installation tool from the seal replacement tool, SPL-14666 kit.

(n) Insert the three pins of the rotating seal housing [21] into the three slots of the housing removal and installation tool.

(o) Rotate the housing removal and installation tool into position to hold rotating seal housing [21].

(p) Install the rotating seal housing [21] into the groove of the main fuel pump pad [4] until the thrust of the mating ring [23].

(q) Remove the housing removal and installation tool from the rotating seal housing [21].

(r) Make sure that the groove for the retaining ring [20] is clean.

(s) Turn the retaining ring [20] until it is installed in its groove.

(t) Push the first end of the retaining ring [20] with a spatula, STD-809.

(u) Use the spatula, STD-809 to complete the installation of the retaining ring [20].

(v) Make sure that the retaining ring [20] is installed in the correct position.

(w) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-64-01-860-004-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-64-01-860-005-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-64-01-860-006-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

J. Accessory Gearbox Rotating-Seals Installation Test

SUBTASK 72-64-01-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE MODULES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fan disk and booster spool replacement

(2) Low Pressure Compressor (LPC) shaft assembly replacement

(3) HPC rotor stage 1 blisk; stage 2 blisk; stage 3-4 blisk; stage 5 blisk; 6-10 spool; and rotating seal replacement

(4) HPT rotor stage 1 disk; HPT rotor stage 2 disk; the forward outer seal; mid seal; and the aft rotating seal replacement

(5) LPT Disks 1-5, LPT Stage 4 Stubshaft Disk and LPT Shaft (Scheduled Maintenance Requirement) Replacement

(6) HPC impeller tube support replacement

(7) Combustion nozzle case assembly replacement

(8) High pressure turbine case replacement.

B. These are scheduled maintenance tasks.

TASK 72-99-99-960-801-G00

2. Fan Disk and Booster Spool (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Fan Disk and Booster Spool Replacement

SUBTASK 72-99-99-960-004-G00

(1) Remove and replace the fan disk and booster spool.

——— END OF TASK ———

TASK 72-99-99-960-809-G00

3. Low Pressure Compressor (LPC) Shaft Assembly (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

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B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Fan Shaft Assembly Replacement

SUBTASK 72-99-99-960-001-G00

(1) Remove and replace the LPC shaft assembly.

——— END OF TASK ———

TASK 72-99-99-960-803-G00

4. HPC Rotor Stage 1 Blisk; Stage 2 Blisk; Stage 3-4 Blisk; Stage 5 Blisk; 6-10 Spool; and Rotating Seal (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. HPC Rotor Stage 1 Blisk; Stage 2 Blisk; Stage 3-4 Blisk; Stage 5 Blisk; 6-10 Spool; and Rotating Seal Replacement

SUBTASK 72-99-99-960-006-G00

(1) Remove and discard the HPC rotor stage 1 blisk.

SUBTASK 72-99-99-960-007-G00

(2) Remove and discard the HPC rotor stage 2 blisk.

SUBTASK 72-99-99-960-008-G00

(3) Remove and discard the HPC rotor stages 3-4 blisk.

SUBTASK 72-99-99-960-009-G00

(4) Remove and discard the HPC rotor stage 5 blisk.

SUBTASK 72-99-99-960-010-G00

(5) Remove and discard the HPC rotor stages 6-10 spool.

SUBTASK 72-99-99-960-011-G00

(6) Remove and discard the rotating seal.

——— END OF TASK ———

TASK 72-99-99-960-804-G00

5. HPT Rotor Stage 1 Disk; HPT Rotor Stage 2 Disk; the Forward Outer Seal; Mid Seal; and the Aft Rotating Seal (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

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A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. HPT Rotor Stage 1 Disk; HPT Rotor Stage 2 Disk; the Forward Outer Seal; Mid Seal; and the Aft Rotating Seal Replacement

SUBTASK 72-99-99-960-012-G00

(1) Remove and discard the HPT rotor stage 1 disk.

SUBTASK 72-99-99-960-013-G00

(2) Remove and discard the HPT rotor disk stage 2 disk.

SUBTASK 72-99-99-960-014-G00

(3) Remove and discard the forward outer seal.

SUBTASK 72-99-99-960-015-G00

(4) Remove and discard the mid seal.

SUBTASK 72-99-99-960-016-G00

(5) Remove and discard the aft rotating seal.

——— END OF TASK ———

TASK 72-99-99-960-811-G00

6. LPT Disks 1-5, LPT Stage 4 Stubshaft Disk and LPT Shaft (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. LPT Disks 1-5, LPT Stage 4 Stubshaft Disk and LPT Shaft (Scheduled Maintenance Requirement) Replacement

SUBTASK 72-99-99-960-003-G00

(1) Replace the Low Pressure Turbine (LPT) disks 1-5, LPT Stage 4 Stubshaft Disk and LPT Shaft as follows:

(a) Remove and replace the LPT stage disk 1.

(b) Remove and replace the LPT stage disk 2.

(c) Remove and replace the LPT stage disk 3.

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(d) Remove and replace the LPT stage disk 5.

(e) Remove and replace the LPT stage 4 stubshaft disk.

(f) Remove and replace the LPT shaft.

——— END OF TASK ———

TASK 72-99-99-960-810-G00

7. HPC Impeller Tube Support (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. HPC Impeller Tube Support Replacement

SUBTASK 72-99-99-960-002-G00

(1) Remove and discard the HPC impeller tube support.

——— END OF TASK ———

TASK 72-99-99-960-807-G00

8. Combustion Case Assembly (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Combustion Case Assembly Replacement

SUBTASK 72-99-99-960-017-G00

(1) Remove and discard the combustion case assembly.

——— END OF TASK ———

TASK 72-99-99-960-808-G00

9. High Pressure Turbine Case (Scheduled Maintenance Requirement) Replacement

NOTE: This procedure is a scheduled maintenance task.

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A. General

(1) This task is for the life limited components and modules that require replacement at scheduled intervals.

(2) This task is a scheduled maintenance task.

B. High Pressure Turbine Case Replacement

SUBTASK 72-99-99-960-005-G00

(1) Remove and discard the high pressure turbine case.

——— END OF TASK ———

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